

# Main Tab

[Click the options below for more Information](#)

## At Parts and Materials View



## At Patterns and Parts View



The Return option returns you to the main S2M CENTER screen.

### Keystroke



## Parts

This is the default opening view for the S2M CENTER.



You will notice in the figure above that the "Finished Left End" has a Red icon next to it on the Left. Any parts in the list denoted with a red icon is a part that will not be optimized. In this case the part is wider than the 48" wide Material that is specified. An icon that displays just a Router icon, such as the "Nailer" part, indicates that this Part is not tagged to Optimize. An icon that displays both the brown sheet and the yellow router, such as the "Finished Right End" part, indicates that this Part is tagged for both Optimize and CNC. See Right Click options later in this topic for more information.

To find the exact meaning of a particular Part icon simply hover over it to receive a description.

By default all items that can be sent to the Optimizer and/or your CNC machine will be sent to them. However, if you just wanted to send certain items to the Optimizer and/or CNC machine you can select them individually. To select a part simply click the icon next to the part. To select more than one simply hold the Ctrl key down while selecting. You can also select a range of parts by selecting the first part in the list you would like to send and then hold the Shift key down and select the last part. This will select all parts in between the first and last parts selected.

You can also change the Material of any part in the list. Simply click on the Material Name and a drop down will appear and allow you to change the current Part to any other Material available in the current PNC file.

Any item in the list can be modified. You can change the Quantity of parts, the parts size, you can even add an Overage using the Over column.

If you have created a Machine Set you will notice the second column in this view is the **Set** column. This column indicates which Machine set the Part will be outputted to. This is a quick method to compare. In the figure below I have three possible Machine Sets, Manual (which is simply Manual output), Saw Set, and Router Set. The column, in order, will show either a | or a .

|.. indicates the Part is going to be output with Manual but not Saw Set or Router Set.

|.| indicates the Part is going to be output with both Manual and the Saw Set.



 The red arrow in the view above is there to let you know you can add a new Part to this Part list.

**Selection Copy**

There are a couple of important features to note when working in List Views throughout S2M CENTER.

When in a List View is possible to copy text from one cell into multiple cells by grabbing the origin cells copy handle and dragging it up or down into other cells. See the example illustrated below.

Here I have several Adjustable Shelves where I would like their Descriptions to all be Adj Shelf. Normally I would need to type the Description for each or use normal Windows Copy and Paste functionality.



Using the Selection Copy functionality I can hover near the bottom right corner of the origin field and will get the copy handle as shown below.



Then simply drag to the last cell I want to copy to as shown below.



Then simply let go of the click and you will be prompted to accept the changes.



Click Yes and you will see the change.



 Cells which are automatically populated or that must be unique will not allow this functionality.

**Cell Hover Hints**

When hovering on a cell which includes a button for selection you will see an outline of the button. Clicking on that outline will activate the button and clicking outside that outline will activate the field. Notice below I am hovering over the first Material field in the illustration below.



**Right Click Options**

Right clicking on a Part in the Part list brings up the following menu:



There are six possible commands available in this Right Click menu:

- ☒ **Copy** - Creates a Copy of the selected Part.
- ☒ [Edit](#)
- ☒ **Delete** - Clicking the Delete command will delete the selected part from the Parts List.
- ☒ **OK to Optimize** - This option allows you to determine if the selected Part is OK to go to the Optimizer/Nester.
  - ☒ If you do not see this option check to make sure Show Parts Not Set for Optimization or Output is enabled in [Preferences](#).
- ☒ **OK to Output CNC** - This option allows you to determine if the selected Part is OK to go to your Secondary Machine.
  - ☒ If you do not see this option check to make sure Show Parts Not Set for Optimization or Output is enabled in [Preferences](#).
- ☒ **Save to Library** - Clicking the Save to Library command will save the selected part to the [Part Library](#) for use in future Jobs.

## Edit



There are three primary Part Edit options in the sidebar:

[Edit Shape](#)



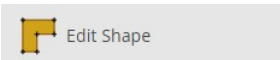
[Edge Band](#)



[Edit Operations](#)

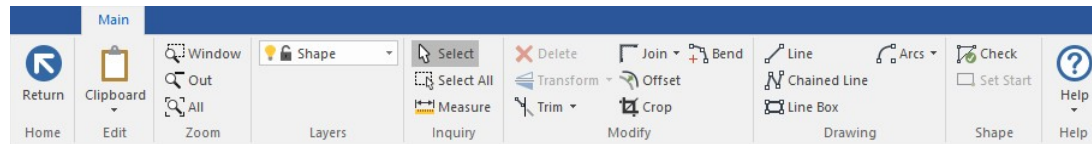


## Edit Shape

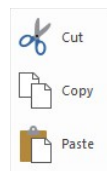


The Shape Editor consists of several tools designed to allow editing the shapes of Parts in the S2M Center.

[Click the ribbonbar options below for more information](#)



The Return option will prompt to Save your Part and return you to the main Edit View.



The Clipboard option has three tools:



The Cut option allows you to "Cut" selected objects to the Windows clipboard for pasting.

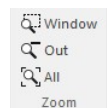


The Copy option will allow you to "Copy" selected objects to the clipboard for pasting.



The Paste allows you to "Paste" an object stored on the Windows Clipboard onto your Part.

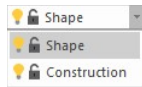
## Zoom Group



See the [Zoom Group](#) topic for more information.

## Layer Selector





There are two options in the Edit Shape Layer selector:

**Shape** - The default Layer and the one all Shapes must be drawn on.

**Construction** - This Layer allows you to draw Construction/Reference Lines to assist in creating your shape.



Depending on which Layer is active some options in the Modify section of the Ribbonbar may or may not be available.



## Select

This option will change the cursor to an arrowhead and allow selection of lines, arcs, etc. After selecting the item, you will be able to change properties by selecting the desired property off the list in the sidebar and selecting the item to apply it to. Lines can be moved, resized, etc.



## Select All

This option will **Select All** objects in the Shape Editor.



## Measure

This option allows you to **Measure** Objects.

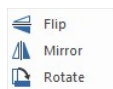


## Delete

This option will allow you to **Delete** the selected CAD Object(s).



## Transform

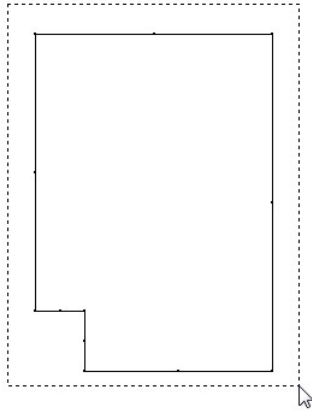


The Transform tools will only be available (not greyed out) when selecting objects which are not Constrained.

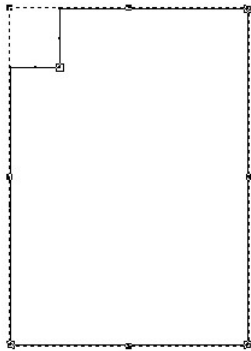


## Flip

This option will Flip the selected object. First select all of the CAD Objects you would like to Flip using the [CAD Selection Options](#).

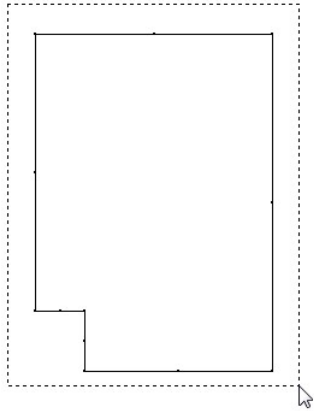


Then click the **Transform|Flip** option in the Ribbonbar.

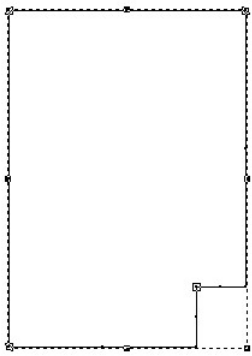


#### Mirror **Mirror**

This option will Mirror the selected object. First select all of the CAD Objects you would like to Mirror using the [CAD Selection Options](#).

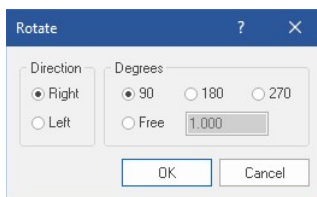


Then click the **Transform**|**Mirror** option in the Ribbonbar.

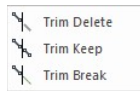


### Rotate **Rotate**

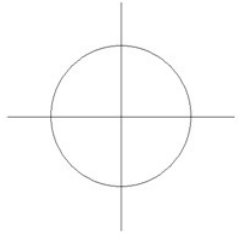
This option will Rotate the selected object. You will see the following dialog box when you select this option to choose how you want to rotate the object.



### Trim **Trim**

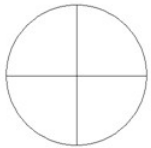


The following drawing will be used to illustrate each of the Trim functions:



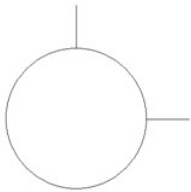
 Trim Delete **Trim Delete**

This option will **Delete** the object you click on, up to the next object it finds. When you click on each of the lines outside of the circle of the above example you will get the following:



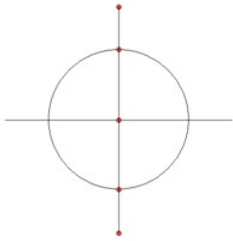
 Trim Keep **Trim Keep**

This option will **Keep** the object you click on, up to the next object it finds and will delete the rest. When you click on the top and right lines outside of the circle of the original example you will get the following:



 Trim Break **Trim Break**

This option will **Break** the object you click on, at the next object it finds. When you click on the top line outside of the circle of the original example you will get the following: (the dots represent the beginning and ending of a line. What was a single line is now “broken” into four.)



## Join



### Connect

This option allows you to Connect two lines together.

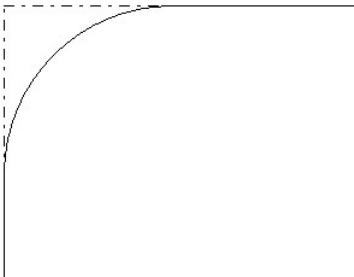
### Fillet

This option allows you to **Fillet** two lines together. Specify the Fillet Radius in the sidebar...

|         |                                |
|---------|--------------------------------|
| Fillet  |                                |
| Radius: | <input type="text" value="5"/> |

...then Left Click the first line then Left Click the second to connect them.

In the following example, two lines were drawn at a 90 degree angle to one another and the radius in the field on the left of the screen was set to 5". The dashed lines represent where the lines used to extend.



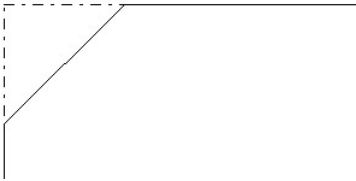
### Chamfer

This option allows you to **Chamfer** two lines together. Specify the Chamfer Length in the sidebar...

|         |                                 |
|---------|---------------------------------|
| Chamfer |                                 |
| Length: | <input type="text" value="10"/> |

...then Left Click the first line then Left Click the second to connect them.

The example below shows a 5" Chamfer, the dashed lines represent where the lines used to extend.



## Offset

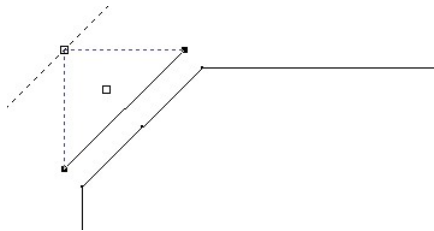
This option allows you to **Offset** a copy of the selected Object a specific Dimension. Simply click the Offset option in the Ribbonbar then select (Left Click) the CAD Object you would like to Offset.

You can enter the Offset Dimension in the sidebar...

|         |                                     |
|---------|-------------------------------------|
| Offset  |                                     |
| Offset: | <input type="text" value="9 5/32"/> |

...or drag your cursor the amount (distance) you would like to Offset.

In the following example, the chamfer line has been offset by  $\frac{3}{4}$ " and is being offset again by  $2\frac{1}{2}$ " as noted in the Offset field on the left. The dashed line parallel to the chamfer is where the offset will land when the left mouse button is clicked.

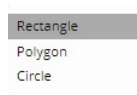


## Crop

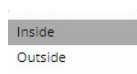
There are 2 main Crop options in the sidebar once the Crop Tool is selected:

|            |                                        |
|------------|----------------------------------------|
| Crop       |                                        |
| Start X:   | <input type="text" value="-9 3/4"/>    |
| Start Y:   | <input type="text" value="27 1/32"/>   |
| Crop Type: | <input type="text" value="Rectangle"/> |
| Crop Area: | <input type="text" value="Inside"/>    |

**Crop Type** - You have three Crop Types available; Rectangle, Circle, and Polygon that determine the shape of the area being cropped.



**Crop Area** - The Inside option will get rid of what is Outside the Boundary and keep what is Inside. The Outside option specifies that you want to keep what is outside the Crop Boundary and get rid of the objects inside the Boundary.



The Inside option can be a bit aggressive as it will remove all Objects in the current view if it is not inside the boundary. To keep this option from being so aggressive you can limit the Objects that are affected by a Crop by selecting the Objects first.



Remember you can always click Undo option in the Quick Access Toolbar or use the  +  Undo shortcut to Undo your changes.



Remember you can select multiple Objects by holding the  key on the keyboard down as you click the Objects.



## Bend

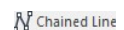
This tool allows you to select a Line and turn it into an Arc by dragging it to **Bend** it.



## Line

This tool allows the creation of **Lines** to define the shape of your Part. After selecting and placing a line, the line properties window will display in the sidebar. Clicking on the items listed in this window will allow modification of that item.

|          |                                       |
|----------|---------------------------------------|
| Line     |                                       |
| Start X: | <input type="text" value="-30 9/16"/> |
| Start Y: | <input type="text" value="29 29/32"/> |
| End X:   | <input type="text" value="-9 5/16"/>  |
| End Y:   | <input type="text" value="29 29/32"/> |
| Length:  | <input type="text" value="21 9/32"/>  |
| Angle:   | <input type="text" value="0.00"/>     |



## Chained Line

This tool allows the continuous creation of connected Lines.

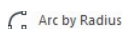


## Line Box

This tool allows the creation of a Rectangle.



## Arcs

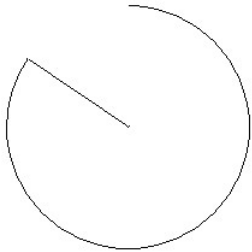


## Arc by Radius

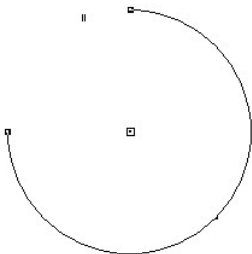
This option allows definition of a point where a radius begins, the radius, and the degrees of radius. The first mouse click marks the center of the radius, pull the radius that you want and click again.



A circle will appear with a radius line within it. Move the cursor and the circle opening becomes an arc as you do so.



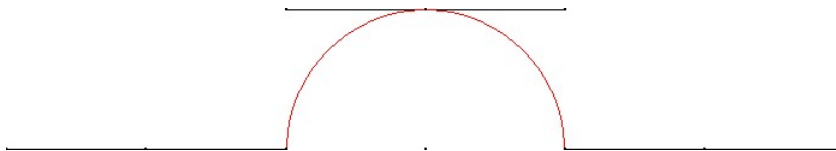
When you reach the correct position, a third click will set the arc at the selected point.



### 3 Points Arc

This option lets you place two points to define the opening of the arc, and the third point pulls the arc increasing the radius. A 3 Points Arc is very handy when you have three known points to Snap to.

In the example below I want to create the Arc shown in red.

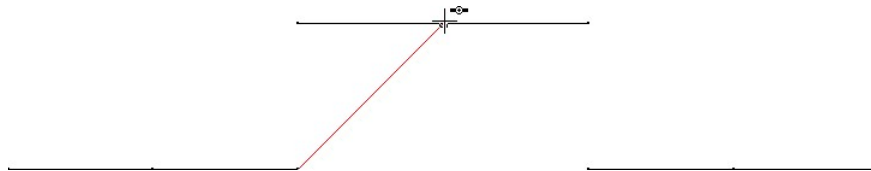


To begin I select the **Arcs|3 Points Arc** option from the Ribbonbar then Left Click the End Point (With Snap To Points on) of the first line.

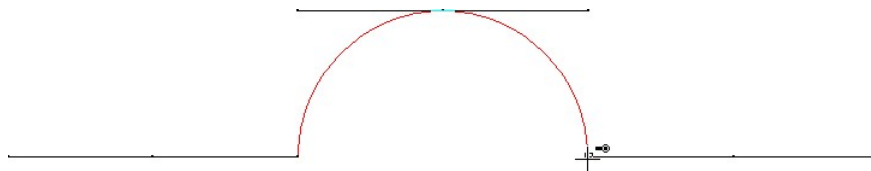




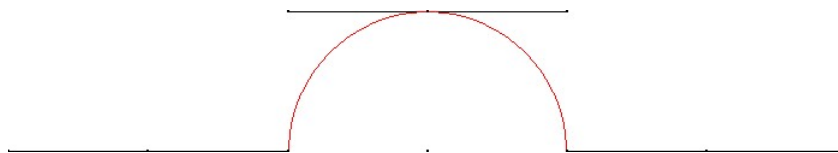
Next I Left Click the Mid Point of the second line.



And the last and third point I Left Click the End point of the third line.



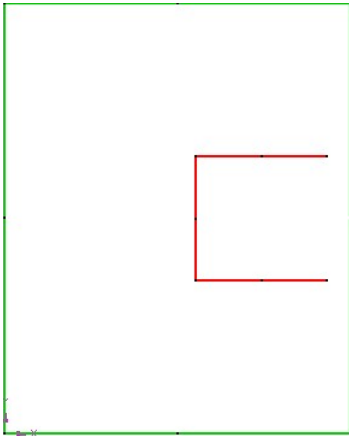
And now I have the Arc.



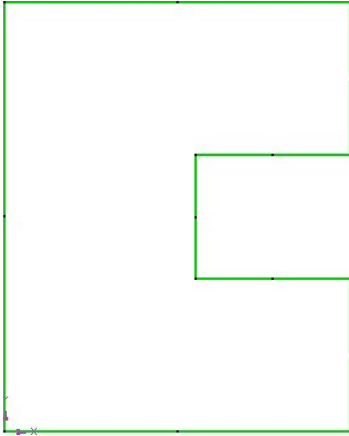
### Check

This option will check your Assembly or Part Shape to make sure it is valid. Objects that are valid will be Green and objects which are not will be Red. Usually a Shape is judged as invalid if it is not a complete enclosed shape. See illustrations below for examples.

### Invalid

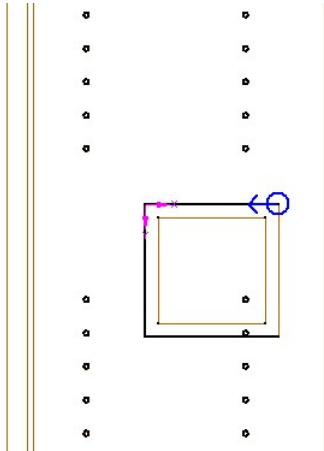


**Valid**

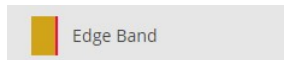


**Set Start** ☐ Set Start

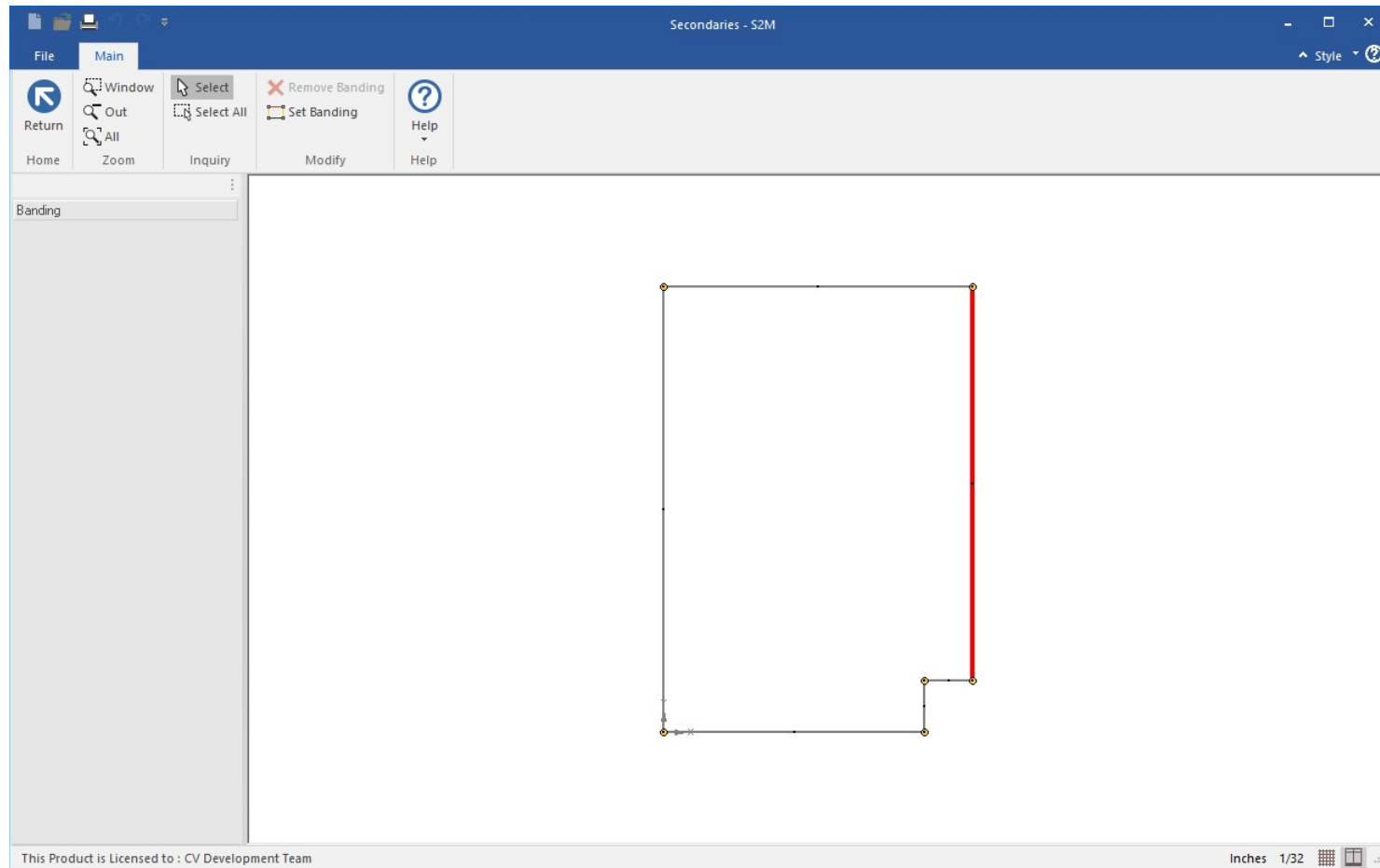
Available only with open Route Paths this option allows you to set the Start Point for the Route. The start point is indicated visually by a blue circle with an arrow pointing in its direction as indicated below.



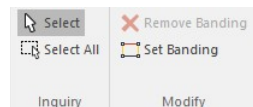
## Edgeband



The Edgeband Editor allows you to assign Edgebanding to your Parts. You must have Edgeband Materials and have setup [Edgeband Indicators](#) in the [Material Catalog](#) in order to fully utilize this facility.



The Edge Banding option allows you define which edge of a Part gets edge banded with which [edgeband indicator](#). Once the Edge Band option is active you will be placed in Edge Band mode with the following 4 options in the ribbonbar.



- **Select** - When entering Edgeband mode this tool is selected by default. Simply click on the edge you would like to apply the Edgeband to and then select its type in the sidebar.





You can use Windows  + Click to select multiple edges.

-  **Select All** - To quickly change all Edges to a specific Edgeband select this option then select their type in the sidebar.
-  **Remove Banding** - Will remove Edgebanding from the selected edge(s).
-  **Set Banding** - Will expose the Edgeband type selector in the sidebar without having to select an edge first.



It is important to note that the default values in the sidebar options are NOT representative of the current Edge Banding on the selected Part. To use this function simply select which Edgebanding type you would like.

Once the Banding options are chosen simply select any of the Part Edges you would like these Edgeband properties assigned to. In the image below the Front Edge of the End has been selected.



Return

Note that the front edge is now red which indicates an Edgeband has been applied. Once you are through defining the Edges simply click the Return option to save the new Edge Properties.

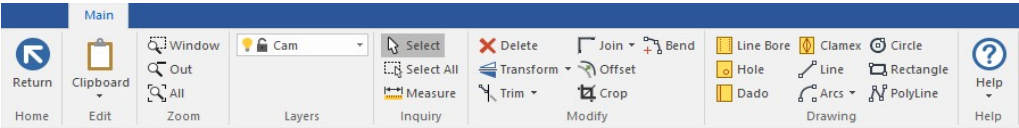
## Edit Operations



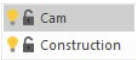
Edit Operations

The Operations Editor consists of several tools designed to allow editing and adding of Operations to Parts in the S2M Center.

[Click the ribbonbar options below for more information](#)



Layer Selector  Cam



There are two options in the Edit Shape Layer selector:

- CAM** - The default Layer and the one all Operations must be drawn on.
- Construction** - This Layer allows you to draw Construction/Reference Lines to assist in creating/editing your Operations.

 Depending on which Layer is active some options in the Modify section of the ribbonbar may or may not be available.

See the [Edit Shape](#) topic for more information on the Zoom, Inquiry and Modify groups.

 Line Bore **Line Bore**

When selecting the Line Bore option you will left click to establish the start point of your row of Line Bores then drag and left click to establish the end point.

When selecting a Line Bore operation there are thirteen options in the sidebar Properties area:

|              |                                              |
|--------------|----------------------------------------------|
| Line Bore    |                                              |
| Pos X:       | <input type="text" value="7 3/16"/>          |
| Pos Y:       | <input type="text" value="9 13/32"/>         |
| Pos Z:       | <input type="text" value="0"/>               |
| End X:       | <input type="text" value="7 3/16"/>          |
| End Y:       | <input type="text" value="27 1/32"/>         |
| Length:      | <input type="text" value="17 5/8"/>          |
| Diameter:    | <input type="text" value="1/2"/>             |
| Depth:       | <input type="text" value="1/2"/>             |
| Angle:       | <input type="text" value="0.00"/>            |
| Spacing:     | <input type="text" value="1 1/4"/>           |
| Repeat:      | <input type="text" value="15"/>              |
| Tool:        | <input type="text" value="Auto Select"/>     |
| Description: | <input type="text" value="Front Line Bore"/> |

- **Pos X** - Establishes the X Position of the start point for the Line Bore.
- **Pos Y** - Establishes the Y Position of the start point for the Line Bore.
- **Pos Z** - Establishes the Z Position for the Line Bore.
- **End X** - Establishes the X Position of the end point for the Line Bore.
- **End Y** - Establishes the Y Position of the end point for the Line Bore.
- **Length** - Establishes the Length of the Line Bore.
- **Diameter** - Establishes the Line Bore Diameter.

- **Depth** - Establishes the Line Bore Depth.
- **Angle** - Establishes the Line Bore angle. Rotation occurs around the start point.
- **Spacing** - Establishes the Line Bore spacing.
- **Repeat** - Establishes the number of Line Bores.
- **Tool** - Allows you to assign a specific tool to the Line Bore.
- **Description** - The Description of this Line Bore.

 Hole **Hole**

When selecting the Hole option you will left click to establish the center-point of the Hole.

When selecting a Hole operation there are seven options in the sidebar Properties area:

|              |                                          |
|--------------|------------------------------------------|
| Hole         |                                          |
| Pos X:       | <input type="text" value="17"/>          |
| Pos Y:       | <input type="text" value="27 9/32"/>     |
| Pos Z:       | <input type="text" value="0"/>           |
| Diameter:    | <input type="text" value="1/2"/>         |
| Depth:       | <input type="text" value="1/2"/>         |
| Tool:        | <input type="text" value="Auto Select"/> |
| Description: | <input type="text" value="Front Hole"/>  |

- **Pos X** - Establishes the X Position of the start point for the Hole.
- **Pos Y** - Establishes the Y Position of the start point for the Hole.
- **Pos Z** - Establishes the Z Position for the Hole.
- **Diameter** - Establishes the Hole Diameter.
- **Depth** - Establishes the Hole Depth.
- **Tool** - Allows you to assign a specific tool to the Hole.
- **Description** - The Description of this Hole.

 Dado **Dado**

When selecting the Dado option you will left click to establish the start point of your Dado then drag and left click to establish the end point.

When selecting a Dado operation there are eleven options in the sidebar Properties area:

| Dado         |             |
|--------------|-------------|
| Pos X:       | 2 25/32     |
| Pos Y:       | 6 7/32      |
| Pos Z:       | 0           |
| End X:       | 21 1/32     |
| End Y:       | 6 7/32      |
| Width:       | 1/2         |
| Length:      | 18 1/4      |
| Depth:       | 1/2         |
| Angle:       | -90.00      |
| Tool:        | Auto Select |
| Description: | Front Dado  |

- **Pos X** - Establishes the X Position of the start point for the Dado.
- **Pos Y** - Establishes the Y Position of the start point for the Dado.
- **Pos Z** - Establishes the Z Position for the Dado.
- **End X** - Establishes the X Position of the end point for the Dado.
- **End Y** - Establishes the Y Position of the end point for the Dado.
- **Width** - Establishes the Width of the Dado.
- **Length** - Establishes the Length of the Dado.
- **Depth** - Establishes the Dado Depth.
- **Angle** - Establishes the Dado angle. Rotation occurs around the start point.
- **Tool** - Allows you to assign a specific tool to the Dado.
- **Description** - The Description of this Dado.

## Clamex

Use this option to place machining for Lamello Clamex™ connectors.

When selecting a Clamex operation there are ten options in the sidebar Properties area:

| Clamex       |                    |
|--------------|--------------------|
| Pos X:       | 19 23/32           |
| Pos Y:       | 19 5/16            |
| Pos Z:       | 0                  |
| Width:       | 3/8                |
| Length:      | 2 13/16            |
| Depth:       | 19/32              |
| Angle:       | 0.00               |
| Type:        | P15                |
| Tool:        | Auto Select        |
| Description: | Front Primary Clam |

- **Pos X** - Establishes the X Position of the Clamex.
- **Pos Y** - Establishes the Y Position of the Clamex.



- **Pos Z** - Establishes the Z Position of the Clamex.
- **Width** - Displays the Width of the Clamex. The Type selected determines its Width.
- **Length** - Displays the Length of the Clamex. The Type selected determines its Length.
- **Depth** - Displays the Clamex Depth. The Type selected determines its Depth.
- **Type** - Establishes the Type of Clamex to machine for. There are four types available:
  - P10
  - P15
  - T14
  - 15/10
  - 14/10
  - P14
  - D18
  - S20
- **Tool** - Allows you to assign a specific tool to the Clamex.
- **Description** - The Description of this Clamex.

## Line

Using the Line option allows you to draw one Line at a time by selecting the start point then the end point. See the Line option in the [Edit Shape](#) topic for more information.

When selecting a Line there are ten options in the sidebar Properties area:

| Line         |             |
|--------------|-------------|
| Pos X:       | 11 3/4      |
| Pos Y:       | 11 3/16     |
| Pos Z:       | 0           |
| End X:       | 11 3/4      |
| End Y:       | 30 1/8      |
| Length:      | 18 15/16    |
| Depth:       | 1/2         |
| Angle:       | 0.00        |
| Cut:         | Center      |
| Tool:        | Auto Select |
| Description: | Front Line  |

- **Pos X** - Establishes the X Position of the start point for the Line.
- **Pos Y** - Establishes the Y Position of the start point for the Line.
- **Pos Z** - Establishes the Z Position for the Line.
- **End X** - Allows you to set the Ending X position for the Line.

- **End Y** - Allows you to set the Ending Y position for the Line.
- **Length** - Establishes the Length of the Line.
- **Depth** - Establishes the Line Depth.
- **Angle** - Establishes the Line angle. Rotation occurs around the start point.
- **Cut** - There are three options for Cut:
  - **Centerline** - Cuts along the Centerline of the Line.
  - **Inside** - Cuts along the Inside of the Line.
  - **Outside** - Cuts along the Inside of the Line.
- **Tool** - Allows you to assign a specific tool to the Line.
- **Description** - The Description of this Line.

 Arcs ▾ **Arcs**



Using the Arc option allows you to draw Arcs. See the Arcs option in the [Edit Shape](#) topic for more information.

When selecting an Arc there are fourteen possible options in the sidebar Properties area:

|              |                                          |
|--------------|------------------------------------------|
| Arc          |                                          |
| Pos X:       | <input type="text" value="-17 21/32"/>   |
| Pos Y:       | <input type="text" value="16 31/32"/>    |
| Pos Z:       | <input type="text" value="0"/>           |
| Radius:      | <input type="text" value="7 9/32"/>      |
| Start X:     | <input type="text" value="-10 3/8"/>     |
| Start Y:     | <input type="text" value="17 7/16"/>     |
| End X:       | <input type="text" value="-23 5/32"/>    |
| End Y:       | <input type="text" value="12 7/32"/>     |
| Begin Angle: | <input type="text" value="3.65"/>        |
| End Angle:   | <input type="text" value="-139.18"/>     |
| Depth:       | <input type="text" value="1/2"/>         |
| Cut:         | <input type="text" value="Center"/>      |
| Tool:        | <input type="text" value="Auto Select"/> |
| Description: | <input type="text" value="Front Arc"/>   |

- **Pos X** - Establishes the X Position of the start point for the Arc.
- **Pos Y** - Establishes the Y Position of the start point for the Arc.
- **Pos Z** - Establishes the Z Position for the Arc.
- **Radius** - Allows you to set the Radius of the Arc.
- **Start X** - Allows you to set the Starting X position of the Arc.
- **Start Y** - Allows you to set the Starting Y position of the Arc.
- **End X** - Allows you to set the Ending X position of the Arc.

- **End Y** - Allows you to set the Ending Y position of the Arc.
- **Begin Angle** - Allows you to set the Begin Angle for the Arc.
- **End Angle** - Allows you to set the End Angle for the Arc.
- **Depth** - Establishes the Arc Depth.
- **Cut** - There are three options for Cut:
  - **Centerline** - Cuts along the Centerline of the Arc.
  - **Inside** - Cuts along the Inside of the Arc.
  - **Outside** - Cuts along the Inside of the Arc.
- **Tool** - Allows you to assign a specific tool to the Arc.
- **Description** - The Description of this Arc.

 Circle **Circle**

This option allows you to create Circles. Click in the place where you want the center of the circle and pull the mouse to one side or the other and a Circle will develop. The area in the sidebar allows modifying the radius by numerically entering a value.

When selecting an Circle there are eight options in the sidebar Properties area:

|              |                                           |
|--------------|-------------------------------------------|
| Circle       |                                           |
| Pos X:       | <input type="text" value="6 7/8"/>        |
| Pos Y:       | <input type="text" value="31 19/32"/>     |
| Pos Z:       | <input type="text" value="0"/>            |
| Diameter:    | <input type="text" value="3 15/32"/>      |
| Depth:       | <input type="text" value="1/2"/>          |
| Cut:         | <input type="text" value="Center"/>       |
| Tool:        | <input type="text" value="Auto Select"/>  |
| Description: | <input type="text" value="Front Circle"/> |

- **Pos X** - Establishes the X Position of the center-point for the Circle.
- **Pos Y** - Establishes the Y Position of the center-point for the Circle.
- **Pos Z** - Establishes the Z Position for the Circle.
- **Diameter** - Establishes the Diameter of the Circle.
- **Depth** - Establishes Circle Depth.
- **Cut** - There are three options for Cut:
- **Tool** - Allows you to assign a specific tool to the Circle.
- **Description** - The Description of this Circle.

 Rectangle **Rectangle**

This tool provides for creating a Rectangle (or square) by clicking the point where it is to start and clicking on the point where it is to finish. It can thereafter be adjusted by using the Select tool. The area in the sidebar is similar to the one for the Line tool.



Pressing



+ Left Mouse Drag will force the Rectangle to be a Square.

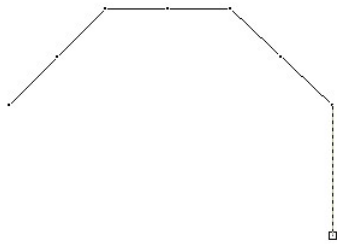
When selecting an Rectangle there are eleven options in the sidebar Properties area:

| Route        |             |
|--------------|-------------|
| Pos X:       | 14 19/32    |
| Pos Y:       | 9 5/16      |
| Pos Z:       | 0           |
| Width:       | 6 21/32     |
| Length:      | 5 1/4       |
| Depth:       | 1/2         |
| Angle:       | 0.00        |
| Type:        | Path        |
| Cut:         | Center      |
| Tool:        | Auto Select |
| Description: | Front Route |

- **Pos X** - Establishes the X Position of the start point for the Rectangle.
- **Pos Y** - Establishes the Y Position of the start point for the Rectangle.
- **Pos Z** - Establishes the Z Position for the Rectangle.
- **Width** - Establishes the Width of the Rectangle.
- **Length** - Establishes the Length of the Rectangle.
- **Depth** - Establishes the Rectangle Depth.
- **Angle** - Establishes the Rectangle angle. Rotation occurs around the start point.
- **Type** - There are two choices for Type:
  - **Outline** - Will cut an Outline of the Rectangle.
  - **Pocket** - Will Pocket out the Rectangle.
- **Cut** - There are three options for Cut:
  - **Centerline** - Cuts along the Centerline of the Rectangle.
  - **Inside** - Cuts along the Inside of the Rectangle.
  - **Outside** - Cuts along the Inside of the Rectangle.
- **Tool** - Allows you to assign a specific tool to the Rectangle.
- **Description** - The Description of this Rectangle.

## Polyline

This option creates a joined line with multiple segments. Left click to start, and at intermediate points. Right click to set the Polyline. The example below shows a polyline that has three line segments that have been set and a fourth being drawn.



When selecting a Polyline there are ten options in the sidebar Properties area:

| Route        |             |
|--------------|-------------|
| Pos X:       | -23 15/32   |
| Pos Y:       | 11 1/32     |
| Pos Z:       | 0           |
| Width:       | 20 23/32    |
| Length:      | 16 25/32    |
| Depth:       | 1/2         |
| Angle:       | 0.00        |
| Cut:         | Center      |
| Tool:        | Auto Select |
| Description: | Front Route |

- **Pos X** - Establishes the X Position of the start point for the Polyline.
- **Pos Y** - Establishes the Y Position of the start point for the Polyline.
- **Pos Z** - Establishes the Z Position for the Polyline.
- **Width** - Establishes the Width of the Polyline.
- **Length** - Establishes the Length of the Polyline.
- **Depth** - Establishes the Polyline Depth.
- **Angle** - Establishes the Polyline angle. Rotation occurs around the start point.
- **Cut** - There are three options for Cut:
  - **Centerline** - Cuts along the Centerline of the Polyline.
  - **Inside** - Cuts along the Inside of the Polyline.
- **Tool** - Allows you to assign a specific tool to the Polyline.
- **Description** - The Description of this Polyline.

## Edit Shape Selection Options

There are several methods to select Edit Shape objects as well as several functions you can perform on Edit Shape Objects once they are selected.

### Left Click Select

Simply Left Click on a single CAD Object to select it.

## Left Click Select +



To select multiple CAD Objects simply hold the



key while Left Clicking.



Left clicking +



on a single CAD entity and selecting a CAD node for it will also put that item into Scale/Stretch/Move mode just like using the multi-select options.

## Window Selection

You can also Left Click the screen and drag a Window to enclose all of the CAD Objects you want to Select.

## Selecting Single Objects



Left Clicking +



on a single entity and selecting a node for it will also put that item into Scale/Stretch/Move mode just like using the multi-select options. See the Selecting a Group of Objects section below for more information on Scale/Stretch/Move.

## Selecting a Line

|          |          |
|----------|----------|
| Line     |          |
| Start X: | -30 9/16 |
| Start Y: | 29 29/32 |
| End X:   | -9 5/16  |
| End Y:   | 29 29/32 |
| Length:  | 21 9/32  |
| Angle:   | 0.00     |

There are six options when selecting a [Line](#).

- **Start X** - Allows you to set the Starting X position of the selected CAD Object.
- **Start Y** - Allows you to set the Starting Y position of the selected CAD Object.
- **End X** - Allows you to set the Ending X position of the selected CAD Object.
- **End Y** - Allows you to set the Ending Y position of the selected CAD Object.
- **Length** - Allows you to set the Length of the selected CAD Object.
- **Angle** - Allows you to set the Angle of the selected CAD Object.

## Selecting an Arc (Arc by Radius)

| Arc          |                                    |
|--------------|------------------------------------|
| Start X:     | -24 1/32                           |
| Start Y:     | 22 3/16                            |
| End X:       | -13 15/32                          |
| End Y:       | 15 7/16                            |
| Arc Point X: | -24 5/16                           |
| Arc Point Y: | 10 3/32                            |
| Center X:    | -20 17/32 <input type="checkbox"/> |
| Center Y:    | 16 1/16 <input type="checkbox"/>   |
| Begin Angle: | 119.97                             |
| End Angle:   | -5.02                              |
| Radius:      | 7 1/16                             |

There are five options when selecting an Arc (Arc by Radius).

- **Center X** - Sets the Center X position of the selected CAD Object.
- **Center Y** - Sets the Center Y position of the selected CAD Object.
- **Begin Angle** - Allows you to set the Begin Angle for the selected CAD Object.
- **End Angle** - Allows you to set the End Angle for the selected CAD Object.
- **Radius** - Allows you to set the Radius of the selected CAD Object.

### Selecting an Arc (3 Point Arc)

| Arc          |           |
|--------------|-----------|
| Start X:     | -7 1/8    |
| Start Y:     | 7 1/8     |
| End X:       | -28 3/16  |
| End Y:       | 30 21/32  |
| Arc Point X: | -12 13/32 |
| Arc Point Y: | 29 31/32  |
| Center X:    | -20 15/16 |
| Center Y:    | 15 31/32  |
| Begin Angle: | -32.65    |
| End Angle:   | 116.24    |
| Radius:      | 16 3/8    |

There are eleven options when selecting an Arc (3 Point Arc).

- **Start X** - [Previously covered](#)
- **Start Y** - [Previously covered](#)
- **End X** - [Previously covered](#)
- **End Y** - [Previously covered](#)
- **Arc Point X** - Allows you set the Arc Point X position for the selected Arc.
- **Arc Point Y** - Allows you set the Arc Point Y position for the selected Arc.
- **Center X** - [Previously covered](#)
- **Center Y** - [Previously covered](#)

- **Begin Angle** - [Previously covered](#)
- **End Angle** - [Previously covered](#)
- **Radius** - [Previously covered](#)

Selecting a Group of Objects

|           |                                     |
|-----------|-------------------------------------|
| Group     |                                     |
| Left:     | <input type="text" value="0"/>      |
| Right:    | <input type="text" value="23 1/4"/> |
| Top:      | <input type="text" value="34 1/2"/> |
| Bottom:   | <input type="text" value="0"/>      |
| Center X: | <input type="text" value="11 5/8"/> |
| Center Y: | <input type="text" value="17 1/4"/> |
| Width:    | <input type="text" value="23 1/4"/> |
| Height:   | <input type="text" value="34 1/2"/> |

There are eight options when selecting a Group of CAD Objects:

- **Left** - Sets the Left position of the CAD Object.
- **Right** - Sets the Right position of the selected CAD Object.
- **Top** - Sets the Top position of the selected CAD Object.
- **Bottom** - Sets the Top position of the selected CAD Object.
- **Center X** - [Previously covered](#)
- **Center Y** - [Previously covered](#)
- **Width** - Sets the Width of the selected CAD Object.
- **Height** - Sets the Height of the selected CAD Object.

There are several options when selecting CAD Objects you can initiate by simply clicking points on screen around the selected CAD Object(s).

Scale

Once a Group of CAD Objects is selected simply left click one of the corner points to Scale the selected Objects. You either eye-ball the Scale or you can use the Tab key to change the values in the sidebar for the selected objects.

|                    |                                       |
|--------------------|---------------------------------------|
| Group              |                                       |
| Right:             | <input type="text" value="26 13/32"/> |
| Bottom:            | <input type="text" value="-4 11/16"/> |
| Width:             | <input type="text" value="26 13/32"/> |
| Height:            | <input type="text" value="39 3/16"/>  |
| Hor. Scale:        | <input type="text" value="1.136"/>    |
| Ver. Scale:        | <input type="text" value="1.136"/>    |
| Hor. Size:         | <input type="text" value="113.6 %"/>  |
| Ver. Size:         | <input type="text" value="113.6 %"/>  |
| Keep Aspect Ratio: | <input type="text" value="Yes"/>      |

Stretch

Once a Group of CAD Objects is selected simply left click one of the mid-points to Stretch the selected Objects. You can either eye-ball the Stretch or you can use the Tab key to change the values in the sidebar for the selected objects.



|             |         |
|-------------|---------|
| Group       |         |
| Right:      | 32 9/16 |
| Width:      | 32 9/16 |
| Hor. Scale: | 1.400   |
| Hor. Size:  | 140.0 % |



You can also Stretch to a [Snap point](#). For example you can pick a single line and grab and end-point and stretch/extend that line to any other Snap point.

## Move

Once a Group of CAD Objects is selected simply left click any point within the object to use that as the Move point, this will allow you to Move an Object or Group of Objects from one known (snap point) position to another (snap point position). You can also Move the selected Group of Objects by using the Tab key to change values in the sidebar.

|        |          |
|--------|----------|
| Group  |          |
| Pos X: | 4 3/32   |
| Pos Y: | 29 29/32 |



Since Scale and Stretch are also available by clicking the corners of selected CAD make sure the proper cursor prompt is used.

↖ Scale/Stretch

+↖ Move from End Point

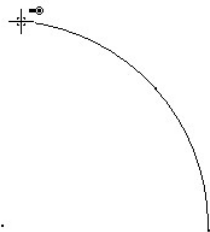


When using Copy and Paste on CAD Objects, upon Pasting the Objects, they remain selected to allow you to use the three (Scale, Stretch and/or Move) options described above.

## Object Snapping

The S2M CENTER provides a special kind of cursor snap in addition to the typical grid snap. This allows the cursor to automatically snap to different hot-points on or adjacent to another object; for example, end-points, mid-points, center-points, and intersections. These locations are commonly referenced when creating or editing geometry. For most functions, the cursor will also snap to the origin of the object.

For example, you may want to create a line that is connected to the end of an arc. Start creating the line by locating the endpoint of the arc. When located, a round [hot-point marker](#) appears on the arc endpoint.



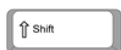
Click the hot-point marker to start the line there. The start of the line snaps precisely to the end of the arc, connecting these objects end-to-end.

You can choose what types of positions and conditions will automatically be located by enabling or disabling options in [Preferences | Tools](#).

## General Snapping Shortcut Keys



Ignores snap to Grid and Points, and leaves the Smart-Lines angle snap on.



Activates the "full grid snap" mode, forcing snap to the Grid over a position just aligned with a Point.

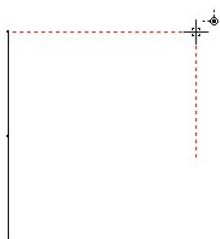


Completely ignores snap, including Smart-Lines.

## Dynamic Snapping

The S2M CENTER also has the ability to Snap Dynamically (also known as Assumptive Snapping). When you are in line with a Snap Point on another Object within the Drawing space the S2M CENTER will project that position to your location. When this projection along with the associated [hot-point marker](#) is displayed you can be assured that the Object will be placed in-line with that Objects Snap Point.

For Example you would like to draw two line Parallel to each other. After drawing the first line simply move your cursor in-line with the first lines Start Point and you will see a Snap Line projection assuring you they are in-line.



## Hot-Point Markers

When the cursor moves over a valid snap location, the located position displays with a hot-point marker and a special cursor appears to identify the type of location.



The system will not display a hot-point marker for a particular location type if it is disabled in [Preferences | Tools](#).

The list below identifies these special cursors:



Cursor is snapping to one of the extremities (End Point) of an object.



Cursor is snapping to the middle position (Mid Point) of an object.



Cursor is snapping to the center of an arc or circle.



Cursor is snapping to the intersection of two objects.



Cursor has one coordinate aligned with any of the snap points.



Editing CAD line is extending along the original angle.



Editing CAD line is snapping to one of the smart line angles.



Editing CAD line has a locked angle.



Cursor is snapping to any of the points in the editing shape.



Cursor is snapping to the grid.



Cursor is snapping to the origin (0,0 position) in the view.

## Materials

This changes the View to a Material List. You can modify the Material, Sheet Size, Cost, and Quantity of materials in this list.



The red arrow in the view above is there to let you know you can add a new Material from the [S2M Material Catalog](#), or a copy of a Material within the current Job with an alternate size, to this Material list.

### Right Click Options

Right clicking on a Material in the Material list brings up the following menu:



There are 4 commands available in this Right Click menu:

1.  **Delete** - Clicking the Delete command will delete the selected material from the Materials List.
2.  **Remove Tool Override** - If you have set an alternate Tool as the Outline Tool, via Material Properties below, for this Material there will be a small green tool " " on the Materials icon. This lets you know that a Tool override is present on the Material. Clicking this option will remove that override allowing the S2M CENTER to select the proper Outline Tool based on your Tool Catalog settings.
3.  **Remove Override** - If you have modified the Optimization/Nesting Properties for a specific Material either from this menu or from the [Patterns Area sidebar](#) there will be a check mark " " on the Materials icon. This lets you know that an override is present on the Material. Clicking this command will remove that override allowing the Material to be Optimized/Nested utilizing the Machines default properties.
4.  **[Properties](#)** - Clicking the Properties command will bring up the optimization/nesting properties for the selected material. This works like an override. The Properties you change here will be for the selected Material only. This will allow you to re-optimize with different results and not adversely affect the other materials for this PNC file.

## Material Properties

Saw as Primary

Parameters : 3/4 Solid Ash

Patterns

Restrictions

Trim

Offcuts

Tools

First Cut

☐ Rip

☐ Cross Cut

☒ Either

Strip Sort

☒ Smallest to Largest

☐ Largest to Smallest

Parts in Strip Sort

☒ Smallest to Largest

☐ Largest to Smallest

Priority

☐ Yield

☒ Labor

☐ Eliminate Z-Cuts

☐ Eliminate T-Cuts

Quality of Optimization

Offcut / Waste Order

☐ First

☒ Last

Save

OK

Cancel

**Router as Primary**

Parameters : 3/4 Solid Ash

Patterns

Trim

Offcuts

Tools

Nest Type

Block

Quality of Optimization

☐ Restrict rotation to 90 degrees

Pack Direction

☒ Horizontal

☐ Vertical

☐ Either

☐ Common Line Cutting

Maximum Retrace

0

☐ Retrace by Small Parts

Save

OK

Cancel

There are 5 possible tabs available in Material Properties. Most of theses are available in the Optimizer Parameters sidebar option on the Parts View. However, when utilizing the Options here they are applied to the selected Material avoiding effecting the Optimization/Nesting Parameters for other Materials in the Job.

[Patterns](#)

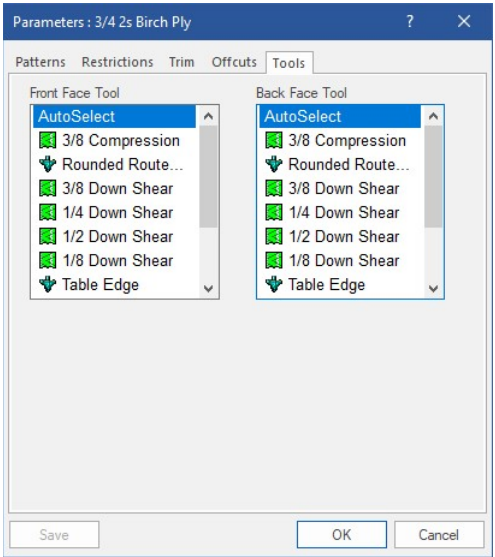
[Restrictions](#)

[Trim](#)

[Offcuts](#)

 The Offcuts Tab in Material Properties is the same as in Optimizer Properties with one exception. Offcuts set differently here will hold with the Material. This is handy when utilizing a high cost Material. You may want to set the Offcut Size (piece you save) to a smaller dimension for your high cost Materials.

**Tools**



The Tools Tab allows you specify specific Tools for the Front and Back Face of the selected Material.

 A Tool Selection override is married to the currently selected Primary Machine. So, if you pick a different Primary Machine the Material will use the Tool Override that has been assigned to it.

 A selection other than Auto Select made here will stick to your Material for all future Jobs utilizing the modified Material with the current Primary Machine.

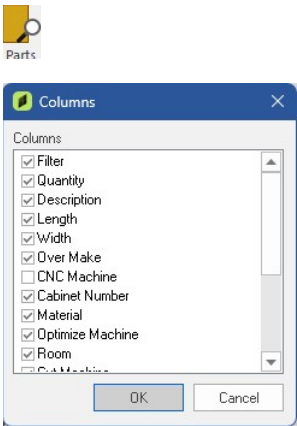


**Set Columns**

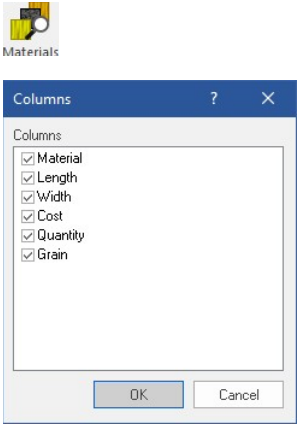
The Set Columns option is available in the Part and Material List area as well as the Pattern View.

**Set Columns - Parts and Material List Area**

The Set Columns option allows you to change which Columns are displayed in the Part List area.

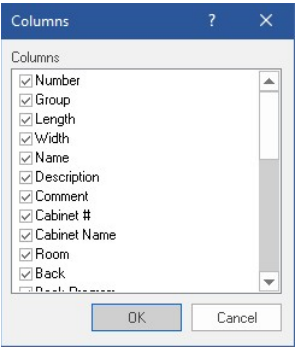


The Show/Hide Columns option also allows you to change which Columns are displayed in the Material List area.



The Show/Hide Columns option also allows you to change which Columns are displayed in the Pattern View.

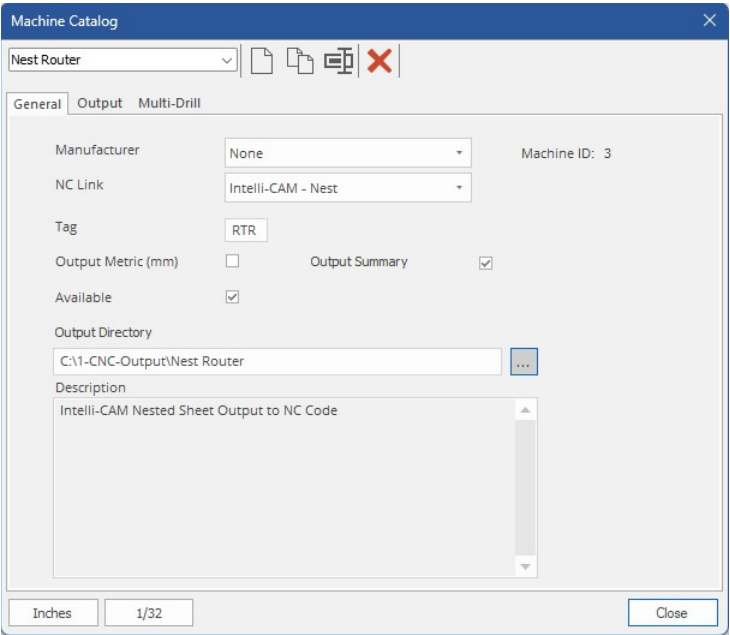




In addition to controlling which items display you can also control the order in which the columns appear. Simply click and drag the item you want to move into its new position.

## Machine Catalog

You must access the Machine Catalog in order to setup your machines. The Machine Catalog has detailed graphics designed to make the setup process intuitive.



## Machine Tab

Following are the tools in the Machine Catalog toolbar:

Nest Router    

Nest Router  **Setup Name** - The name of the currently selected Machine (Setup).

 **Create New Machine** - Allows you to Create a New Machine (Setup). There are eight possible Machine Types depending on your level and which add-ons you may have for the S2M CENTER.

Add New Machine

Machine Name:

Type: Saw

- [Saw](#)
- [Sheet Router](#)
- [P2P](#)
- [Part Router](#)
- [Drill and Dowel](#)
- [Chop Saw](#)
- [Other](#)
- [Machine Set](#)
- [Material Handling](#)

 **Copy Selected Machine** - Allows you to Copy the currently selected Machine (Setup).

Copy Machine ? 

New Name for Copy

Copy of Nest Router

 **Change Machine Name** - Allows you to Rename the currently selected Machine (Setup).

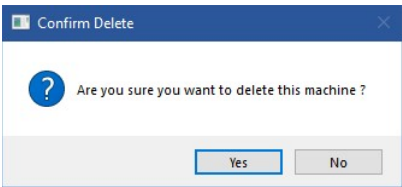
Rename Machine ? 

New Name

Nest Router

 **Delete Selected Machine** - Allows you to Delete the selected Machine (Setup).





When selecting a normal (not a [Machine Set](#)) Setup you have the following possible options on the Machine tab:

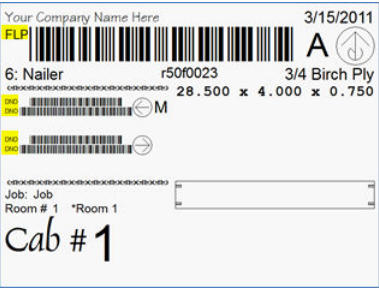
|                    |                                     |                                                    |
|--------------------|-------------------------------------|----------------------------------------------------|
| Manufacturer       | None                                |                                                    |
| NC Link            | Intelli-CAM - Nest                  |                                                    |
| Tag                | RTR                                 |                                                    |
| Output Metric (mm) | <input type="checkbox"/>            | Output Summary <input checked="" type="checkbox"/> |
| Available          | <input checked="" type="checkbox"/> |                                                    |
| Output Directory   | C:\1-CNC-Output\Nest Router\        |                                                    |

**Manufacturer** - Use this option to specify the Manufacturer of your Machine, if not using Intelli-CAM output.

**NC Link** - Use this option to specify the NC Link to use with the current Machine (Setup). See the Topic on [NC Links](#) for more information on the types of Links and there use.

**Tag** - This is a short name for the currently selected Machine (Setup). This option helps a machine operator identify which Barcode on a Label goes with which Machine (Setup). For example the following three examples demonstrate how it is used (displayed). The tags are highlighted in yellow.

Label for one Secondary Machine

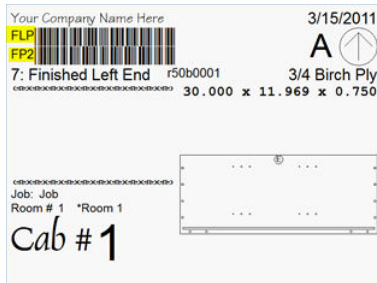


Same Barcode Sent to Two Different Machines (Machine Set)



Your Company Name Here 3/15/2011  
 FLP  
 15: Top r50b0006 3/4 Birch Ply  
 28.500 x 11.969 x 0.750  
 M  
 Job: Job  
 Room # 1 \*Room 1  
 Cab # 1  
 PDP

#### Different Barcode to Two Different Machines (Machine Set)



Your Company Name Here 3/15/2011  
 FLP  
 7: Finished Left End r50b0001 3/4 Birch Ply  
 30.000 x 11.969 x 0.750  
 Job: Job  
 Room # 1 \*Room 1  
 Cab # 1

**Output Metric (mm)** - Use this option if the Output to your Machine must be Metric (mm).

**Output Summary** - Determines whether the [CNCRun](#) is generated for the selected Machine.

**Available** - Determines whether the Machine is available for Output.

**Output Directory** - Specify the Directory where you want the code for the Machine output to. Utilize the Browse  button in the field to set the Output Directory for your Machine. You may also use a UNC path for the Output Directory.

To create a new Machine click the  "Create New Machine" tool in the toolbar .

Type in a descriptive name for the Machine and select its type.

#### [Click the Machine Types below for Setup Tab Options](#)

- Type - [Machine Set](#)
- Type - [Material Handling](#)
- Type - **Sheet** (Use this type for Panel Saws and Nested Based Routers)
  - [Saw](#) - Creates a new NC Panel Saw for use with the Optimizer.
  - [Router](#) - Creates a new CNC Router. Select this option to setup an Intelli-CAM Nested Based Machine.
- Type - **Part** (Use this type for Point-to-Point machines or for your Secondary Operations on your Nested based Router)
  - [P2P](#) - Creates a new Point-to-Point Machine. Select this option to setup an Intelli-CAM Parts Machine.
  - [Router](#) - Creates a new Parts Machine for your Router. Select this option to setup an Intelli-CAM Parts Machine.

- Type - **Specialized** (Use this type for Drill & Dowel and Chop Saws)
  - [Drill and Dowel](#) - Creates a new Drill and Dowel Machine. Select this option to setup a CNC Drill and Dowel Inserter.
  - [Chop Saw](#) - Creates a new Chop Saw. Select this option to setup an Automated Chop Saw system.
  - [Other](#) - Creates machine types that don't fit into the other categories. At this time the only Machine that needs to be setup as this type is the Hoffmann/RazorGage Haunching System.

## Output Tab

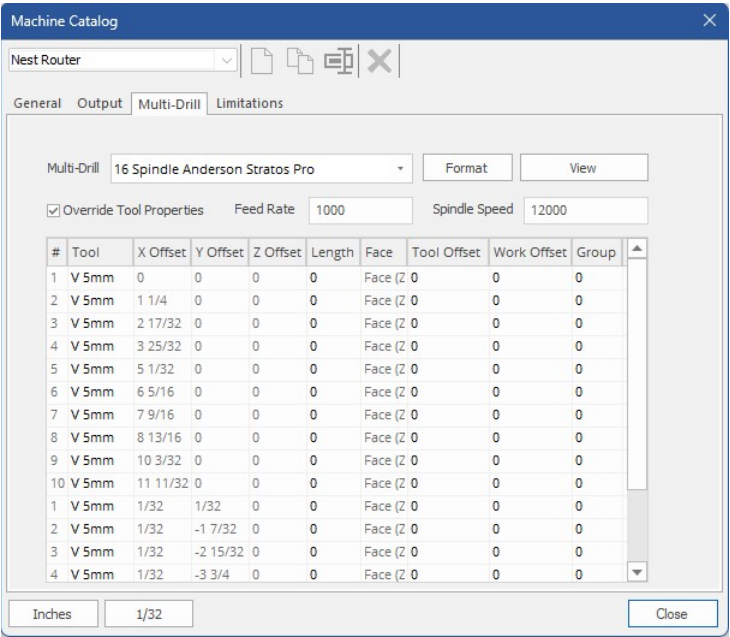
The options on the Output tab are different depending upon the Machine Type. See the following topics for information on the Output tab for each Machine Type:

- [Material Handling](#)
- [Sheet - Saw](#)
- [Sheet - Router](#)
- [Part - P2P](#)
- [Part - Router](#)
- [Specialized Drill and Dowel](#)
- [Specialized Chop Saw](#)
- [Specialized Other](#)

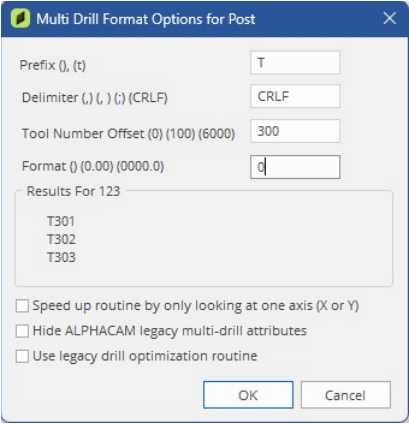
## Multi-Drill Tab

This tab contains Multi-Drill head Setups for machines using Intelli-CAM for G-Code generation.

Simply select the Multi-Drill head to use with your machine. Make sure to look at the X & Y offsets for the tools from the Master (X=0/Y=0) spindle to verify orientation is the same as your Machine.



The **Format** button brings up options you can set for your multi-drill if you are using an ALPHACAM post that may not have been specifically written for S2M CENTER output.



Following are the options here:

- **Prefix** - Establishes the tool number prefix.
- **Delimiter** - Establishes the delimiter for the tool numbers
- **Tool Number Offset** - Sets the numeric offset for the tool numbers.
- **Format** - Sets the numeric format for the tool numbers.

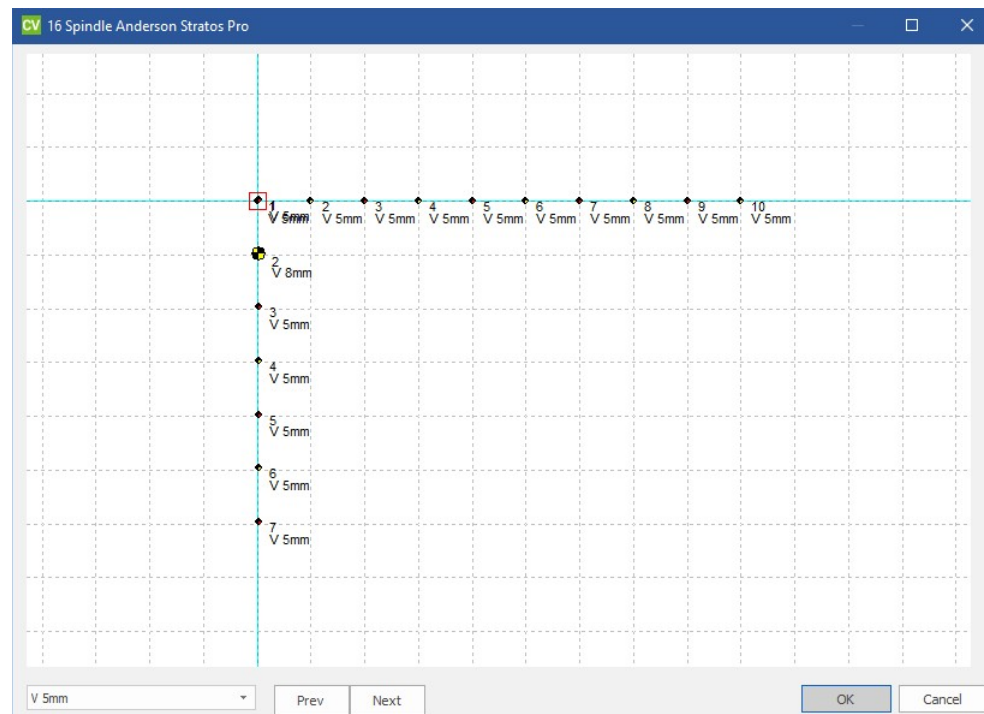
As you set the values for the four fields above you will see the results in the Results For 123 area.

- Speed up routine by only looking at one axis (X or Y)
- Hide ALPHACAM legacy multi-drill attributes
- Use legacy drill optimization routine

The **Override Tool Properties** check box, when enabled, will ignore your Feed Rate and Spindle Speed settings on your individual Drills utilizing instead the values entered here. When disabled Intelli-CAM will look at the individual drills assigned here for Feeds and Speeds.

To assign tools to the spindles simply click in the Tool field representing the appropriate spindle. A drop down box will appear listing the appropriate Tools for use in the spindle.

To view a layout and graphically assign bits for the spindles click the View button.

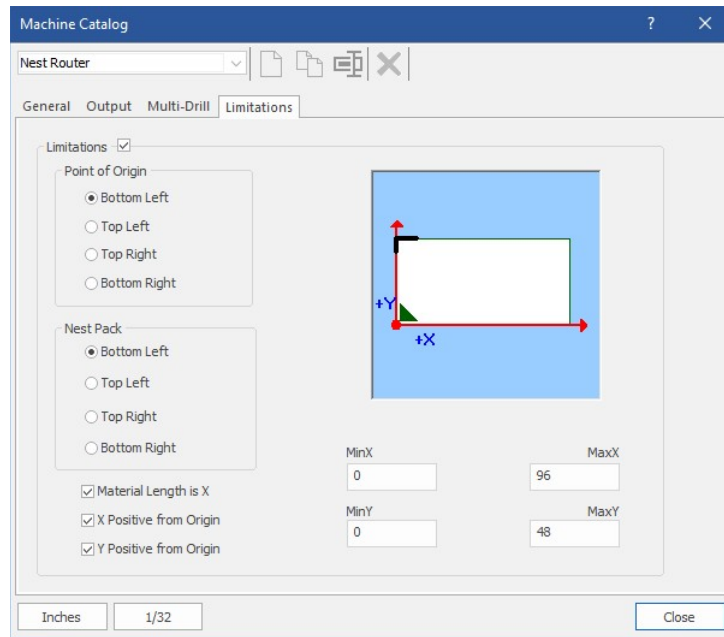


The Spindle that is active is highlighted with a red square. You can assign a bit to this Spindle by clicking the drop down at the lower left and select your Drill. To change to a different Spindle click the Next Spindle or Prev Spindle buttons.

To discontinue the viewing/editing of the Multi-drill head simply click OK to Save your changes.

## Limitations Tab

This tab was put in specifically for those CNC Machines that have Boring bits in their Multi-Drill head that do not reach all areas of the Machine Table. When Limitations are activated Intelli-CAM looks to the Min X, Max X, Min Y, and Max Y fields for the Minimum and Maximum reach for the **Master Spindle** in the Multi-Drill head.



There are several other options on this Tab:

1. **Point of Origin** - This option allows you to control where the Origin Point for your Machine is located.
2. **Nest Pack** - This option allows you to control how parts are packed (placed) on the Nested Sheet.
3. **Material Length is X** - When disabled the Material Length is placed along the Y-Axis.
4. **X Positive from Origin** - When de-selected the X is calculated as a Negative value from the Origin point.



This option does not change the G-Code Output the Post will require modification to achieve a change in output results.

3. **Y Positive from Origin** - When de-selected the Y is calculated as a Negative value from the Origin point.



This option does not change the G-Code Output the Post will require modification to achieve a change in output results.

## CUT Fields Tab

See the Custom [CUT File Saw Link](#) topic for information on the CUT Fields tab.

## Machine Set

Machine Sets allow you to group together Machines you have created in the Machine Catalog. Machine Sets are specifically designed for an "automated" factory where operations are being performed in line on multiple machines and operations are being consumed until no operations remain. Machine Sets are also a very flexible solution even if you are not running a fully automated factory allowing you to set machine sets to focus certain Materials, Part Types, Operations, etc... onto multiple machines using the Filtering capability. You can also use Machine Sets to send Programs to different Machines and based on workload decide which machine to work them through (see Example 3 below).

### Example 1

Let's say you have a Saw, D&D, and a P2P. The P2P has 5mm Horizontal Bores and the D&D has 8mm Horizontal Bores.

A Part with only 8mm Horizontal Bores will never get to the P2P because the D&D will consume them first.

A Part with both 5mm and 8mm Horizontal bores will get a program for the D&D which consumes the 8mm Horizontal bores and then get a program for the P2P which consumes the 5mm Horizontal bores.

### **Example 2**

If you had three machines where machine 1 has 3mm HBores, machine 2 has 5mm HBores, and machine 3 has 8mm HBores.

A Part with 3mm, 5mm, and 8mm HBores would have programs generated for all 3 Machines.

### **Example 3**

Let's say you have a Saw and 2 Point-to-Point Machines and you want to spread production load across the 2 Point-to-Point machines and let the shop make the decision for which Point-to-Point to run the programs on. Simply add your 2 Point-to-Point Machines to the Machine Set with your Saw and enable the "**Operations Pass Through**" option for the first of the two Point-to-Point Machines and programs will be created for the job for both Point-to-Point Machines.

Having each Machine within a Machine Set uniquely Tagged is extremely important especially to get good workflow reporting. The workflow string is the series of machines the part needs to move through. The workflow string is available in [xLabel](#) and for Reports in the [PSNC-xx.mdb](#). It combines the machine tags of each machine in the workflow process some examples are SAW-DND-P2P, SAW-(PP1/PP2)-(DD1/DD2). Machines in () means the part can be milled from any machine in the brackets. If a part normally can go to a (group) but not all the operations can be milled at each machine in the group then the machines that can't do all the operations will be eliminated. If the only way to get all the operations milled in a (group) is to mill some operations at one machine and other operations at another then the machines will no longer be grouped and only the 1st instance of the operation will be milled.

See the [Workflow](#) topic for more information.

When selecting a Machine Set you have the following options on the Machine tab:



**Tag** - This is a short name for the currently selected Machine Set. This option helps a machine operator identify which Barcode on a Label goes with which Machine (Setup). For example the following three examples demonstrate how it is used (displayed). The tags are highlighted in yellow.

Label for one Secondary Machine



Same Barcode Sent to Two Different Machines (Machine Set)



Different Barcode to Two Different Machines (Machine Set)



**Available** - Determines whether the Machine Set is available for Output.

**Machines In Set** - This area is where you assign Machines to your Machine Set. A Machine set can only have one Primary Machine but any number of the other Machine Types. Once you have a Primary Machine included the --BAND PARTS-- option is added to the list and can be used to establish when in the Machine Set process edgebanding is applied to your Parts. This lets the Machine Set know which Machines need to be compensated for Banding and which do not. Those Machines after the --BAND PARTS-- sequence need to be machined with Edgeband included.

**Available Machines** - These are all the Available Machines in your Machine Catalog. Simply highlight the Machine you want to add to the Machine Set and click the left arrow button to add it.

**Operations Pass Through** - The default action of a Machine Set is to consume operations and once consumed not be passed to the next Machine. With this option enabled the Operations will pass through to all the Machines within the Machine Set. This is a good solution if you want to output programs to multiple Machines and then allow the shop to choose which machine they want to run the programs on.

**Filters** - Any [Filters](#) you have saved can be applied to a Machine Set in the Filters area. Simply select your saved Filter from the selector (highlighted yellow below) then click the Add button to make it part of the Machine Set.



## Workflow

### Objectives

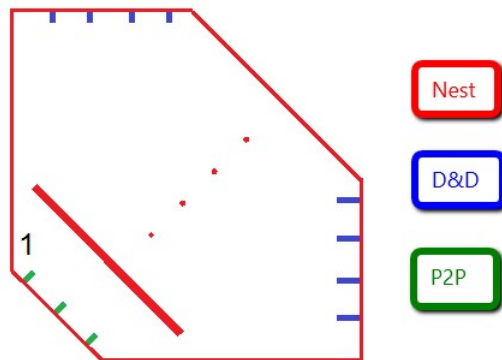
1. Allow for different work flow scenarios where operations may be milled on multiple machines.
2. Report work flow for a particular part.
3. Disable warnings of an operation could not be performed on a part if it can be picked up by another machine down the chain.
4. Provide a fail-safe program for damaged parts.

### Manual (non-Machine Set) Workflow

In a Manual (non-Machine Set) Workflow scenario, operations start at the Primary Machine (Saw/Nest). If an operation can be performed on the machine then it is marked as being complete and is not available on the next machine. This works great with the following machine setup.



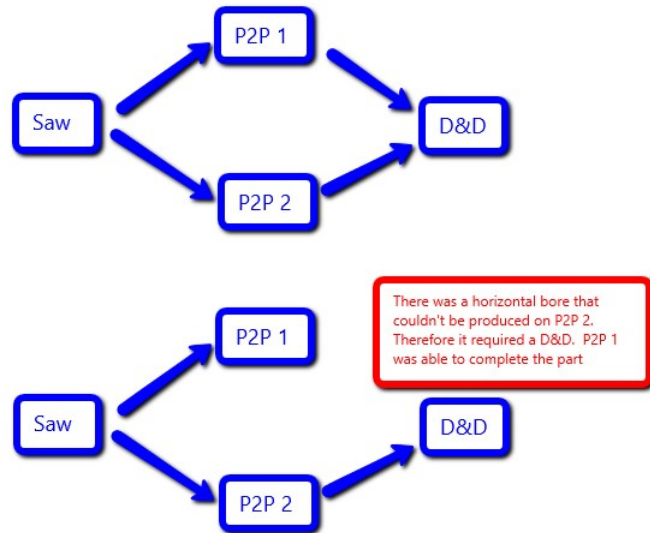
The following shows how the operations would be divided for this part with the nest creating the outline and vertical operations. The D&D taking the boring on the cardinal edges and the P2P getting whatever is left over.



### Work Flow Option #1a

**More than one P2P machine available:** In this scenario, after the part is cut at the Saw or Nest the part could be brought to either P2P for finishing. All operations that were still present after the Saw/Nest would be performed on either machine:

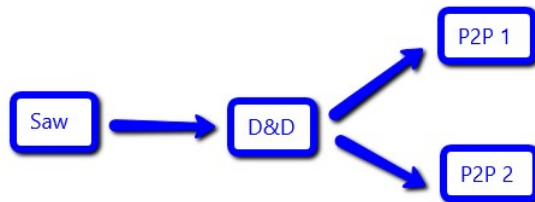




**i** If the operations are completely consumed prior to the next machine the work flow would be complete

### Work Flow Option #1b

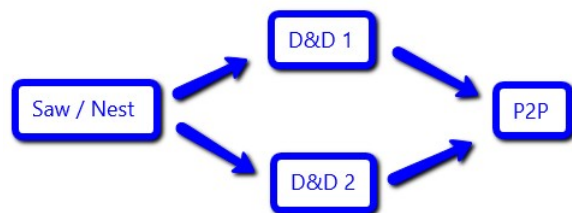
This is a similar scenario except that the D&D machine is used prior to the P2P



**i** In this scenario if there are no D&D operations, that machine is skipped.

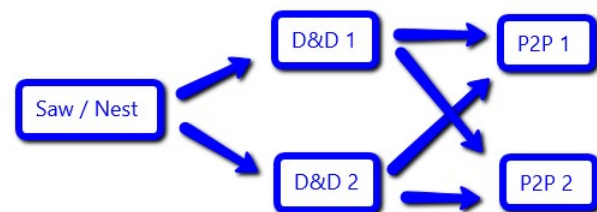
### Work Flow Option #1c

Another variation of option #1 with multiple D&D machines, you could use either one



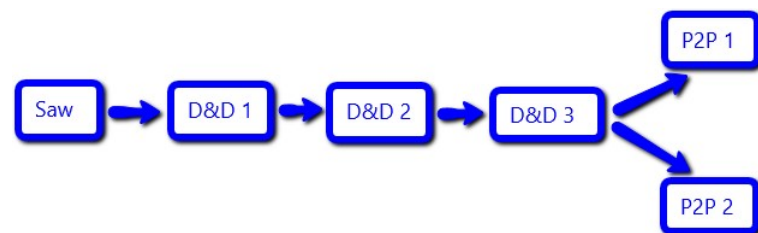
### Work Flow Option #1d

Final variation of option #1: Multiple D&D and P2Ps available. You can decide where to mill the part based on current machine availability.



### Work Flow Option 2

This scenario account for having a line of D&D machines, each doing specific tasks. Unlike the previous scenario where the P2P operations went to each machine, in this case the D&D operations would be consumed by a specific machine and not repeated on another.



### Work Flow Option 3

In this scenario maybe you never want to mill any bore holes on the Nest machine. You want to do all boring on the secondary. However, all other operations are to be performed by the primary machine.

When the S2M Center encounters a situation like this where operations have been designated on a machine that is further down the work flow chain, we don't want any error/warning messages about the operations not being performed on the Nest machine.



## User Interface

The Machine Set interface will show the order in which the Machines would be processed. Each Machine would decide if the operations are consumed by that Machine or pass-through to the next Machine. These two icons is used to give you some feedback as to whether the operations being consumed or pass-through:



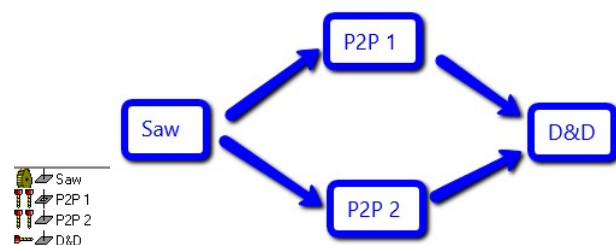
The icon on the left indicates a pass-through. Any operation performed on this Machine will still be available on the next Machine.

The icon on the right indicates a stop. Any operation performed on this Machine will consume the operation and it will not be available for any other Machine in the set.

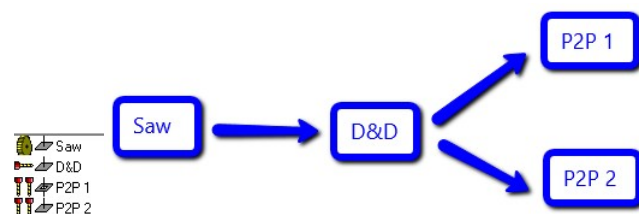
Here are some sample setups for scenarios listed above:



All Stops: Each machine consumes any operation it performs



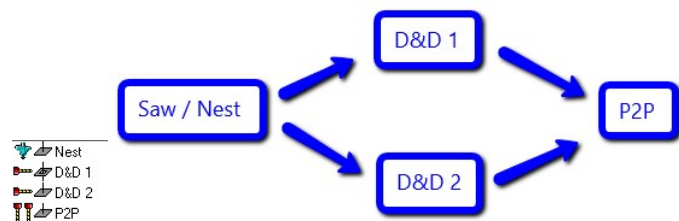
Since we want the operations to be valid for either P2P, we allow the operations to pass-through P2P-1, however, we consume them on P2P -2 so they don't get passed down to the D&D.



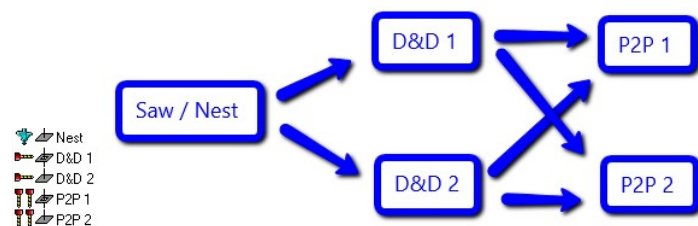
This is close to the traditional output except it can go to either P2P machine; therefore we allow all operations to pass-through P2P-1.



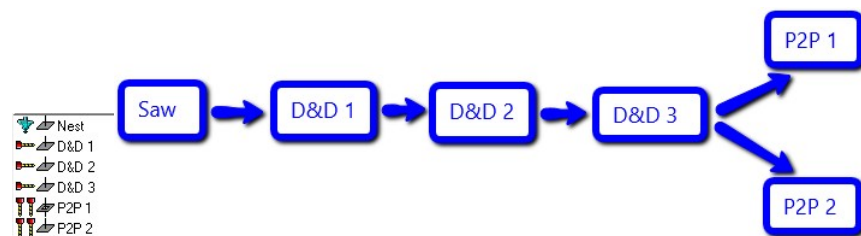
If we had set the D&D as a pass-through, the horizontal boring would have been passed to the P2Ps.



By setting D&D 1 as a pass-through, you can do the horizontal boring at either D&D 1 or 2. Note that D&D 2 is set to stop, otherwise the horizontal boring would have been passed along to the P2P machine.



This is setup so that either D&D can be used, and either P2P can be used on what is left over. Note that we put a stop on the last of each machine type.

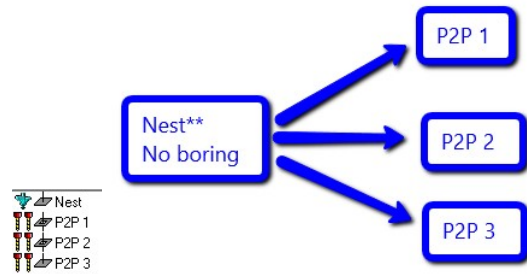


Since each D&D does specific operations, we don't want any pass-through to occur on the D&Ds, each has been set to a stop. However, whatever operations are left over can go to either P2P so we set the P2P-1 as a pass-through so any leftover operations from D&D would be available for P2P-2 as well.

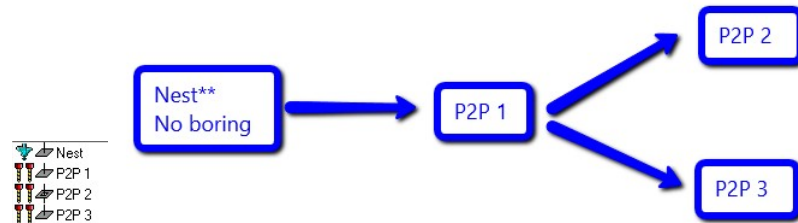


Even though we intend to have the boring output on the P2P we DO NOT want to set the Nest as a pass-through, this would cause all the other operations (route, dado, and outline) to pass-through to the P2P as well. So this would be a typical "no pass-through" setup with the boring holes assigned to the P2P machine by assigning them to a ToolID that was only available on the P2P.

## Special Situations

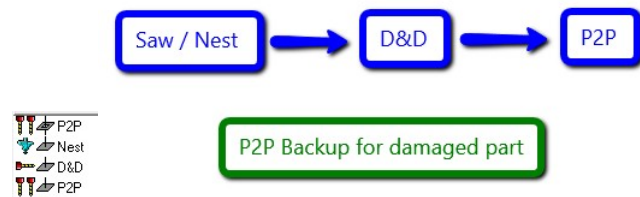


In a modification to the above scenario, we have the boring assigned to a ToolID that is available on each P2P machine. We have set both P2P1 and P2P2 as pass-through so all operations are available on any of the P2P machines.



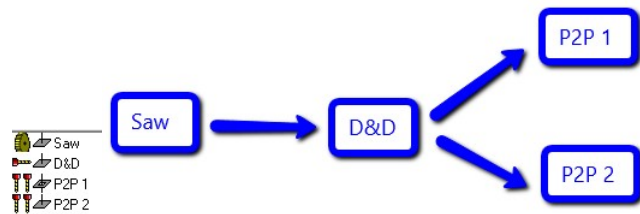
In this scenario we have setup some special large boring tools on P2P 1. We want those operations done with those boring bits and we don't want them passed along to P2P2 & P2P3 so we set a stop after P2P1.

### The Safety Net



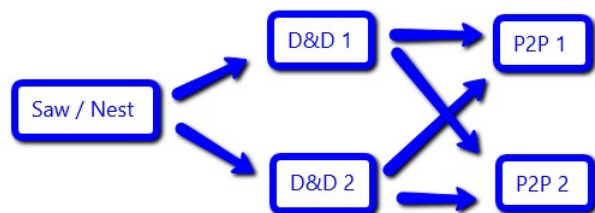
This scenario has a normal workflow, but it has added a safety net by 1<sup>st</sup> outputting to a "Full Outline" P2P machine that is set to pass-through. Since it is a pass-through all operations will go onto the nest as normal, but if the part is damaged for any reason, a new part can be cut to an approximate size and using the barcode for the 1<sup>st</sup> P2P machine you can completely Re-Mill the damaged part.

### Pitfall – Not all paths produce the same results

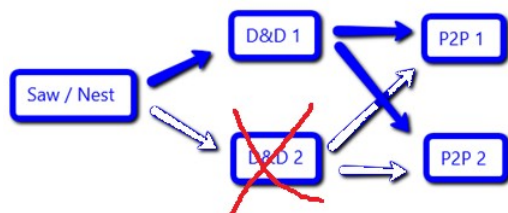


The S2M Center can anticipate scenarios where due to the machine limitations an operation can only be performed on one of the two P2P machines. In this case you really wouldn't have the option to take the part to either P2P machine.

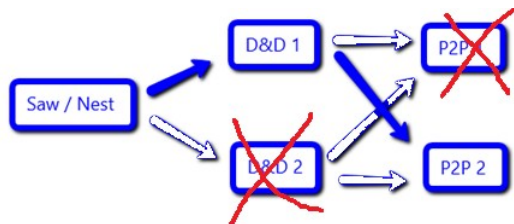
It is suggested that at any shared set of machines, if ALL the machines cannot perform the same amount of operations then you eliminate that path. Using the following arrangement as an example:



If D&D 1 could perform more operations than D&D 2 then we would only output to D&D 1 for this part. But you could still use either P2P.



If P2P 2 performs more operations than P2P 1 then P2P 1 is not acceptable output either. The only work-flow for this part is Saw/Nest→D&D 1→ P2P 2



## Material Handling

This Machine Type is specifically for automated Material Handling Systems. Following are the Output Tab options for Material Handlers:

### Output Tab - Material Handling

To change information in these fields, highlight the field name at the left. Then click on the data at the right and enter the new values. Some of the options below require that you select a specific Machine prior to seeing the option.

#### Import Group

- **Import File** - Utilize the Browse  button in the field to set the Import File and its location for your Material Handling system. Available with the [Flexstore](#), [Intellistore](#), and [Winstore](#) links.

Properties

☐ Import

|                       |    |
|-----------------------|----|
| Import File           |    |
| Auto Sync on Optimize | No |
|                       |    |
|                       |    |
|                       |    |
|                       |    |
|                       |    |
|                       |    |
|                       |    |
|                       |    |
|                       |    |

Set location for Import File

◦ **Auto Sync on Optimize** - If this is set true the Sync will occur each time you Create CNC / Optimize. If false the you would need to click [Add Offcuts](#) to Sync the offcuts. Available with the [Flexstore](#), [Intellistore](#), and [Winstore](#) links.

Properties

☐ Import

|                       |    |
|-----------------------|----|
| Import File           |    |
| Auto Sync on Optimize | No |
|                       |    |
|                       |    |
|                       |    |
|                       |    |
|                       |    |
|                       |    |
|                       |    |
|                       |    |
|                       |    |

Auto Sync on Optimize

◦ **Create Import File Command** - Option for the [Intellistore](#) link.

Properties

☐ Import

|                            |    |
|----------------------------|----|
| Import File                |    |
| Auto Sync on Optimize      | No |
| Create Import File Command |    |

☐ Parameters

|  |  |
|--|--|
|  |  |
|  |  |
|  |  |
|  |  |
|  |  |
|  |  |
|  |  |
|  |  |
|  |  |
|  |  |

Create Import File Command

- **Offcut Import File** - Utilize the Browse  button in the field to set the Offcut Import File and its location for your Material Handling system. Available for the [Winstore](#) link.

Properties

[-] Import

Import File

Auto Sync on OptimizeNo

Offcut Import File

[+] Parameters

Offcut Import File

Parameters Group

Following are the special Parameters available for Material Handling Systems, these are typically unique to each type of Materail Handling System.

[Barbaric Material Handler](#) Parameters

Properties

[-] Import

[-] Parameters

Single PanelNo

Use Material ParametersNo

Material ID Key0

Global Directory0

Label Directory0

Add Output ExtensionNo

PNC name as order nameNo

- **Single Panel** - When enabled, this will set the panel qty as 1 instead of the book number.
- **Use Material Parameters** - When enabled, this will use the material data from the material parameters set in CABINET VISION.
- **Material ID Key** - This allows you to set the ID value for the material ID in the material parameters.
- **Global Directory** - The directory, on the machine that the files will reference in this output.
- **Label Directory** - The directory, on the machine, where the label files are stored.
- **Add Output Extension** - When enabled, this will add the program file extension to the end of program entries in the file output.
- **PNC name as order name** - When enabled, this will use the PNC file name as the order name.

[Intellistore](#) Parameters



| Properties          |    |
|---------------------|----|
| [-] Import          |    |
| [-] Parameters      |    |
| Long material names | No |
|                     |    |
|                     |    |
|                     |    |
|                     |    |
|                     |    |
|                     |    |
|                     |    |
|                     |    |
|                     |    |
|                     |    |
|                     |    |
|                     |    |
|                     |    |

- **Long material names** - When enabled the maximum material name length is Increased from 25 to 50.

[Storemaster](#) Parameters

| Properties       |    |
|------------------|----|
| [-] Import       |    |
| [-] Parameters   |    |
| Global Directory | 0  |
| Label Directory  | 0  |
| Output Extension | 0  |
| Mirror Label Y   | No |
|                  |    |
|                  |    |
|                  |    |
|                  |    |
|                  |    |
|                  |    |
|                  |    |
|                  |    |
|                  |    |

- **Global Directory** - The directory, on the machine, where the file information is found.
- **Label Directory** - The directory, on the machine, where the label information is found.
- **Output Extension** - The file extension for the output.
- **Mirror Label Y** - When enabled, mirrors the Y position for the label placement.

[Winstore](#) Parameters

Properties

+

 Import

-

 Parameters

|                           |    |
|---------------------------|----|
| Connect with SQL Server   | No |
| SQL Connect Text          | 0  |
| Offcut Minimum Index      | 0  |
| Offcut Maximum Index      | 0  |
| Description As FileName   | No |
| Winstore Sync Description | No |
|                           |    |
|                           |    |
|                           |    |
|                           |    |
|                           |    |
|                           |    |
|                           |    |
|                           |    |

- **Connect with SQL Server** - If enabled, connect to the SQL data store specified in SQL Connect text.
- **SQL Connect Text** - Sample: Provider=SQLOLEDB;Data Source=ABC\CV;User Id=CV\_MSSQL\_User;Initial Catalog=WINSTORE\_EXCHANGE;
- **Offcut Minimum Index** - Sets an offcut index range minimum (default is 12000000)
- **Offcut Maximum Index** - Sets an offcut index range maximum (default is 129999999)
- **Description As FileName** - Enabling this option uses the material Description as xml filename instead of material Name
- **Winstore Sync Description** - Enabling this option uses the material Description to Sync to Winstore instead of material Name

More Information

Program Flow Diagram



The screenshot shows the "Offcut Manager" application window. At the top is a blue title bar with the text "Offcut Manager", a question mark icon, and a close button (X). Below the title bar is a toolbar containing four icons: a printer, two document icons, and a grid icon. The main area features two tabs, "Offcuts" and "Bins", with "Offcuts" currently selected. Below the tabs is a large table with five columns: "ID", "Material", "Width", "Length", and "Qty". The first row under the "ID" column contains a red triangle icon. The table has approximately 10 rows. At the bottom left is a "Delete" button. At the bottom right is a yellow bar containing a checked checkbox labeled "Sync with:", a dropdown menu showing "Material Handling", and a "Sync Now" button.

Simply select your Material Handling System and enable the **Sync with** option. You can test the link by clicking the Sync Now button.

If [Auto Sync](#) is disabled you will need to click the [Add Offcuts](#) sidebar option prior to clicking [Optimize](#).

If [Auto Sync](#) is enabled simply click the [Optimize](#) sidebar option and the Offcuts are automatically added.

If the import file can not be found you will be prompted to select a file.

## Sheet Saw

This Machine Type is specifically for Saws. Following are the Setup Tab options for Sheet Saws:

### Setup Tab - Sheet Saw

#### Blade Group

- **Rip Cut Kerf** - Use this value to specify the Kerf width of the Rip Blade on your Saw.

All

Properties

Blade

|                      |     |
|----------------------|-----|
| Rip Cut Kerf         | 1/4 |
| Cross Cut Kerf       | 1/4 |
| Cut Clearance Travel | 8   |

Booking

Label IT - Label

Label IT - Image

Label IT - Machine Settings

Optimization - Complexity

Optimization - Sorting

Parameters

Restrictions - Strips

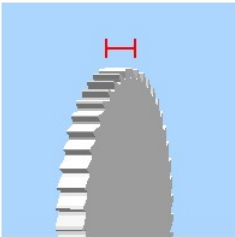
Restrictions - Z-Cuts

Restrictions - Sheet Size

Speed

Times

Trim



- **Cross Cut Kerf** - Use this value to specify the Kerf width of the Cross Cut Blade on your Saw.

All

Properties

Blade

|                      |     |
|----------------------|-----|
| Rip Cut Kerf         | 1/4 |
| Cross Cut Kerf       | 1/4 |
| Cut Clearance Travel | 8   |

Booking

Label IT - Label

Label IT - Image

Label IT - Machine Settings

Optimization - Complexity

Optimization - Sorting

Parameters

Restrictions - Strips

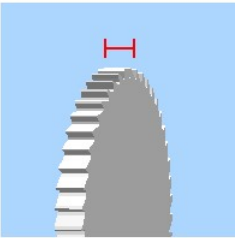
Restrictions - Z-Cuts

Restrictions - Sheet Size

Speed

Times

Trim



- **Cut Clearance Travel** - Specifies how far the Saw needs to move past the Edge of the Material to complete the cut. Most Saw Links do NOT require this information.

All

Properties

Blade

|                      |     |
|----------------------|-----|
| Rip Cut Kerf         | 1/4 |
| Cross Cut Kerf       | 1/4 |
| Cut Clearance Travel | 8   |

Booking

Label IT - Label

Label IT - Image

Label IT - Machine Settings

Optimization - Complexity

Optimization - Sorting

Parameters

Restrictions - Strips

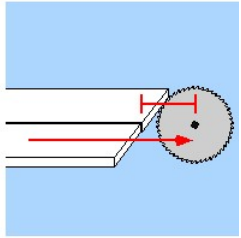
Restrictions - Z-Cuts

Restrictions - Sheet Size

Speed

Times

Trim

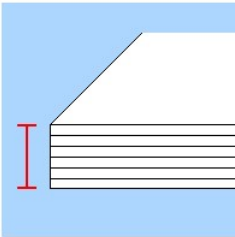


Booking Group

- **Booking Max Height** - Specifies the Maximum Thickness of Material your Saw will accept.

All

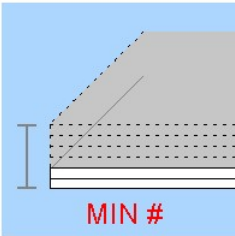
|                             |   |
|-----------------------------|---|
| Properties                  |   |
| Blade                       |   |
| Booking                     |   |
| Booking Max Height          | 3 |
| Booking Min Sheets          | 1 |
| Booking Max Sheets          | 6 |
| Label IT - Label            |   |
| Label IT - Image            |   |
| Label IT - Machine Settings |   |
| Optimization - Complexity   |   |
| Optimization - Sorting      |   |
| Parameters                  |   |
| Restrictions - Strips       |   |
| Restrictions - Z-Cuts       |   |
| Restrictions - Sheet Size   |   |
| Speed                       |   |
| Times                       |   |
| Trim                        |   |



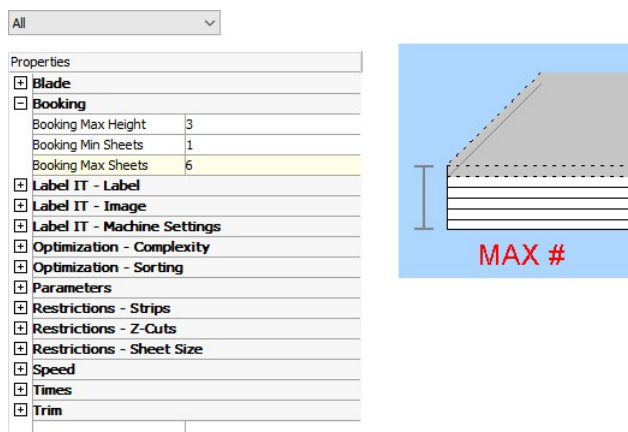
- **Booking Min Sheets** - Specifies the Minimum number of Sheets allowed in a Book (Stack) on your Saw.

All

|                             |   |
|-----------------------------|---|
| Properties                  |   |
| Blade                       |   |
| Booking                     |   |
| Booking Max Height          | 3 |
| Booking Min Sheets          | 1 |
| Booking Max Sheets          | 6 |
| Label IT - Label            |   |
| Label IT - Image            |   |
| Label IT - Machine Settings |   |
| Optimization - Complexity   |   |
| Optimization - Sorting      |   |
| Parameters                  |   |
| Restrictions - Strips       |   |
| Restrictions - Z-Cuts       |   |
| Restrictions - Sheet Size   |   |
| Speed                       |   |
| Times                       |   |
| Trim                        |   |



- **Booking Max Sheets** - Specifies the Maximum number of Sheets allowed in a Book (Stack) on your Saw.



#### +Label - Label Group

[See Sheet Router](#)

#### +Label - Image Group

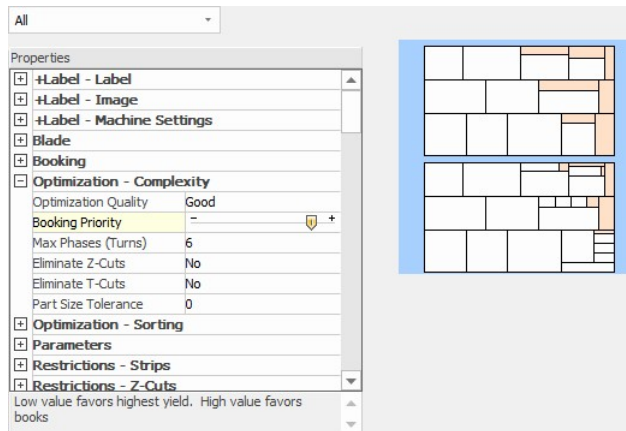
[See Sheet Router](#)

#### Optimization - Complexity Group

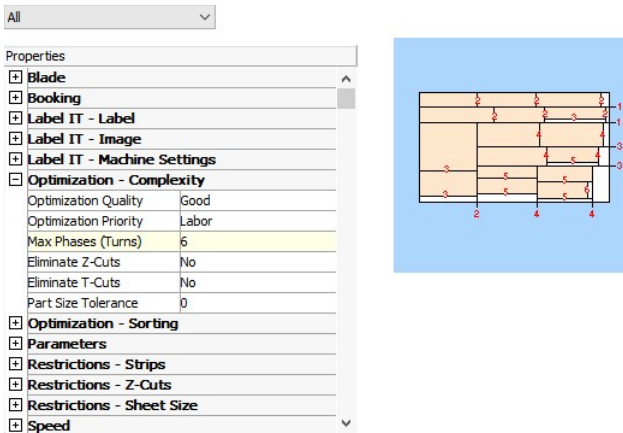
- **Optimization Quality** - Controls the Quality of the Optimization of your Parts.
  - **Fastest** - This setting offers the fastest Nesting at the computer, sacrificing Yield.
  - **Fast** - This setting offers a slightly slower speed at the computer improving yield slightly.
  - **Good** - This setting offers a balance between speed at the computer and Yield.
  - **Better** - This offers slower speed at the computer with increased improvement to Yield.
  - **Best** - This offers the slowest speed at the computer with the best Yield results sacrificing speed at the computer.
- **Booking Priority** - Sliding to the left favors Yield (less books) while sliding to the right favors Labor (more books).



Make certain to set up [Booking](#) for your Saw in the Machine Catalog Setup Tab.



- **Max Phases (Turns)** - Sets the maximum number of Phases your Saw can handle.



Most older saws have a maximum of 3 phases. Make sure to check with your Saw Manufacturer to find out how many Phases (Turns) your Saw will support.

- **Eliminate Z-Cuts** - Allows you to eliminate Z-Cuts. A Z-Cut is the "Third Phase" cut on any given strip. A "Z-Cut" and "Third Phase Cut" are the same as long as they are cut from the 1st strip set in a pattern. However, we allow the strip series to change directions a maximum of 6 times, the Z-Cut could occur on the 3rd, 4th, 5th, 6th, 7th, or 8th phase. Allowing Z-Cuts can cause a dramatic improvement in Yield but at the cost of time spent on the Saw. This can cause the operator to make multiple cuts to bring a part to its finished size.



All

Properties

Blade

Booking

Label IT - Label

Label IT - Image

Label IT - Machine Settings

Optimization - Complexity

Optimization - Sorting

Parameters

Restrictions - Strips

Restrictions - Z-Cuts

Restrictions - Sheet Size

Speed

|                       |       |
|-----------------------|-------|
| Optimization Quality  | Good  |
| Optimization Priority | Labor |
| Max Phases (Turns)    | 6     |
| Eliminate Z-Cuts      | No    |
| Eliminate T-Cuts      | No    |
| Part Size Tolerance   | 0     |

A diagram of a rectangular sheet with a grid of internal lines. A blue border surrounds the sheet. A small orange-shaded rectangular area is located in the top right corner. The grid lines represent potential cut paths.

 Eliminating Z-Cuts will affect the type of patterns that are generated but it will not reduce the phase count. Make certain to set up the [Phase Count](#) for your Saw in the Machine Catalog Setup Tab.

- **Eliminate T-Cuts** - Allows you to eliminate T-Cuts. T-Cuts are cuts made to break a sheet apart prior to Ripping and/or Cross-Cutting. Allowing T-Cuts can cause an improvement in Yield but like Z-Cuts can also cost time spent on the Saw.

All

Properties

Blade

Booking

Label IT - Label

Label IT - Image

Label IT - Machine Settings

Optimization - Complexity

Optimization - Sorting

Parameters

Restrictions - Strips

Restrictions - Z-Cuts

Restrictions - Sheet Size

Speed

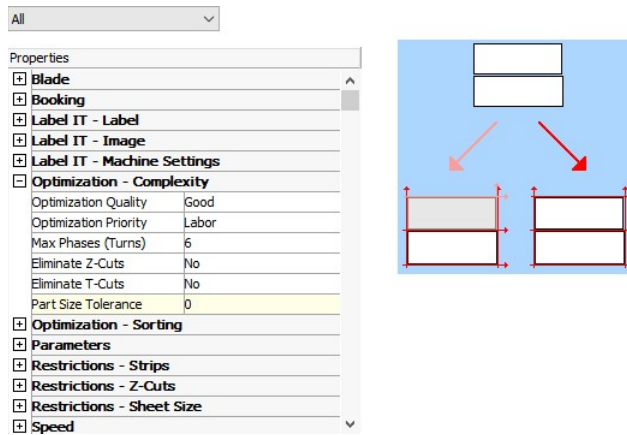
|                       |       |
|-----------------------|-------|
| Optimization Quality  | Good  |
| Optimization Priority | Labor |
| Max Phases (Turns)    | 6     |
| Eliminate Z-Cuts      | No    |
| Eliminate T-Cuts      | No    |
| Part Size Tolerance   | 0     |

A diagram of a rectangular sheet with a grid of internal lines. A blue border surrounds the sheet. A large orange-shaded rectangular area covers the bottom half of the sheet. The grid lines represent potential cut paths.

- **Part Size Tolerance** - Depending on possible unit conversion variances during transfer to the S2M Center from your design/engineering product you may have parts you consider the same size but may have a slight variance in size. This allows you to set a Tolerance for which the S2M Center will consider the Parts the same size.

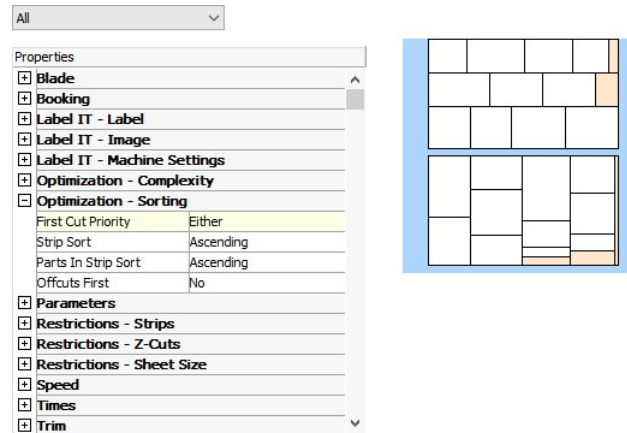
file:///C:/Users/celav/AppData/Local/Temp/~hhEA7D.htm

9/25/2026

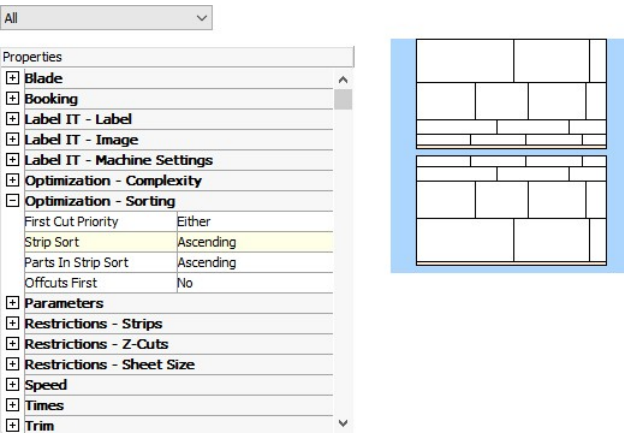


### Optimization - Sorting Group

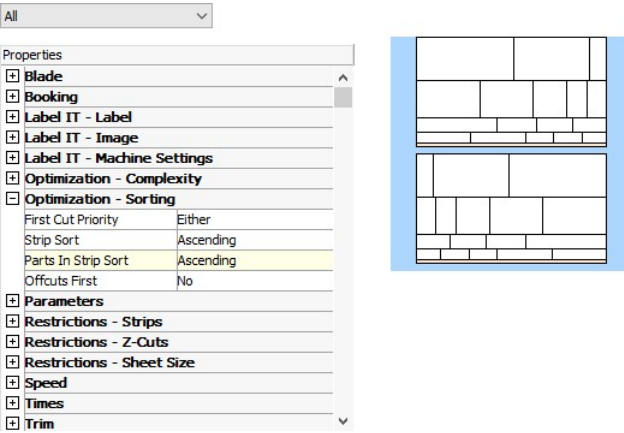
- **First Cut Priority** - Allows you to control whether you want your first cut on the saw to be a Rip (Cut with the grain), a Cross Cut (Cut across the grain), or Either.



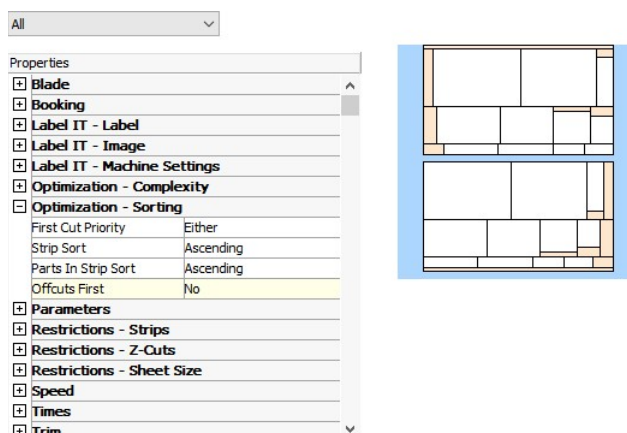
- **Strip Sort** - Allows you to control how you want your Strips sorted within a [Strip Set](#).



- **Parts in Strip Sort** - Allows you to control how you want your Parts within each Strip to be sorted.



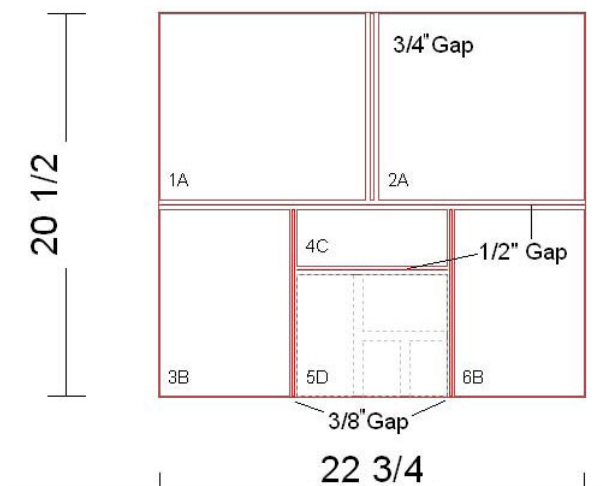
- **Offcuts First** - Allows you to specify that Offcuts should come off the Strips first rather than last.



### Parameters Group

- **Min pos of crosscut fence** - Allows you to establish the Minimum Position of the Crosscut Fence with the [Altendorf Elmo](#) or [Martin T74](#) Saw Link. Set this to match the setting on your Saw.
- **Height of blade over mat** - Allows you to establish the Height of the Saw Blade over the Material on the [Altendorf Elmo](#) or [Martin T74](#) Saw Link. Set this to match the setting on your Saw.
- **Max pos of rip fence** - Allows you to establish the Maximum Position of the Rip Fence with the [Altendorf Elmo](#) or [Martin T74](#) Saw Link. Set this to match the setting on your Saw.
- **Max l/w ratio for X-cut** - Allows you to establish the Maximum Length/Width Ratio for cross cuts on the [Altendorf Elmo](#) or [Martin T74](#) Saw Link. Set this to match the setting on your Saw. Set this to match the setting on your Saw.
  - Excerpt explaining the Ratio value from Altendorf:  
If the ratio length / width of the part exceeds a value of 4 or 5, its width has to be cut at the rip fence, otherwise the crosscut fence could give you no exact 90 degree angle.
- Examples:**
  - The material is sized 700mm x 500mm, the workpiece has to be 660mm x 480mm:
    - Both cuts are performed at the crosscut fence.
  - The material is sized 1200mm x 200mm, the workpiece has to be 1150mm x 190mm:
    - The length cut of 1150mm is done with the crosscut fence, but the width of 190mm is cut at the rip fence, because the 200mm are too narrow to give an exact 90 degree position for the cut. Therefore the postprocessor parameter “ratio length/width workpiece”.
    - Some identical parts have to be cut out of the rest of a stripe: For instance 7 identical parts have to be cut. For 6 parts the rip fence gives the measure, the 7th part is turned around and cut at the crosscut fence. An exception might be that the 7th workpiece is smaller than 146mm or the length/width ratio forces the part to the rip fence.
- **Favor rip fence for rips** - Allows you to establish that you favor the Rip Fence for Rip cuts with the [Altendorf Elmo](#) or [Martin T74](#) Saw Link. Set this to match the setting on your Saw.
- **Favor rip fence for X-cut** - Allows you to establish that you favor the Rip Fence for Cross Cuts with the [Altendorf Elmo](#) or [Martin T74](#) Saw Link. Set this to match the setting on your Saw.
- **Favor rip fence for last** - Allows you to establish that you favor the Rip Fence for your Last Cut with the [Altendorf Elmo](#) or [Martin T74](#) Saw Link. Set this to match the setting on your Saw.
- **Allow Negative Waste** - This option will appear when a [Casadei CN04](#), [CPOUT](#), [Gabbiani Prima](#), [Homag Espana CH03](#), [Macmazza SEZ2000](#), [Macmazza Tork MP70](#), [SCMI Sigma 65](#), [SCMI x915](#), [SCMI z32](#), or [SCMI z45](#) NC Links are selected. Negative Waste can be created when Grain Matching.
  - Each cut in the figure below is considered a “Gap Cut”; a gap cut is a cut required to maintain the gap between 2 parts. This machine’s kerf is set to 1/4”. Therefore the cut instruction for the 3/4” gap cut would be a 1/4” waste piece. When part 1A is cut it leaves a 1/2” gap before 2A. To remove the 1/2” the machine must cut a 1/4” piece of waste (1/4” waste + the 1/4” kerf = the 1/2” gap).

- In the figure below the 1/2" and 3/8" gap cuts are considered Negative Waste. Negative waste is output red because it requires special saw instructions. Using the logic described in Gap Cuts, the 1/2" gap requires a 0" cut and the 3/8" gap requires a 1/8" cut. Since the typical saw controller cannot accept either a 0 or negative cut these have to be handled with special cutting instructions. Since only some saws can process negative waste this feature is turned off by default. For saws that cannot process these instructions the negative waste gap cuts are reduced to equal the saw kerf. This would affect the 1/2" and 3/8" gaps above, but not the 3/4" gap.



- **Subtract Kerf from Trim** - This option will appear when a [CPOUT](#) NC Link is selected. Some Machines that except CPOUT files require that the Kerf be subtracted from output. Check with your Machine Manufacturer to see whether or not this setting is necessary. The default for most Saws is No.
- **File Name** - This option allows you to establish the [CPOUT](#) file name. If blank the output name will be CPOUT.nnn. Some Saw require other names such as PRV.nnn. Typing PRV here will output this way.
- **Label Image in PRT1,8** - This option will place the Label Image name in PRT1,8 in [CPOUT](#) output.
- **Set Banding to 1111** - This option will appear when a [CPOUT](#) NC Link is selected. This option instructs the S2M Center to output 1111 for Banding to overcome a limitation with HolzHer CUT85 software. Leave this set to No unless you are using CUT85.
- **Saw Type** - This option will appear when a [CPOUT](#) NC Link is selected. This option establishes the Saw Type. There are 4 valid options:
  - 0 = no saw type specified
  - 1 = single axis saw
  - 2 = two axis saw with single feeder
  - 3 = two axis saw with multiple feeder
- **Label Image in PRT1,9** - This option will place the Label Image name in PRT1,9 in [CPOUT](#) output.
- **Label Image in PRT2,2** - This option will place the Label Image name in PRT2,2 in [CPOUT](#) output.
- **Ignore Kerf Adjust on GMB** - This option will appear when a [CPOUT](#) NC Link is selected. This option instructs the S2M Center to Ignore Kerf Adjustments on Grain Match Blocks.
- **Band order W1L1W2L2** - This option will appear when a [CPOUT](#) NC Link is selected. This option instructs the S2M Center ensures the edgebanding instructions are in a specific ordering depending on the CPOUT implementation for your Saw.
- **Output Image Name Only** - This option will appear when a [CPOUT](#) NC Link is selected. This option allows you to specify that you want only the image name output (i.e. C:\Junk\myimage.bmp becomes myimage.bmp).
- **Enhanced CPOUT** - This option will appear when a [CPOUT](#) or [Eaton Saw](#) NC Link is selected. This allows the output of an enhanced CPOUT file, which increases the amount of characters in several fields as well as addition PRT records with additional information. The following changes are made to the CPOUT.

- Add additional columns to the INV3 record (all are empty)
- Increase ORD2 columns 2 and 3 to 40 characters each
- Set PRT1 columns 6,7 and 8 to 40, 40, and 30 characters, respectively
- Add additional empty records to the PRT1 record
- Add additional empty columns to the PRT3 record
- Add additional empty columns to the PRT4 column
- Add a new PRT6, PRT7, and PRT8 record that will contain edgebanding information (PRT7 is entirely empty).
- **Include Offcuts in ORD** - This option allows you to specify that offcut pieces should get ORD records also.
- **Output 2 for No Grain** - This option will appear when the [CUT File](#) NC Link is selected. This option will allow you to send instructions for Grained Parts in the CUT file.
- **Limit to 32 Fields** - This option will appear when the [CUT File](#) NC Link is selected. This option will allow you to output a CUT file that is limited to 32 fields.
- **Add Banding Fields** - This option will appear when the [CUT File](#) NC Link is selected. This option will add your Edgebanding instructions to the CUT File.
- **Output Optimized Only** - This option will appear when the [CUT File](#) NC Link is selected. This option will allow you to only output Parts marked as Optimized.
- **Output Panel Stock** - This option will appear when the [CUT File](#) NC Link is selected. This option will send all Material classified as Panel Stock to the CUT File.
- **Output Board Stock** - This option will appear when the [CUT File](#) NC Link is selected. This option will send all Material classified as Board Stock to the CUT File.
- **Output Laminate** - This option will appear when the [CUT File](#) NC Link is selected. This option will send all Material classified as Laminate to the CUT File.
- **Output Misc Material** - This option will appear when the [CUT File](#) NC Link is selected. This option will send all Material classified as Misc to the CUT File.
- **Output Tops** - This option will appear when the [CUT File](#) NC Link is selected. This option will send all Counter Top Parts to the CUT File.
- **Output Part's Image Name?** - This option will appear when a [CUT File](#) or [PNL File](#) NC Link is selected. This option allows you to specify the Image file name is included in the CUT file for Real-time Labeling.
- **Output Part's Image Path?** - This option will appear when a [CUT File](#) or [PNL File](#) NC Link is selected. This option allows you to specify the Image's path name is included in the CUT file for Real-time Labeling.
- **Grain Match Data** - This option will appear when a [CUT File](#) NC Link is selected. This option allows you to specify if Grain match data is included in the CUT File.
- **Scoring** - This option will appear when a [Elcon Quadra](#) NC Link is selected. Establishes whether the Saw's Scoring function should be activated.
- **Settings File** - This option will appear when a [Elcon Quadra](#) NC Link is selected. The location of the Elcon Saw Settings file.
- **No Limit on File Size** - This option will appear when an [Excitech EP](#), [SCM Ottimo](#), or [Holzma](#) NC Link is selected. This option can be set if your Holzma Saw has a file size limit.
- **Show Band Code** - This option will appear when an [Excitech EP](#) or [Holzma](#) NC Link is selected. This option determines if the Band Code information is to be output.
- **Show Cabinet Name** - This option will appear when an [Excitech EP](#) or [Holzma](#) NC Link is selected. This option determines if the Cabinet Name is to be output.
- **Version** - This option will appear when an [Excitech EP](#), [SCM Ottimo](#), or [Holzma](#) NC Link is selected. If left blank it will output in the pre-v1.06 PTX format. If you are using v1.06 or higher you will want to enter 1.06 in this field.
- **Optimizing Parameter** - This option will appear when an [Excitech EP](#), [SCM Ottimo](#), or [Holzma](#) NC Link is selected. This option specifies the Optimizing Parameter to use.
- **Saw Parameter** - This option will appear when an [Excitech EP](#), [SCM Ottimo](#), or [Holzma](#) NC Link is selected. This option specifies the Saw Parameter to use.
- **Include Pallet Field** - This option will appear when an [Excitech EP](#) or [Holzma](#) NC Link is selected. This option specifies whether you want the Pallet Field output.
- **Extra ',' After PROD\_INFO** - This option will appear when an [Excitech EP](#) or [Holzma](#) NC Link is selected. Some saws require this will shift the fields after PROD\_INFO in the PTX file.
- **Min retrim allowance** - This option will appear when an [Excitech EP](#), [SCM Ottimo](#), or [Holzma](#) NC Link is selected. This option specifies the minimum retrim allowance for the saw.
- **Max Lines in One File** - This option will appear when the [Gabbiani Prima](#) NC Link is selected. This option allows you to set a line limit for a single output file.

- **Decimal Precision** - This option will appear when a [Holzer saw with Tria Controller](#) is selected. This option allows you to establish the number of Decimal places the machine will accept. This can vary depending on whether the Saw is setup in Imperial or Metric.
- **Origin** - This option will appear when a [Holzma](#), [SCM Ottimo](#), or [SCM PTX](#) is selected. This option controls the origin for the saw.  
**Output to \NPSPRG & DIST** - This option will appear when a [Homag Espana CH12](#) NC Link is selected. See Saw documentation for more information.
- **Saw Blade Height** - This option will appear when a [Kappa Automatic](#) NC Link is selected. Enables/disables the automatic blade height carriage. Not all of the Kappa saws will have this option.
- **Add Label Data** - This option will appear when a [Macmazza SEZ2000](#) NC Link is selected. Includes Label Data as specified by Macmazza.
- **Add Edge Barcodes** - This option will appear when a [Mayer](#) NC Link is selected. This option will allow you to output Edge Barcodes to the Mayer saw. Currently the only Edge Programs (Barcodes) generated are when utilizing a Drill and Dowel Machine.
- **Add Face/Back Barcodes** - This option will appear when a [Mayer](#) NC Link is selected. This parameter will add the front and back barcodes to the MA015 part line (if they exist).
- **Rest-sectional area** - This option will appear when a [PIOS](#) NC Link is selected. Please see the PIOS documentation for information on this Parameter and its use.
- **Side pressure collision** - This option will appear when a [PIOS](#) NC Link is selected. Please see the PIOS documentation for information on this Parameter and its use.
- **Side pressure width** - This option will appear when a [PIOS](#) NC Link is selected. Please see the PIOS documentation for information on this Parameter and its use.
- **Clamp close time** - This option will appear when a [PIOS](#) NC Link is selected. Please see the PIOS documentation for information on this Parameter and its use.
- **Clamp open time** - This option will appear when a [PIOS](#) NC Link is selected. Please see the PIOS documentation for information on this Parameter and its use.
- **Pressure bar up/down time** - This option will appear when a [PIOS](#) NC Link is selected. Please see the PIOS documentation for information on this Parameter and its use.
- **Saw carriage time** - This option will appear when a [PIOS](#) NC Link is selected. Please see the PIOS documentation for information on this Parameter and its use.
- **Side pressure inc time** - This option will appear when a [PIOS](#) NC Link is selected. Please see the PIOS documentation for information on this Parameter and its use.
- **Side pressure dec time** - This option will appear when a [PIOS](#) NC Link is selected. Please see the PIOS documentation for information on this Parameter and its use.
- **Alignment time** - This option will appear when a [PIOS](#) NC Link is selected. Please see the PIOS documentation for information on this Parameter and its use.
- **Sectional are rest time** - This option will appear when a [PIOS](#) NC Link is selected. Please see the PIOS documentation for information on this Parameter and its use.
- **Positioning time** - This option will appear when a [PIOS](#) NC Link is selected. Please see the PIOS documentation for information on this Parameter and its use.
- **Max cutting height** - This option will appear when a [PIOS](#) NC Link is selected. Please see the PIOS documentation for information on this Parameter and its use.
- **Max cutting depth** - This option will appear when a [PIOS](#) NC Link is selected. Please see the PIOS documentation for information on this Parameter and its use.
- **Push out automatically** - This option will appear when a [PIOS](#) NC Link is selected. Please see the PIOS documentation for information on this Parameter and its use.
- **Strip aligner** - This option will appear when a [PIOS](#) NC Link is selected. Please see the PIOS documentation for information on this Parameter and its use.
- **Finger** - This option will appear when a [PIOS](#) NC Link is selected. Please see the PIOS documentation for information on this Parameter and its use.
- **Remainder cut printing** - This option will appear when a [PIOS](#) NC Link is selected. Please see the PIOS documentation for information on this Parameter and its use.
- **Cut Height Index (0-5)** - This option will appear when a [PIOS](#) NC Link is selected. Please see the PIOS documentation for information on this Parameter and its use.
- **Format printing** - This option will appear when a [PIOS](#) NC Link is selected. Please see the PIOS documentation for information on this Parameter and its use.
- **Cross aligner** - This option will appear when a [PIOS](#) NC Link is selected. Please see the PIOS documentation for information on this Parameter and its use.
- **Scoring saw len and cross** - This option will appear when a [PIOS](#) NC Link is selected. Please see the PIOS documentation for information on this Parameter and its use.
- **Softforming** - This option will appear when a [PIOS](#) NC Link is selected. Please see the PIOS documentation for information on this Parameter and its use.
- **Clamps** - This option will appear when a [PIOS](#) NC Link is selected. Please see the PIOS documentation for information on this Parameter and its use.

- **Feed type** - This option will appear when a [PIOS](#) NC Link is selected. Please see the PIOS documentation for information on this Parameter and its use.
- **Align manually** - This option will appear when a [PIOS](#) NC Link is selected. Please see the PIOS documentation for information on this Parameter and its use.
- **Format cut printing** - This option will appear when a [PIOS](#) NC Link is selected. Please see the PIOS documentation for information on this Parameter and its use.
- **Length aligner** - This option will appear when a [PIOS](#) NC Link is selected. Please see the PIOS documentation for information on this Parameter and its use.
- **Home position** - This option will appear when a [PIOS](#) NC Link is selected. Please see the PIOS documentation for information on this Parameter and its use.
- **Automatic cutting height** - This option will appear when a [PIOS](#) NC Link is selected. Please see the PIOS documentation for information on this Parameter and its use.
- **Return saw carriage** - This option will appear when a [PIOS](#) NC Link is selected. Please see the PIOS documentation for information on this Parameter and its use.
- **Scoring saw cross** - This option will appear when a [PIOS](#) NC Link is selected. Please see the PIOS documentation for information on this Parameter and its use.
- **Finish-cut selection** - This option will appear when a [PIOS](#) NC Link is selected. Please see the PIOS documentation for information on this Parameter and its use.
- **Defined cutting speed** - This option will appear when a [PIOS](#) NC Link is selected. Please see the PIOS documentation for information on this Parameter and its use.
- **Defined saw blade RPM** - This option will appear when a [PIOS](#) NC Link is selected. Please see the PIOS documentation for information on this Parameter and its use.
- **Label printing mode** - This option will appear when a [PIOS](#) NC Link is selected. Please see the PIOS documentation for information on this Parameter and its use.
- **Waste flap** - This option will appear when a [PIOS](#) NC Link is selected. Please see the PIOS documentation for information on this Parameter and its use.
- **Blade cooling** - This option will appear when a [PIOS](#) NC Link is selected. Please see the PIOS documentation for information on this Parameter and its use.
- **Material on stock** - This option will appear when a [PIOS](#) NC Link is selected. Please see the PIOS documentation for information on this Parameter and its use.
- **Grooving with oversize** - This option will appear when a [PIOS](#) NC Link is selected. Please see the PIOS documentation for information on this Parameter and its use.
- **Saw blade RPM** - This option will appear when a [PIOS](#) NC Link is selected. Please see the PIOS documentation for information on this Parameter and its use.
- **Score blade RPM** - This option will appear when a [PIOS](#) NC Link is selected. Please see the PIOS documentation for information on this Parameter and its use.
- **Output Room # for Cabinet #** - This option will appear when a [PNL File](#) NC Link is selected. This option allows you to Output the Room number for the Cabinet Number.
- **Add Room & Cabinet Names** - This option will appear when a [PNL File](#) NC Link is selected. This option allows you to add these two additional fields to the PNL file.
- **Output Board Database** - Determines whether or not to output the Board database with the [Schelling HPO](#) Saw Link.
- **Custom Text F10 - F29** - Defines the content of the custom text fields with the [Schelling HPO](#) Saw Link.
- **Remove Line/Position** - Determines whether to remove the line/position with the [Schelling HPO](#) Saw Link.
- **Banded Part Size** - Determines whether to show final part size with or without banding.
- **Label Directory** - This option will appear when a [SCM MaestroCUT](#) NC Link is selected. The directory, on the machine, where label information is found.
- **3rd Phase Trim** - Allows you to establish if you Trim your 3rd phase cuts with the [SCMI Sezione](#) and [SCMI WinCut](#) Saw Link.
- **Output Drill Programs** - Allows you to output drill programs with the [SCMI Sezione](#) Saw Link.
- **Clamp Reach** - This option will appear when a [SCMI WinCut](#) NC Link is selected. This option allows you specify the Clamp Reach of the Saw.
- **List Path Prefix** - This option will appear when a [SCMI WinCut](#) NC Link is selected. This option allow you to insert the path to your List files location into the output file.
- **Reverse Cut Order** - This option will appear when a [SCMI WinCut](#) NC Link is selected. This option allows you to reverse the cut order for Offcuts.
- **3rd Phase Incremental** - This option will appear when a [SCMI x915](#), [SCMI z32](#), or [SCMI z45](#) NC Link is selected.
- **Use Material ID as Name** - This option will appear when a [SCMI x915](#), [SCMI z32](#), or [SCMI z45](#) NC Link is selected. Will use the Material ID for the Material Name.
- **Run # as File Name** - This option will appear when a [SCMI x915](#), [SCMI z32](#), or [SCMI z45](#) NC Link is selected. Will use the Run Number as the file name.



- **Multiply Output by** - This option will appear when a [SCMI x915](#) NC Link is selected. All output for this saw must be in whole numbers. Depending on the Unit of Measure the control is expecting the multiplier may need to be changed.
- **Draw\_1 to match Desc1** - This option will appear when a [Selco XML](#) NC Link is selected. This option allows you to specify that the Draw\_1 attribute of the Part record should have identical output to the Desc1 attribute
- **Ignore Primary Edge** - This option will appear when a [Selco XML](#) NC Link is selected. This option allows you to specify that the NC output should ignore the primary edge assignment when determining where to place edge banding information. When set to no, it uses the primary edge to determine where to start banding and move from there. When set to yes, it uses Cabinet Visions default edge
- **Omit Drops** - This option will appear when a [Selco XML](#) NC Link is selected.
- **Long material names** - This option will appear when a [Selco XML](#) NC Link is selected.
- **Capital L in Cutlist** - This option will appear when a [Selco XML](#) NC Link is selected.
- **Swap BrdCode for BrdInfo** - This option will appear when a [Selco XML](#) NC Link is selected.
- **DrpInfo = A Minimum Width** - This option will appear when a [Selco XML](#) NC Link is selected.
- **DrpInfo = A Minimum Length** - This option will appear when a [Selco XML](#) NC Link is selected.
- **Split XML Output** - This option will appear when a [Selco XML](#) NC Link is selected. Creates 1 XML file per pattern.
- **File name in CutList Line** - This option will appear when a [Selco XML](#) NC Link is selected.

The following Selco XML parameters can be populated with information from the Parts table, Materials Table and the PartRunData table. Information about each of these can be found in the [psnc-ac-cv-cw-nc & +Label Databases](#) topic.

To define these parameters values, note the table you want the information from and then note the name of the Field name that you wish to use. To add the information to Draw\_1 use the following syntax {Table Name.Field Name}. Example Image name would be written as {PartRunData.ImageName}

Any field with a value of 0 will output a blank.

- **Draw\_1** - This option will appear when a [Selco XML](#) NC Link is selected. See explanation above.
- **Draw\_2** - This option will appear when a [Selco XML](#) NC Link is selected. See explanation above.
- **Draw\_3** - This option will appear when a [Selco XML](#) NC Link is selected. See explanation above.
- **Draw\_4** - This option will appear when a [Selco XML](#) NC Link is selected. See explanation above.
- **Draw\_5** - This option will appear when a [Selco XML](#) NC Link is selected. See explanation above.
- **InfoDraw\_1** - This option will appear when a [Selco XML](#) NC Link is selected. See explanation above.
- **InfoDraw\_2** - This option will appear when a [Selco XML](#) NC Link is selected. See explanation above.
- **InfoDraw\_3** - This option will appear when a [Selco XML](#) NC Link is selected. See explanation above.
- **InfoDraw\_4** - This option will appear when a [Selco XML](#) NC Link is selected. See explanation above.

### Restrictions - Strips Group

- **Min Strip Length** - Allows you to specify the Minimum Strip Length your Saw can handle.

All

Properties

Blade

Booking

Label IT - Label

Label IT - Image

Label IT - Machine Settings

Optimization - Complexity

Optimization - Sorting

Parameters

Restrictions - Strips

Min Strip Length0

Max Strip Length144

Min Strip Width0

Max Strip Width144

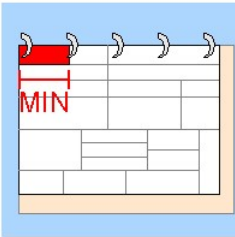
Restrictions - Z-Cuts

Restrictions - Sheet Size

Speed

Times

Trim



- **Max Strip Length** - Allows you to specify the Maximum Strip Length your Saw can handle.

All

Properties

Blade

Booking

Label IT - Label

Label IT - Image

Label IT - Machine Settings

Optimization - Complexity

Optimization - Sorting

Parameters

Restrictions - Strips

Min Strip Length0

Max Strip Length144

Min Strip Width0

Max Strip Width144

Restrictions - Z-Cuts

Restrictions - Sheet Size

Speed

Times

Trim



- **Min Strip Width** - Allows you to specify the Minimum Strip Width your Saw can handle.

All

Properties

Blade

Booking

Label IT - Label

Label IT - Image

Label IT - Machine Settings

Optimization - Complexity

Optimization - Sorting

Parameters

Restrictions - Strips

Restrictions - Z-Cuts

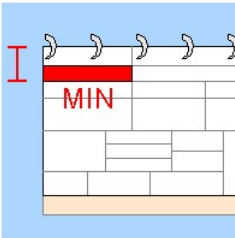
Restrictions - Sheet Size

Speed

Times

Trim

|                  |     |
|------------------|-----|
| Min Strip Length | 0   |
| Max Strip Length | 144 |
| Min Strip Width  | 0   |
| Max Strip Width  | 144 |



- **Max Strip Width** - Allows you to specify the Maximum Strip Width Your Saw can handle.

All

Properties

Blade

Booking

Label IT - Label

Label IT - Image

Label IT - Machine Settings

Optimization - Complexity

Optimization - Sorting

Parameters

Restrictions - Strips

Restrictions - Z-Cuts

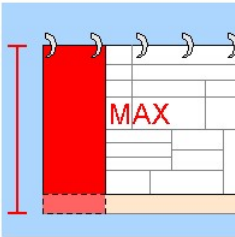
Restrictions - Sheet Size

Speed

Times

Trim

|                  |     |
|------------------|-----|
| Min Strip Length | 0   |
| Max Strip Length | 144 |
| Min Strip Width  | 0   |
| Max Strip Width  | 144 |



**Restrictions - Z-Cuts Group**

- **Min Z-Cut Width** - Allows you to specify the Minimum Z-Cut Width your Saw can handle.

All

Properties

Blade

Booking

Label IT - Label

Label IT - Image

Label IT - Machine Settings

Optimization - Complexity

Optimization - Sorting

Parameters

Restrictions - Strips

Restrictions - Z-Cuts

Min Z-Cut Width0

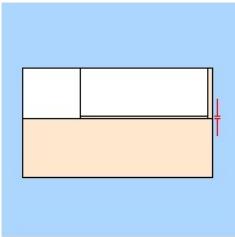
Max Z-Cut Length999

Restrictions - Sheet Size

Speed

Times

Trim



- **Max Z-Cut Length** - Allows you to specify the Maximum Z-Cut Length your Saw can handle.

All

Properties

Blade

Booking

Label IT - Label

Label IT - Image

Label IT - Machine Settings

Optimization - Complexity

Optimization - Sorting

Parameters

Restrictions - Strips

Restrictions - Z-Cuts

Min Z-Cut Width0

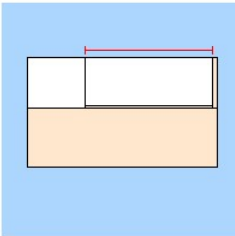
Max Z-Cut Length999

Restrictions - Sheet Size

Speed

Times

Trim



**Restrictions - Sheet Size Group**

- **Max Sheet Width** - Allows you to specify the Maximum Sheet Width your Saw can handle.
- **Max Sheet Length** - Allows you to specify the Maximum Sheet Length your Saw can handle.

**Speed Group**

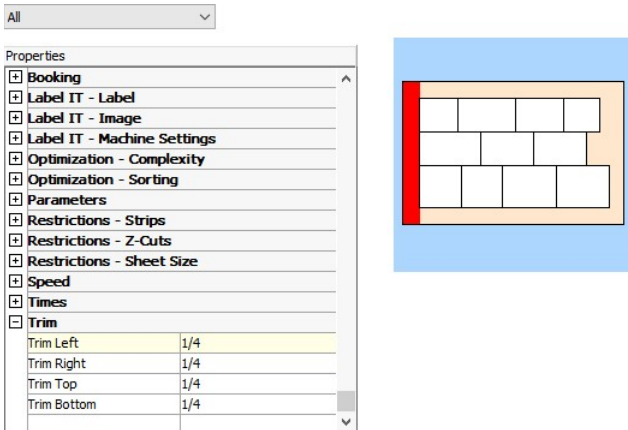
- **Rip Cuts in/sec** - Allows you to enter an amount inches per second for your Saw to make a Rip cut.
- **Head Cuts in/sec** - Allows you to enter an amount in inches per second for your Saw to make a Head cut.
- **Rip Cut Retrack in/sec** - Allows you to enter an amount in inches per second for your Saw to retract back to make its next Rip cut.
- **Head Cut Retrack in/sec** - Allows you to enter an amount in inches per second for your Saw to retract back to make its next Head cut.

Times Group

- **Panel Load** - Allows you to specify a time in seconds it takes to load a single sheet onto your Saw.
- **Pattern Setup** - Allows you to specify a time in seconds it takes to Setup to cut a Pattern on your Saw.
- **2nd Phase Turns** - Allows you to specify a time in seconds it takes to make a 2nd Phase turn on your Saw.
- **3rd Phase Turns** - Allows you to specify a time in seconds it takes to make a 3rd Phase turn on your Saw.
- **Booking Overhead** - Allows you to specify a time in seconds that Booking adds to processing on your Saw.
- **Saw Cost /hr** - Allows you to specify an hourly cost to run your Saw.
- **Overmake Penalty** - Allows you to specify a time in seconds to allow for each over make part in your Patterns.

Trim Group

- **Trim** - Specifies the amount to Trim off (Dust Cut) your Material.



Sheet Router

This Machine Type is specifically for Nested based Routers. Following are the Output Tab options for Sheet Routers:

Output Tab - Sheet Router

To change information in these fields, highlight the field name at the left. Then click on the data at the right and enter the new values. Some of the options below require that you select a Machine prior to seeing the option, Small Parts is an example, and it appears on the second tab after selection of a machine.

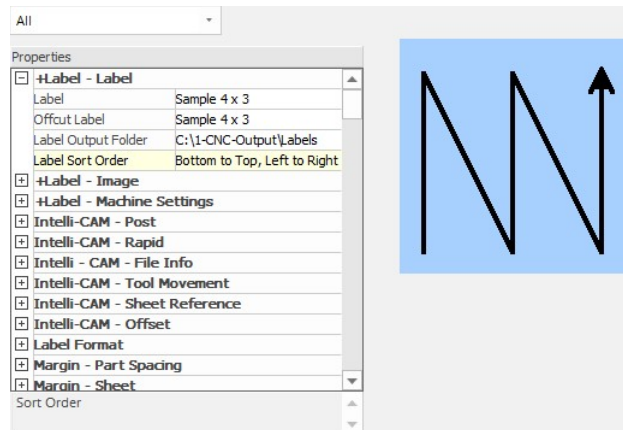
+Label - Label Group

- **Label** - If you are using +Label select the Label Style you want to use with the selected Machine. Use the [+Label Label Designer](#) to create your own Label styles for selection here.
- **Offcut Label** - If you are using +Label select the Label Style that you want to use for Offcuts.
- **Label Output Folder** - Select the Default location for your +Label files to be placed.
- **Label Sort Order** - Select the Default Sort Order for your Labels to print at the +Label station.

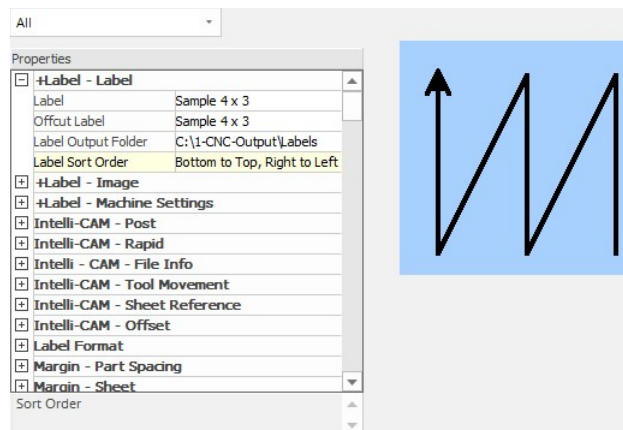


Sort Order is calculated based on the Part's Origin. The quickest way to identify the Part's Origin in +Label is to note the Part Number's location. The Part Number's location is at the Part Origin.

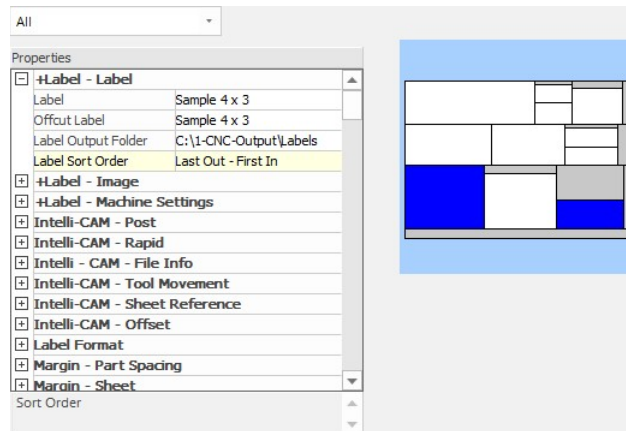
- **Bottom to Top, Left to Right** - Orders the parts bottom to top, left to right.



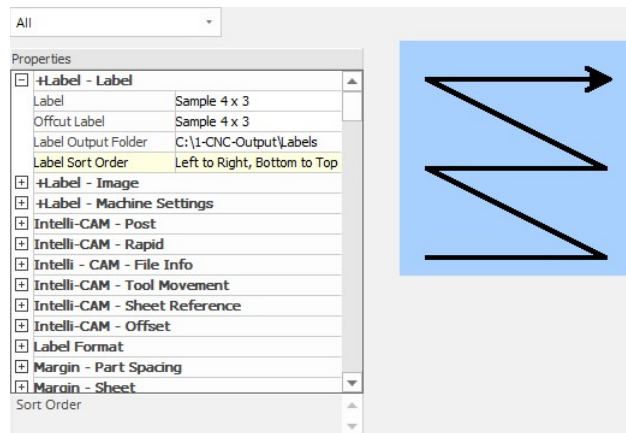
- **Bottom to Top, Right to Left** - Orders the parts bottom to top, right to left.



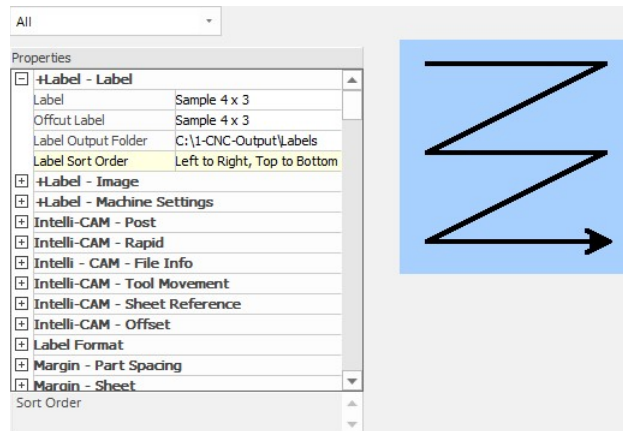
- **Last Out - First In** - This order is a common order when cutting parts with a beam saw. All rips in a sheet are done before any cross cuts. The last rip cut is the first rip to feed in for cross cuts. All cross cuts for that rip are done before any 3rd phase cuts. The last cross cut will be the first to feed for 3rd phase cuts.



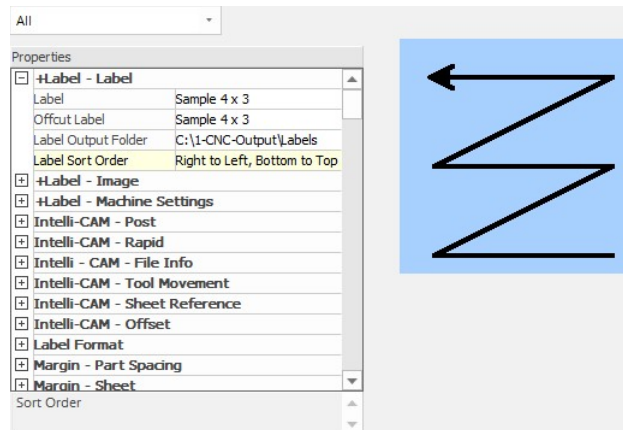
- **Last Out - First In - Reverse** - Reverse of Last Out - First In above.
- **Left to Right, Bottom to Top** - Orders the parts left to right, bottom to top.



- **Left to Right, Top to Bottom** - Orders the parts left to right, top to bottom.

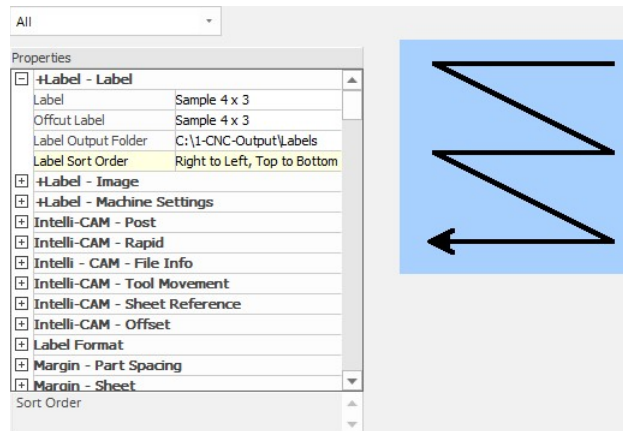


- **Right to Left, Bottom to Top** - Orders the parts right to left, bottom to top.

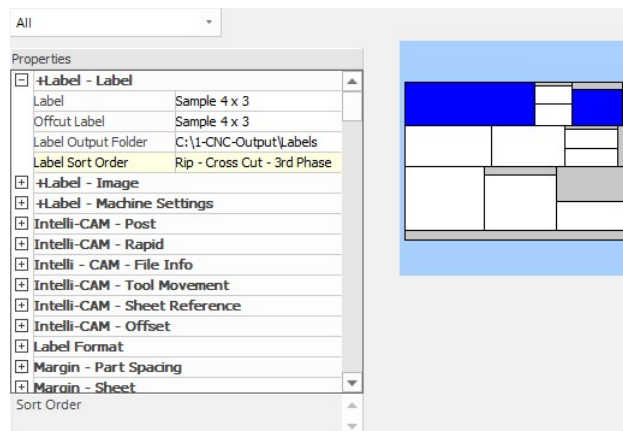


- **Right to Left, Top to Bottom** - Orders the parts right to left, top to bottom.

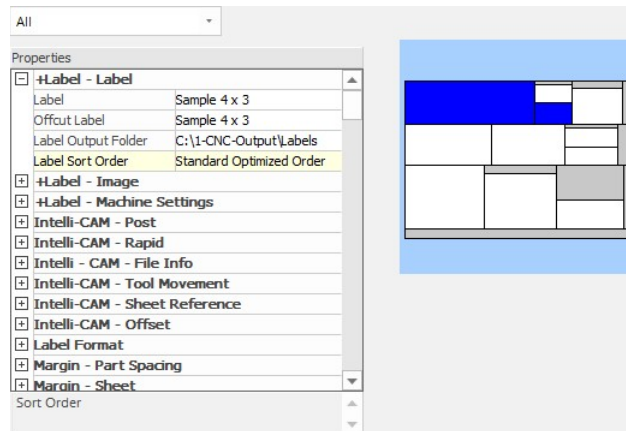




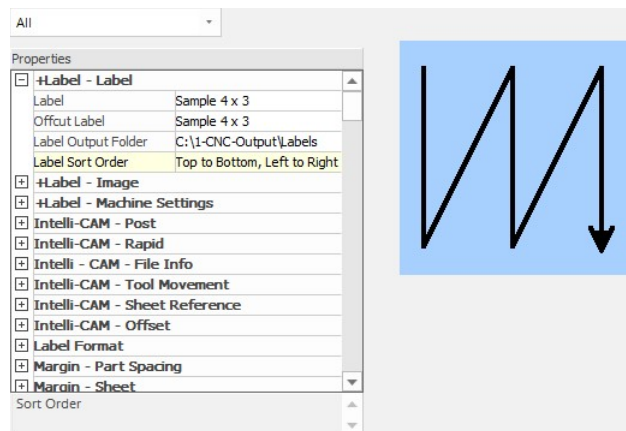
- **Rip - Cross Cut - 3rd Phase** - This order is a common order when cutting parts with a table saw. Every part in the first rip will be processed before going to the second rip. All cross cuts will be processed in the rip prior to processing any 3rd phase cuts.



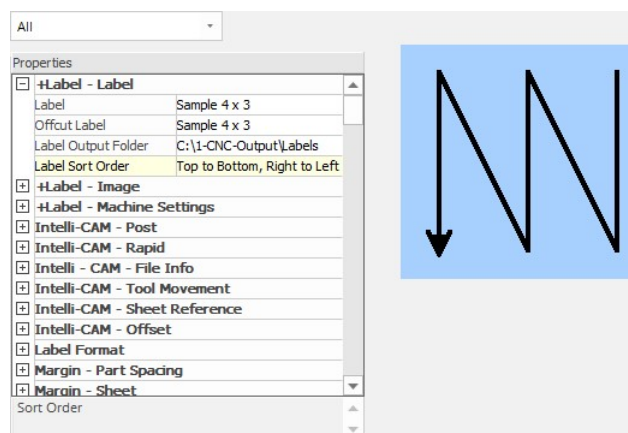
- **Standard Optimized Order** - This order will match the order from the S2M Center Optimizer. Every part in the first rip will be processed before going to second rip. 3rd phase cuts will immediately follow the preceding cross cut.



- 
- **Top to Bottom, Left to Right** - Orders the parts top to bottom, left to right.



- **Top to Bottom, Right to Left** - Orders the parts top to bottom, right to left.



### +Label - Image Group

- **Image Type** - This field allows you to control the Part image type that is output for 3rd Party Labeling programs. Following are the options:

- 16 Color Bitmap
- 24 bit Color Bitmap
- 256 Color Bitmap
- EMF
- GIF
- Jpeg
- Monochrome Bitmap
- PNG
- ZPL Automatic
- ZPL Condensed
- ZPL Neat
- ZPL Raw
- ZPL Standard

**Pixel Width** - Allows you to control the Output Image's Width in Pixels. This value is typically a scaled match to the Label Graphic size. For example if you design a 4" x 3" Label Graphic the Width specified here would be 400.

**Pixel Height** - Allows you to control the Output Image's Height in Pixels. This value is typically a scaled match to the Label Graphic size. For example if you design a 4" x 3" Label Graphic the Height specified here would be 300.

**Image Output Folder** - This field allows you to Browse to and select the location to output the Label Graphic.

**Full Image Path** - Specifies whether simply the Graphic name or the entire path to the Graphic name for Real-Time Labels is included in the output.

**Images for Offcuts** - Specifies if you need to output a Real-Time Label image for Offcuts. Offcut Labeling is not supported by all Manufacturers, consult with your Machine Manufacturer to find out if it is supported.

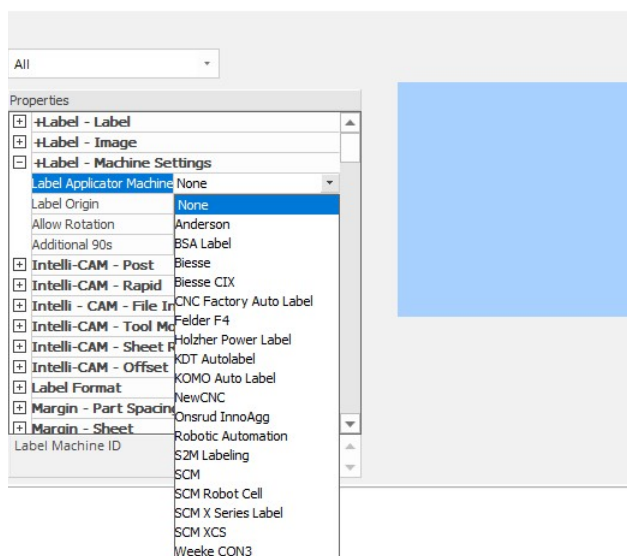
**ZPL Template Folder** - Specifies the folder where the ZPL Template resides. This field is only active if the Image Type is set to one of the ZPL options. See the [Real-Time Labeling](#) topic for more information on the .ZPL file format.



There is a sample ZPL Template file as well as a handy ZPL Label Preview application in the Support Folder on the Installation DVD.

### +Label - Machine Settings Group

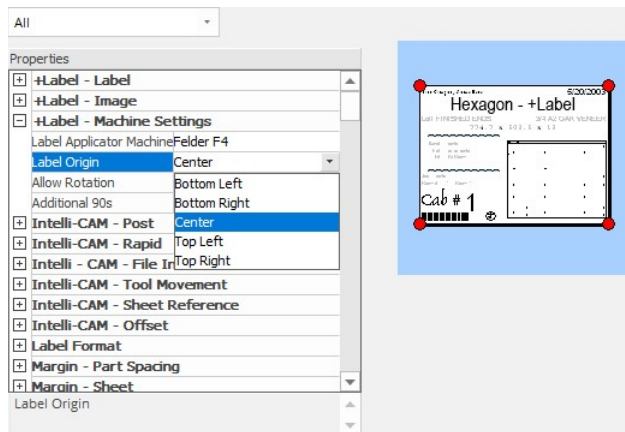
- **Label Applicator Machine** - If your Router has a built in Label Applicator this option allows you to choose which one you will be using so the data is properly output for it. See the [Real-Time Labeling](#) topic for more information on the Label Applicator Machines.



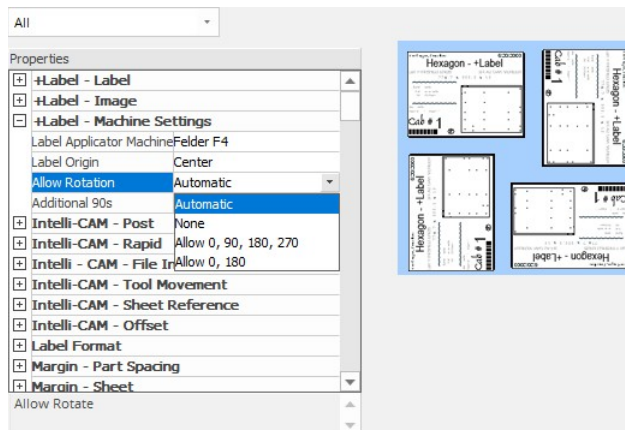
- Anderson - No +Label Machine Parameters.
- BSA Label - No +Label Machine Parameters.
- Biesse - Following are the +Label - Machine Parameters available for a Biesse Label Applicator Machine.
  - Label X Max
  - Label X Min
  - Label Y Max
  - Label Y Min
  - Remove STRCUST Slash
- Biesse CIX - Following are the +Label Machine Parameters available for a Biesse CIX Label Applicator Machine.
  - Label X Max
  - Label X Min
  - Label Y Max
  - Label Y Min
  - Set ROT Value
  - Remove STRCUST Slash
  - Worklist Folder

- Capitalize ID - This allows you to specify that the ID attribute of the Part node should be capitalized.
  - Label Layout - This allows you to specify a label layout that will create a new 'Labeling' node. If left blank, this will not create the new node
  - Names in English - This allows you to specify that certain XML node names are in English instead of Italian. Currently only effects the Etichette XML node.
  - Set CRN Value
  - Offcut Counter as ID
  - Use Part.ID as Index
  - Print Drop
- CNC Factory Auto Label - Following are the +Label Machine Parameters available for a CNC Factory Auto Label Applicator Machine.
  - Mirror X Position
  - Mirror Y Position
  - Flip X and Y
- Felder F4 - No +Label Machine Parameters.
- Holzher Power Label - No +Label Machine Parameters.
- KDT Autolabel
  - Use Sheet Origin
- Komo Auto Label - Following are the +Label Machine Parameters available for a Komo Auto Label Applicator Machine.
  - Compression Value
  - Increment Value
  - Machine Image Path
- NewCNC - No +Label Machine Parameters.
- Onsrud InnoAgg - Following are the +Label Machine parameters available for a Onsrud InnoAgg Label Applicator Machine.
  - PLO Folder
- Robotic Automation - No +Label Machine Parameters.
- S2M Labeling - No +Label Machine Parameters.
- SCM - Following are the +Label Machine parameters available for a SCM Label Applicator Machine.
  - Mirror Y Position
  - Mirror X Position
  - Add Pipe Symbols
  - Single RLL File
- SCM Robot Cell - No +Label Machine Parameters.
- SCM X Series Label
  - Label Template Name
  - Label Image Path
  - Mirror X Position
  - Mirror Y Position
- SCM XCS
  - Label Directory
  - Label Template

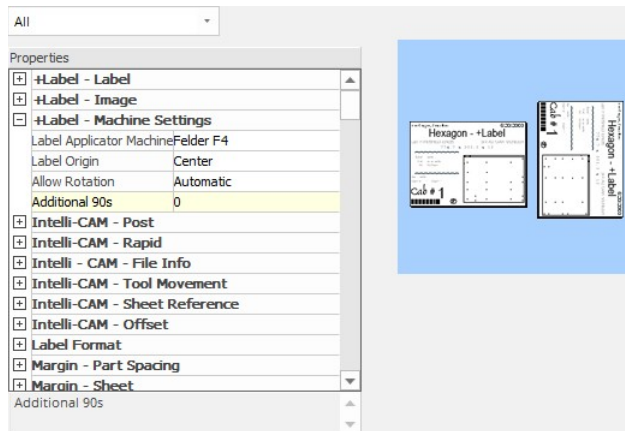
- Weeke CON3 - No +Label Machine Parameters.
  - Non-Stiles Machine - When turned on, a simplified .ptx file will be created when outputting the other label applicator files (usually used for Intellistore ingest).
  - Sub Folder - When a non-empty value is placed here, all files for the Weeke Con3 Label Applicator will be placed in a sub folder in the output directory (this should be the folder name, not an entire path)
- **Label Origin** - Allows you to set the Origin for the Label. This setting is used when utilizing an on-board Label Applicator and determines where the X/Y Position of the Label is based on the selected Label's size.



- **Allow Rotation** - Specifies whether Rotation is allowed and on what increments.



- **Additional 90s** - The S2M CENTER has a Default orientation it outputs a Label Image to the Printer. However, the Printer may be oriented in a fixed position that is different than the default output. This setting allows you to change the Graphic orientation to match the fixed position of the Printer. The value can be 1 (90 degrees), 2 (180 degrees), or 3 (270 degrees).

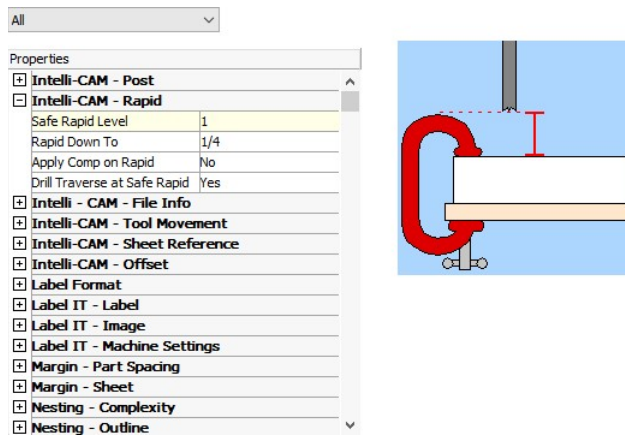


### Intelli-CAM - Post Group

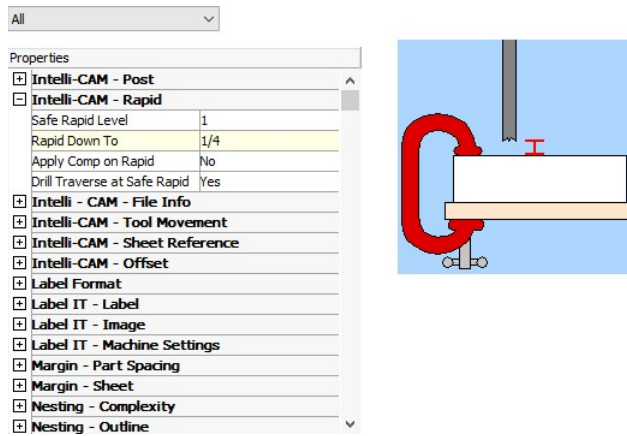
- **Post Processor** - Utilize the Browse  button in the field to set the Post and its location for your Machine.

### Intelli-CAM - Rapid Group

- **Safe Rapid Level** - The safe distance above the material the tool will move to the next operation.

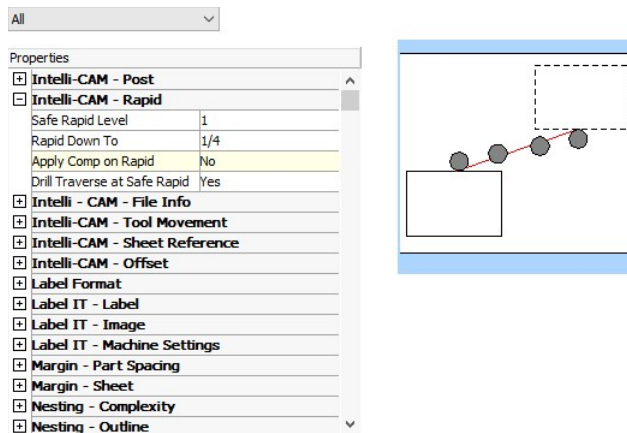


- **Rapid Down To** - When drilling you can retract to safe level or this level. The point where the rapid, between operation movements, of the machine slows to or approaches the machining speed.



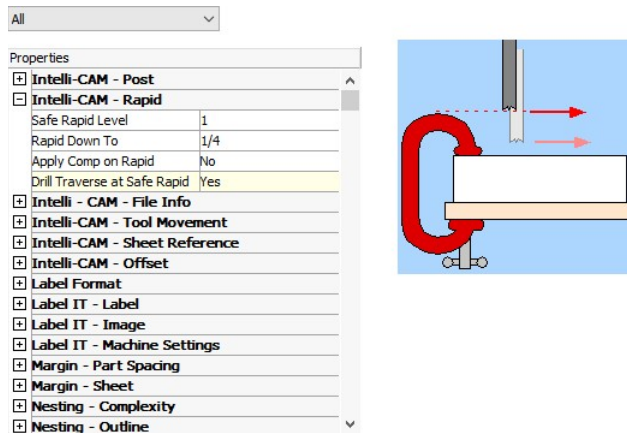
The Safe Rapid and Rapid Down To values must be different if even only slightly. If the values are the same you will get either Errors or at the very least undesirable results.

- **Apply Comp on Rapid** - **Only works on some machines.** This value instructs the Machine to perform its Cutter Compensation during the safe rapid move.



- **Drill traverse at safe rapid** - See Safe Rapid





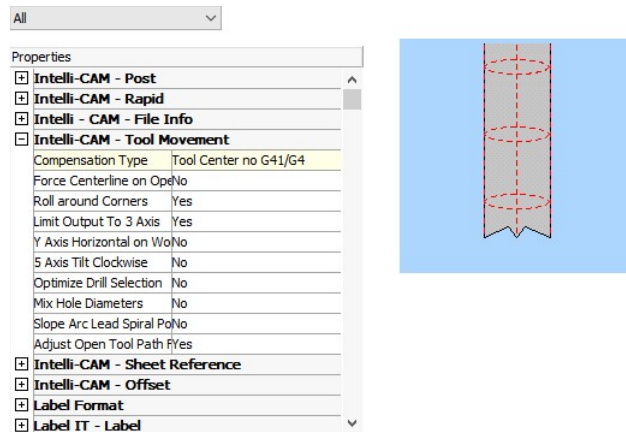
### Intelli-CAM - File Info

- **File Extension** - Enter the File Extension your Machine requires, here.
- **Save Drawing File** - Setting this to True will create an .ARD file for each G-Code file created for any given Job. ARD files are native ALPHACAM Router Drawing files.
- **Output Comment Character(s)** - The characters representing comments in the G-Code output.

### Intelli-CAM - Tool Movement

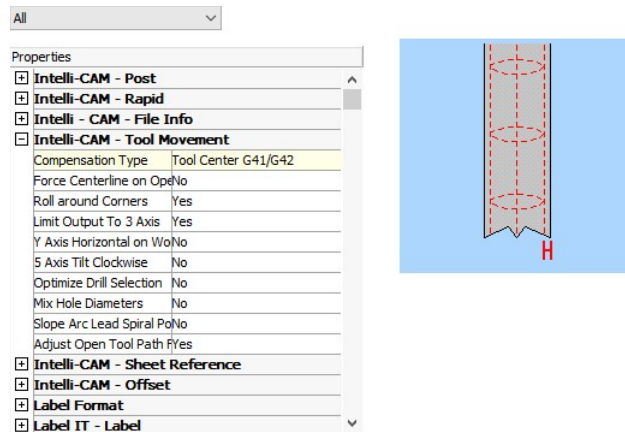
- **Compensation Type** - (Compensation refers to an adjustment in the path of the tool to compensate for a change in tool size due to tool wear. This is referred to as Tool Comp. or Tool Compensation). See the document on [Output Path](#) for more information.

**Tool Center No G41/G42** - Tool Comp is accommodated strictly through the tool path. It is offset from the actual cut path and doesn't look to a register in the machine's control to adjust the path. The outline path of the part is larger than the part's geometry by the tool radius. This path is to be followed by the center of the tool.

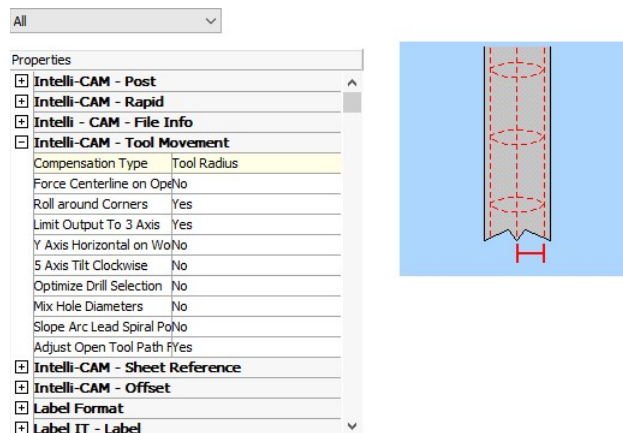


- **Tool Center G41/G42** - The outline path is identical to Tool Center no G41/G42. The difference is it triggers to look up tool compensation values on the machine's control. Tool comp is important when you start sharpening tools. When you sharpen a tool it gets smaller. Tool comp allows you to inform the control that the bit has changed size and to compensate for the change. For example: If you use a 1/2" router bit (.25 radius) and as a result of the bit being sharpened it has a new size (.23 radius). You need to account for this tool size change somewhere.

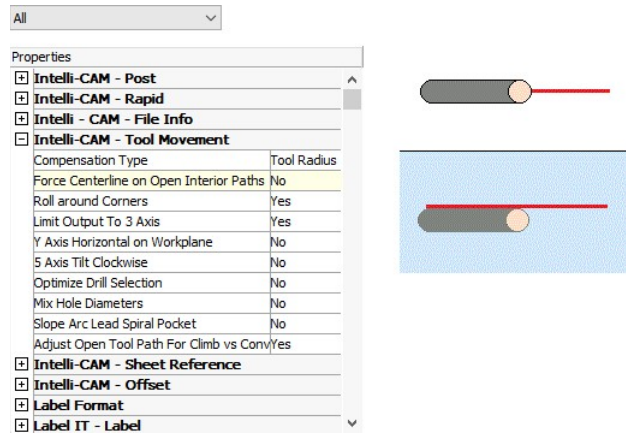
- With Centerline - Tool Center no G41/G42 you would go back to the Tool Catalog in the S2M Center and enter the new size of the tool (.46 diameter) and re-output the program. Otherwise your part would cut out larger than it should be.
- With Centerline - Tool Center G41/G42 you can enter a value of .02 at the machine's control ( $.25 - .23 = .02$ ) and the machine's control will offset the tool path by .02 and correctly cut out the part.
- It boils down to user preference whether the sharpened tool size is accounted for in the S2M Center or at the machine's control although you have advantages at the machine since if you need to swap bits you can compensate for different sizes on the fly rather than having to re-output from the S2M Center.



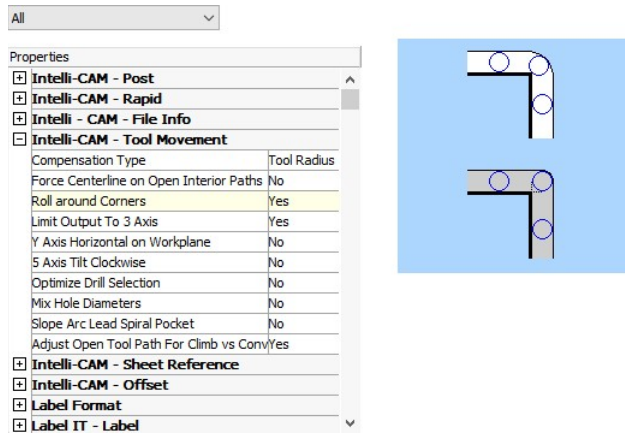
- **Tool Radius** - This path follows the outline of the part exactly. Thus if the router were to follow the path centerline, the part would be too small. This path must be compensated for at the machine's control. If you are using a 1/2 bit, the machine's control will need an offset of .25 (radius). If you sharpen the bit you would need to change the comp value at the machine's control to (.23). In either case the machine's control will offset the path of the tool by the tool radius resulting in the part being milled to the correct size.



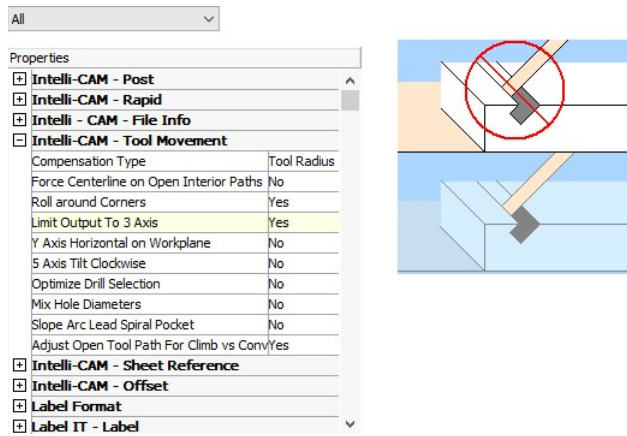
- **Force Centerline on Open Interior Paths** - Forces open Route paths inside the Part to centerline.



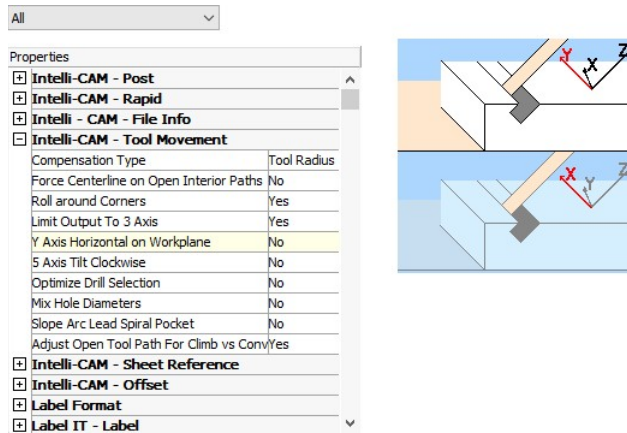
- **Roll Around Corners** - Breaks 90 degree corners into a tiny radius. For many controllers this will give them a hint to slow down before they reach the corner.



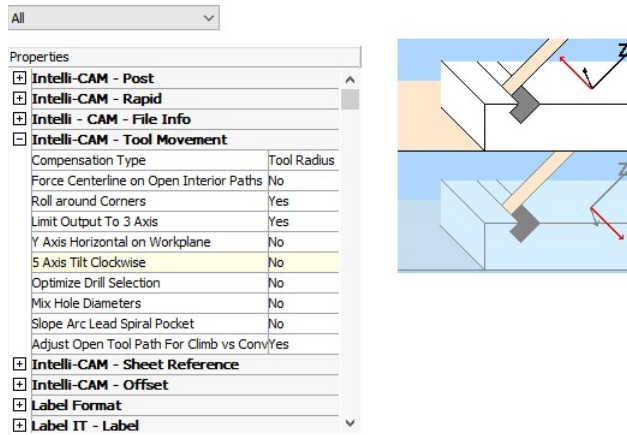
- **Limit Output to 3 Axis** - Do not set this to No unless your machine has C-Axis capabilities.



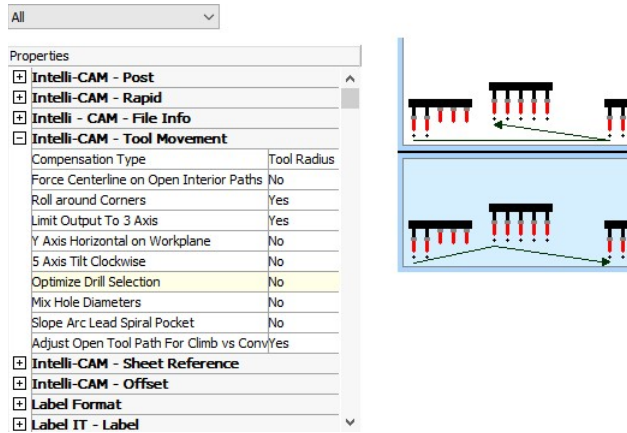
- **Y Axis Horizontal on Workplane** - Determines the orientation of the Y-Axis on the machining workplane.

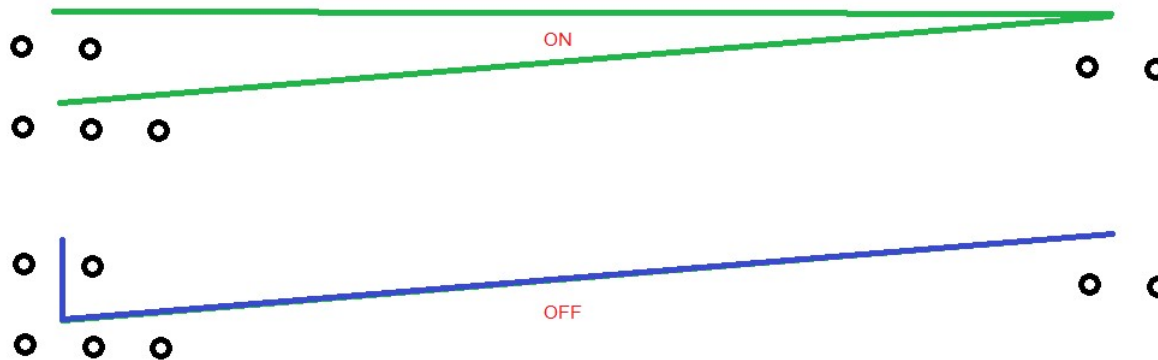


- **5 Axis Tilt Clockwise** - Establishes the 5 Axis tilt direction.

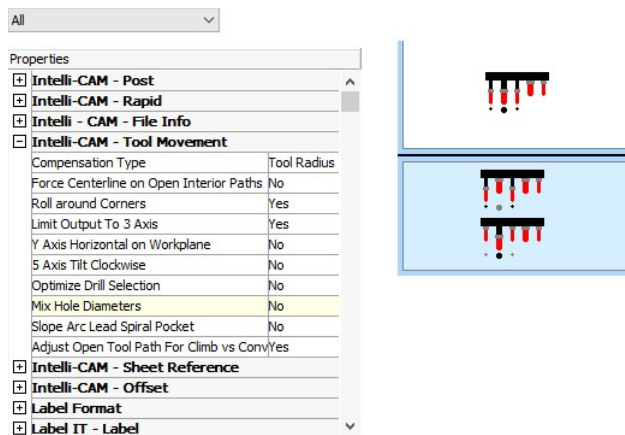


- **Optimize Drill Selection** - Some older machines and entry level machines can take a long time to change spindle drops on a gang drill head. This option optimizes the tool path to minimize the number of gang drill head dropped spindle configurations.

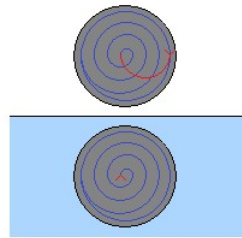
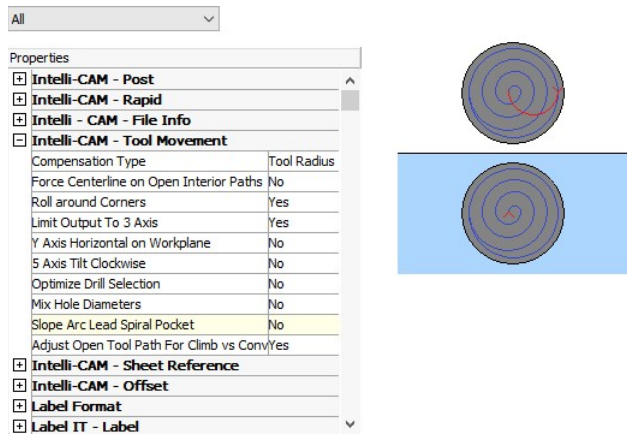




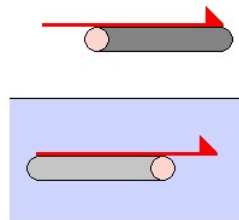
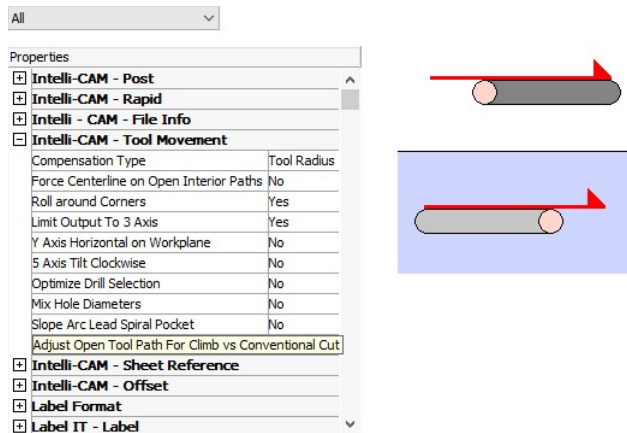
- **Mix Hole Diameters** - If enabled will mix different Drill Diameters into a spindle drop if possible based on spacing.



- **Slope Arc Lead Spiral Pocket** - If enabled will perform a Spiral Lead for Pockets as illustrated.

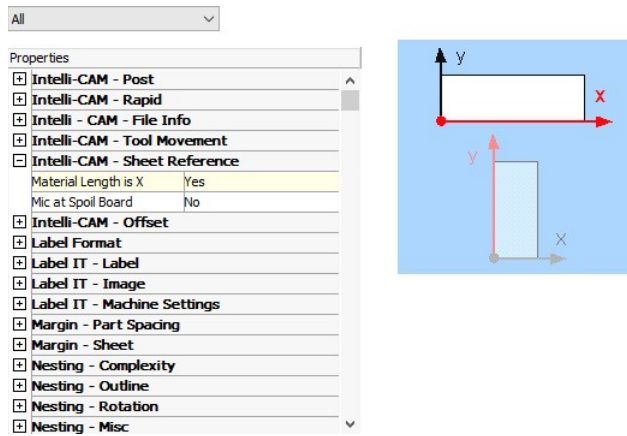


- **Hover Tool Comp on Arc** - This is for controllers that restrict tool comp during an arc. When this option is enabled a line will be added prior to the arc above the material for compensation.
- **Adjust Open Tool Path For Climb vs Conventional Cut** - If enabled will adjust open tool paths to account for Climb versus Conventional cuts.

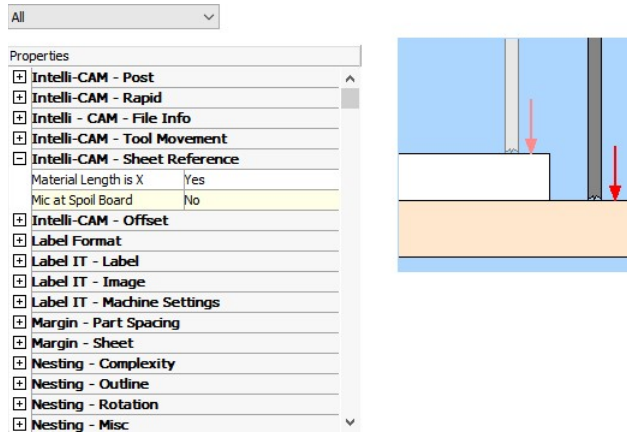


### Intelli-CAM - Sheet Reference Group

- **Material Length is X** - If the machine(s) longer dimension runs along the X-axis, Intelli-CAM rotates the sheet so that the sheets axis matches the machines axis. If you require this option, set the option to TRUE (your machine sets length dimensions along the X axis). A False setting will rotate the material from the normal X-axis for length.



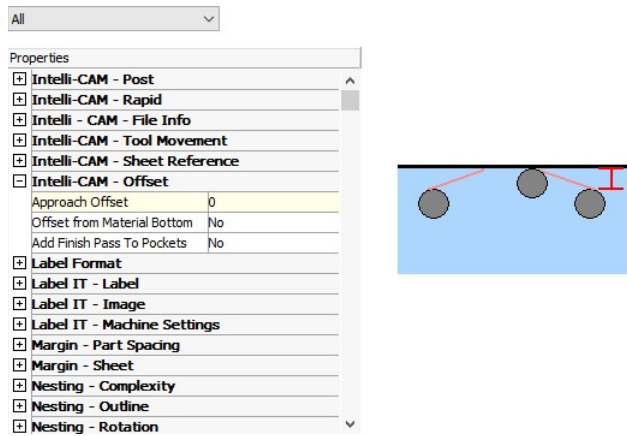
- **Mic at Spoil Board** - Select Yes or No here based on how you Touch Off your Tools on your Machine. A Yes here indicates that you touch your Tools off at the top of the spoil board. A No indicates you touch off your Tools at the Top of your Material.



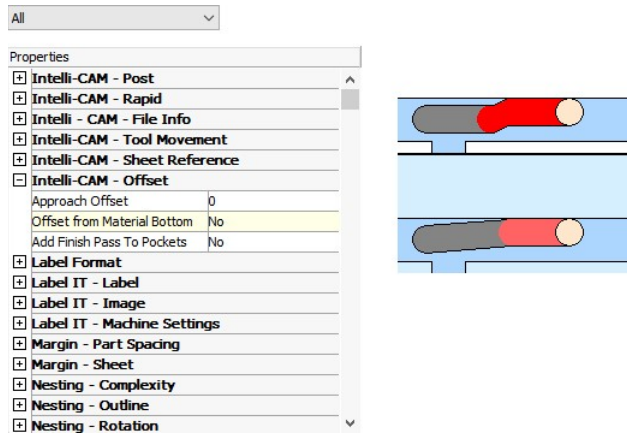
### Intelli-CAM - Offset Group

- **Approach Offset** - This controls the Approach Offset away from the top of the Part Edge on Sloping Leads.

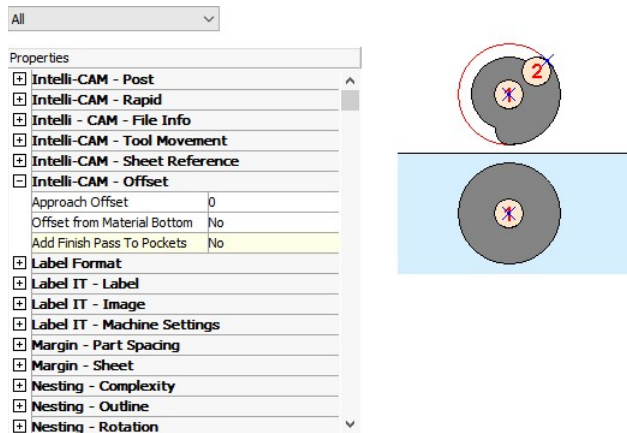




- **Offset from Material Bottom** - This controls the Offset of the Tool away from the Bottom of the Part Edge on a Lead-In. This option can be used to reduce chipping of the Material on the bottom side of your Parts.

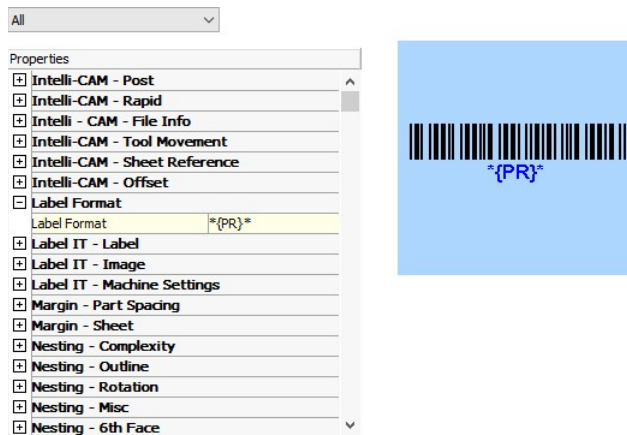


- **Add Finish Pass to Pockets** - This determines whether you would like a Compensated Finish Pass to Pockets. Pockets generated through the S2M Center are Spiral Pockets with a Centerline Path and can not be compensated. You must set this option to Yes if you want a compensated final pass.



### Label Format Group

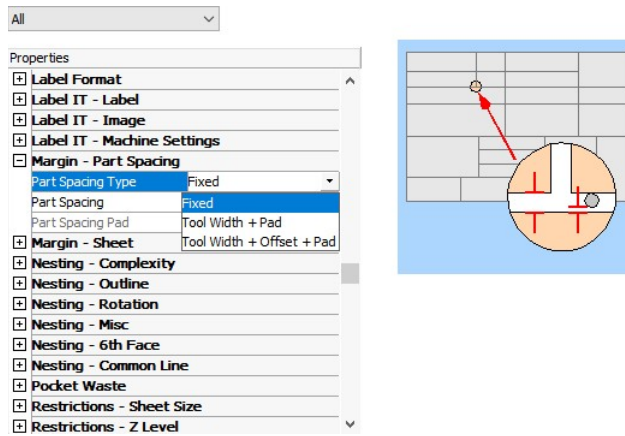
- **Label Format** - Defines the formatting for your barcode as required by your machine and/or bar-code scanner manufacturer.



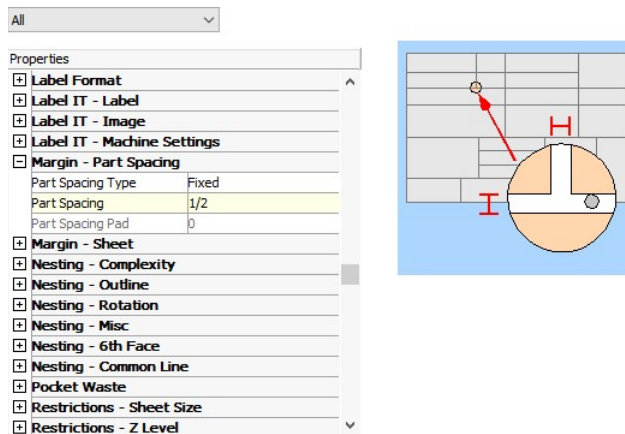
You can use text variables to define your Label Format. Variables include {ZO} for Zone/Run Field and {PR} for Program Name.

### Margin - Part Spacing Group

- **Part Spacing Type** - There are three options to control the space between parts.



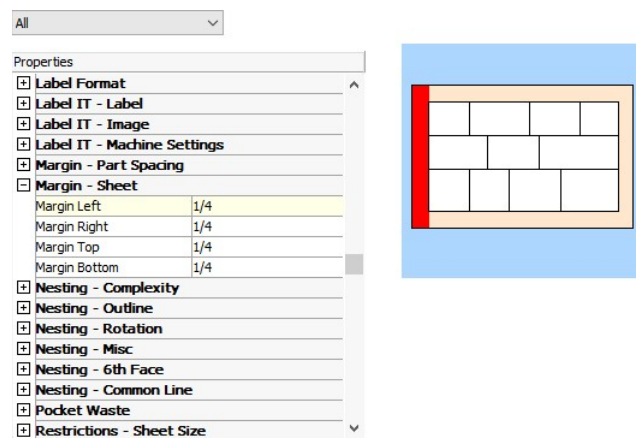
- **Fixed** - A fixed Part Spacing based on the Part Spacing value.
- **Tool Width + Pad** - This option will space the Parts by the Tool's Diameter plus the Part Spacing Pad value.
- **Tool Width + Offset + Pad** - This option will space the Parts by the Tool's Diameter plus the First Pass Offset value plus the Part Spacing Pad value.
- **Part Spacing** - The value for Part Spacing. This value is used depending upon the setting for the Part Spacing Type option.



- **Part Spacing Pad** - This field allows you to set a Pad to your Part Spacing.

### Margin - Sheet Group

- **Margins** - Typically, parts are never placed against the raw edges of a panel. There is a "margin", or small portion of the panel next to the edge that is removed in order to insure a clean part edge. The graphic shows the margin on the four edges of the panel, and requires that you input the amount you desire for this margin.

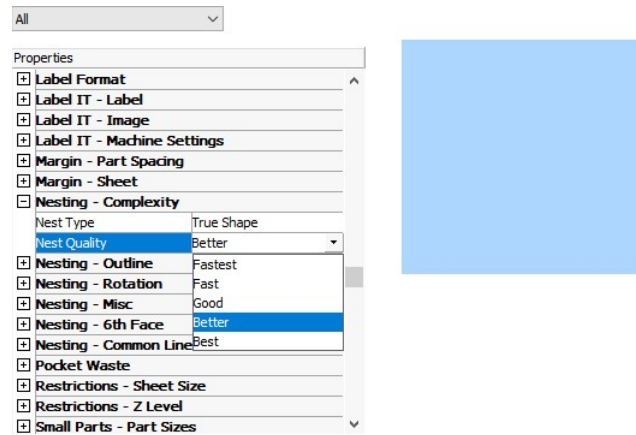


Margin - Offcut Group

- **Margins** - Typically, parts are never placed against the raw edges of a panel. There is a "margin", or small portion of the offcut next to the edge that is removed in order to insure a clean part edge. Here you define the margin on the four edges of the offcut.

Nesting - Complexity Group

- **Nest Type** - You can select either Block or True Shape as your Nest type.
- **Nest Quality** - Controls the Quality of the Optimization/Nesting of your Parts

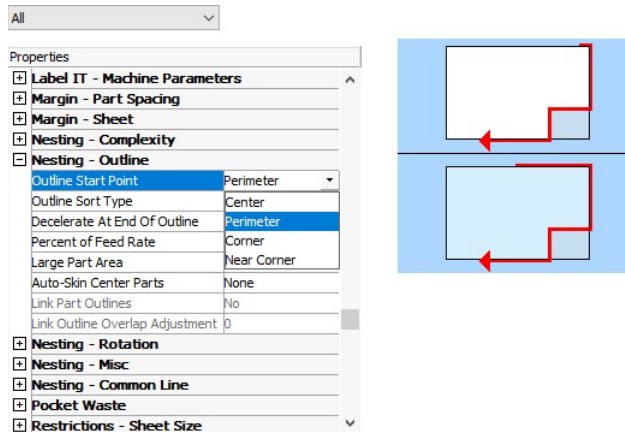


- Fastest - This setting offers the fastest Nesting at the computer, sacrificing Yield.
- Fast - This setting offers a slightly slower speed at the computer improving yield slightly.
- Good - This setting offers a balance between speed at the computer and Yield.
- Better - This offers slower speed at the computer with increased improvement to Yield.

- Best - This offers the slowest speed at the computer with the best Yield results sacrificing speed at the computer.

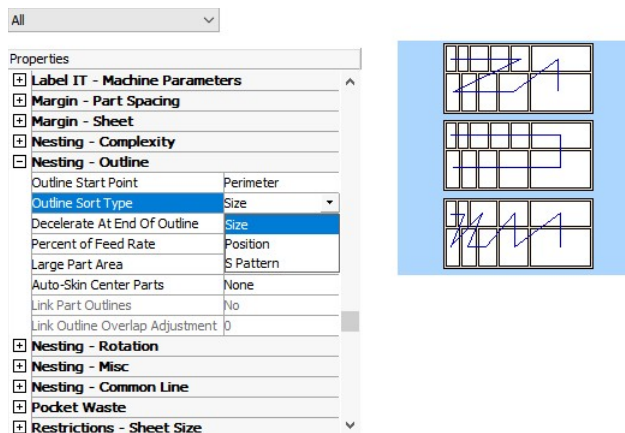
### Nesting - Outline Group

- **Outline Start Point** - This option allows you to specify the Outline start point type.



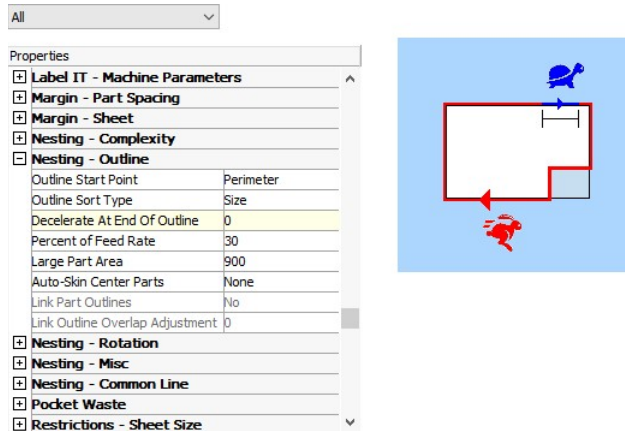
There are three options here:

- **Center** - The standard S2M start point along the mid point of the edge of the Part.
  - **Perimeter** - An innovative S2M Center start point methodology. The concept is cutting the parts from the outside of the sheet first moving towards the center with the point of origin always being towards the center of the sheet. Using this pattern each part is attached to a very large section of the sheet when its outline is cut so that its holding power is not just from the size of the part but the size of all other uncut parts. We have found this to be an excellent method with superior results for minimizing part movement with smaller Parts in most cases not requiring any special small Part Handling (Onion Skins, etc...). There are several other options which are available below when this Outline Start Point type is selected.
  - **Corner** - An alternative start point method for use on Machines where witness marks on along the start/end points are a problem. This method leads in and out at the corners of the part rather than the mid point of the edge.
- **Outline Sort Type** - This option allows you to control the general order which Parts are machined in the Nest.

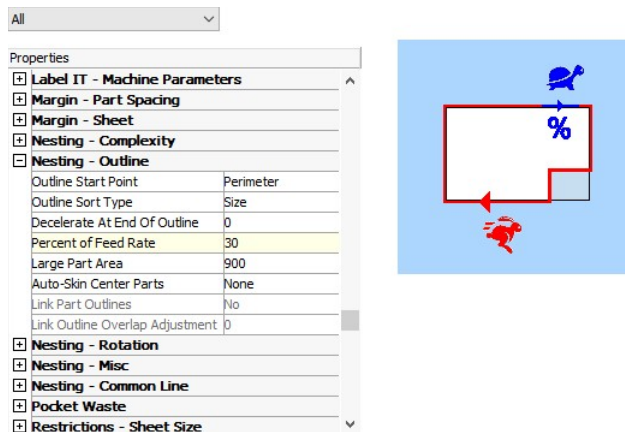


There are three options here:

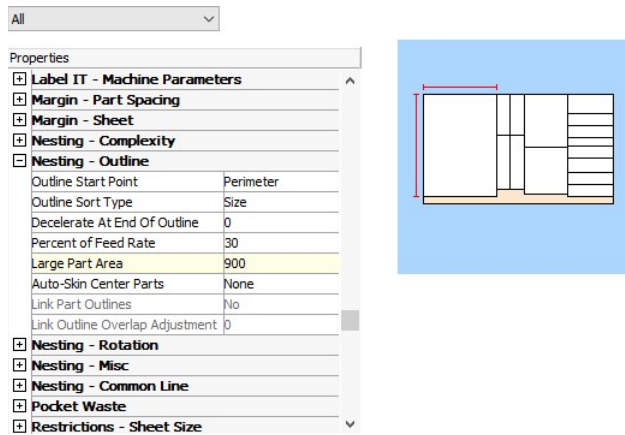
- **Size** - Sorts by size smallest to largest. This is a good strategy to avoid Part movement making sure the smaller Parts are cut first while the maximum vacuum is available.
  - **Position** - Follows a logical sequence around the sheet.
  - **S Pattern** - Follows an S Pattern on the sheet.
- **Decelerate At End Of Outline** - If you want the tool to decelerate at the end of the outline you can set the value (measurement) here for when the deceleration should begin. This can minimize Part shifting on the final release of the Part.



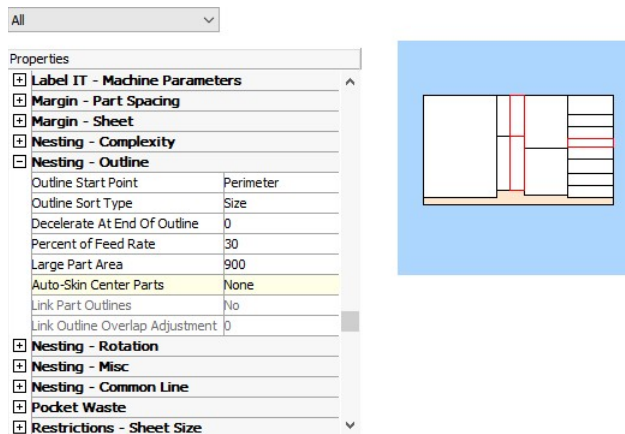
- **Percent of Feed Rate** - The percentage of the current feed rate you would like to use on the decelerate above.



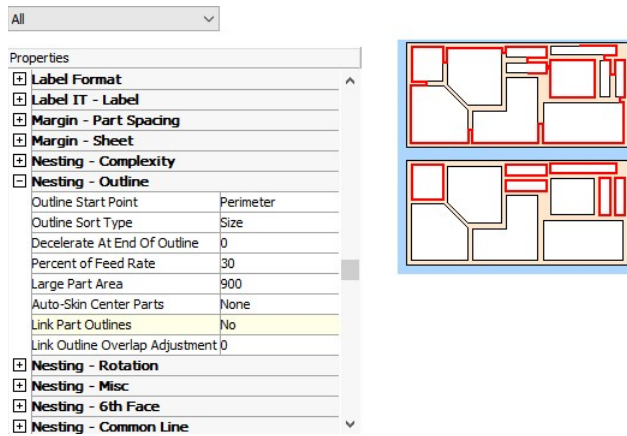
- **Large Part Area** - Once defined, any large parts are removed from this routine, they are cut last since they are large enough to have sufficient hold down in themselves. This will increase the hold down of all other parts in the process.



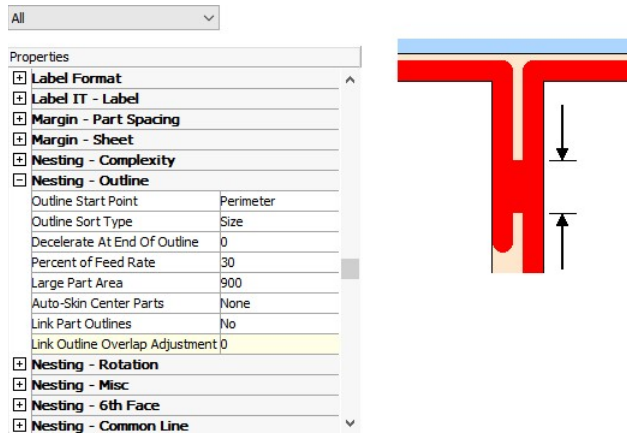
- **Auto-Skin Center Parts** - With this feature on, any small part or very small part will automatically get a return onion skin if it is placed in the center of the sheet. Since the center of the board is cut last these parts have a greater possibility for movement. With this feature on you can have Ramp-In/Out set for all parts types and only those parts susceptible to movement will receive a return onion skin, optimizing both part hold down and overall cycle time.



- **Link Part Outlines** - This option allows you to turn on Part Bridging in your Nests. Bridging is when all Parts within the Nest are cut with a single Lead-In and Lead-Out for the Sheet. This option can improve Sheet cycle time by reducing the number of times the Tool Lead In/Out on each individual Part.



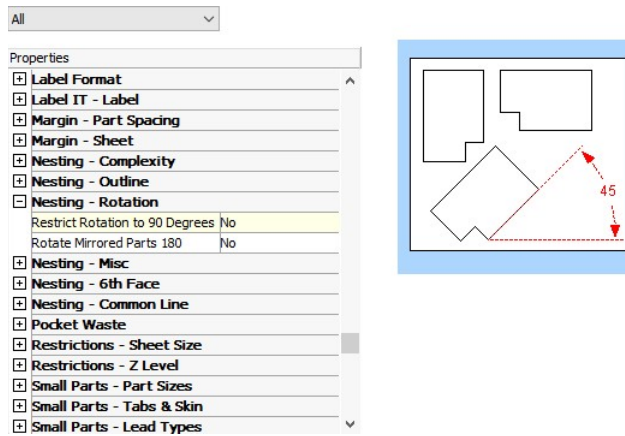
- **Link Outline Overlap Adjustment** - This option allows you to adjust the Overlap where bridges occur.



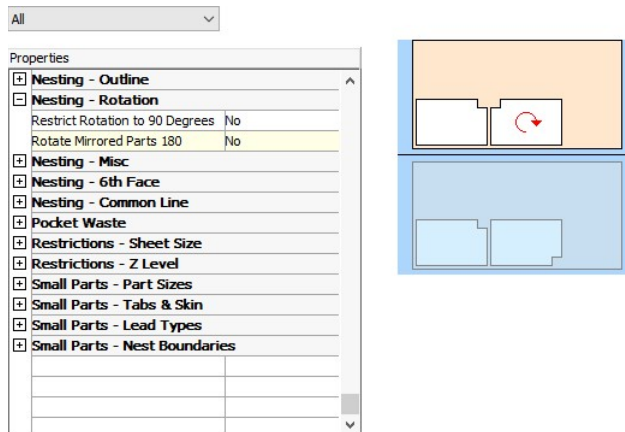
### Nesting - Rotation Group

- **Restrict Rotation to 90 Degrees** - Restricts Part Rotation in TrueShape Nests to 90 degree increments.



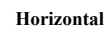


- **Rotate Mirrored Parts 180** - Allows you to determine if Mirrored Parts (Such as Right Ends) are rotate 180 degrees on the sheet. This can help the Machine Operator more easily visually identify Left and Right Ends in the Nest.



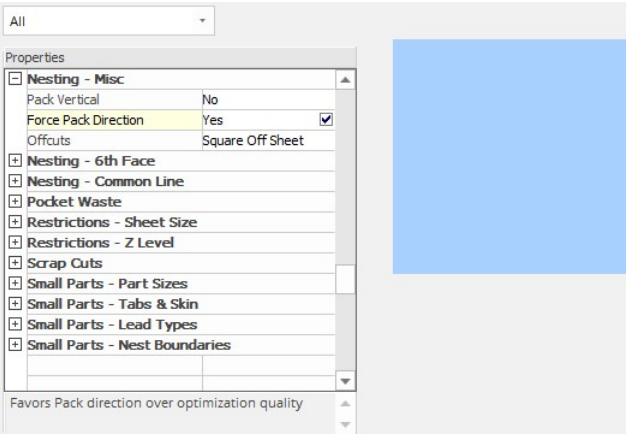
### Nesting - Misc Group

- **Pack Vertical** - Allows you to control how Parts are packed on the nest. Vertically or Horizontally.

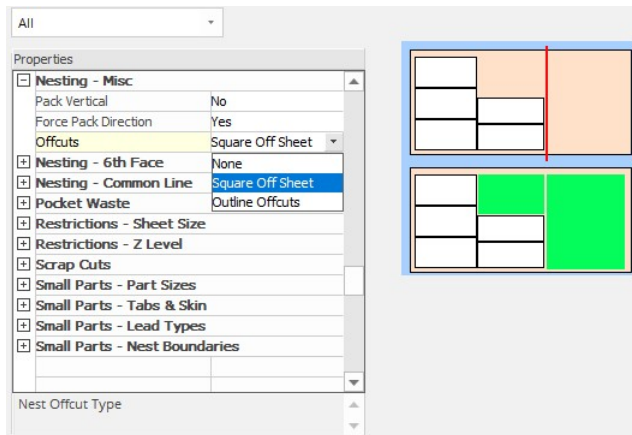




- **Force Pack Direction** - Favors Pack Direction over Optimization Quality.



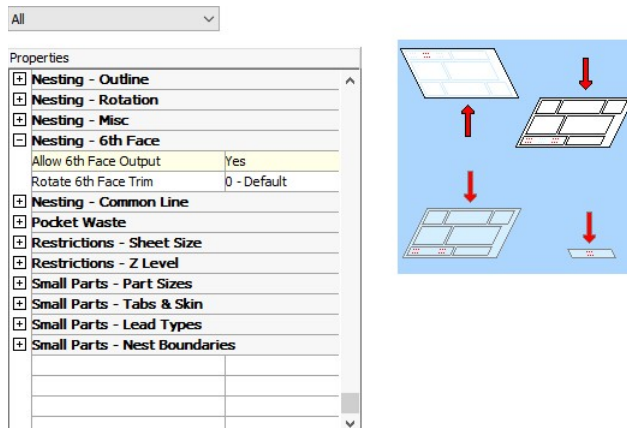
- **Offcuts** - There are three options.



- **None** - Selecting None here will ignore creating offcuts from your nests.
- **Square Off Sheet** - This will add an additional Route operation to your Nests to Square off your Sheet leaving a single Offcut.
- **Outline Offcuts** - This option will locate available Offcuts within the sheet largest to smallest and outline cut them into the largest possible Offcuts honoring your machine and Offcut settings.

### Nesting - 6th Face Group

- **Allow 6th Face Output** - When using this option you will be able to actually machine your Secondary (Flip Side/Face 6) operations on the reverse side of the Sheet prior to machining your Operations and Cutting out your Parts on the Primary Face. This option will use your outline tool to clean up one long edge and one short edge of your sheet (by one half of your Trim Cut) to give you a clean edge for alignment to 0/0 after flipping the Sheet to finish your nest.



When Nest Programs are output they will have one of three suffixes helping the CNC Operator identify 6th Face Output.

- N = Nested sheet with no corresponding 6th face program.
- A = The back-side (6th face) program to run before Z program.

- Z = Main program with nest operations and part outlines; just like N.

- **Rotate 6th Face Trim** - Depending on which corner of your machine your 0/0 Reference Pins reside you may need to rotate the trim cuts that are run along one long and one short edge. This options allows you to control where these cuts are performed.

All

Properties

+

Nesting - Outline

+

Nesting - Rotation

+

Nesting - Misc

+

Nesting - 6th Face

Allow 6th Face Output

Yes

Rotate 6th Face Trim

0 - Default

+

Nesting - Common Line

+

Pocket Waste

+

Restrictions - Sheet Size

+

Restrictions - Z Level

+

Small Parts - Part Sizes

+

Small Parts - Tabs & Skin

+

Small Parts - Lead Types

+

Small Parts - Nest Boundaries

0

90

270

180

Nesting - Common Line Group

- **Common Line Cutting** - This option will turn Common Line Cutting on or off. Common Line Cutting output tightly packs the parts in a nest so that the outline path can cut two parts with a single pass. Although this type of output has a better yield on material and allows for a longer tool life it has its downsides. A toolpath only has one clean edge. Based on the tools rotation and the material climb vs. conventional cut the S2M Center instructs the tool to cut around the parts in the optimal direction. With common line cutting there is no way to do this. With a common cut one part is going to get a clean edge and the other will experience some blowout. You should be working with sharp tools to minimize the blowout. In addition common line cutting uses a centerline path with no compensation. It is very important to have the correct tool size in the Tool Catalog to ensure the parts are the correct size. For this reason it is recommended that common line cutting only be used for non-essential parts and less than premium Materials. Because of this the common line settings are both per machine and per material based.

All

Properties

+

Label IT - Machine Settings

+

Margin - Part Spacing

+

Margin - Sheet

+

Nesting - Complexity

+

Nesting - Outline

+

Nesting - Rotation

+

Nesting - Misc

+

Nesting - 6th Face

+

Nesting - Common Line

Common Line Cutting

No

Common Line Maximum Retrace

0

Common Line Retrace by Small Parts

No

+

Pocket Waste

+

Restrictions - Sheet Size

+

Restrictions - Z Level

+

Small Parts - Part Sizes

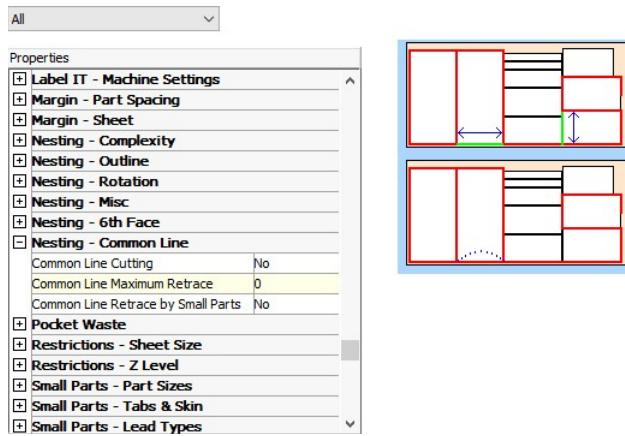
+

Small Parts - Tabs & Skin

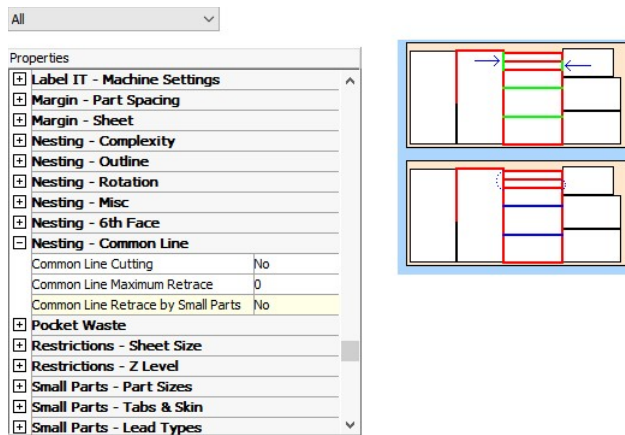
+

Small Parts - Lead Types

- **Common Line Maximum Retrace** - This value will affect the final tool path for the outlines. If you specify a retrace value that allows the outline path to move alongside a path that has already been cut resulting in fewer rapid moves. You should base this preference on how fast you machine transitions from rapid to machine moves.

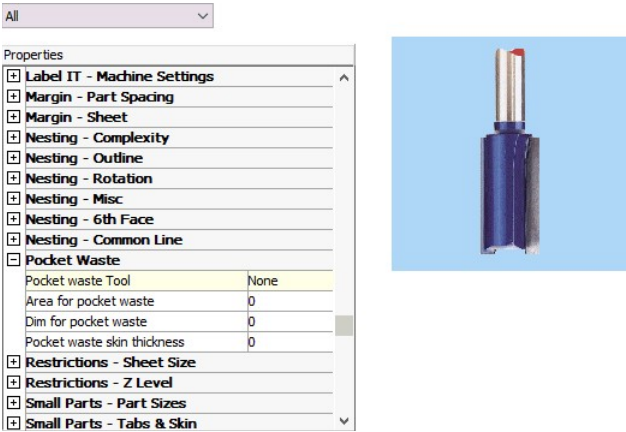


- **Common Line Retrace by Small Parts** - If you are allowing retracing you may want to ensure that retracing does not occur near small parts since they may have shifted after being cut. The S2M Center automatically avoids retracing near very small parts. This setting will ensure that retracing also avoids the small parts. Setting this value true will result in more rapid moves so it should be set to false if you are not having problems with your small parts shifting.

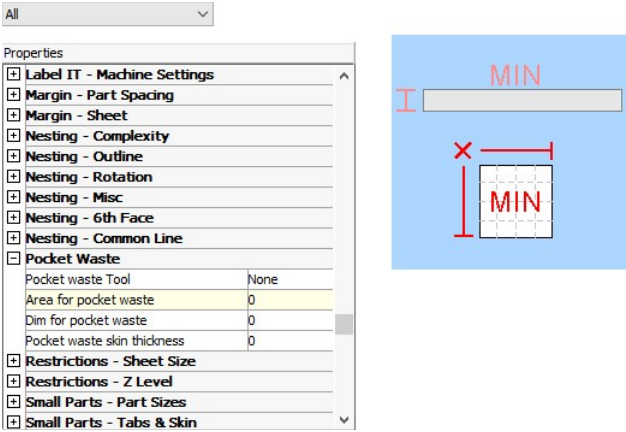


### Pocket Waste Group

- **Pocket Waste Tool** - Select a Tool to use to Pocket out any Waste Material in your Nest. By leaving this selection at None you are indicating not to Pocket Waste. Note: This does Pocket out Waste but does not Pocket out valid Offcuts.



- **Area for Pocket Waste** - Specifies the Minimum Area for Pocket Waste.



- **Dim for Pocket Waste** - Specifies the Minimum Dimension for Pocket Waste.

All

Properties

Label IT - Machine Settings

Margin - Part Spacing

Margin - Sheet

Nesting - Complexity

Nesting - Outline

Nesting - Rotation

Nesting - Misc

Nesting - 6th Face

Nesting - Common Line

Pocket Waste

Restrictions - Sheet Size

Restrictions - Z Level

Small Parts - Part Sizes

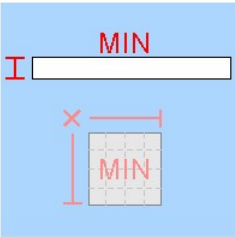
Small Parts - Tabs & Skin

Pocket waste Tool

Area for pocket waste

Dim for pocket waste

Pocket waste skin thickness



- **Pocket Waste Skin Thickness** - If you need to maximize Vacuum Clamping Pressure you may want to establish a Pocket Waste Skin Thickness here.

All

Properties

Label IT - Machine Settings

Margin - Part Spacing

Margin - Sheet

Nesting - Complexity

Nesting - Outline

Nesting - Rotation

Nesting - Misc

Nesting - 6th Face

Nesting - Common Line

Pocket Waste

Restrictions - Sheet Size

Restrictions - Z Level

Small Parts - Part Sizes

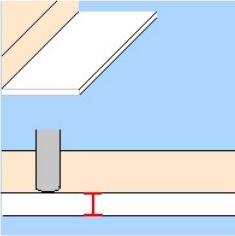
Small Parts - Tabs & Skin

Pocket waste Tool

Area for pocket waste

Dim for pocket waste

Pocket waste skin thickness



Restrictions - Sheet Size Group

- **Max Sheet Width** - Establishes a Maximum Sheet Width that can be used on the selected Machine.



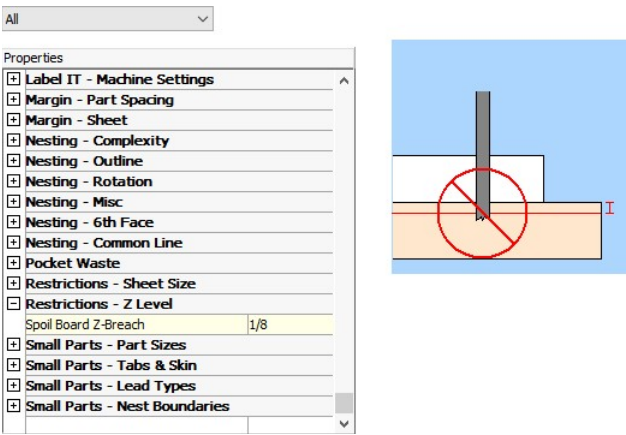
This option can be very important when using the Alphacam S2M Center or S2M Center Ultimate. The Alphacam S2M Center and S2M Center Ultimate allows multiple sheet sizes. It will look at these settings for the Machine to make sure a sheet that is too big is not utilized.

- **Max Sheet Length** - Establishes a Maximum Sheet Length that can be used on the selected Machine.

Restrictions - Z Level Group

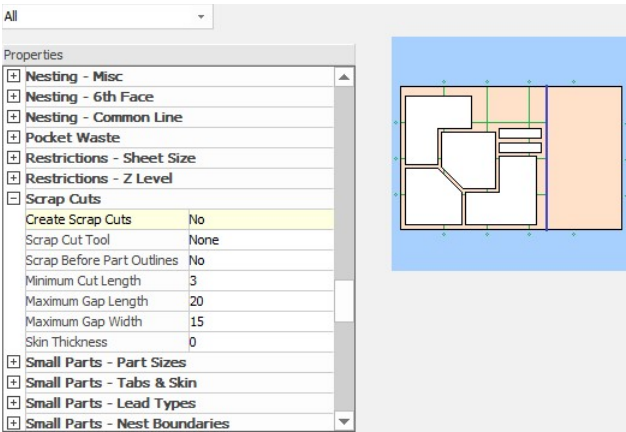
- **Spoil Board Z-Breach** - Use this value to set a threshold for Z Level checking. This allows you to assure that you do not breach too deeply into your Spoil Board/Machine.



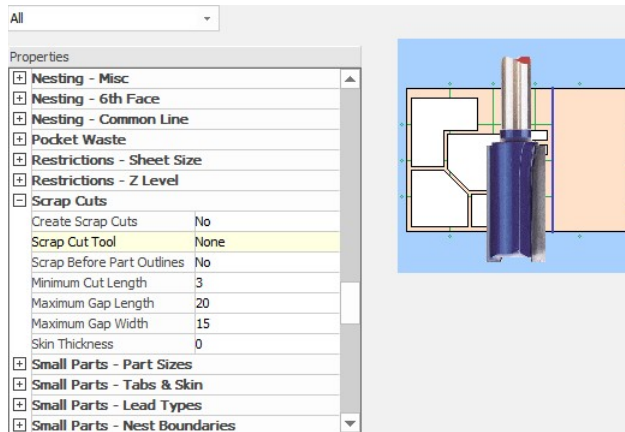


Scrap Cuts

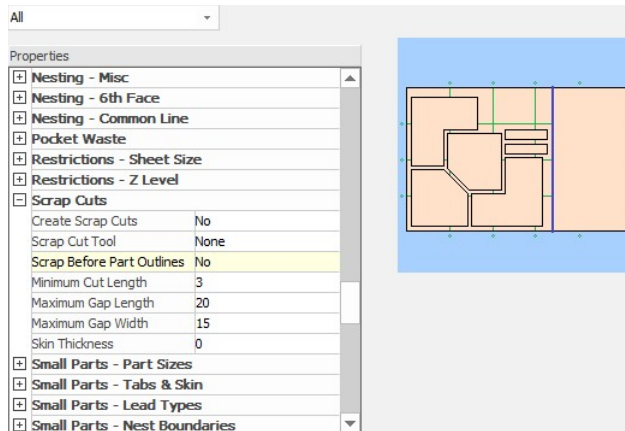
- **Create Scrap Cuts** - Use this option to turn on Scrap Cuts. This option gives you the ability to cut up the scrap from your nest into smaller pieces to make it easier to dispose of them.



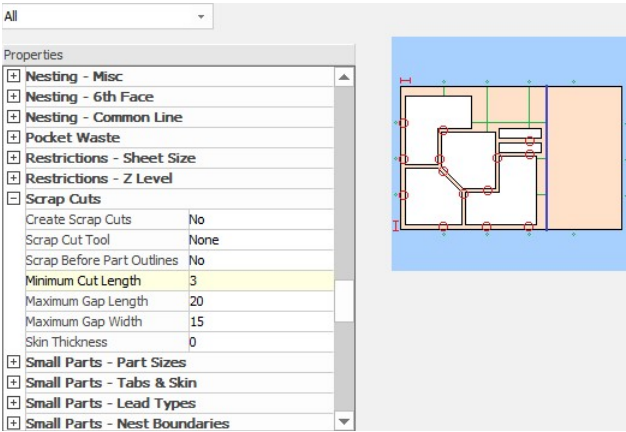
- **Scrap Cut Tool** - Choose the tool to use to make the Scrap Cuts.



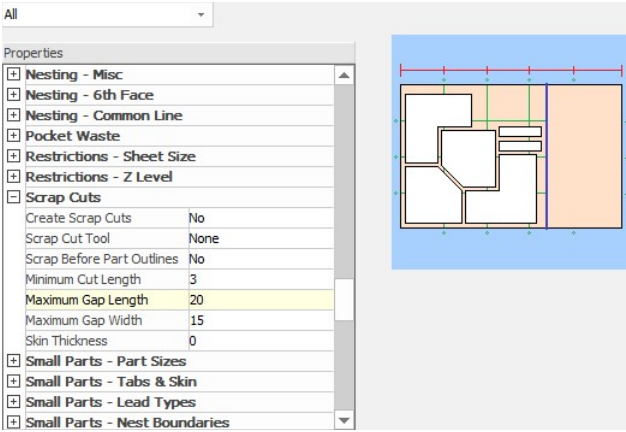
- **Scrap Before Part Outlines** - Determine if you want the Scrap Cuts performed before the Part Outlines.



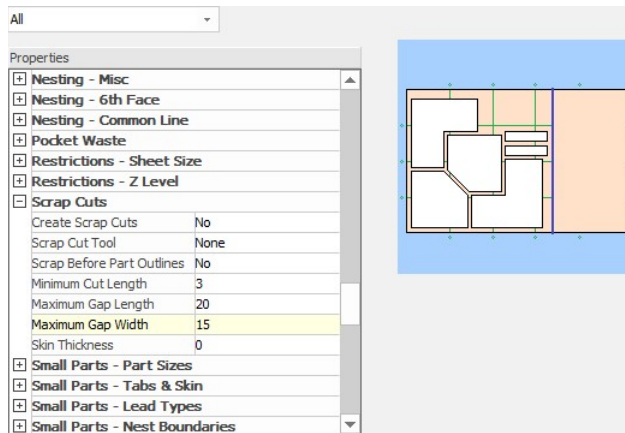
- **Minimum Cut Length** - If a scrap is less than this amount in either dimension it will not be Scrap Cut.



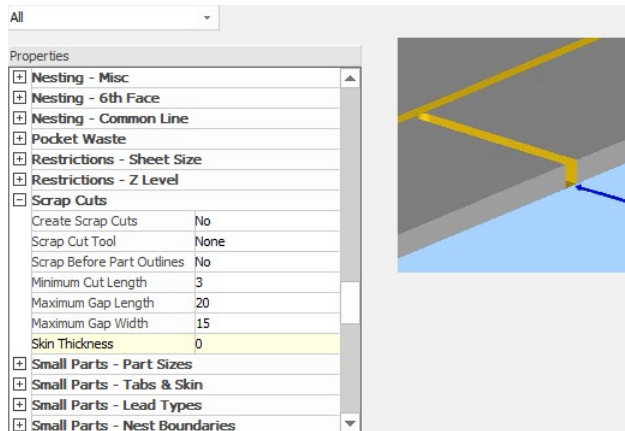
- **Maximum Gap Length** - The maximum length for the Scrap Cut.



- **Maximum Gap Width** - The maximum width for the Scrap Cut.



- **Skin Thickness** - If you want to Skin the Scrap Cut enter the Skin Thickness here.



### Small Parts - Part Sizes Group

- **Area and Dim for Small Parts** - The dimensions you provide here will define a small Part. Small Parts may require special procedures to securely hold them on the work surface during the machining process. Area defines the Square Inch or Square Millimeter area of a Small Part. Dim defines the dimension in Width or Length that determines a Small Part overriding the Area setting that defines a Small Part. Consult your Machine documentation for information regarding this item.

All

Properties

Nesting - Complexity

Nesting - Outline

Nesting - Rotation

Nesting - Misc

Nesting - 6th Face

Nesting - Common Line

Pocket Waste

Restrictions - Sheet Size

Restrictions - Z Level

Small Parts - Part Sizes

Area for Small Part60

Dim for Small Part6

Area for Very Small Part40

Dim for Very Small Part3

Small Parts - Tabs & Skin

Small Parts - Lead Types

Small Parts - Nest Boundaries

- **Area and Dim for Very Small Parts** - A very small Part is smaller than a small Part. The real difference is in how the Parts are secured prior to being cut free.

All

Properties

Nesting - Complexity

Nesting - Outline

Nesting - Rotation

Nesting - Misc

Nesting - 6th Face

Nesting - Common Line

Pocket Waste

Restrictions - Sheet Size

Restrictions - Z Level

Small Parts - Part Sizes

Area for Small Part60

Dim for Small Part6

Area for Very Small Part40

Dim for Very Small Part3

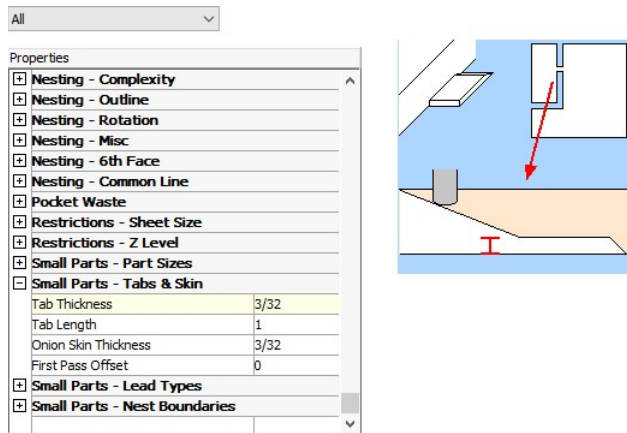
Small Parts - Tabs & Skin

Small Parts - Lead Types

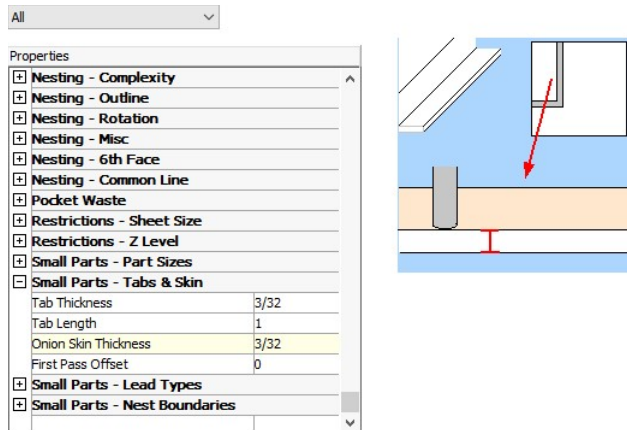
Small Parts - Nest Boundaries

Small Parts - Tabs & Skin Group

- **Tabs** - Tabs can be employed to help hold a "Small" or "Very Small" Part. They are areas where the router cuts at less than full depth for a short distance during the lead in thus leaving a "bridge". The lead out of the router removes this "Tab" over the area. During the time the router is cutting the Part free, the Tab provides the security that can help keep the Part from shifting.



- **Onion Skin Thickness** - Allows you to set the Onion Skin Thickness when using the Onion Skin Lead Type.



- **First Pass Offset** - If you are using an Onion Skin Pass this option allows you to have the first pass cut the Part oversized (rough cut) by the specified amount.

All

Properties

+

Nesting - Complexity

+

Nesting - Outline

+

Nesting - Rotation

+

Nesting - Misc

+

Nesting - 6th Face

+

Nesting - Common Line

+

Pocket Waste

+

Restrictions - Sheet Size

+

Restrictions - Z Level

+

Small Parts - Part Sizes

-

Small Parts - Tabs & Skin

+

Small Parts - Lead Types

+

Small Parts - Nest Boundaries

|                      |      |
|----------------------|------|
| Tab Thickness        | 3/32 |
| Tab Length           | 1    |
| Onion Skin Thickness | 3/32 |
| First Pass Offset    | 0    |

Small Parts - Lead Types

- **Lead Types** - The manner in which the Tool enters and exits the operation. There are five Lead Types:
  1. **None** - A simple tool plunge where the bit plunges straight into the Material (least favored). Cutter Compensation will not be applied.

All

Properties

+

Nesting - Complexity

+

Nesting - Outline

+

Nesting - Rotation

+

Nesting - Misc

+

Nesting - 6th Face

+

Nesting - Common Line

+

Pocket Waste

+

Restrictions - Sheet Size

+

Restrictions - Z Level

+

Small Parts - Part Sizes

-

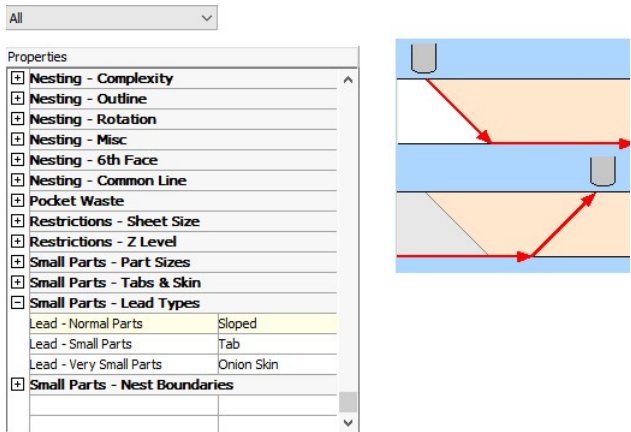
Small Parts - Lead Types

+

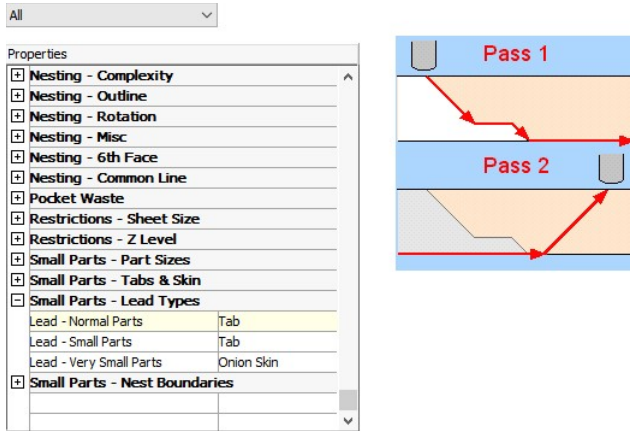
Small Parts - Nest Boundaries

|                         |            |
|-------------------------|------------|
| Lead - Normal Parts     | None       |
| Lead - Small Parts      | Tab        |
| Lead - Very Small Parts | Onion Skin |

2. **Sloped** - The most common Lead Type. This lead allows for less strain on the bit and provides cleaner machining of the Material. Sloped is also a required Lead on most Machines if Cutter Compensation (G41/G42) is being utilized. The Cutter Compensation is actually applied by the Machine during the Lead In.



3. **Tab** - Tabs can be employed to help hold a "Small" or "Very Small" Part. They are areas where the router cuts at less than full depth for a short distance thus leaving a "bridge". A second pass of the router removes this "Tab" over the area. During the time the router is cutting the Part free, the Tab provides the security that can help keep the Part from shifting.



4. **Onion Skin** - Typically used for small parts, the path defining the part shape is cut to a depth slightly less than the material thickness. After the first pass, there remains a thin "Onion Skin" of material that anchors the part in place. The router then repeats the path, this time removing the onionskin, and freeing the part.



All

Properties

Nesting - Complexity

Nesting - Outline

Nesting - Rotation

Nesting - Misc

Nesting - 6th Face

Nesting - Common Line

Pocket Waste

Restrictions - Sheet Size

Restrictions - Z Level

Small Parts - Part Sizes

Small Parts - Tabs & Skin

Small Parts - Lead Types

Small Parts - Nest Boundaries

|                         |            |
|-------------------------|------------|
| Lead - Normal Parts     | Onion Skin |
| Lead - Small Parts      | Tab        |
| Lead - Very Small Parts | Onion Skin |

Pass 1

Pass 2

5. **Return Onion Skin** - Similar to Onion Skins, except ALL Return Onion Skin parts receive the first pass prior to any Return Onion Skin Part receives the second (or "liberating") pass.

All

Properties

Nesting - Complexity

Nesting - Outline

Nesting - Rotation

Nesting - Misc

Nesting - 6th Face

Nesting - Common Line

Pocket Waste

Restrictions - Sheet Size

Restrictions - Z Level

Small Parts - Part Sizes

Small Parts - Tabs & Skin

Small Parts - Lead Types

Small Parts - Nest Boundaries

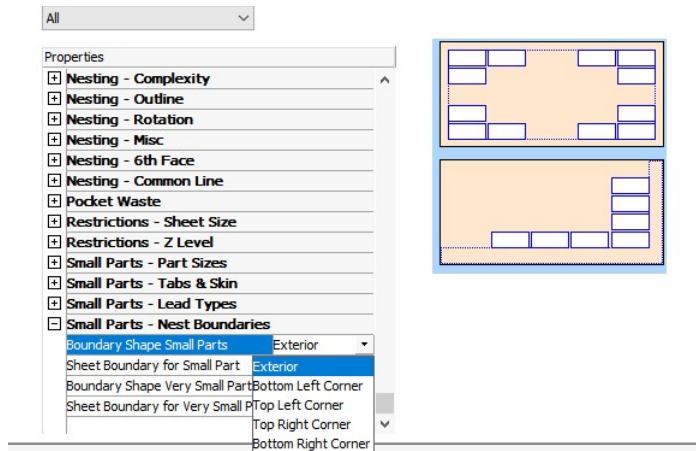
|                         |                   |
|-------------------------|-------------------|
| Lead - Normal Parts     | Return Onion Skin |
| Lead - Small Parts      | Tab               |
| Lead - Very Small Parts | Onion Skin        |

Pass 1

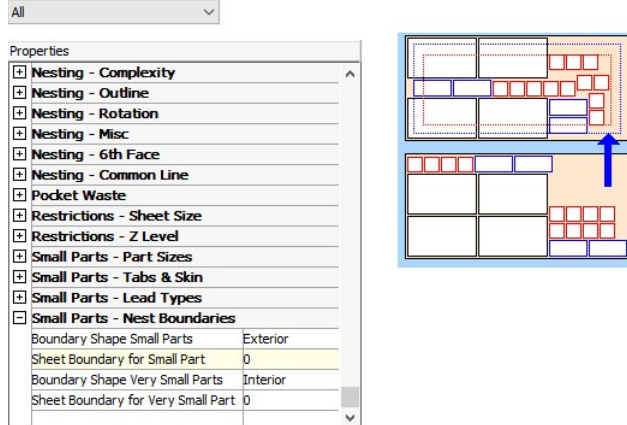
Pass 2

Small Parts - Nest Boundaries Group

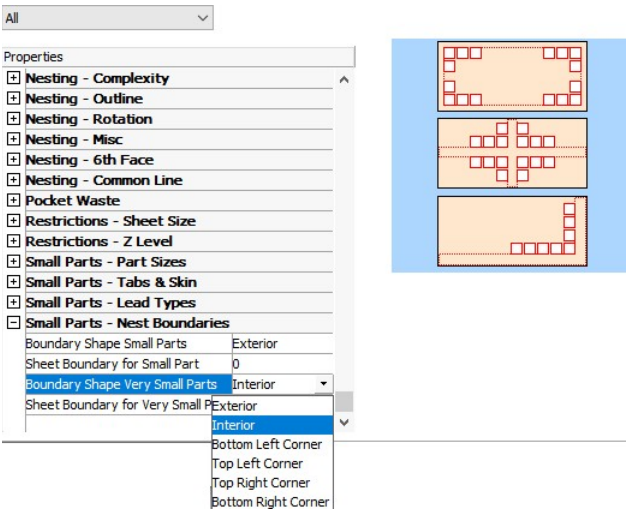
- **Boundary Shape Small Parts** - There are several options here:



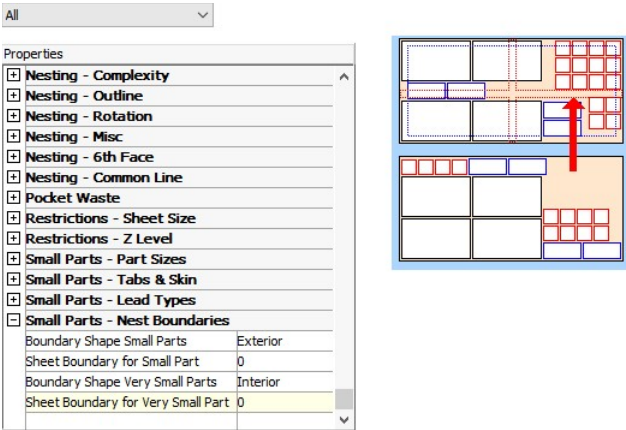
- **Exterior** - Keeps Small Parts away from the sheet edge.
  - **Bottom Left Corner** - Keeps Small Parts away from the Bottom Left Corner.
  - **Top Left Corner** - Keeps Small Parts away from the Top Left Corner.
  - **Top Right Corner** - Keeps Small Parts away from the Top Right Corner.
  - **Bottom Right Corner** - Keeps Small Parts away from the Bottom Right Corner.
- **Sheet Boundary for Small Part** - This setting uses your definition for a "Small Part" to establish a Boundary around the edge of your Sheet where "Small Parts" are **not** allowed to be placed. It basically establishes a "Zone" within the center of the Sheet where "Small Parts" can be placed.



- **Boundary Shape Very Small Parts** - There are several options here:



- **Exterior** - Keeps Very Small Parts away from the sheet edge.
  - **Interior** - This setting uses your definition for a "Very Small Part" to establish a Boundary at the Center of the Sheet where "Very Small Parts" are **not** allowed to be placed. It basically establishes a "Zone" within the center of the Sheet where "Very Small Parts" should not be placed. This is a good setting to use in conjunction with [Perimeter Nesting](#).
  - **Bottom Left Corner** - Keeps Very Small Parts away from the Bottom Left Corner.
  - **Top Left Corner** - Keeps Very Small Parts away from the Top Left Corner.
  - **Top Right Corner** - Keeps Very Small Parts away from the Top Right Corner.
  - **Bottom Right Corner** - Keeps Very Small Parts away from the Bottom Right Corner.
7. Sheet Boundary for Very Small Part - This setting uses your definition for a "Very Small Part" to establish a Boundary around the edge of your Sheet where "Very Small Parts" are **not** allowed to be placed.



**i** The following items should be setup by the integrator. Once these items are set correctly, they probably will never need to be reset or adjusted.

- Post Processor

- Apply Comp on Rapid
- Roll around corners
- Material Length is X
- Tool Compensation

## Part P2P

This Machine Type is specifically for Point to Point Machining Centers. Following are the Setup Tab options for Part P2P:

### Setup Tab - Part Point-to-Point



If you are using a DXF NC Link see the [STD DXF Setup Tab](#) topic.

- **Intelli-CAM - Post** - [See Sheet Router](#)
- **Intelli-CAM - Rapid** - [See Sheet Router](#)
- **Intelli-CAM - File Info** - [See Sheet Router](#)
- **Intelli-CAM - Tool Movement** - [See Part Router](#)
- **Intelli-CAM - Sheet Reference** - [See Sheet Router](#)
- **Intelli-CAM - Offset** - [See Sheet Router](#)
- **Intelli-CAM - Banding Operation** - [See Part Router](#)
- **Label Format** - [See Part Router](#)
- **Restrictions** - [See Sheet Router](#)
- **Zone Sizes** - [See Part Router](#)
- **Front and Rear Zones** - [See Part Router](#)
- **Dimensions** - [See Part Router](#)
- **Alternate Zones** - [See Part Router](#)

## Part Router

This Machine Type is specifically for Part Routers. Following are the Setup Tab options for Part Routers:

### Setup Tab - Part Router

#### Intelli-CAM - Post Group

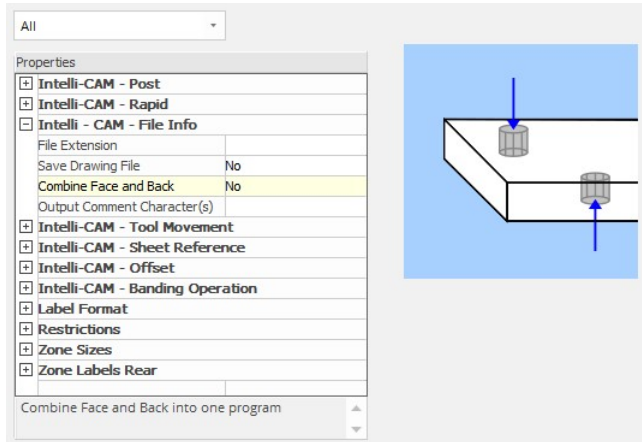
[See Sheet Router](#)

#### Intelli-CAM - Rapid Group

[See Sheet Router](#)

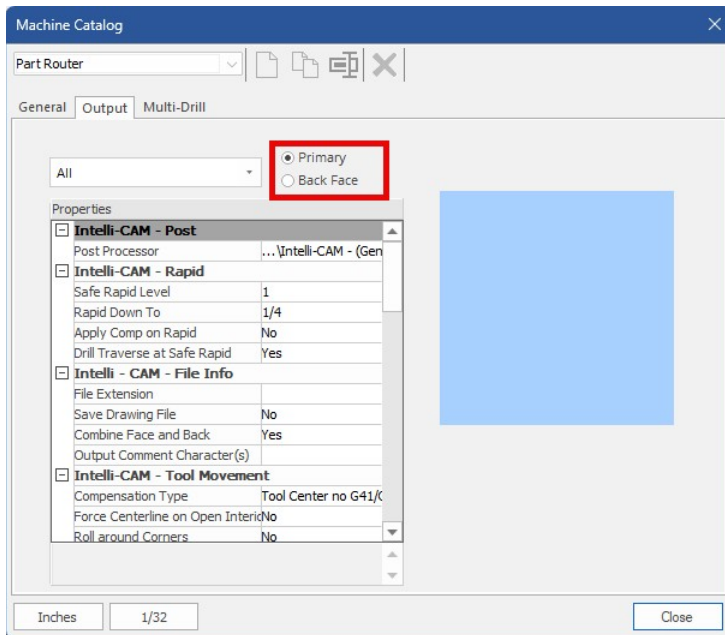
## Intelli-CAM - File Info Group

- **Combine Face and Back** - Use this setting to define a machine which can machine from both the Face and Back sides simultaneously.



If you set Combine Face and Back to Yes you get several new options in the Machine Catalog.

At the top of the Output Tab you will see a Primary and Back Face option.



Selecting each allows you to specify different machining options for the Primary and Back Faces. The following options are available when the Back Face is selected.

Intelli-CAM - Rapid

[See Sheet Router](#)

Intelli-CAM - File Info

- **Face Up Indicator** - A single character used in the file output name indicating the orientation of the part wen running the program. The default value is +.
- **Back Up Indicator** - A single character used in the file output name indicating the orientation of the part wen running the program. The default value is -.

Intelli-CAM - Tool Movement

See Intelli-CAM Tool Movement Group section below

 Also on the Multi-Drill tab you will get a Primary and Back Face option. This allows you to define a different Multi-Drill head for the Primary versus Back face. Keep in mind when defining these that the Multi-Drill definition is always a top down view from the machine persepective.

[See Sheet Router for other settings](#)

Intelli-CAM - Tool Movement Group

- **No Sloping Lead In/Outs** - With this option enabled Sloping Lead Ins and Outs will not be generated.

All

Properties

Intelli-CAM - Post

Intelli-CAM - Rapid

Intelli - CAM - File Info

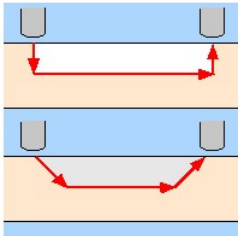
Intelli-CAM - Tool Movement

|                                         |             |
|-----------------------------------------|-------------|
| Compensation Type                       | Tool Center |
| Force Centerline on Open Interior Paths | No          |
| Roll around Corners                     | Yes         |
| Limit Output To 3 Axis                  | Yes         |
| Y Axis Horizontal on Workplane          | No          |
| 5 Axis Tilt Clockwise                   | No          |
| Optimize Drill Selection                | No          |
| Mix Hole Diameters                      | No          |
| Slope Arc Lead Spiral Pocket            | No          |
| No Sloping Lead In/Outs                 | No          |
| Adjust Open Tool Path For Climb vs Conv | Yes         |

Intelli-CAM - Sheet Reference

Intelli-CAM - Offset

Intelli-CAM - Banding Operation



[See Sheet Router for other settings](#)

Intelli-CAM - Sheet Reference Group

[See Sheet Router](#)

Intelli-CAM - Offset Group

- **Offset by 1/2 EDX/EDY** - This option is important when using EDX/EDY to oversize a Part and the Part Router is setup to Route the full outline. Use this option to make sure the over-sized amount is compensated for.

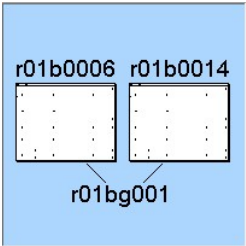
[See Sheet Router for other options](#)

Intelli-CAM - Banding Operation

- **Banding Tool** - Select the Tool that applies (represents) the Banding Operation. You post-processor must be setup properly in order to use this option.

Intelli-CAM - Label Format Group

- **Label Format** - Defines the formatting for your barcode as required by your machine and/or bar-code scanner manufacturer.
- **Match Programs** - With Match Programs enabled the S2M Center will identify Parts which are the same and generate a single Program for them versus unique programs. This is considered a group program and will have a "g" designator within the program name as illustrated in the Machine Catalog image below.



Zone Sizes Group

- **Front and Rear Zones** - Allows you to specify whether or not your Machine has both Front and Rear Zones.

All

Properties

Intelli-CAM - Post

Intelli-CAM - Rapid

Intelli - CAM - File Info

Intelli-CAM - Tool Movement

Intelli-CAM - Sheet Reference

Intelli-CAM - Offset

Intelli-CAM - Banding Operation

Label Format

Restrictions

Zone Sizes

Front and Rear Zones

Dimension 'A'

Dimension 'B'

Dimension 'C'

Alternate Zones

Zone Labels Front

Zone Labels Rear

Yes

0

0

0

No

A diagram on a light blue background showing a 3x3 grid. The grid is divided into nine squares. The top row is labeled '1', '2', '3' from left to right. The middle row is labeled '4', '5', '6' from left to right. The bottom row is labeled '7', '8', '9' from left to right. Dimension 'A' is indicated by a horizontal line above the grid, spanning the width of the first two columns. Dimension 'B' is indicated by a vertical line to the left of the grid, spanning the height of the first two rows. Dimension 'C' is indicated by a vertical line to the left of the grid, spanning the height of all three rows.

- **Dimension 'A'** - Allows you to specify the Dimension on your Machine denoted by "A" in the graphic.

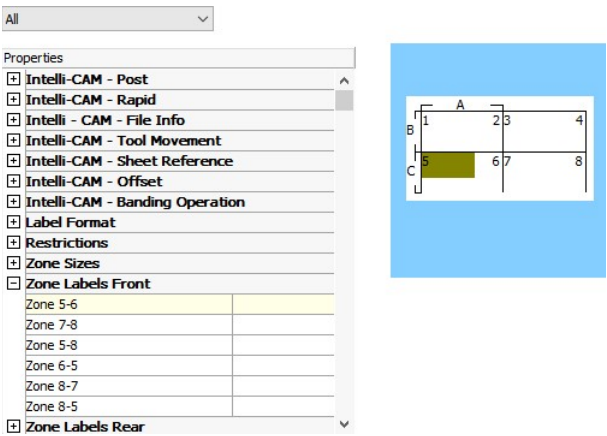
**Dimension 'B'** - Allows you to specify the Dimension on your Machine denoted by "B" in the graphic.

- **Dimension 'C'** - Allows you to specify the Dimension on your Machine denoted by "C" in the graphic.

**Alternate Zones** - Specifies whether you want the S2M Center to between Zones. For example if a Part is determined to be a Zone 1-2 Part and the next Part also is determined to be a Zone 1-2 Part, with this option on it would send the first Part to Zone 1-2 and the next to Zone 3-4 then the next to Zone 1-2. Alternating Zones allows the machine to process one part while loading another into the alternative zone. This can be a real time saver if you have a multi-zone machine.

Zone Labels Front Group

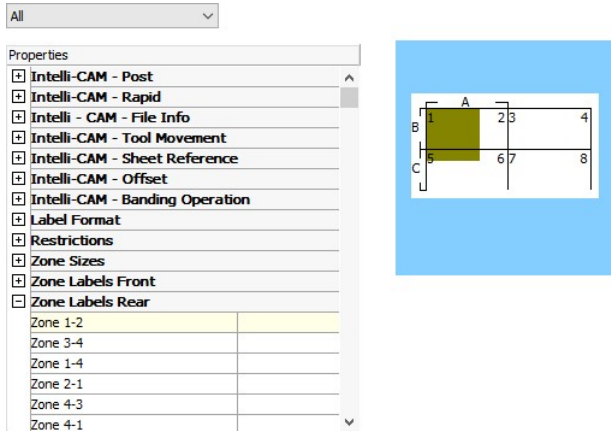
- **Zone 5-6** - Allows you specify the character set your machine uses to identify this Zone.



- **Zone 7-8** - Allows you specify the character set your machine uses to identify this Zone.
- **Zone 5-8** - Allows you specify the character set your machine uses to identify this Zone.
- **Zone 6-5** - Allows you specify the character set your machine uses to identify this Zone.
- **Zone 8-7** - Allows you specify the character set your machine uses to identify this Zone.
- **Zone 8-5** - Allows you specify the character set your machine uses to identify this Zone.

Zone Labels Rear Group

- **Zone 1-2** - Allows you specify the character set your machine uses to identify this Zone.



**i** Use the ^ character to pad any zone names for bar code scanners that require fixed length field names.

- **Zone 3-4** - Allows you specify the character set your machine uses to identify this Zone.
- **Zone 1-4** - Allows you specify the character set your machine uses to identify this Zone.



- **Zone 2-1** - Allows you specify the character set your machine uses to identify this Zone.
- **Zone 4-3** - Allows you specify the character set your machine uses to identify this Zone.
- **Zone 4-1** - Allows you specify the character set your machine uses to identify this Zone.

## Specialized Drill and Dowel

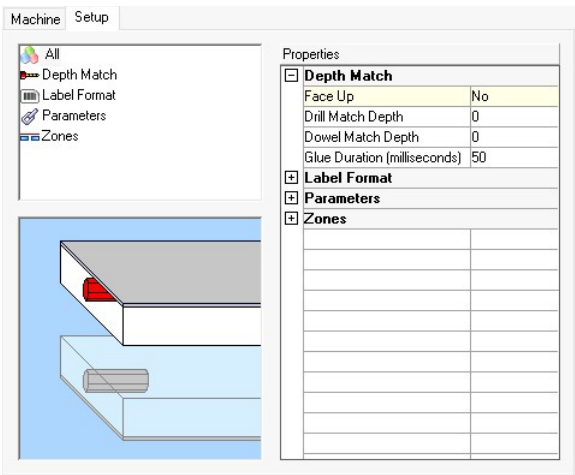
This Machine Type is specifically for Drill and Dowel Machines. Following are the Setup Tab options for Specialized Drill and Dowel:

### Setup Tab - Specialized Drill & Dowel

The fields defined below will only be available if your Drill & Dowel supports the feature the field defines.

#### Depth Match Group

- **Face Up** - This setting will force the program to be written either face up or down regardless of if the part is face up/down on the nest or saw.



 You can override the up/down flag on an individual part with the DNDFLIP parameter.

#### Face Up with Label-IT

Below is sample output for 2 different edge machines. Each machine has a different Face Up setting.

The arrows will indicate how the part should be fed into the machine and since this part was on the saw Face Down that the Face Up program(s) require the part to be flipped.

When two programs share the same name but different Face Up/Down requirements then multiple arrows will be shown to accommodate. The arrows will coincide with the machine tags field.

Note that when this part is BU (back up) it has a Left and Bottom program. When it is flipped Face Up, it has a Right and Bottom program. The L,R,B,T programs are always relative to the primary edge of the part and if the parts is Face Up/Down.

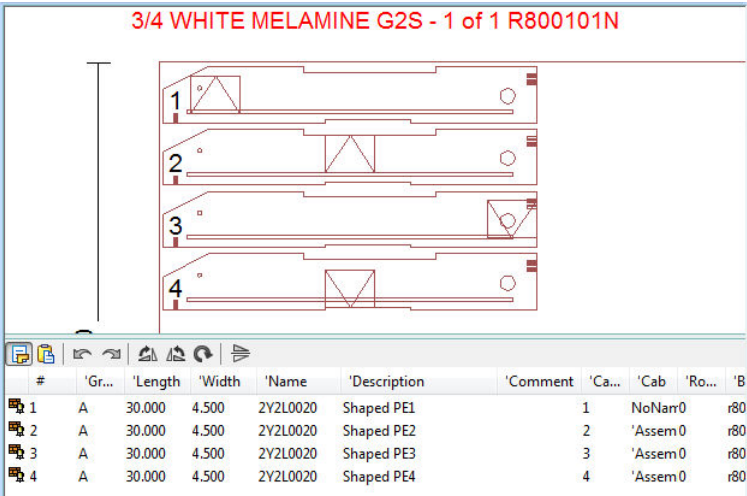
|                                                    |            |                                                                                            |
|----------------------------------------------------|------------|--------------------------------------------------------------------------------------------|
| Your Company Name Here                             |            | 2/21/2012                                                                                  |
| <b>*r75b0021*</b>                                  |            | Zone5-6  |
| 1: Shaped Back PE1 r75b0021 3/4 WHITE MELAMINE G2S |            |                                                                                            |
| 30.000 x 4.500 x 0.750                             |            |                                                                                            |
| BU                                                 | *r750021L* |  M        |
| FU                                                 | *r750021R* |           |
| FU, BU                                             | *r750021B* |  M        |
| Job: OMAL H_Bore                                   |            |          |
| Room # 25 'Room 1                                  |            |                                                                                            |
| <b>Cab # 1</b>                                     |            |                                                                                            |

In the past the Label was always rotated so that the Arrow pointed to the primary edge of the part. However there was a disconnect with this rotation and how the part appears on the Label in Label-IT. In Label-IT the part is always rotated 90 degrees since parts are typically longer then they are wide to maximize the part image.

Now the label is always rotated so that it points to a part length.

In the image depicted below we have the same part with 4 different primary edges. The primary edge corresponds with the part number. In the past the 2nd parts label would point to the right and the 4th parts label would point left. In the new scheme the part with the primary edge of 2 or 4 have the same rotation as a part with 1 or 3 respectively.

Note that the position of the label has not changed, only the rotation.



**Silent Mirrored Programs**

If a part is shaped there may be difficulty lining up the part on the fence of the machine on its shaped side. We now output a silent mirrored program for all shaped parts. The mirrored program can be run on the opposite fence if needed.

Note that the mirrored program will still have the same Face Up/Down orientation as the primary program. In essence only the X-Position is mirrored (not the Z).

These additional programs will only occur for shaped parts.

The Barcodes on the Label Image will be for the primary edge program. The operator will need to be trained/informed that there is an additional program if necessary.

The edge program has a new naming convention:

**Previously**

R010011L (Run Number R01) (PartID 0011) (Edge L = Left)

**New**

R01P00BL (Run Number R01) (Primary or Mirrored P/M) (PartID in base 36 00B) (Edge L = Left)

R01M00BL (Run Number R01) (Primary or Mirrored P/M) (PartID in base 36 00B) (Edge L = Left)

With Base 36, the three 3 characters reserved for PartID can now store 46,655 unique IDs from 000 to ZZZ

If the part is shaped the program will output a R01M\* program in addition to the R01P\*

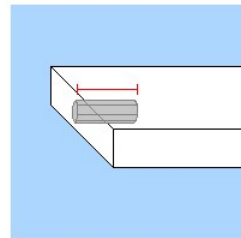
All D & D output except for the [Weeke BEK](#) that describes the entire part now feature the mirrored programs.

**EXAMPLE OUTPUT**

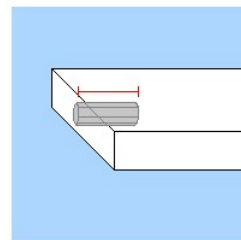
R78P00DL:  
Edge Width  
4.50000  
Number Of Holes  
2  
Hole 1  
0.75000 1 0 1  
Hole 2

R78M00DL:  
Edge Width  
4.50000  
Number Of Holes  
2  
Hole 1  
3.75000 1 0 1  
Hole 2  
3.25000 1 0 1

- [illegible]



- [illegible]



- file:///C:/Users/celav/AppData/Local/Temp/~hhEA7D.htm

All

Properties

Depth Match

Face Up

No

Drill Match Depth

20

KD Match Depth

0

Glue Match Depth

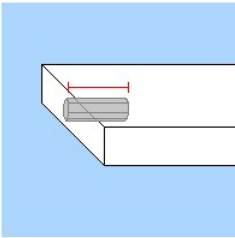
0

Glue Duration (milliseconds)

50

Label Format

Parameters



◦ **Glue Match Depth** - Specifies to the S2M Center when to send a Drill, Glue, and Dowel code to your Drill and Dowel. See Match Depth explanation below.

All

Properties

Depth Match

Face Up

No

Drill Match Depth

20

KD Match Depth

0

Glue Match Depth

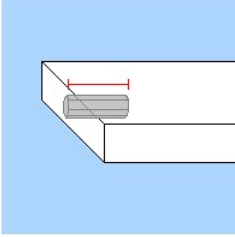
0

Glue Duration (milliseconds)

50

Label Format

Parameters



Match Depth Explanation

In the grid below I have defined 4 different Match Depths (Triggers) for each Operation Type. The dimension used here are a bit extreme. You would probably define them with about .001” difference when you actually utilize it.

|                   |       |
|-------------------|-------|
| Drill Match Depth | 0.125 |
| Dowel Match Depth | 0.25  |
| KD Match Depth    | 0.375 |
| Glue Match Depth  | 0.5   |

In this situation if I passed a Horizontal Bore Operation through the S2M Center that was 1/8” Deep the Machine would Drill Only.

If I passed a ¼” Deep Operation it would Drill and Insert a Dowel. This would be used for inserting glue-less dowels.

If I passed a 3/8” Deep Operation it would Drill and Insert a KD Fitting (If your Machine supports it, otherwise this field will not be available.)

If I passed a ½” Deep Operation it would Drill, Inject Glue, and Insert a Dowel.

- **Glue Duration (milliseconds)** - Allows you to specify the Glue Duration on your Drill and Dowel Machine in milliseconds.

All

Properties

Depth Match

Face Up

Drill Match Depth

KD Match Depth

Glue Match Depth

Glue Duration (milliseconds)

No

20

0

0

50

Label Format

Parameters



Label Format Group

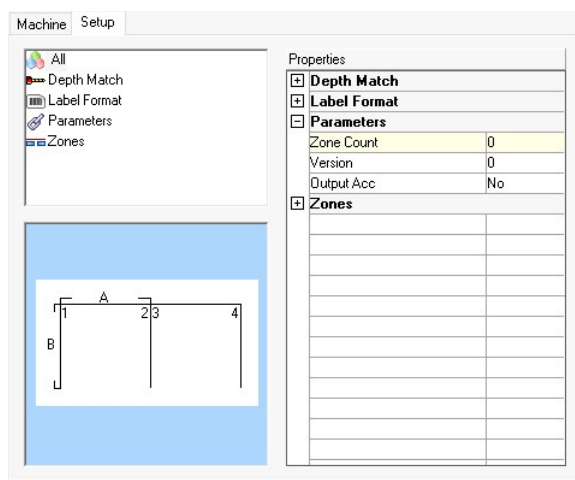
[See Sheet Router](#)

Output Options

- **Allow Horizontal Dado** - Specifies if the Machine is setup (allowed) to machine Horizontal Dadoes with the [Gannomat Index Pro](#) and [Omal Frame TpaEdi32](#) Drill and Dowel Link.
- **Allow Horizontal Route** - Specifies if the Machine is setup (allowed) to machine Horizontal Routes with the [Gannomat Index Pro](#) Drill and Dowel Link.
- **Allow Vertical Bore** - Specifies if the Machine is setup (allowed) to machine Vertical Bores with the [Gannomat Index Pro](#), [Koch Sprint Plus](#), [Omal TPAEdi32](#), and [Weeke ABD](#) Drill and Dowel Link.
- **Allow Vertical Dado** - Specifies if the Machine is setup (allowed) to machine Vertical Dadoes with the [Accu-Systems HPJ XML](#) and [Gannomat Index Pro](#) Drill and Dowel Link..

Parameters Group

- **Zone Count** - Specifies the number of available Zones with the [Accu-Systems](#), [Accu-Systems HPJ](#), [Accu-Systems HVVP](#), and the [Accu-Systems Junior Series](#) Drill and Dowel Link.



- **Version** - Allows you to specify the control version for the [Accu-Systems](#), [Accu-Systems HPJ](#), [Accu-Systems HPMTH](#), [Accu-Systems HVVP](#), and [Accu-Systems Junior Series](#) Drill and Dowel Link.
- **Output Acc** - Sets ACC as the output type for the [Accu-Systems](#) and [Accu-Systems Junior Series](#) Drill and Dowel Link.
- **Base Zone# On Part Size** - Sets the Zone assignment to be based on the Part size for the [Accu-Systems HPJ](#), [Accu-Systems HVVP](#), [Omal TPAEdi32](#), and [Vitap Thunder](#) Drill and Dowel Link.
- **Dimension 'A'** - Sets the width of Zone A for the [Accu-Systems HPJ](#), [Accu-Systems HVVP](#), [Omal TPAEdi32](#), and [Vitap Thunder](#) Drill and Dowel Link.
- **Drill Match Depth 2** - Specifies to the S2M Center when to send a Drill 2 only code to your [Accu-Systems HPJ](#) and [Gannomat Index](#) Drill and Dowel. Match Depths are used to specify a trigger to output the correct codes to your Machine. See [Match Depth](#) explanation above.
- **Output Rcm** - Sets RCM as the output type for the [Accu-Systems Junior Series](#) Drill and Dowel Link.
- **Output Mcm** - Sets MCM as the output type for the [Accu-Systems Junior Series](#) Drill and Dowel Link.
- **Name Output [JobName].mdb** - Specifies whether your [Gannomat Index](#) supports unique output names.
- **Output Drill Depth** - Allows you to Output the Drill Depth for the [Gannomat Index](#) Drill and Dowel Link.
- **Use Zone 2 and 3** - Allows you to disable Zones 2 & 3 for the [Gannomat Index](#) and [Gannomat Index Pro](#) Drill and Dowel Link.
- **Verticals In Pos Flat** - Output vertical operations in PositionsFlat rather than PositionsFront for the [Gannomat Index](#).
- **8mm Dowel Length** - Allows you to set the 8mm Dowel Length for the [Gannomat Index](#) Drill and Dowel Link.
- **6mm Dowel Length** - Allows you to set the 6mm Dowel Length for the [Gannomat Index](#) Drill and Dowel Link.
- **Glue Gap** - Allows you to set the Glue Gap for the [Gannomat Index](#) and [Gannomat Index Pro](#) Drill and Dowel Link.
- **Glue Match Depth 2** - Specifies to the S2M Center when to send a Glue 2 only code to your [Gannomat Index](#) Drill and Dowel. Match Depths are used to specify a trigger to output the correct codes to your Machine. See [Match Depth](#) explanation above.
- **Fixed Dowel Outstanding** - If 0 the output value will be based on the depth of hole, dowel length and glue gap for your [Gannomat Index](#) Drill and Dowel. Otherwise enter the required value.
- **Dowel Diameter** - Allows you to set the Dowel Diameter for the [Gannomat Index Pro](#) Drill and Dowel Link.
- **Dowel Length** - Allows you to set the Dowel Length for the [Gannomat Index Pro](#) Drill and Dowel Link.
- **Saw Diameter** - Allows you to set the Saw Diameter for the [Gannomat Index Pro](#) Drill and Dowel Link. You will only see the Output Options for this machine if the Saw has a valid (non zero) Diameter.

- **One Program Per Part** - Allows you to output one program per Part with the [Koch Sprint Plus](#), [Laguna Lockdowel](#), [Weeke ABD](#), and [Weeke BEK](#) Drill and Dowel Link.
- **Output P1** - Allows you to output P1 with the [Omal](#) Drill and Dowel Link.
- **Read Easy** - Allows Read Easy output for the [Omal](#) Drill and Dowel Link.
- **Use Number From Tool Cat** - Use the tool number from the tool catalog for the value of #8530 on the [Omal Albatros](#).
- **Barcode Format** - Allows you to set the Barcode Format for the [Omal Frame](#) Drill and Dowel Link.
- **Interior Zone Max** - Allows you to set the Interior Zone Max for the [Omal Frame](#) Drill and Dowel Link.
- **Zone 2-1** - Allows you to set Zone 2-1 for the [Omal Frame](#) Drill and Dowel Link.
- **Zone 3-4** - Allows you to set Zone 3-4 for the [Omal Frame](#) Drill and Dowel Link.
- **Vertical Bore Reach** - Allows you to set the Vertical Bores reach for the [Omal Frame](#) Drill and Dowel Link.
- **Path to Subs** - Allows you to specify the Path To Subs for the [Omal Frame TpaEdi32](#) Drill and Dowel Link.
- **Drill** – Applied to a tool that is assigned to the [Omal Velox](#) machine, this specifies that this tool should include a drilling operation.
- **Glue** – Applied to a tool that is assigned to the [Omal Velox](#) machine, this specifies that this tool should include a gluing operation.
- **Dowel** – Applied to a tool that is assigned to the [Omal Velox](#) machine, this specifies that this tool should include a dowel insertion operation.
- **Insert Dowel** - Allows you to enable/disable dowel insertion on the [ProEdge Lockdowel](#) machine.
- **Remove Z Pos** - Allows you to Remove the Z Position value for the [Sprint](#) Drill and Dowel Link.
- **Use Part Len as Prog Len** - Allows you to set the Program Length to match the Parts Length for the [Sprint PTP2 Beckhoff](#) Drill and Dowel Link.
- **One .dat File Per Part** - Allows you to output one program per Part with the [Sprint PTP2 Beckhoff](#) Drill and Dowel Link.

## Zones Group

[See Parts Router](#)

## Specialized Chop Saw

This Machine Type is specifically for Chop Saws with automated fence systems. Following are the Setup Tab options for Specialized Chop Saw:

## Setup Tab - Chop Saw

### Parameters Group

- **Output for T421 SNC** - Establishes if the output needs to be for a T4212 SNC when the [Omga T 421/521](#) NC Link is selected.
- **Output Material Name** - Establishes whether the Material Name should be output when the [Pistorius BF-A](#) NC Link is selected.
- **Mirror Output** - Establishes whether Mirrored Output should be produced when the [Pistorius BF-A](#) NC Link is selected.
- **Output to Hoffmann CNC** - Enable this option if you are outputting to the Hoffmann CNC with the [RazorGage](#) link.
- **Swap A/B Notches** - Enable this option to Swap the A/B Notches for the [RazorGage](#) link.
- **Add Part ID** - Establishes whether the Part ID is output when the [RazorGage](#) NC Link is selected.
- **Add Job Description** - Establishes whether the Job Description is output when the [RazorGage](#) NC Link is selected.
- **Add Program Name** - Establishes whether the Program Name is output when the [RazorGage](#) NC Link is selected.



- **Strip Quotes** - Enable this option to remove Quotation Marks as the Text Qualifier in output when using the [TigerStop](#) NC Link.

## Specialized Other

This Machine Type is specifically for other woodworking machines which do not fit in the other Machine Type Categories. Following are the Setup Tab options for Specialized Chop Saw:

### Setup Tab - Other

#### Parameters Group

- **Barcode Format** - This option will appear when an [Accu-Systems MTH](#) or [Weinig French Miter](#) NC Link is selected. [See Sheet Router](#) for information on this field.
- **Tenon Thickness Clearance** - This option will appear when an [Accu-Systems Universal](#) NC Link is selected. Sets the clearance required for the Tenon Thickness.
- **Tenon Width Clearance** - This option will appear when an [Accu-Systems Universal](#) NC Link is selected. Sets the clearance required for the Tenon Width.
- **ID Parameter Name** - This option will appear when the [Anderson MLS](#) NC Link is selected. This allows you to specify the name of the unique material id parameter that will identify this material to the lift system.
- **Swap A/B Notches** - This option will appear when the [Hoffman Beaded Frame System](#) NC Link is selected. Allows you to swap the A/B notches.
- **Mirror Output** - This option will appear when the [Hoffman Beaded Frame System](#) NC Link is selected. Enables Mirrored Output.
- **Feed Speed** - This option will appear when the [Homag Edgebander](#) NC Link is selected. Allows you to set the machine feed speed.
- **Oversize** - This option will appear when the [Homag Edgebander](#) NC Link is selected. Allows you to set the oversize value.
- **Glue Type** - This option will appear when the [Homag Edgebander](#) NC Link is selected. Allows you to set the glue type.

## NC Links

There are four different types of machines that utilize NC Links:

### Saw

The currently available Panel/Beam Saw NC Links are listed below. Simply click on the NC Link below for more information.

[Alpha Beam Saw](#)

[Altendorf Elmo](#)

[Ayza Spear Lanza P5](#)

[BeamBoard](#)

[Casadei CN04](#)

[Casadei MX39](#)

[CPOUT](#)

[CPOUT - Mayer](#)

[Custom CUT File](#)

[CUT File](#)

[Elcon Quadra](#)

[Etalon Saw](#)

[Excitech EP](#)[Fimal Concept 350](#)[Fimal P3200](#)[Gabbiani Prima](#)[Giben Prismatic](#)[Holz-Her Kvara](#)[Holz-Her Tria](#)[Holzma](#)[Homag España CH03](#)[Homag España CH03+](#)[Homag España CH12](#)[Kappa Automatic](#)[KDT XML](#)[Macmazza On-Board](#)[Macmazza SEZ2000](#)[Macmazza Tork MP70](#)[Martin T74](#)[Mayer](#)[Maver PC6200 controller](#)[Maver PC7000 controller](#)[NIMAC Ilenia](#)[PIOS](#)[PNL File](#)[Scheer DNC - AT](#)[Schelling HPO](#)[SCM MaestroCUT](#)[SCM Ottimo](#)[SCM PTX](#)[SCMI Sezionia](#)[SCMI Sigma 65](#)[SCMI WinCut](#)[SCMI x915](#)[SCMI z32](#)

[SCMI z45](#)[Selco XML](#)

## Sheet Router

The currently available Sheet Router NC Links are listed below. Simply click on the NC Link below for more information.

[CimTech](#)[IntelliCAM - Nest](#)

## Parts Router/Point to Point

The currently available Parts Router/Point to Point NC Links are listed below. Simply click on the NC Link below for more information.

[AlphaCAM Nest](#)[AlphaCAM P2P](#)[Auto Link](#)[Biesse Rover](#)[Building Blocks](#)[Building Blocks Nest](#)[CAD Code](#)[CAD Code Nest](#)[Cam 2](#)[Cam 3](#)[CimTech Nest](#)[CimTech P2P](#)[EnRoute](#)[Intelli-CAM - All Parts Full Outline](#)[Intelli-CAM - Parts \(Mirror\)](#)[Intelli-CAM - Parts \(No Mirror\)](#)[Intelli-CAM - Parts Full Outline](#)[Intelli-CAM - All Parts Full Outline \(Mirror\)](#)[Intelli-CAM - Parts Full Outline \(Mirror\)](#)[Speedy Link](#)[Standard DXF](#)[TASK](#)[TASK Version 3](#)[TASK Version 5](#)

[Thermwood](#)

[TwinCAM 32](#)

[WoodWop 25](#)

[WoodWop 4](#)

[WoodWop 5](#)

## Specialized - Chop Saw

The currently available Chop Saw NC Links are listed below. Simply click on the NC Link below for more information.

[Dimter S90](#)

[Grecon Dimter Opticut](#)

[Koch Maxi 5](#)

[OMGA T 421/521](#)

[Pistorius BF-A](#)

[RazorGage](#)

[Supercut](#)

[TigerStop](#)

## Specialized - Drill & Dowel

The currently available Drill & Dowel NC Links are listed below. Simply click on the NC Link below for more information.

- [Accu-Systems](#)
- [Accu-Systems HPJ](#)
- [Accu-Systems HPJ XML](#)
- [Accu-Systems HPMTH](#)
- [Accu-Systems HVVP](#)
- [Accu-Systems Junior Series](#)
- [Biesse ELIX](#)
- [Biesse ELIX B](#)
- [CNC Factory Scorpion](#)
- [Gannomat Express S1](#)
- [Gannomat Index](#)
- [Gannomat Index Pro](#)
- [Koch Sprint Plus](#)
- [Koch Sprint Plus II/L](#)
- [Laguna Lockdowel](#)

- [Omal](#)
- [Omal Albatros](#)
- [Omal Frame](#)
- [Omal Frame TpaEdi32](#)
- [Omal Lockdowel](#)
- [Omal TPAEdi32](#)
- [Omal Velox](#)
- [Pillar Lockdowel](#)
- [ProEdge Lockdowel](#)
- [Scheer Woodpecker](#)
- [Sprint](#)
- [Sprint PTP2 Beckhoff](#)
- [T.O.P. Albatros](#)
- [TpaCAD](#)
- [Vitap Thunder](#)
- [Vitap Thunder CID3](#)
- [Weeke ABD](#)
- [Weeke BEK](#)

## Specialized - Other

The currently available Other NC Links are listed below. Simply click on the NC Link below for more information.

[Accu-Systems 2S](#)

[Accu-Systems MTH](#)

[Accu-Systems Universal](#)

[Anderson MLS](#)

[Biesse Door Machine](#)

[Hoffmann Beaded Frame System](#)

[Homag Edgebander](#)

[Omal MTH](#)

[OSync](#)

[Pillar MTH](#)

[Pillar PLT](#)

[Voorwood Shape and Sand](#)

[Weinig French Miter](#)

## Material Handling

- [Barbaric Material Handler](#)
- [Flexstore](#)
- [Intellistore](#)
- [Storemaster](#)
- [Winstore](#)

## Miscellaneous

- [Populate Database Only](#)
- [Script Test](#)

## Dimter S90

This is a Chop Saw Machine Link for the Weinig Dimter S90 Cut Saw.

Only selected parts made of board stock are output.

## Parameters

Box Alignment - Contact Weinig for information on this Parameter.

Box Assignment - Contact Weinig for information on this Parameter.

Box Type - Contact Weinig for information on this Parameter.

## Grecon Dimter Opticut

This is a Chop Saw Machine Link for the Weinig Grecon Dimter Opticut 104 Optimizing Cut Saw.

Only selected parts made of board stock are output.

## Koch Maxi 5

This is a Chop Saw Machine Link.

Output for Koch/Paul Maxi 5 Premium Control.

Only selected parts made of board stock are output.

## Structure of Cutting List To Be Imported into MAXI 5 Premium Control

The file to be imported must start with the order details comprising 9 lines, each with a maximum of 100 characters. This line will start with #; and end with ;

Then follow the individual lists.

Each list may contain a 'project' line with a maximum of 100 characters. The project name will start with #; and end with ;

Each list may contain max. 400 sizes.

Data have to be in ASCII code, i.e. ASCII characters, hex 20 to hex 7E. Not allowed are: \$ , @ , \ .

Decimals to be separated by a point, not by a comma, each line to end with <CRLF> (Carriage return / line feed).

File name: nnnnnnnnnnn.stk . Any name with the ending stk.

Line structure:

P xx; Active; Control; Length; Number of Pieces; Sorting Station; Priority; Printing Code; <CRLF>

Thereby:

P xx: Cutting list number 1 .. 60

Active: If the lengths in this line are to be used, set this value to 1, otherwise to 0

Control: For special applications, always set to 1

Length: in millimeters with one decimal place, e.g. 1000.2 mm, min. 20.0 mm, max. 7000.0 mm

Number of pcs. 1 to 5 000000

Sorting station: 1 to 48

Priority: 1 to 300 in %, standard value is 100%

Printing code: If not used, then send “ ;“ (2 blanks and semicolon).

Printing code may comprise max. 30 ASCII characters (no semicolon, no @ within the character string)

Example: Test cutting list\_220304.stk

#; 'Order' line1;

#; Client: Muller Timber;

#; Species: Oak;

#; Timber section 33\*45;

#; Completion. Xx.yy.zzzz;

#; STATISTICS: 3345;

#;

#; Further information;

#; Line 9;

P1; #; 'Project' text cutting list 1, max. 100 characters;

P1; 1; 1; 567.3; 125; 2; 100; 70.47.45 42 front jamb;

P1; 1; 1; 825.6; 222; 3; 95; 10.39.45 651 rear jamb ;

P1; 1; 1; 925.6; 322; 4; 105; 71.43.45 600 upright ;

P2; #;'Project' text 2;

P2; 1; 1; 456.7; 222; 4; 100; TTTT ;

P2; 1; 1; 625.6; 347; 3; 95; rail Q2 ;

P3; #; Additional components Q3;

P3; 1; 1; 555.7; 225; 2; 90; GHZF;

P3; 1; 1; 721.6; 1347; 9; 100; batten 567.34 ;

...

...

P60;

## Omga T 421/521

This is a Chop Saw Machine Link.

Only selected parts made of board stock are output.

### File Protocol T 421/521 OPT

The Structure must be the following

- ‘S’ ; = start transmission
  1. ‘P’ ; = cycle number (max 3 characters)
  2. ‘.’ ; = (optional line) id , cycle alphanumeric name (max 8 characters)
  3. ‘X’ ; = (optional line) n° box ( max 3 characters)
  4. ‘Y’ ; = (if first grade, optional line; if omitted, grade =1) grade values = only 1 or 2
  5. ‘Q’...’T’ ; = length (max 3 numbers+ 3 after the decimal sign), number of repetitions (Limit of 500 Records per File)
- ‘E’ ; = end transmission

The cycles sent as per above will be written in the programs memory with the same rules as function FUN1; therefore no cycle number grater than the next cycle available will be accepted.

By transmitting cycle number already present in the controller, the cycle on the controller will be overwritten.

#### IMPORTANT:

Because of programming restrictions it is necessary to:

1. close each line with a semi-colon’;
2. start the line with the character identifying the data that will follow (see above)
3. use only UPPER-CASE characters for the letters preceding the data on each line

Example of ASCII file with unit of measure Inches (controller memory empty) :

```
S                ;start transmission

P1                ;cycle #1
:first            ;id = 'first'
X1                ;box # =1
```



|             |                                                            |
|-------------|------------------------------------------------------------|
| Y1          | ;grade 1                                                   |
| Q25.125 T5  | ;piece length =25 1/8” , Number of pieces (repetitions) 5  |
| P2          | ;cycle #2                                                  |
| Q12.500 T8  | ;piece length =12 1/2” , Number of pieces (repetitions) 8  |
| P2          | ;cycle #2                                                  |
| Y2          | ;grade 2                                                   |
| Q10.750 T10 | ;piece length =10 3/4” , Number of pieces (repetitions) 10 |
| P3          | ;cycle #3                                                  |
| Q30.750 T12 | ;piece length =30 3/4” , Number of pieces (repetitions) 12 |
| Q15.500 T5  | ;piece length =15 1/2” , Number of pieces (repetitions) 5  |
| E           | ;end transmission                                          |

**Note 1:** new cycles begin each time the Material or Width changes

**Note 2:** in order to make the data sequence more understandable, it is allowed to:

- insert blank lines in between the data lines
- insert spaces (only after the numbers and not the TAB character)
- insert comments on each line after the semi-colon ';' character

**Note 3:** the unit of measures will be the one selected on the controller with the 'UN\_MIS' parameter

**Note 4:** by not transmitting the optional lines those field will be set to ZERO (i.e. ID, width, box, grade).

## Pistorius BF-A

This is a Chop Saw Machine Link.

Output for Pistorius BF-A CNC Sawing and Notching (Haunching) Machine. Only parts made of board stock are output. Output can include machining mortise and tenons for stiles and rails when operations are linked to parts using a specific tool (see the eSupport Tech Packages area for examples of using this NeLink with Cabinet Vision Solid).

Valid feed rates for the tool are 1 - 5 in/sec.

### Output File Specifications

The are 3 sections in the file, Cabinets, Parts and Materials:

#### [Cabinets]

The Cabinets section is not used.

#### [Parts]

The Parts section has the following structure:

**PartID** - Unique identifier for the part (Data type: long integer)

**CabinetID** - Not used. Must be 0.

**PartName** - Name of part (not the part type) (Data type: string 50 characters max)

**MaterialID** - Unique identifier from [Materials] section that defines the parts material (Data type: long integer)

**PartTypeID** - Identifies the part type (Data type: long integer, 1-Right style, 2-Bottom rail, 3-Left style, 4-Top rail, 5-Mid style, 6-Mid rail)

**ParentID** - Not used. Must be 0.

**ChildID** - Not used. Must be 0.

**Side1Data** - Notch locations (Data type: string 255 characters max, see notes below)

**Side2Data** - Notch locations (Data type: string 255 characters max, see notes below)

**PartNumber** - Not used. Must be 0.

**Left** - Not used. Must be 0.

**Right** - Set to length of part not including extra length required due to bead (Data type: string 16 characters max)

**Bottom** - Set to length of part not including extra length required due to bead (Data type: string 16 characters max)

**Top** - Always set to zero for your import file

#### **Notes on format of Side1Data and Side2Data:**

Each notch on the side has the following format

";MaterialID:NotchStart,NotchEnd;"

Each entry is surrounded by a semicolon. If no notching is required there will just be a ; in the field.

The MaterialID comes from Materials section with width of the material matching the width of the notch.

### **[Materials]**

The Materials section has the following structure:

**MaterialID** - Unique identifier for the material (Data type: long integer)

**MaterialName** - Name of material (Data type: string 20 characters max)

**NumberOfBeads** - 0, 1, or 2 (Data type: integer)

**Width** - Overall width of material including bead dimension (Data type: string 10 characters max)

**Bead** - Size of bead (Data type: single)

**CutSpeed** - Speed material will be cut. (Data type: string 10 characters max, 1-5in/sec 3 seems to work with everything)

## **RazorGage**

This is a Chop Saw Machine Link.

Only parts made of board stock are output. Output can include scribe marks for positioning mid stiles and rails.



If you are using Solid to output to the S2M Center and you need the output for Scribe Marks included you will want to download the RazorGage UCS/Intelli-Joint Package from the Downloads section of eSupport.

## Razor Gauge File Format

The following is the field information for the Razor Gauge Cutoff Saw.

Material Name  
Material ID  
Part Quantity  
Part Name  
Part Width  
Part Length  
Part Thickness  
Room Name  
Room ID  
Cabinet Name  
Cabinet ID  
Scribe Marks

The scribe marks are identified in a string formatted using the pipe symbol “|” as the delimiter:

A part that needs scribe marks at 10, 12, 12.5, 18, and 19.75 would be output as:

“10|2|.5|5.5|1.75”

A part that had no scribe mark would be “ “

Currently the output is restricted to imperial

The data in the file is already sorted by:

- 1. Material
- 2. Part Width (descending)
- 3. Part Length (descending)

Materials that are not board stock are not filtered out.

## Supercut

This is a Chop Saw Machine Link.

This link will create the .asc files that are used by the Salvador SuperCut Series of Chop Saws. There are a number of parameters that are used to generate the output for this link and details on what each parameter means is below.

|      |                                                                                                                                                                            |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| JN   | Adds the job name to the file that is created for this output                                                                                                              |
| LAB  | This is a label that will be applied to all parts, this can contain variable text (i.e. {PartName}). The full variable list will be provided at the end of this text file. |
| KICK | The kicker list, must follow this format: [max length (mm - in)],[kicker (#)];etc. This would look like this: 36,1;59.9,2;75,13                                            |
| PM   | Printing Mode (0 = None)                                                                                                                                                   |
| SP   | Speed (0 - 100%)                                                                                                                                                           |
| TCL  | Trim Cut Length (mm - in)                                                                                                                                                  |
| ECL  | End Cut (mm - in)                                                                                                                                                          |
| PRI  | Priority (0 = Length, 1 = Fixed, 3 = Value)                                                                                                                                |
| FO   | Fast Optimization                                                                                                                                                          |
| DMC  | Delete cut between two pieces if their sum is less longer than this value (mm - in)                                                                                        |
| SGT  | Shattering - If the waste is more than (mm)                                                                                                                                |
| SLT  | Shattering - and less than (mm)                                                                                                                                            |
| SHA  | Shattering - cut it in pieces of (mm)                                                                                                                                      |
| WO   | Cutting Waste Optimization                                                                                                                                                 |
| DQ2  | Disable quality from 2nd                                                                                                                                                   |
| WGT  | Waste Selection - If waste is more than (mm - in)                                                                                                                          |
| WEJ  | Waste Selection - eject it on ejector n                                                                                                                                    |

|        |                                                          |
|--------|----------------------------------------------------------|
| CMEL   | Cut On Mark - Eject only pieces longer than (mm - in)    |
| CMP    | Cut On Mark - Print Length                               |
| ER     | Ejectors work for length range                           |
| CSP    | Cut Speed (% of max cut speed)                           |
| MWP    | Maximum waste allowed                                    |
| LAW    | Least absolute Waste (mm - in)                           |
| MAW    | Maximum absolute waste (mm - in)                         |
| WST    | If waste is longer than (mm - in)                        |
| WDB    | go into width database                                   |
| QDB    | and quality database                                     |
| RPW    | Recover measure on previous waste                        |
| RMPWM  | Recover measure on previous waste for max (0 = all; mm)  |
| RMNWM  | Recover measure on next waste for max (0 = all; mm)      |
| COMEE  | Cut only on mark and eject on ejector n                  |
| FJWLT  | Finger Joint - If the waste is longer than (mm - in)     |
| FJIW   | Finger Joint - increase the waste up to (mm - in)        |
| FJPWM  | Finger Joint - to produce waste max (mm - in)            |
| FJEO   | Finger Joint - and eject on n                            |
| FJPLR  | Print length on recovers for finger joint                |
| COMWGT | If waste is greater than allowed waste, cut only on mark |
| COMEON | and eject on ejector n                                   |
| PRLN   | Print Length                                             |
| PREFIX | Printing prefix                                          |
| SUFFIX | Printing suffix                                          |
| TPWM   | To produce waste max (mm - in)                           |

## Label Variables

|              |                                    |
|--------------|------------------------------------|
| AsmComment   | The comment applied to the cabinet |
| AsmDepth     | The depth of the cabinet           |
| AsmDesc      | The Description of the cabinet     |
| AsmHeight    | The height of the cabinet          |
| AsmNo        | The cabinet id                     |
| AsmName      | The name of the cabinet            |
| AsmWidth     | The Width of the cabinet           |
| InternalName | The part internal name             |
| JobDesc      | The job description                |
| JobName      | The job name                       |
| MatDesc      | The material description           |
| MatLength    | The material length                |
| MatName      | The material name                  |
| MatWidth     | The material width                 |
| PartComment  | The comment added to the part      |
| PartLength   | The length of the part             |
| PartName     | The name of the part               |
| PartNLength  | The parts nominal length           |
| PartNWidth   | The parts nominal width            |
| PartThick    | The parts thickness                |
| PartWidth    | The parts Width                    |
| PrimaryEdge  | The primary edge of the part       |
| RoomName     | The name of the room               |
| WallNo       | The wall number                    |

# TigerStop

This is a Chop Saw Machine Link.

Only selected parts made of board stock are output.

## File Format

| Field # | Description                                                          |
|---------|----------------------------------------------------------------------|
| 1       | Job Number (string)                                                  |
| 2       | Wall Number                                                          |
| 3       | Room Number                                                          |
| 4       | Cabinet Number                                                       |
| 5       | Quantity of Part                                                     |
| 6       | Width of Part                                                        |
| 7       | String Width of Part                                                 |
| 8       | Length of Part                                                       |
| 9       | String Length of Part                                                |
| 10      | Thickness                                                            |
| 11      | String Thickness                                                     |
| 12      | Part Number                                                          |
| 13      | Part Name                                                            |
| 14      | Material Number                                                      |
| 15      | Material Name                                                        |
| 16      | Old Mat Type (1 = Panel, 2 = Board, 3 = Frame, 4 = Door, 5 = Drawer) |
| 17      | Grain Dependent (1 = Yes, 2 = No)                                    |
| 18      | Optimize (1 = Yes, 0 = No)                                           |
| 19      | Stock Part (1 = Yes, 2 = No)                                         |
| 20      | Waste Percentage                                                     |
| 21      | Cost                                                                 |
| 22      | Cost Unit                                                            |
| 23      | Labor in Minutes                                                     |
| 24      | Spare Number 1                                                       |
| 25      | Spare Number 2                                                       |
| 26      | Spare Number 3                                                       |
| 27      | Spare Number 4                                                       |
| 28      | Cabinet Name                                                         |
| 29      | Part Comment                                                         |
| 30      | Spare String 1                                                       |
| 31      | Spare String 2                                                       |
| 32      | Spare String 3                                                       |

# Accu-Systems

This is a Drill and Dowel Machine Link.

This output will create program files for the Accu-System Drill / Dowel machines. Each program will represent one side of a part. When running these programs place the part interior - up, exterior -down.

## Accu-Systems HPJ

This is a Drill and Dowel Machine Link.

This output will create program files for Accu-Systems HPJ. Each program will represent one side of a part.

## Accu-Systems HPJ XML

This is a Drill and Dowel Machine Link.

This output will create program files for the Accu-Systemss HPJ Drill / Dowel machines with Vertical Routing. Drill and Dowel operations are determined by matching the horizontal bore depth. Assign tools to this machine and to a dado operation to create vertical grooving output. Additional parameters can be set on tools assigned to this machine in tool catalog.

## Accu-Systems HPMTH

Accu-Systems HPMTH. This machine can haunch, mortise, and tenon. Each program will represent one part. Tools should be assigned to this Machine then linked to an operation.

## Accu-Systems HVVP

This is a Drill and Dowel Machine Link.

This machine has 2 horizontal drills, 1 vertical drill and can insert either a 6mm or 8mm dowel. In addition it can create a horizontal slot with a 6mm router bit.

## Accu-Systems Junior Series

This is a Drill and Dowel Machine Link.

This output will create program files for the Accu-System Drill / Dowel machines. Each program will represent one side of a part. When running these programs place the part interior - up, exterior -down.

## Biesse ELIX

This is a Drill and Dowel Machine Link.

This output will create program files for the Biesse ELIX Drill / Dowel machines. Each program will represent one side of a part.

## Biesse ELIX B

This is a Drill and Dowel Machine Link.

This output will create program files for the Biesse ELIX B Drill / Dowel machines. Each program will represent one side of a part.

## CNC Factory Scorpion

This is a Drill and Dowel Machine Link.

This output will create program files for the CNC Factory Scorpion CNC Drill & Dowel Machine. Assign tools to this machine to create the correct output. In the tool parameters you can provide information such as tool height offset id, tool number, whether or not the tool is an insertion tool, as well as the insertion depth, insertion offset id, and the insertion gun offset.

When assigning a Tool to a CNC Factory Scorpion Machine you will get the following additional Tool options:

- **Tool Number**

- **Height Offset ID**
- **Descent Rate**

They will also have the following Parameters added:

- **Insert Dowel (Yes/No)** - Denotes this tool as an insertion tool
- **Insert Depth (Measurement)** - Provides the insertion Depth for this tool
- **Insert Offset (Integer)** - The insertion offset id for the tool
- **Gun Offset (Integer)** - The gun insertion offset id
- **Safe Rapid Level (Measurement)** - Provide safe z level for rapid movement

## Gannomat Express S1

Gannomat Express S1. Machine can drill as Vertical boring. Assign tools to this machine to create output. Assign tools to this machine with tool catalog to set Drill or Drill and Insert.

## Gannomat Index

This is a Drill and Dowel Machine Link.

This output will create a DataBackEnd.mdb file for producing Drill / Dowel Output for the GannoMAT Index & Matrix machines.

## Gannomat Index Pro

Gannomat Index Pro. Machine can drill, glue, dowel as well as perform vertical boring, horizontal and vertical grooving and horizontal routing. Assign tools to this machine to create output. Assign Tools to this Machine with in the Tool catalog to set additional parameters.

## Koch Sprint Plus

This is a Drill and Dowel Machine Link.

This output will create program files for the Koch Sprint-Plus and Sprint-Plus-L machines with Vertical Boring. Assign tools to this machine and to the boring operations to create output. Additional instructions such as dowel and glue can be set on tools assigned to this machine in tool catalog. One program will be created for each part or per edge depending on your setup.

## Koch Sprint Plus II/L

This is a Drill & Dowel Link

Has the following machine parameters:

- **One program per part (Boolean)** - Output one program per part rather than per edge
- **Use part len as prog len (Boolean)** - Use part's length as program length rather than edge against the fence. Without this set L & R's length = part width
- **Display RPM (Boolean)** - Display RPM (Spindle Speed) in output. Without this, the spindle speed is set to 0
- **Display Operation Depth (Boolean)** - Display operation depth in output. Without this, the depth is set to 0

Has the following tool parameters:

- **Canned Cycle Number (string)** - the canned cycle number to use with this tool
- **Operation Type (string)** - The type of operation that this tool represents

## Laguna Lockdowel

This is a Drill and Dowel Machine Link.

This output will create program files for Laguna Tools Drill / Dowel machines. Assign tools to this machine and to the operations to create output. It is specific to Lockdowel setup with support for EClips and Spring Pins. Each program will represent one side of a part.

## Omal

This is a Drill and Dowel Machine Link.

This output will create program files for the OMAL Drill / Dowel machines. Each program will represent one side of a part.

## Omal Albatros

This is a Drill and Dowel Machine Link.

This output will create program files for Omal with the Albatros control. Each program will represent one side of a part. Assign Tools to this Machine with Tool Catalog to set additional parameters.

## Omal Frame

This is a Drill and Dowel Machine Link.

Output for Omal Frame Machine. This machine can dowel edges and the back of the part. Output is one program per edge. Tools should be assigned to this machine then linked to an operation. Additional parameters are required in the tool catalog

## Omal Frame TpaEdi32

This is a Drill and Dowel Machine Link.

This output will create program files for mortise, tenon, and haunching using Omal using TpaEdi32. Assign tools to this machine and to the mortise/tenon operations to create output. See the OmalFrame.pkg in the eSupport Cabinet Vision > Files > Extras area for setup. One program will be created per edge.

## Omal Lockdowel

This is a Drill & Dowel Link.

Has the following tool parameter added:

- **Macro** - The macro to be assigned when using this tool

## Omal TPAEdi32

This is a Drill and Dowel Machine Link.

This output will create program files for Omal using TpaEdi32. The output support Horizontal Drill / Glue / Dowel and Vertical Boring. Assign tools to this machine and to the boring operations to create output. Additional instructions such as dowel, glue and macro name can be set on tools assigned to this machine in tool catalog. One program will be created per edge.

## Omal Velox

This is a Drill and Dowel Machine Link.



Output for the Omal Velox Drill & Dowel machine

## Pillar Lockdowel

This is a Drill and Dowel Machine Link.

Output for the Pillar Lockdowel machine.

When a Tool in the Tool Catalog is assigned to this machine you will get two Parameters:

- **Drill** (\_hmInsert) - When set to Yes indicates the tool can be used to Drill.
- **Insert** (\_hmDrill) - When set to Yes indicated the tool can be used to Insert.

## ProEdge Lockdowel

The ProEdge Lockdowel machine link works with the Lockdowel connectors which can be created in the Material manager. The output of this machine link is controlled by several parameters on the Material connectors that can be edited if required. The parameters that control the link start with LD\_ followed by the machine parameter.

## Scheer Woodpecker

This output will create program files for the Scheer Woodpecker Drill / Dowel machines. Each program will represent one side of a part. Assign tools to this machine with tool catalog for additional parameters to determine if operation will dowel and glue.

## Sprint

This is a Drill and Dowel Machine Link.

This output will create program files for the Sprint Drill / Dowel machines.

## Sprint PTP2 Beckhoff

This is a Drill and Dowel Machine Link.

This output will create program files for the Sprint Beckhoff Drill / Dowel machines. Each program will represent one side of a part.

## T.O.P. Albatros

This is a Drill and Dowel Machine Link.

Output for the T.O.P Drill & Dowel machine.

When a Tool in the Tool Catalog is assigned to this machine you will get the following Parameters:

- **Drill** (\_hmDrill) – Allows you to specify that this tool should perform a drill operation
- **Insert** (\_hmInsert) – Allows you to specify that this tool should perform an insertion operation
- **Macro** (\_hmMacro) - Macro name to use for the insertion operation

## TpaCAD

This is a Drill and Dowel Machine Link.

This output will create program files for Drill and Dowel Machines using TpaCAD. The output supports Horizontal Drill / Glue / Dowel and Vertical Boring. Assign tools to this machine and to the boring operations to create output. Additional instructions such as dowel, glue, macro name, programmable depth, and tool number can be set on tools assigned to this machine in tool catalog. One program will be created per edge.

If you need to use the pocket screw attachment for this link, create a new horizontal tool, then assign it to the machine using this link, then set the Macro property to pk.tcn (this is case insensitivite). When that is done, the neccessary changes to the output will be made when the pocket screw attachment is needed.

### Tool Properties

PSHOWTOOLPROP\_ToolNumber

### Link Parameters

- **Add Glue - (\_hmGlue) [Bool/Tool]:** Add glue when using this tool
- **Add Dowel - (\_hmDowel) [Bool/Tool]:** Add dowel when using this tool
- **Macro - (\_hmMacro) [String/Tool]:** The macro name to use for Glue/Dowel operation: forainsert.tmc
- **Base zone # on Part Size - (\_hmZoneOnPrtSize) [Bool/Machine]:** Option to uset he smaller inside zones if the part will fit
- **Dimension 'A' - (\_hmMaxSizeForZone) [Measurement/Machine]:** Maximum size for the small zone
- **Programmable Depth - (\_hmProgDepth) [Bool/Tool]:** When true, this will output operation depth for this tool, when false, depth will equal 0

## Vitap Thunder

Machine setup is for 2 horizontal drills, 1 vertical drill, horizontal glue and dowel insertion. Assign tools to this machine with Tool Catalog to set additional parameters.

## Vitap Thunder CID3

This is a Drill and Dowel Machine Link.

Vitap Thunder CID3. Machine for boring, gluing and dowel insertion.

## Weeke ABD

This is a Drill and Dowel Machine Link.

This output will create program files for the Weeke ABD Drill / Dowel machines with Vertical Boring. Assign tools to this machine and to the boring operations to create output. Additional instructions such as dowel and glue can be set on tools assigned to this machine in tool catalog. One program will be created for each part or per edge depending on your setup.

## Weeke BEK

This is a Drill and Dowel Machine Link.

This output will create program files for the Weeke BEK. One program will be created for each part or per edge depending on your setup.

## Barbaric Material Handler

This link will create a .csv file, for each material used, that the Barbaric Material Handler can read to help with loading of materials onto a machine. All materials that are to be used will need to have a unique ID assigned to them in the Material parameters of CABINET VISION. By default the name of the ID parameter is \_mhid, but you can change this to whatever you want it to be and then just define what the parameter name should be in this links output section.

## Flexstore

Output for the SCM Flexstore Material Handling system.

SCM Mahros Material Handling system. Materials can by synchronized with this system in the [Offcut Manager](#).

## Intellistore

Output for the Homag Intellistore Material Handling system.

Homag Intellistore Material Handling system. Materials can by synchronized with this system in the [Offcut Manager](#).

## Storemaster

Output for the Holz-Her Storemaster material handler.

Materials can by synchronized with this system in the [Offcut Manager](#).

## Winstore

Output for the Biesse Winstore Material Handling system.

Biesse Winstore Material Handling system. Materials can by synchronized with this system in the [Offcut Manager](#).

## Populate Database Only

The Ardis Optimizer (a third party Optimizer) can directly read the PSNC-NC (-CV-AC).MDB file which is populated with machining information. This link will populate the data for the Ardis Optimizer or any other 3rd Party Application to read.

## Script Test

This is a link demonstrating the use of a Visual Basic Script file. To view or add a Visual Basic Script to a link you are utilizing you must open the PSNC.MDB file and go to the Links table. In the Links table there is a field named Program this is where a script would need to be placed. If you are not familiar with Visual Basic Script there are several publications on creating them which are available from most major book sellers.

## CimTech

This is a Primary Machine Link.

Use this output when using CimTech as a post when CNC Center has nested the parts. CimTech knowledge must be setup to match the layer names output by CNC Center. To ensure proper output, the router diameters in the tool catalog should be slightly less then the required dado width. Tool compensation is recommended for this output. The following information from the tool catalog can not be used by CimTech and is therefore ignored: Feed rate, descent rate, and maximum depth per pass.

## IntelliCAM - Nest

This is a Primary Machine Link.

IntelliCAM Nested Sheet Output to NC Code.

## Accu-Systems 2S

Output for Accu-Systems 2S Machine.

Output is one program per job. The 2S is a multi axis, multi function CNC horizontal machine, designed to size & square edge glued panels.

## Accu-Systems MTH

Output for the Accu-Systems MTH.

Output for Accu-Systems MTH frame Mortise/Tenon/Haunching machine thru AccuCom. This machine can haunch stiles and rails. Output is one program per job. Tools should be assigned to this machine then linked to an operation.

### Specifications for Accu-Systems MTH variable

The MTH machine manufactures face frame and door components. These components are relational in their intersections, and units. This relational information is usually not provided in the standard CSV exports provided by design software. This document is designed to outline the data, and format that is needed from the design software in order to process the data for proper machining on the MTH machine.

A normal record from a CSV file would appear as follows:

*Description,Width,Thickness,Length,Qunatity,Job,Material,Assembly,Comment1,2,3  
Stile,2.5,0.75,29.5,4,JOB,Red Oak,Unit1,1,0,Profile1*

The record provides items such as width, length, and other component specific data used for cutting and labeling. What is missing is any data related to an intersecting part like you would find in a door, or a face frame. Also profiling data for the sides of each part is required to detail the component. For design software to coordinate this data, additional information is needed. A new variable that breaks down all relating intersecting parts, and profiling can be added to provide this data. It requires several things:

- 1. Unique Unit names for assemblies per job
- 2. Unique part names per unit
- 3. Center point intersection
- 4. Side distinct referencing
- 5. Profile name for Side 0
- 6. Profile name for Side 1
- 7. Profile name for End 0
- 8. Profile name for End 1

An example of this would be:

*Stile-1,2.5,0.75,29.5,4,JOB,Red Oak,Unit1,1,0, Profile1,(1~2~Rail-1|1~28.375~Rail-2),ProfS0,ProfS1,ProfE0,ProfE1*

*Stile-2,2.5,0.75,29.5,4, JOB,Red Oak, Unit1,1,0, Profile15,(0~2~Rail-1|0~28.375~Rail-2),ProfS0,ProfS1,ProfE0,ProfE1*

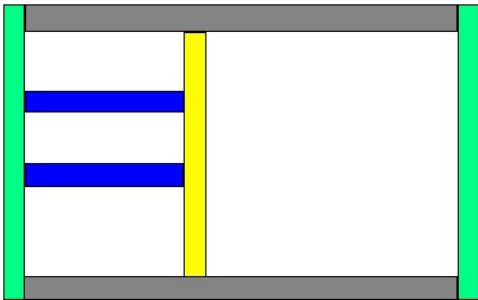
*Rail-1,4,0.75,52.875,2, JOB,Red Oak, Unit1,1,1, Profile1,,ProfS0,ProfS1,ProfE0,ProfE1*

*Rail-2,2.5,0.75,52.875,2, JOB,Red Oak, Unit1,1,1, Profile1,,ProfS0,ProfS1,ProfE0,ProfE1*

In this example, each part within the unit is defined as Stile-1 or Stile-2 etc. within Unit1. The last field *(0~2~Rail-1|0~28.375~Rail-2)* references:

- 1. The side of the part where the intersection occurs
  - a. 0 references the left side of a vertical part
  - b. 0 references the top of a horizontal part
  - c. 1 references the right side of a vertical part
  - d. 1 references the bottom side of a horizontal part
- 2. The center point of the intersection
- 3. The part name of the intersecting part
- 4. “|” separates the intersections within the field

In this way complex face frames can be manufactured quickly and easily. The following section demonstrates in detail.



With this example there are two stiles, rails, mid-rails, and one mullion. The naming nomenclature to make the two parts unique is unimportant as long as the names are different. In this example they will be specified with a “-1” or “-2”.

The left stile would be detailed in the CSV as follows:

```
Stile-1,2.5,0.75,29.5,1,JOB,Red Oak,Unit1,1,0, Profile1,
(1~1.25~Rail-1|1~28.25~Rail-2|1~13~MidRail-1|1~18~MidRail-2),ProfS0,ProfS1,ProfE0,ProfE1
```

Since all of the intersections of Stile-1 are on the right side, all references are “1”. The rails (Rail-1-Rail-2) in this instance are 2.5 inches wide causing the intersection to be centered at 1.25 inches and 28.25 respectively.

The right stile would be detailed in the CSV as follows:

```
Stile-2,2.5,0.75,29.5,1,JOB,Red Oak,Unit1,1,0, Profile1,
(0~1.25~Rail-1|0~28.375~Rail-2),ProfS0,ProfS1,ProfE0,ProfE1
```

The Mid-Rail has no intersections so its record would simple be void of the variable:

```
MidRail-1,2.5,0.75,17.875,1,JOB,Red Oak,Unit1,1,0, Profile1, ,ProfS0,ProfS1,ProfE0,ProfE1
```

The Mullion would be defined much like the Right Stile:

```
Mullion-1,2.5,0.75,24.5,1,JOB,Red Oak,Unit1,1,0, Profile1,
(0~10.5~ MidRail -1|0~15.5~ MidRail -2) ,ProfS0,ProfS1,ProfE0,ProfE1
```

The Top and Bottom Rails in this case would be identical with the exception of machining side orientation:

```
Rail-1,2.5,0.75,52.5,1, JOB,Red Oak,Unit1,1,0, Profile1,( 1~20.375~Mullion-1),ProfS0,ProfS1,ProfE0,ProfE1
```

```
Rail-2,2.5,0.75,52.5,1, JOB,Red Oak,Unit1,1,0, Profile1,( 0~20.375~Mullion-1),ProfS0,ProfS1,ProfE0,ProfE1
```

# Accu-Systems Universal

Universal output for various Accu-Systems Drill/Dowel/Mortise/Tenon/Haunching machines. Output is one program per job. Tools should be assigned to this machine then linked to an operation.

# Anderson MLS

This link will create a .csv file, for each material used, that the Anderson Material List can read to hep with loading of materials onto a machine. All materials that are to be used will need to have a unique ID assigned to them in the Material parameters of CABINET VISION. By default the name of the ID parameter is \_mhid, but you can change this to whatever you want it to be and then just define what the parameter name should be in this links output section.

If you do not provide a material ID, <invalid> will appear in the file instead

## Biesse Door Machine

Output for Biesse Door Machine. Output is one program per job. Doors require additional parameters for proper output: \_BD\_BIESSEDOOR, \_BD\_HINGETYPE, \_BD\_PROFILES, \_BD\_TOPHCUP, \_BD\_BOTHCUP, \_BD\_PULLPOX, \_BD\_PULLPOSY, \_BD\_PULLDESC, \_BD\_DOORNOTES

### Sample Output

```
[DISTINTA : CAMPI]
campoD0=LABL,A,4,4,NULL,0,0
campoD1=PROG,A,256,11,NULL,0,2
campoD2=QNTA,U,4,4,NULL,0,0
campoD3=CONT,U,4,4,NULL,0,0
campoD4=COMM,A,256,20,NULL,0,0
[DISTINTA : RIGHE]
,ASSEMBLEDDOOR.BPP,1,0,LPX=1200.00 LPY=400.00 PROFILE=EDGE1 TOPHCUP=60 BOTHCUP=65 HINGETYPE=HETTICH HANDLET=3 HANDLEPOX=102.46 HANDLEPOSY=102.46 SPECIES=CHERRY
ARCH=2,
,00000000PANEL.BPP,1,0,LPX=1200.00 LPY=400.00 PROFILE=EDGE1 SPECIES=MAPLE,
,00ARCHTOPRAIL.BPP,1,0,LPX=1200.00 LPY=400.00 PROFILE=EDGE1 SPECIES=MAPLE TYPEARCH=2,
****
****
```

---

This part of the topic is a description of the above:

Line beginning with AssembledDoor.bpp,1,0 (is the Name of the assembled Door but has to have the .bpp,1,0)

Followed by the finished size of the door and the profile name for the outside edge, followed by the centerline of each hinge assembly and hinge name, then handle/knob type/name and X Y location of center line of handle, then material name. All of descriptions below per the sample above are separated by spaces.

#### Assembled door operation

- Size & profile outside edges (same tool)
- Sand profile
- Drill for hinges (custom locations for base doors)
- Drill for door pulls
- Drill for bumper pads (not necessary)

The next line contains the Panel Details.

#### Door panel operation (before assembly)

- Size panel edges
- Profile outside edges (multiple passes depending on wood species)
- Sand profile (multiple passes depending on wood species)

The last line contains the Top Rail Shape which is not currently supported by the S2M Center.

## Hoffmann Beaded Frame System

Output for Hoffmann CNC Notching Machine. This machine can haunch stiles and rails. Each program will represent one part. Tools should be assigned to this machine then linked to an operation. (see the eSupport Tech Packages area for examples of using this NcLink with Cabinet Vision Solid)

## Homag Edgebander

This link will create a .csv file that can be used with the Homag CNC Edgebander.

## Omal MTH

Output for Omal MTH frame Mortise/Tenon/Haunching machine. This machine can haunch stiles and rails. Output is one program per frame. (see [OmalMTH.pkg](#) for example).

## OSync

Output for the C.R. Onsrud OSync machine.

## Pillar MTH

Output for Pillar MTH Machine. This machine can haunch stiles and rails. Each program will represent one edge of the part. Tools should be assigned to this machine then linked to an operation. (see [PillarMTH.pkg](#) for examples).

## Pillar PLT

This outptus an xml file that can be used with the Pillar PLT machine.

Following are the Link Parameters:

- Print data 1 variable - (`_varPrintData1`) [String/Machine]: Variable text for print data field 1
- Print data 2 variable - (`_varPrintData2`) [String/Machine]: Variable text for print data field 2
- Print data 3 variable - (`_varPrintData3`) [String/Machine]: Variable text for print data field 3
- Print data 4 variable - (`_varPrintData4`) [String/Machine]: Variable text for print data field 4
- Print data 5 variable - (`_varPrintData5`) [String/Machine]: Variable text for print data field 5
- Print data 6 variable - (`_varPrintData6`) [String/Machine]: Variable text for print data field 6
- Print data 7 variable - (`_varPrintData7`) [String/Machine]: Variable text for print data field 7
- Print data 8 variable - (`_varPrintData8`) [String/Machine]: Variable text for print data field 8
- Print data 9 variable - (`_varPrintData9`) [String/Machine]: Variable text for print data field 9
- Print data 10 variable - (`_varPrintData10`) [String/Machine]: Variable text for print data field 10

Following are the valid Variables to use:

Each variable should be enclosed in a set of curly braces (`{ }`) otherwise the text will be considered a string literal.

| Name          | Type         | Size |
|---------------|--------------|------|
| ID            | Long Integer | 4    |
| JobID         | Long Integer | 4    |
| MatDBID       | Long Integer | 4    |
| Name          | Short Text   | 50   |
| InternalName  | Short Text   | 25   |
| Optimize      | Yes/No       | 1    |
| CNC           | Yes/No       | 1    |
| Width         | Double       | 8    |
| Length        | Double       | 8    |
| WidthString   | Short Text   | 12   |
| LengthString  | Short Text   | 12   |
| NominalWidth  | Double       | 8    |
| NominalLength | Double       | 8    |
| Comment       | Long Text    | N/A  |
|               |              |      |

|                  |              |     |
|------------------|--------------|-----|
| Qty              | Integer      | 2   |
| OverMake         | Integer      | 2   |
| RoomID           | Long Integer | 4   |
| AssemblyID       | Long Integer | 4   |
| PrimaryEdge      | Byte         | 1   |
| Mirror           | Yes/No       | 1   |
| Shaped           | Yes/No       | 1   |
| Outline          | Long Text    | N/A |
| OutlineToolID    | Long Integer | 4   |
| GangBore         | Yes/No       | 1   |
| HorizontalBore   | Yes/No       | 1   |
| PriMachineID     | Long Integer | 4   |
| SecMachineID     | Long Integer | 4   |
| HorMachineID     | Long Integer | 4   |
| CutMachineID     | Long Integer | 4   |
| PrefPriMachineID | Long Integer | 4   |
| PrefSecMachineID | Long Integer | 4   |
| Margin           | Double       | 8   |
| HasBanding       | Yes/No       | 1   |
| Banding          | Short Text   | 4   |
| BandingCode      | Short Text   | 50  |
| BandingFlags     | Short Text   | 50  |
| FinishFace       | Short Text   | 25  |
| FinishBack       | Short Text   | 25  |
| FinishEdge       | Short Text   | 25  |
| Parameters       | Long Text    | N/A |
| Composite        | Yes/No       | 1   |
| CompositeID      | Long Integer | 4   |
| MatName          | Short Text   | 60  |
| Thick            | Double       | 8   |
| JobName          | Short Text   | 50  |
| JobDesc          | Short Text   | 255 |
| RunID            | Long Integer | 4   |
| PartID           | Long Integer | 4   |
| Output           | Long Integer | 4   |
| GroupID          | Long Integer | 4   |
| FrProgram        | Short Text   | 8   |
| BaProgram        | Short Text   | 8   |
| FrMirror         | Yes/No       | 1   |
| BaMirror         | Yes/No       | 1   |
| FrBarCode        | Short Text   | 20  |
| BaBarCode        | Short Text   | 20  |
| FrWorkZone       | Short Text   | 10  |
| BaWorkZone       | Short Text   | 10  |
| LeBarCode        | Short Text   | 20  |



|                |              |     |
|----------------|--------------|-----|
| RiBarCode      | Short Text   | 20  |
| ToBarCode      | Short Text   | 20  |
| BoBarCode      | Short Text   | 20  |
| IntBandBarCode | Short Text   | 15  |
| ExtBandBarCode | Short Text   | 15  |
| Image          | OLE Object   | N/A |
| ImageName      | Short Text   | 255 |
| RunMargin      | Double       | 8   |
| RunPrimaryEdge | Byte         | 1   |
| RunQty         | Integer      | 2   |
| CutWidth       | Double       | 8   |
| CutLength      | Double       | 8   |
| Orientation    | Long Integer | 4   |
| BandedWidth    | Double       | 8   |
| BandedLength   | Double       | 8   |

## Voorwood Shape and Sand

This will create the output for the Voorwood Shape and Sand machine. This will be a .csv text file that can be used at the machine. This link requires that the report.mdb be populated prior to output. You can accomplish this by setting the CNCPOPULATEREPORT registry setting for CABINET VISION. Please look to your CABINET VISION help files for more information on how to set this value.

## Weinig French Miter

Output for Weinig French Miter Mortise/Tenon/Haunching machine. This machine can haunch stiles and rails. Output is one program per job. Tools should be assigned to this machine then linked to an operation.

## Alphacam Nest

This is a Parts Router and/or Point-to-Point Link.

Use this output when using Alphacam to nest parts on a sheet. Alphacam supports tool compensation for routers and saws. The following information from the Tool Catalog can not be used by Alphacam and is therefore ignored: Descent Rate and Maximum Depth Per Pass. Layer names from the Tool Catalog are ignored and will be generated automatically to the default settings:

1. Panel Dimensions: PL + Part Thickness
2. Routing: RT\_Z + Depth + \_T + Tool Number + \_D + Tool Diameter + \_F + Feed Rate + Tool Comp
3. Grooving Operations: SW\_Z + Depth + \_T + Tool Number + \_D + Kerf + \_F + Feed Rate + Tool Comp
4. Vertical Boring: VD\_Z + Depth + \_T + Tool Number + \_D + Tool Diameter
5. Horizontal Boring On Left: HL\_Z + Z Position + \_T + Tool Number + \_D + Tool Size
6. Horizontal Boring On Right: HR\_Z + Z Position + \_T + Tool Number + \_D + Tool Size
7. Horizontal Boring On Top: HT\_Z + Z Position + \_T + Tool Number + \_D + Tool Size
8. Horizontal Boring On Bottom: HB\_Z + Z Position + \_T + Tool Number + \_D + Tool Size

Tool selection for routing is made by Tool Number. Be sure to have accurate tool numbers for all Flat and Shaped Router Bits in the Tool Catalog.

## Alphacam P2P

This is a Parts Router and/or Point-to-Point Link.

Use this output when using Alphacam with a Point to Point machine. Alphacam supports tool compensation for routers and saws. The following information from the Tool Catalog can not be used by Alphacam and is therefore ignored: Descent rate and Maximum Depth Per Pass. Layer names from the Tool Catalog are ignored and will be generated automatically to the default settings:

1. Panel Dimensions: PL + Part Thickness
2. Routing: RT\_Z + Depth + \_T + Tool Number + \_D + Tool Diameter + \_F + Feed Rate + Tool Comp
3. Grooving Operations: SW\_Z + Depth + \_T + Tool Number + \_D + Kerf + \_F + Feed Rate + Tool Comp
4. Vertical Boring: VD\_Z + Depth + \_T + Tool Number + \_D + Tool Diameter
5. Horizontal Boring On Left: HL\_Z + Z Position + \_T + Tool Number + \_D + Tool Size
6. Horizontal Boring On Right: HR\_Z + Z Position + \_T + Tool Number + \_D + Tool Size
7. Horizontal Boring On Top: HT\_Z + Z Position + \_T + Tool Number + \_D + Tool Size
8. Horizontal Boring On Bottom: HB\_Z + Z Position + \_T + Tool Number + \_D + Tool Size

Tool selection for routing is made by Tool Number. Be sure to have accurate tool numbers for all Flat and Shaped Router Bits in the Tool Catalog.

## Auto Link

This is a Parts Router and/or Point-to-Point Link.

AutoLink supports tool compensation for routers, Tool Numbers, Feed Rate, Descent Rate & Maximum Depth Per Pass.

Saws should be set to output Center Line paths. One Saw may be setup with multiple kerf sizes. If a tool exists in the Tool Catalog with a kerf that matches the dado width the Planit CNC Center will output a single polyline with width. The kerf in AutoLink's configuration setup will determine the number of passes the saw takes to create the dado width.

Tool selection, for all tools, is made by Tool Number. If the Tool Number is 0, then AutoLink will attempt to do a tool match automatically.

Each Tool is interrogated for it's maximum depth for a single pass. If an operation exceeds the maximum depth per pass, multiple passes will be generated automatically.

Routes are output different for Open and Closed Paths. Open paths are output as line segments. Closed paths are output as polylines. This enables a feature to be used with AutoLink where it splits the distance of the 1st line in a polyline and starts the cut here. This eliminates a common problem of overshot in the corners. Closed Path Route Output adds \_RI\_RO to the layer name to communicate to AutoLink to synthesize the Lead In / Out and Ramp In / Out.

A part description and dimension has been added to a description layer. This description will show next to the file name on the controller for easy identification of the part. The batch.lst is generated so that AutoLink can process the files automatically.

Layer Name & Pen Number from the Tool Catalog will be ignored. The DXF files created for AutoLink have the following characteristics:

1. Border: Border + Part Thickness + LL orientation
2. Description: Stores part information. This information will appear in the comment section of Winner90 and Genesis
3. Vertical Boring: Vbore + Depth + Tool Number (If not 0) + Multiple passes
4. Horizontal Boring: Hbore + Z position + Tool number (If not 0) + Multiple passes
5. Grooving: Groove + Depth + Tool Number (If not 0) + Multiple passes
6. Routing: Route + Depth + Tool number (If not 0) + Tool compensation + Feed Rate (If not 0) + Descent Rate (If not 0) + Multiple passes + Ramp In/Out (For closed path routing)

Each unique layer gets a consecutive pen number.

All entities are output at 0z with No Thickness.

A\_Mn (where n is the number of passes) is added to the layer name automatically to process the instruction with multiple passes.

Parts are automatically output in the Top Left quadrant + X , +Y to match the interface in AutoLink.

Dimensions in the layer name use p for a decimal place. They output a maximum of 2 places in the layer name for metric and 5 places for imperial .

Horizontal boring is created with 3 line segments open to the edge of the part the hole will be generated on. Warning messages will be produced for horizontal boring that is not on the edge of part.

Tool Number, Feed Rate, Descent Rate & Tool Compensation from the Tool Catalog - will only be used in the layer name if they are not 0.

## Biesse Rover

This is a Parts Router and/or Point-to-Point Link.

On the NC4x [x](#) only, starting from the release 4.24, it is possible to add more information to the DXF. These DXF files will support tool compensation for routers and selection of routers by tool names. This output does not support tool compensation for saws. All saw tools should be set to output Center Line paths. The following information from the Tool Catalog can not be used in this output and are therefore ignored: Tool Numbers, Feed Rate, Descent Rate, Maximum Depth Per Pass. Layer names will be ignored for all tools except routers. Router tools should have a layer name that matches the tool name on the machine. The following are the standard layer names:

1. Panel dimensions: PANEL + Part Thickness
2. Vertical drills: VERTICAL + Depth
3. Horizontal drills on left side: SIDEA + Z Position
4. Horizontal drills on front side: SIDEB + Z Position
5. Horizontal drills on right side: SIDEC + Z Position
6. Horizontal drills on back side: Sided + Z Position
7. Grooving operations: BLADE + Depth
8. Generic routing operations: Layer Name + Depth + Tool Comp + Lead In/Out

Tool selection, for all tools but routers, is made by a size match between the tool and the operation. Lead In / Out will be added based on tool compensation and if the router operation is on the interior or exterior of the part.

## Building Blocks

This is a Parts Router and/or Point-to-Point Link.

The ASCII output for Building Blocks follows the AutoSoftware version 2.0 file format. Tool compensation for router paths are supported. All saws should be set to output Center Line paths. Tool Number, Feed Rate, Layer Name & Pen Number from the Tool Catalog will be ignored. Tool selection is made by a diameter match between the tool and the operation. Since some tools are selected by matching the width of the tool and the operation, the output unit should match that of your machines tool catalog. This is typically metric.

## Building Blocks Nest

This is a Parts Router and/or Point-to-Point Link.

The ASCII output for Building Blocks follows the AutoSoftware version 2.0 file format. Tool compensation for router paths are supported. All saws should be set to output Center Line paths. Tool Number, Feed Rate, Layer Name & Pen Number from the Tool Catalog will be ignored. Tool selection is made by a diameter match between the tool and the operation. Since some tools are selected by matching the width of the tool and the operation, the output unit should match that of your machines tool catalog. This is typically metric.

## CAD Code

This is a Parts Router and/or Point-to-Point Link.

CADCode does not support tool compensation for saws. All saw tools should be set to output Center Line paths. CADCode does support tool compensation for routers and recommends setting routers to output with tool compensation. The following information from the Tool Catalog can not be used by CADCode and is therefore ignored: Descent Rate and Maximum Depth Per Pass. Layer names will be generated automatically to the following settings:

1. Panel Dimensions: PANEL
2. Routing: ROUTE\_T + Tool Number + F + Feed Rate + Tool Comp
3. Grooving Operations: GROOVE\_T + Tool Number (If not 0)
4. Vertical Boring: VBORE\_T + Tool Number (If Not 0)
5. Horizontal Boring: HBORE\_T + Tool Number (If Not 0)

Routers and Saws are selected by the Tool Number in the layer name. Be sure to have accurate tool numbers for all routers and saws in the Tool Catalog. If the Tool Number for a drill is 0, tool selection is made be a size match with a tool Diameter. You can call any tool by number in CADCode, including drills. However, there are several reasons to use Diameter rather than number, but there are occasions where you may want a specific drill number or sequence. In a layered file, each unique drill would require a separate layer, i.e. "VBORE\_T12." Since you are specifying a specific tool CADCode cannot optimize the sequence or multiple drops and this tool is therefore removed from the optimization process. One reason you might wish a specific tool would be if you have a 5mm through bit or countersink that you want instead of a dowel bit. Tool depth has been eliminated from the layer name since CADCode reads the information directly from the 3D position of each entity. Additional description layers have been added for labeling capabilities.

## CADCode Nest

This is a Parts Router and/or Point-to-Point Link.

CADCode does not support tool compensation for saws. All saw tools should be set to output Center Line paths. CADCode does support tool compensation for routers and recommends setting routers to output with tool compensation. The following information from the Tool Catalog can not be used by CADCode and is therefore ignored: Descent Rate and Maximum Depth Per Pass. Layer names will be generated automatically to the following settings:

1. Panel Dimensions: PANEL
2. Routing: ROUTE\_T + Tool Number + F + Feed Rate + Tool Comp
3. Grooving Operations: GROOVE\_T + Tool Number (If not 0)
4. Vertical Boring: VBORE\_T + Tool Number (If Not 0)
5. Horizontal Boring: HBORE\_T + Tool Number (If Not 0)

Routers and Saws are selected by the Tool Number in the layer name. Be sure to have accurate tool numbers for all routers and saws in the Tool Catalog. If the Tool Number for a drill is 0, tool selection is made be a size match with a tool Diameter. You can call any tool by number in CADCode, including drills. However, there are several reasons to use Diameter rather than number, but there are occasions where you may want a specific drill number or sequence. In a layered file, each unique drill would require a separate layer, i.e. "VBORE\_T12." Since you are specifying a specific tool CADCode cannot optimize the sequence or multiple drops and this tool is therefore removed from the optimization process. One reason you might wish a specific tool would be if you have a 5mm through bit or countersink that you want instead of a dowel bit. Tool depth has been eliminated from the layer name since CADCode reads the information directly from the 3D position of each entity. Additional description layers have been added for labeling capabilities.

## Cam 2

This is a Parts Router and/or Point-to-Point Link.

Biesse Cam2 DXF files do not support tool compensation. All router and saw tools should be set to output Center Line paths. The following information from the Tool Catalog can not be used by Cam2 and are therefore ignored: Tool Numbers, Tool Compensation, Feed Rate, Descent Rate, Maximum Depth Per Pass. Layer names will be generated automatically to the default settings:

1. Panel dimensions: PANEL + Depth
2. Vertical drills: VERTICAL + Depth
3. Horizontal drills on left side: SIDEA + Z Position
4. Horizontal drills on front side: SIDEB + Z Position
5. Horizontal drills on right side: SIDEC + Z Position
6. Horizontal drills on back side: SIDED + Z Position

7. Grooving operations: BLADE + Depth
8. Generic routing operations: ROUTER + Depth

Tool selection is made by a size match between the tool and the operation. To differentiate between multiple routers bits that have the same diameter, enter slightly different tool diameters for each tool and have those diameters match the tool setup in Cam2.

## Cam 3

This is a Parts Router and/or Point-to-Point Link.

Biesse Cam3 DXF files do not support tool compensation. All router and saw tools should be set to output Center Line paths. The following information from the Tool Catalog can not be used by Cam3 and is therefore ignored: Tool Numbers, Tool Compensation, Feed Rate, Descent Rate, Maximum Depth Per Pass. Layer names will be generated automatically to the default settings:

1. Panel dimensions: PANEL + Depth
2. Vertical drills: VERTICAL + Depth
3. Horizontal drills on left side: SIDEA + Z Position
4. Horizontal drills on front side: SIDEB + Z Position
5. Horizontal drills on right side: SIDEC + Z Position
6. Horizontal drills on back side: SIDED + Z Position
7. Grooving operations: BLADE + Depth
8. Generic routing operations: ROUTER + Depth

Tool selection is made by a size match between the tool and the operation. To differentiate between multiple routers bits that have the same diameter, enter slightly different tool diameters for each tool and have those diameters match the tool setup in Cam3.

## CimTech Nest

This is a Parts Router and/or Point-to-Point Link.

Use this output when using CimTech to nest parts on a sheet. CimTech knowledge must be setup to match the layer names output by Solid. To ensure proper output, the router diameters in the Tool Catalog should be slightly less then the required dado width. Tool compensation is recommended for this output. The following information from the Tool Catalog can not be used by CimTech and is therefore ignored: Feed Rate, Descent Rate, and Maximum Depth Per Pass.

## CimTech P2P

This is a Parts Router and/or Point-to-Point Link.

Use this output when using CimTech with a Point to Point machine. CimTech knowledge must be setup to match the layer names output by the S2M Center. To ensure proper output, the router diameters in the Tool Catalog should be slightly less then the required dado width. Tool compensation is recommended for this output. The following information from the Tool Catalog can not be used by CimTech and is therefore ignored: Feed Rate, Descent Rate, and Maximum Depth Per Pass.

## EnRoute

This is a Parts Router and/or Point-to-Point Link.

Use this output with Scanvec EnRoute. EnRoute supports tool compensation for routers and saws. The following information from the tool catalog can not be used by EnRoute and is therefore ignored: Descent rate and maximum depth per pass. Layer names from the tool catalog are ignored and will be generated automatically to the default settings:

- Panel Dimensions: BD

Outline: OT\_T + Tool Number + D + Tool Diameter + F + Feed Rate + Tool Comp

Routing: RT\_T + Tool Number + D + Tool Diameter + F + Feed Rate + Tool Comp

Grooving: SW\_T + Tool Number

Vertical Boring: VF\_T + Tool Number

Through Boring: VT\_T + Tool Number

Horizontal Boring On Left: HL\_T + Tool Number

## Intelli-CAM - All Parts Full Outline

This is a Parts Router and/or Point-to-Point Link.

Intelli-CAM - Parts Full Outline - This is the preferred method if ALL of your Parts were cut on a Panel Saw oversized and you want to make a perimeter clean-up cut on the Part.

## Intelli-CAM - All Parts Full Outline (Mirror)

This is a Parts Router and/or Point-to-Point Link.

Intelli-CAM - Parts Full Outline (Mirror) - This is the preferred method if ALL of your Parts were cut on a Panel Saw oversized and you want to make a perimeter clean-up cut on the Part and you want to take advantage of Mirroring.

## Intelli-CAM - Parts (Mirror)

This is a Parts Router and/or Point-to-Point Link.

Intelli-CAM Parts (Mirror) - Some parts will be mirrored. This is the preferred method if your parts were milled with a saw the sent to a P2P machining center.

## Intelli-CAM - Parts (No Mirror)

This is a Parts Router and/or Point-to-Point Link.

Intelli-CAM Parts (No Mirror) - No parts will be mirrored. This is the preferred method if your parts were milled in a routed nest and any secondary operations will be sent to a router.

## Intelli-CAM - Parts Full Outline

This is a Parts Router and/or Point-to-Point Link.

Intelli-CAM - Parts Full Outline - This is the preferred method if your Parts were cut on a Panel Saw oversized and you want to make a perimeter clean-up cut on the Part.

## Intelli-CAM - Parts Full Outline (Mirror)

This is a Parts Router and/or Point-to-Point Link.

Intelli-CAM - Parts Full Outline (Mirror) - This is the preferred method if your Parts were cut on a Panel Saw oversized and you want to make a perimeter clean-up cut on the Part and take advantage of Mirroring.

## Speedy Link

This is a Parts Router and/or Point-to-Point Link.

SpeedyLink supports Tool Compensation for routers, Tool Numbers, Feed Rate, Descent Rate & Maximum Depth Per Pass.

Saws should be set to output Center Line paths. One saw may be setup with multiple kerf sizes. If a tool exists in the Tool Catalog with a kerf that matches the dado width the Planit CNC Center will output a single polyline with width. The kerf in SpeedyLink's configuration setup will determine the number of passes the saw takes to create the dado width.

Tool selection, for all tools, is made by Tool Number. If the Tool Number is 0, then SpeedyLink will try to do a tool match automatically.

Each Tool is interrogated for it's Maximum Depth for a single pass. If an operation exceeds the Maximum Depth Per Pass, multiple passes will be generated automatically.

Routes are output different for Open and Closed Paths. Open paths are output as line segments. Closed paths are output as polylines. This enables a feature to be used with SpeedyLink where it splits the distance of the 1st line in a polyline and starts the cut here. This eliminates a common problem of overshoot in the corners. Closed Path Route Output adds \_RI\_RO to the layer name to communicate to SpeedyLink to synthesize the Lead In / Out and Ramp In / Out.

A part description and dimension has been added to a description layer. This description will show next to the file name on the controller for easy identification of the part. The batch.lst is generated so that SpeedyLink can process the files automatically.

Layer Name & Pen Number from the Tool Catalog will be ignored. The DXF files created for SpeedyLink have the following characteristics:

1. Border: Border + Part Thickness + LL orientation
2. Description: Stores part information
3. Vertical Boring: Vbore + Depth + Tool Number (If not 0) + Multiple passes
4. Horizontal Boring: Hbore + Z position + Tool number (If not 0) + Multiple passes
5. Grooving: Groove + Depth + Tool Number (If not 0) + Multiple passes
6. Routing: Route + Depth + Tool number (If not 0) + Tool compensation + Feed Rate (If not 0) + Descent Rate (If not 0) + Multiple passes + Ramp In/Out (For closed path routing)

Each unique layer gets a consecutive pen number.

All entities are output at 0z with No Thickness.

A \_Mn (where n is the number of passes) is added to the layer name automatically to process the instruction with multiple passes.

Parts are automatically output in the Top Left quadrant + [X](#) , +Y to match the interface in SpeedyLink.

Dimensions in the layer name use p for a decimal place. They output a maximum of 2 places in the layer name for metric and 5 places for [imperial](#) .

Horizontal boring is created with 3 line segments open to the edge of the part the hole will be generated on. Warning messages will be produced for horizontal boring that is not on the edge of part.

Tool Number, Feed Rate, Descent Rate & Tool Compensation from the Tool Catalog - will only be used in the layer name if they are not 0.

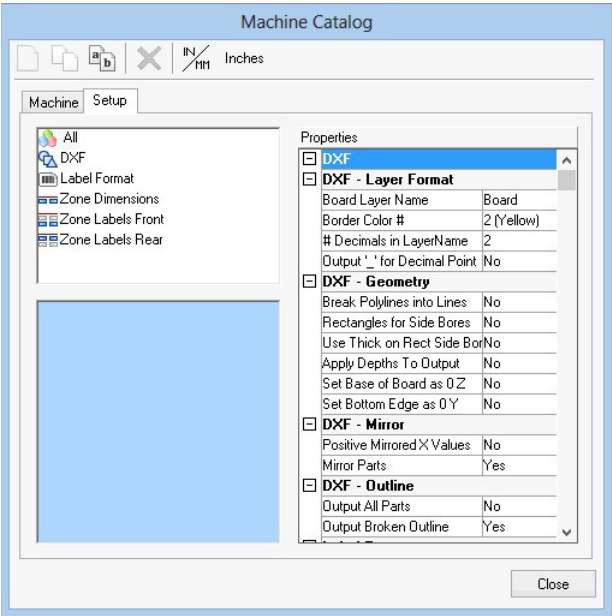
## Standard DXF

This is a Parts Router and/or Point-to-Point Link.

The standard DXF file can be output in multiple variations. This is a good option when there is no preset link to your machine and CAM package. Be sure to select the [Setup tab](#) to make your selection on additional settings. The standard DXF file will use the Layer Names and Pen Numbers from each Tool's settings in the [Tool Catalog](#).

## Standard DXF Machine Catalog Setup Tab

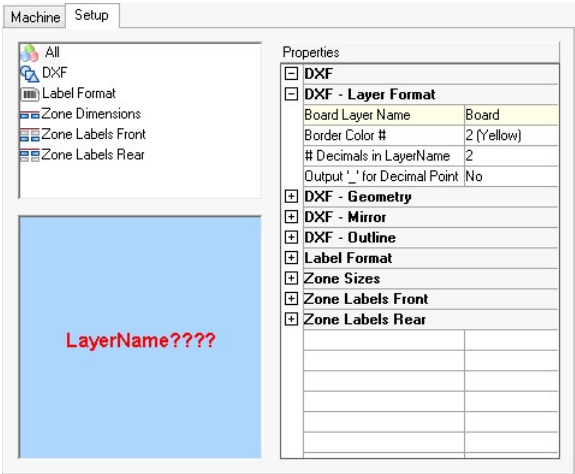
Use this tab to customize a DXF file for your particular CAM package. The Standard DXF is available for both Point to Point machines and Routers.



There are several options available when setting up the Standard DXF Link for a Point to Point and/or Parts Router.

DXF - Layer Format Group

- **Board Layer Name** - Define the name of the Layer representing the rough size or border of the part being cut.

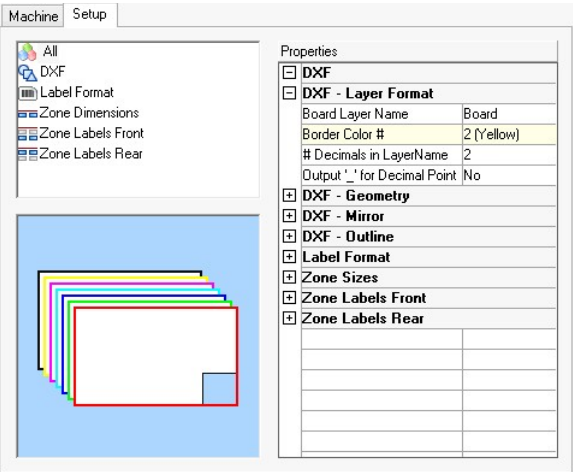


 You can use text variables to define your Layer Name.

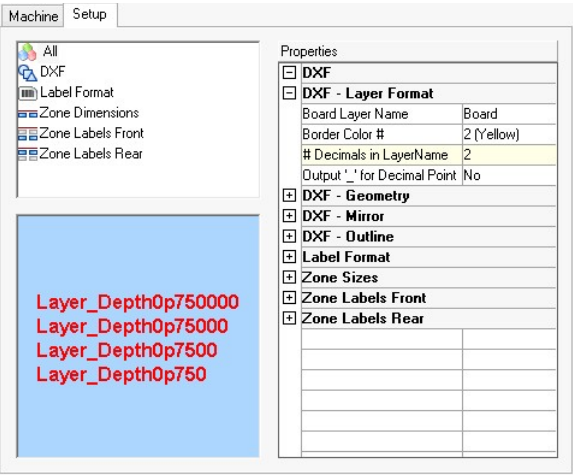


| Variable | Description                                            |
|----------|--------------------------------------------------------|
| {TS}     | Tool Size                                              |
| {DZ}     | Tool Depth                                             |
| {ZP}     | Tool Z Pos (measured from work piece top/front face)   |
| {ZB}     | Tool Z Pos (measured from work piece bottom/back face) |
| {TC}     | Tool Comp                                              |
| {TN}     | Tool Number                                            |
| {PN}     | Pen Number                                             |
| {FD}     | Feed Rate                                              |
| {DR}     | Descent Rate                                           |
| {PR}     | CNC Program                                            |
| {ZO}     | Zone/Run Field                                         |

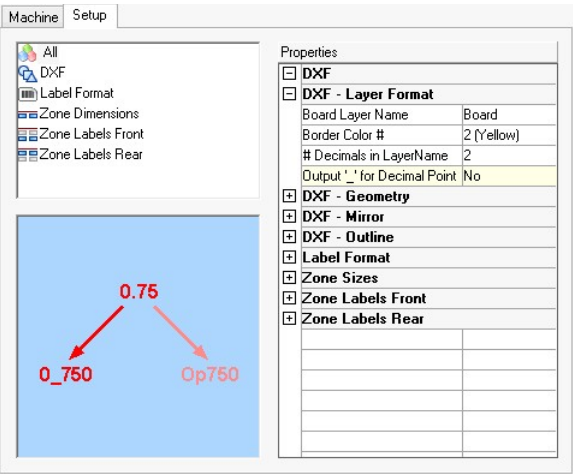
- **Border Color #** - Enter the color number you would like to use for Board's Layer.



- **# Decimals in Layer Name** - Enter the number of decimal places you would like exported in the Layer names to your CAM package.

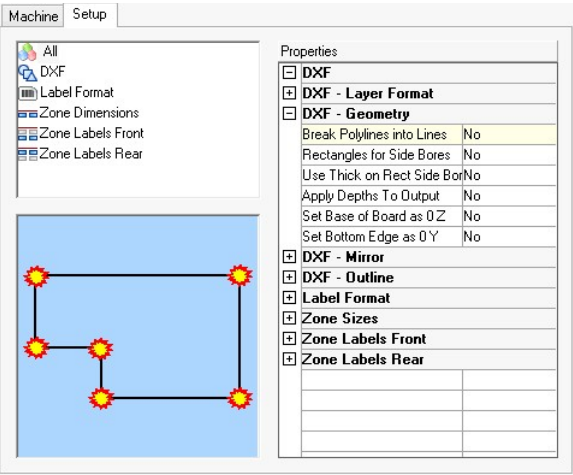


- **Output "\_" for Decimal Point** - A "Yes" here will output an underscore "\_" to represent the decimal place, a "No" here will output a letter "p" to represent the decimal place.

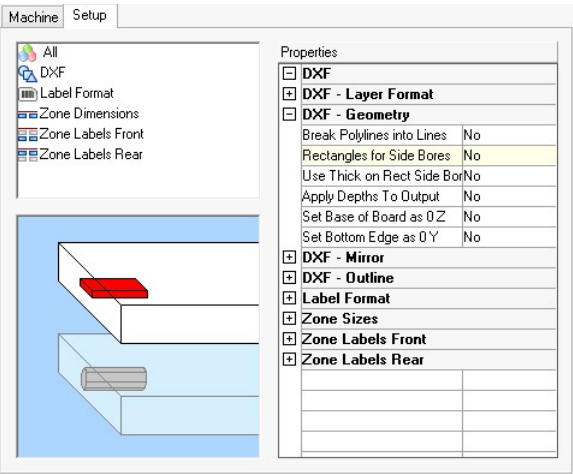


## DXF - Geometry Group

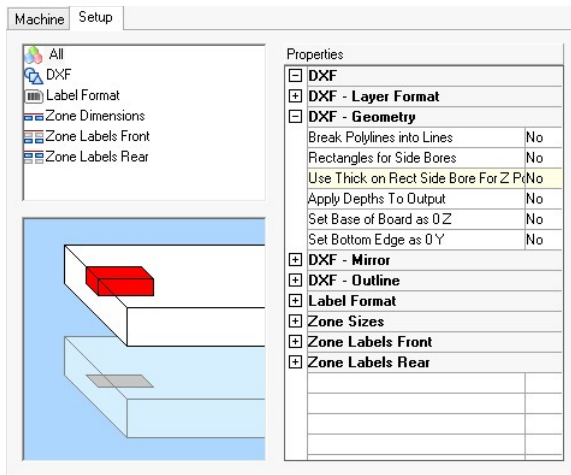
- **Break Polylines into Lines** - A "Yes" here will break route operations into separate lines, a "No" will output the route operations as a continuous polyline.



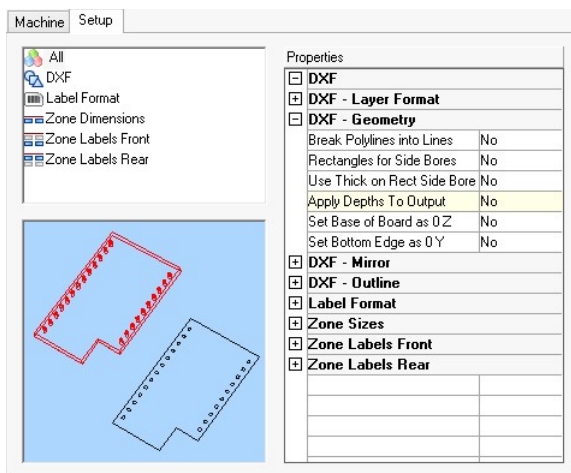
- **Rectangles for Side Bores** - A "Yes" here will output Horizontal boring as rectangles, a "No" will output them as circles.



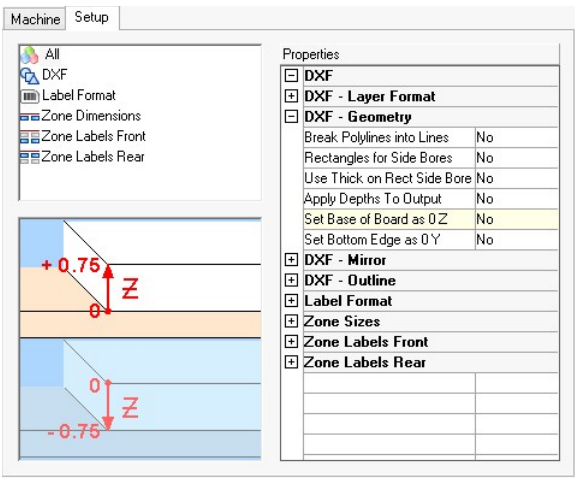
- **Use Thick on Rect Side Bore For Z Position** - A "Yes" here will use the Thickness of the Rectangle Side Bore for the Z Position of the side bore.



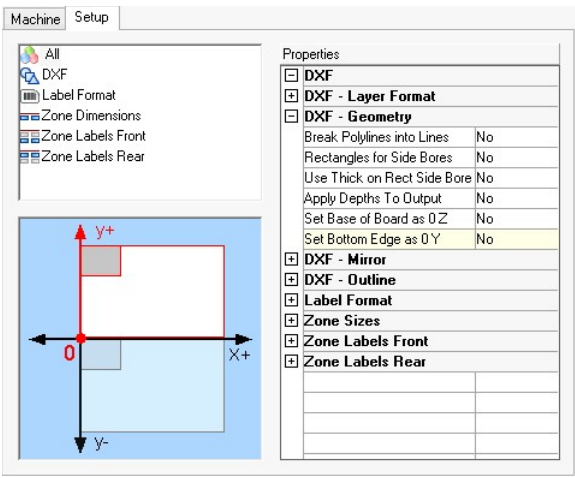
- **Apply Depths to Output** - A "Yes" here will output your DXF's with depths, a "No" will output a 2D representation of your machining.



- **Set Base of Board as 0 Z** - A "Yes" here will set the base of your outputted board to a Z of 0 (Mic @ Spoil Board), a "No" here will set the top of your outputted board to the top of your material.

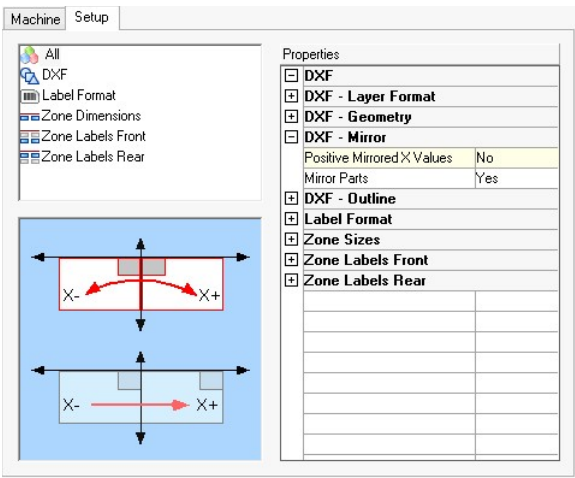


- **Set Bottom Edge as 0 Y** - A "Yes" here will set the bottom edge of your outputted parts at a Y of 0, a "No" will set the top edge to 0.

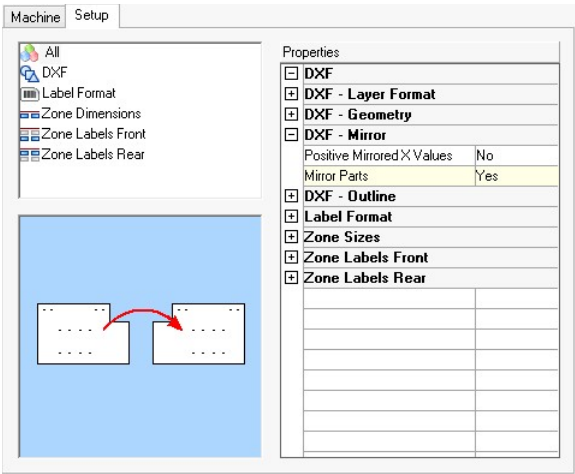


**DXF - Mirror Group**

- **Positive Mirrored X Value** - See image below.

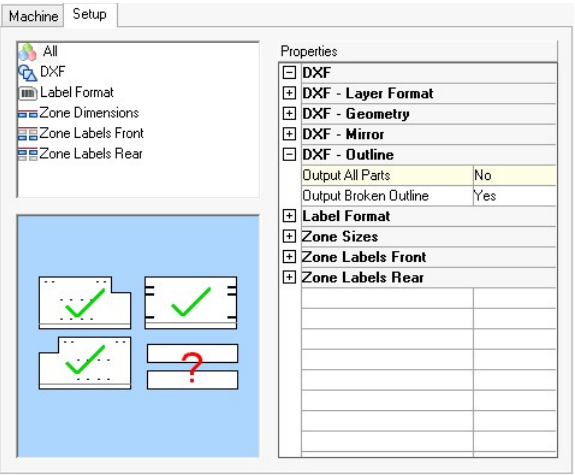


- **Mirror Parts** - A "Yes" here turns on part mirroring.

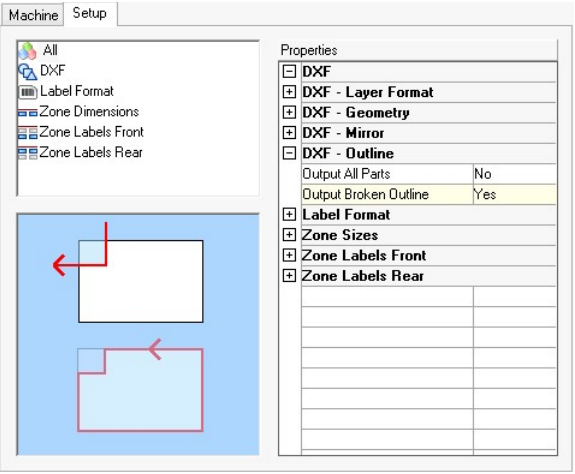


**DXF - Outline Group**

- **Output All Parts** - A "Yes" here tells Solid to output "All Parts" to your machine. Utilize this for nesting all of your parts.



- **Output Broken Outline** - See image below.



[Label Format Group](#)

[Zone Sizes Group](#)

[Zone Labels Front](#)

[Zone Labels Rear](#)

**TASK**

This is a Parts Router and/or Point-to-Point Link.

The TASK output is an [ASCII](#) file format designed with the AutoSoftware 2.0 file specifications. This format only supports Center Line paths for saw. Depending on the post-processor, Feed Rate, Tool Number & Tool Compensation may be ignored. Valid Feed Rate values are 0 - 9. Layer Name & Pen Number from the Tool Catalog are always ignored. Tool selection for boring is made by a diameter match in the tool and operation. Tool selection for saws & routers are made by either Tool Number or Tool Diameter depending on the post-processor. Since some tools are selected by matching the width of the tool and the operation, the output unit should match that of your machines tool catalog. This is typically metric.

## TASK Version 3

This is a Parts Router and/or Point-to-Point Link.

TASK version 3 and earlier do not support Tool Compensation or Tool Number. All saws and routers should be set to output Center Line paths. Feed Rate, Tool Number, Tool Compensation, Layer Name & Pen Number from the Tool Catalog will be ignored. Tool selection is made by a diameter match between the tool and the operation. Since some tools are selected by matching the width of the tool and the operation, the output unit should match that of your machines tool catalog. This is typically metric.

## TASK Version 5

This is a Parts Router and/or Point-to-Point Link.

TASK version 5 supports both Tool Compensation for routers and Tool Numbers for saws and routers. All saws should be set to output Center Line paths. Feed Rate, Layer Name & Pen Number from the Tool Catalog will be ignored. Tool selection will be made by Tool Number. If the Tool Number is 0, then tool selection is made by a diameter match between the tool and the operation. Since some tools are selected by matching the width of the tool and the operation, the output unit should match that of your machines tool catalog. This is typically metric.

## Thermwood

This is a Parts Router and/or Point-to-Point Link.

Select this link to create output for use with the Thermwood Control Nesting software.

## TwinCAM 32

This is a Parts Router and/or Point-to-Point Link.

TwinCAM 32 does not support Tool Compensation for saws. All saw tools should be set to output Center Line paths. Routers can be output with tool compensation. The following information from the Tool Catalog can not be used by TwinCAM and is therefore ignored: Feed Rate, Descent Rate, Maximum Depth Per Pass. Pen and Layer names will be generated automatically to the default settings:

1. Panel Dimensions: PL (White)
2. Routing: RT\_TT + Tool Number + Tool Comp (Red)
3. Grooving Operations: RT\_TT90 (Yellow)
4. Vertical Boring: VD\_TT1 (Blue)
5. Vertical Through Boring: VD\_TT2 (Blue)
6. Horizontal Boring: H\_TT1 (Green)

Tool selection for routing is made by Tool Number. Be sure to have accurate tool numbers for all routers in the Tool Catalog. Tool selection for boring is made by a size match between the tool and the operation.

## WoodWop 25

This is a Parts Router and/or Point-to-Point Link.



WoodWop version 2.5 does not support Tool Compensation. All router and saw tools should be set to output Center Line paths. The following information from the Tool Catalog can not be used by WoodWop 2.5 and is therefore ignored: Tool Compensation, Feed Rate, Descent Rate, Maximum Depth Per Pass. Layer names will be generated automatically to the default settings:

1. Panel Dimensions: S\_ + Part Thickness
2. Routing: FZ + Distance Off Table + T + Tool Number
3. Grooving Operations: NZ + Depth
4. Vertical Boring: ZG + Depth
5. Vertical Through Boring: ZD + Depth
6. Horizontal Boring: HO + Z Position

Tool selection for routing is made by Tool Number. Be sure to have accurate tool numbers for all routers in the Tool Catalog. Tool selection for boring is made by a size match between the tool and the operation. This output will create a JobName\_25.bat file that can be used to automate the process of running BPP and converting the DXF files to the WoodWop file format.

## WoodWop 4

This is a Parts Router and/or Point-to-Point Link.

WoodWop version 4 does not support Tool Compensation for saws. All saw tools should be set to output Center Line paths. Routers can be output with Tool Compensation. The following information from the Tool Catalog can not be used by WoodWop 4 and is therefore ignored: Feed Rate, Descent Rate, Maximum Depth Per Pass. Layer names will be generated automatically to the default settings:

1. Panel Dimensions: S\_ + Part Thickness
2. Routing: FK + Distance Off Table + T + Tool Number + Tool Comp
3. Grooving Operations: NZ + Depth
4. Vertical Boring: ZG + Depth
5. Vertical Through Boring: ZD + Depth
6. Horizontal Boring: HO + Z Position

Tool selection for routing is made by Tool Number. Be sure to have accurate tool numbers for all routers in the Tool Catalog. Tool selection for boring is made by a size match between the tool and the operation. This output will create a JobName\_40.bat file that can be used to automate the process of running BPP and converting the DXF files to the WoodWop file format.

## WoodWop 5

This is a Parts Router and/or Point-to-Point Link.

This link does not support tool compensation for saws. All saw tools should be set to output center line paths. Routers can be output with tool compensation. The following information from the tool catalog can not be used by this link and is therefore ignored: Feed rate, descent rate, maximum depth per pass.

Layer names will be generated automatically to the default settings:

- Panel Dimensions: ProcPart\_ + Part Thickness
- Routing: V\_Trim\_ + Distance Off Table + T + Tool Number + Tool Comp
- Grooving Operations: V\_Saw0\_ + Depth
- Vertical Boring: V\_DrillSF\_ + Depth
- Vertical Through Boring: V\_DrillSFS
- Horizontal Boring: H\_Drill\_ + Z Position

Tool selection for routing is made by tool number. Be sure to have accurate tool numbers for all routers in the tool catalog. Tool selection for boring is made by a size match between the tool and the operation. This output will create a RunNumber\_50.bat file that can be used to automate the process of running BPP and converting the DXF files to the WoodWop file format.

## Setting Up Real Time Labels for Holz-Her Saws

See the Machine Catalog [Saw topic](#) for Label Setup information.

### Real time Label Printing Using a S2M Center CPOUT file and a Label picture

#### Basic setup

To implement this on a CUT 85 saw the design software needs to produce 2 items. These items are a CPOUT file and label pictures. The CPOUT files should be placed in the C:\CPOutPfad folder, and the label pictures in the C:\CV\_Pictures folder. Make these folders if they do not exist. Next check that the saw is set to look at the CPOUT folder as in the example below (figure 1). The folders can have any name you want, you just need to enter the proper names in the proper places.

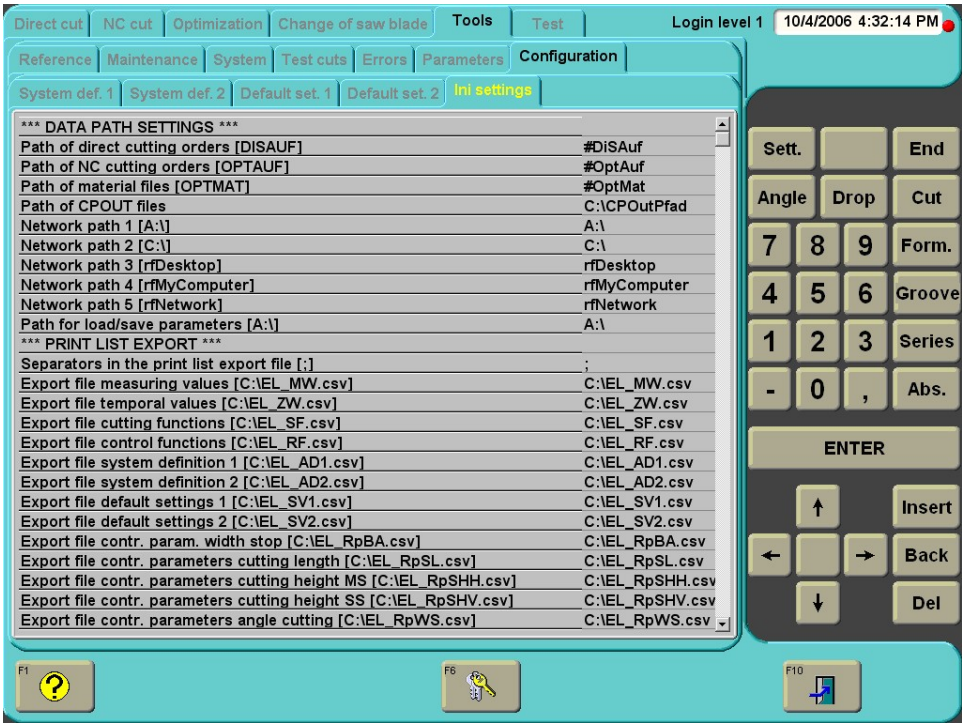


Figure 1 - CPOUT file location

#### Label design

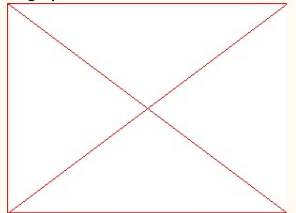
The next item to do is to design the label file at the saw. Set the label size by selecting "F5 Size". Enter the label width and length to match the S2M Center Label Designer label size. In "Line definition" set "Type" to "Variable picture". Set "X-Pos" and "Y-Pos" to "0.000" and width and height to match the label size. The next item "Variable:" needs to match the location of the picture file name in the CPOUT file. Use "14 Element identifier" if the value is in the "PRT1 partno" field. Use "62 Element info 1" if the value is in the "PRT1 drillpat" field. Set "Format:" to 8 to strip out any extra characters. Last set "Path:" to your selected picture folder. Also add "#.XXX" with the XXX being your picture file format (\*.bmp, \*.ico, \*.jpg, \*.jpeg, \*.emf, \*.wmf). Below is the screen shot of the label design screen in the CUT 85 software (Figure 2).

Direct cut NC cut Optimization Change of saw blade Tools Test Login level 1 10/4/2006 5:08:59 PM

Reference Maintenance System Test cuts Errors Parameters Configuration

### Labels

**Label graphics**



**Label designation**

**Raw cut label**

Width: 4.000 inch

Length: 3.000 inch

**Line definition**

Type: Variable picture

X-Pos: 0.000 inch Y-Pos: 0.000 inch

Width: 4.000 inch Height: 3.000 inch

Variable: 14 Element identifier

Format: 8

Path: C:\CV\_Pictures\#.jpg

**Definition list**

|   | Type     | X-Pos | Y-Pos | Text/variable/pict |
|---|----------|-------|-------|--------------------|
| 1 | VPicture | 0.000 | 0.000 | EElem              |
| 2 |          |       |       |                    |

F1 ? F2 Label type F3 Label... F4 F5 Size F6 Test print F7 Printer F10

**Figure 2 – Label Format**

### Select label type

Select “F2 Label type” and double click on “Raw cut label” from the selection box (figure 3). Save the label by selecting “F3 Label...” and select “Save as...”. Type a file name that describes the label type and select “OK” (Figure 4)

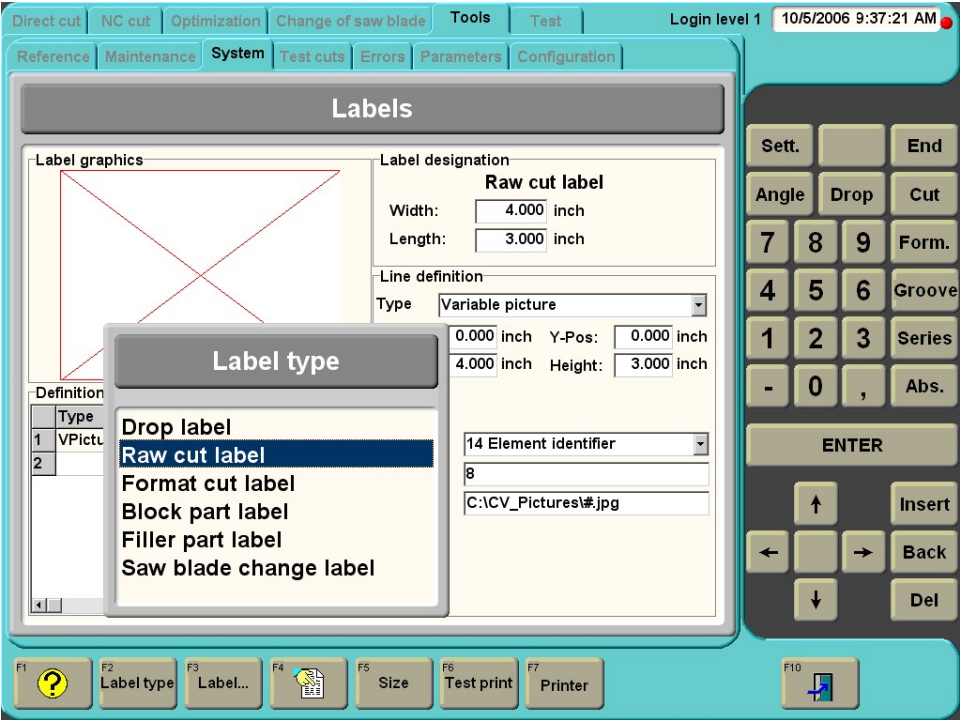
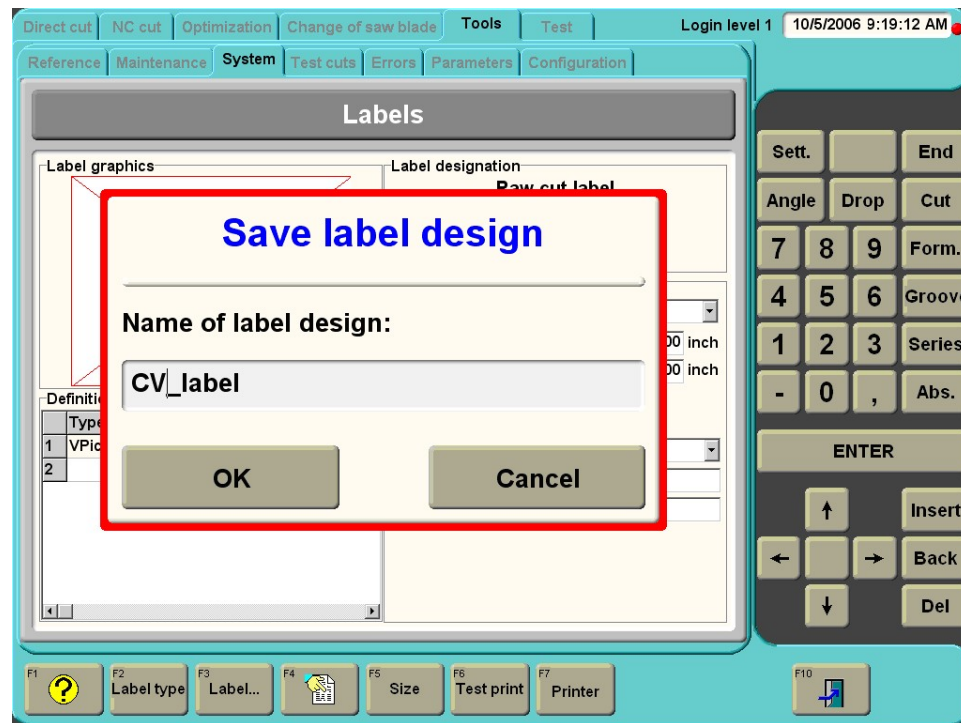


Figure 3 Label type



**Figure 4 Save label**

#### Printer Selection

Now that the label style is set and saved, select "F7 Printer" and select the proper printer for your labels (Figure 5). Once the printer is selected Select "F3 Label..." and this time select "Save"

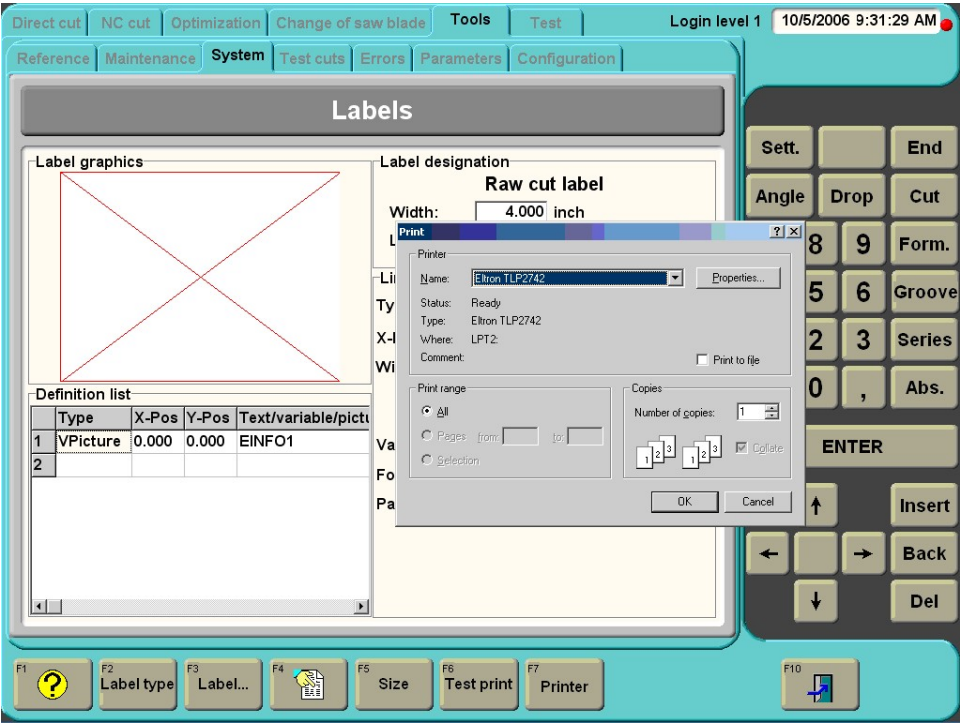


Figure 5 Printer selection

# Alpha Beam Saw

Output for AES Alpha Beam Saw.

# Altendorf Elmo

Saw output for the Altendorf Elmo Control.

Optimization priority must be set to Labor to ensure optimization does not exceed the control's limitations.

# Ayza Spear Lanza P5

Saw output for the Ayza Mizrak Machine with the Ayza Spear Lanza P5 controller.

# BeamBoard

Saw link to create Beamboard XML

## Casadei CN04

This is a Panel Saw Link.

This link is for the Casadei rear loading beam saw for CN04 and similar controllers.

## Casadei MX39

This is a Panel Saw Link.

This link is for the Casadei MX39 using the NPS400 Controller.

## CPOUT

This is a Panel Saw Link.

This link creates a standard CPOUT file with additional formatting conventions as outlined by CADCode. Following is the file specification:

### Pattern Export File Specification (CPOUT)

The first three lines describe the header information as follows:

The **HDR1** line contains the cutlist number, Job name and Materials in the following format:

|                         |                                                                                                                       |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>HDR1,cnum,jobmat</b> |                                                                                                                       |
| <b>HDR1</b>             | is the line descriptor                                                                                                |
| <b>cnum</b>             | indicates the cutlist number, a string with leading zeros included.                                                   |
| <b>jobmat</b>           | indicates the cutlist title, an ASCII string of length 50 or less, composed of the Material Name and Job Information. |

The **HDR2** line contains information regarding the date and time the cutlist was optimized in the following format:

|                       |                                                                                        |
|-----------------------|----------------------------------------------------------------------------------------|
| <b>HDR2,date,time</b> |                                                                                        |
| <b>HDR2</b>           | is the line descriptor                                                                 |
| <b>date</b>           | indicates the date of optimization. It is an ASCII string of length 12. [Dec 08, 2006] |
| <b>time</b>           | indicates the time of optimization. It is an ASCII string of length 8. [01:12 PM]      |

The **HDR3** line contains information about the product that produced the pattern data file in the following format:

|                                  |                                                                                    |
|----------------------------------|------------------------------------------------------------------------------------|
| <b>HDR3,bnum,ver,pcode,sawid</b> |                                                                                    |
| <b>HDR3</b>                      | is the line descriptor                                                             |
| <b>bnum</b>                      | indicates the Build number of the product used. It is an ASCII string of length 6. |
| <b>ver</b>                       | CPOUT Version [2.01]                                                               |
| <b>pcode</b>                     | indicates product code [CVI]                                                       |
| <b>sawid</b>                     | indicates the SawLink ID used [0009]                                               |

The next three lines describe the controls used in the pattern making process.

The **CTL1** line contains the dimensional unit of the cutlist, the output unit, the grain indicator, area reporting, and saw type in the following format:

**CTL1,unit1,unit2,grain,areafac,sawtype**

- CTL1** is the line descriptor.
- unit1** indicates dimensional Unit of Measure. It is a one character field as follows:
- F = Fractional Units
  - D = Decimal Inches
  - M = Millimeters
- unit2** indicates cutting instruction Unit of Measure. It is a one character field as described for unit1 and also always matches unit1.
- grain** indicates the graining control. It is a one character field as follows:
- Y = Cutlist is grained
  - N = Cutlist is not grained
- areafac** indicates the area unit throughout the file. \*Planit always outputs 1. 1 = SqFT in Imperial and SqM in Mtric based on the **unit1** setting.
- sawtype** indicates the type of saw specified. It is a one character field as follows:
- 0 = no saw type specified
  - 1 = single axis saw
  - 2 = two axis saw with single feeder
  - 3 = two axis saw with multiple feeder

The **CTL2** line describes dimensional content in the following format:

**CTL2,rkerf,xkerf,maxsw,minsw,pwadj,pladj**

- CTL2** is the line descriptor
- rkerf** indicates the rip cut kerf
- xkerf** indicates the cross cut kerf
- maxsw** indicates the maximum strip section width
- minsw** indicates the minimum strip section width
- pwadj** indicates the part width adjustment. \*Planit always outputs 0.
- pladj** indicates the part length adjustment. \*Planit always outputs 0.

The **CTL3** line describes the Front Trim value in the following format:

**CTL3,fttrim**

- CTL3** is the line descriptor
- fttrim** indicates the value for the Front Trim.

The **THK1** line specifies the sheet stock thickness and stack (book) height in the following format:

**THK1,sthick,stackh**

- THK1** is the line descriptor



**sthick** indicates the sheet stock thickness

**stackh** indicates the maximum stack height. This is a measurement not the number of sheets.

The next two lines describe cutting statistics.

The **STA1** line specifies the total number of cuts, grouped by each phase type, as indicated below. Each field is right justified and has a length of 6 digits, and is in the following format:

**STA1,nrcut,nxcut,n3cut**

**STA1** is the line descriptor

**nrcut** indicates the number of rip cuts in the cutlist and includes the leading trim cut in the calculation.

**nxcut** indicates the number of cross cuts in the cutlist and includes the trim cut. If there are identical strips then the cross cuts are not counted on the additional strips.

**n3cut** indicates the number of third or greater phase cuts in the cut list.

The **STA2** line specifies the total distances cut, grouped by each phase type. The distances are a function of the dimensional units (unit1). If the dimensional units are fractional or decimal inches, the distances are in feet, If the dimensional units are millimeters, the distances are in meters. Each field is right justified to eleven characters and has the form 'xxxxxxx.yyy'.

The **STA2** line has the following format:

**STA2,drcut,dxcut,d3cut**

**STA2** is the line descriptor

**drcut** indicates the total distance of rip cuts in the cutlist including the leading trim cut.

**dxcut** indicates the total distance of cross cuts in the cutlist including the trim cut. Unlike the nxcut field in STA1, the cross cut distance does not skip identical strips.

**d3cut** indicates the total distance of third or greater phase cuts in the cutlist.

The next three lines describe sheet stock records.

The **INV1** record for each sheet stock item has the following format:

**INV1,supply,used,ucost,width,length,trim1,trim2,trim3,trim4**

**INV1** is the line descriptor

**supply** indicates the available supply of the sheet. It is a right justified numeric field of length 5. A supply of '\*\*\*\*\*' indicates an infinite supply.

**used** indicates the quantity of the sheet used in the cutlist. It is a right justified numeric field of length 5.

**ucost** indicates the unit cost per square foot/meter. It is a right justified field of length 9 in the form 'xxx.yyyy'.

**width** indicates the width of the sheet.

**length** indicates the length of the sheet.

**trim1** indicates the leading width trim.

**trim2** indicates the leading length trim.

**trim3** indicates the trailing width trim.

**trim4** indicates the trailing length trim.

The **INV2** record for each sheet stock item has the following format:

**INV2,desc,areacut,finparts,lpfin**

**INV2** is the line descriptor

|                 |                                                                               |
|-----------------|-------------------------------------------------------------------------------|
| <b>desc</b>     | indicates the description of the sheet. It is a character field of length 24. |
| <b>areacut</b>  | indicates the total area cut of the sheet. [xxxxxxxx.yyyy]                    |
| <b>finparts</b> | indicates the total area finished (parts in sheet). [xxxxxxxx.yyyy]           |
| <b>lpfin</b>    | indicates the total low priority area finished. *Planit always outputs 0.     |

The **INV3** record is used to describe extra information about the sheet stock and has the following format:

**INV3,vendor,location,ori**

|                 |                                                                                                                                                                                                   |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>INV3</b>     | is the line descriptor                                                                                                                                                                            |
| <b>vendor</b>   | indicates the vendor from which the sheet stock was purchased. *Planit does not use this field.                                                                                                   |
| <b>location</b> | indicates the physical location in the shop for the sheet stock. *Planit does not use this field.                                                                                                 |
| <b>ori</b>      | is the orientation that the user prefers for the first cut on the sheet. It is a one character field as follows:<br><br>0 = don't care<br>1 = first cut along length<br>2 = first cut along width |

The next two lines describe part cut sizes.

The **ORD1** record for each cut size has the following format:

**ORD1,demand,supply,width,length,prio,grain**

|               |                                                                                                                                                                            |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ORD1</b>   | is the line descriptor                                                                                                                                                     |
| <b>demand</b> | indicates the quantity requested for the cut size. It is a right justified field of length 5.                                                                              |
| <b>supply</b> | indicates the quantity supplied/produced of the cut size. It is a right justified field of length 5.                                                                       |
| <b>width</b>  | indicates the width.                                                                                                                                                       |
| <b>length</b> | indicates the length.                                                                                                                                                      |
| <b>prio</b>   | indicates the priority of the part. It is a one character field where:<br><br>H or Blank indicates high priority (NcCenter always outputs an H<br>L indicates low priority |
| <b>grain</b>  | *Planit leaves blank because it is indicated in CTL1.                                                                                                                      |

The **ORD2** record for each cut size has the following format:

**ORD2,pcost,desc1,desc2**

|              |                                                                                               |
|--------------|-----------------------------------------------------------------------------------------------|
| <b>ORD2</b>  | is the line descriptor                                                                        |
| <b>pcost</b> | cost per part to 1/10 of a cent. [dddd.ccc]                                                   |
| <b>desc1</b> | indicates the primary part description. *Planit leaves blank because it is described in PRT   |
| <b>desc2</b> | indicates the secondary part description. *Planit leaves blank because it is described in PRT |

The **ORD3** record is used to access the part information for each cut size and has the following format:

**ORD3,nparts,index**

- ORD3** is the line descriptor
- nparts** This is the number of parts that have been combined into this cut size.
- index** This is the index into the list of parts where this cut size begins.

The next five records are for the parts that make up the above cut sizes.

The **PRT1** record for each part has the following format:

**PRT1,dmnd,ordno,width,length,dolabel,prtname,jobname,partcom,  
cncprg**

- PRT1** is the line descriptor
- dmnd** is the quantity the user specified for this part
- ordno** is the index or offset into the cut sizes specified above
- width** is the width for this part
- length** is the length for this part
- dolabel** indicates if user wants a label made for this Part. \*Planit always outputs 1 (True).
- prtname** is the name of this part
- jobname** is the name of the Job this part is in
- partcom** is the name of the Label Graphic for Real-Time Label printing
- cncprg** CNC Program Name – 07/06/06(date) – Primary Barcode

The **PRT2** record for each part has the following format:

**PRT2,postlam,secprog,roomname**

- PRT2** is the line descriptor
- postlam** indicates post laminating instructions. \*Planit always outputs 0.
- secprog** is the Secondary CNC Program Name
- rmname** is the name of the Room the part is in

The **PRT3** record for each part has the following format:

**PRT3,edges,prprog,wmatl1,wcolor1,prizone,lmatl1,lcolor1,secprog,wmatl2,wcolor2,seczone,lmatl2,lcolor2**

- PRT3** is the line descriptor
- edges** is the edge banding codes for the Part [W1,L1,W2,L2]
  - 0 = No Banding
  - 1 = Door Banding
  - 2 = Exterior Banding
  - 3 = Interior Banding

- 4 = BandingB
- 5 = BandingC
- 6 = BandingF
- 7 = BandingG
- 8 = BandingH
- 9 = BandingJ



Other Banding letters are ignored as CPOUT only has placeholders for 0 - 9 identifiers.

- priprog** is the Primary CNC Program Name
- wmatl1** is the Edge Band Material for W1
- wcolor1** is the Edge Band Finish Color for W1
- prizone** is the Primary CNC Program Zone
- lmatl1** is the Edge Band Material Name for L1
- lcolor1** is the Edge Band Finish Color for L1
- secprog** is the Secondary CNC Program Name
- wmatl2** is the Edge Band Material for W2
- wcolor2** is the Edge Band Finish Color for W2
- seczone** is the Secondary CNC Program Zone
- lmatl2** is the Edge Band Material for L2
- lcolor2** is the Edge Band Finish Color for L2

The **PRT4** record for each part has the following format:

**PRT4,dname,cwidth,height,cdepth,ddesc,comments**

- PRT4** is the line descriptor
- cname** is the name of the Cabinet this part is in
- cwidth** is the width of the Cabinet this part is in
- height** is the height of the Cabinet this part is in
- cdepth** is the depth of the Cabinet this part is in
- ddesc** is the description of the Cabinet this part is in
- comments** are the comments on the Cabinet this part is in

The **PRT5** record for each part has the following format:

**PRT5,prio,grain,qty,qtyover**

- PRT5** is the line descriptor
- prio** is the priority which is always set to high (H)

- grain** indicates grain [Y or N] – Blank indicates use the global setting as defined in **CTL1**
- qty** this parts quantity
- qtyover** this parts quantity plus overmake

Following the part records are the drop records. There is one record for each drop. The number of the drop items what will appear in the file is specified in the second line of the file as described above.

The format of the drop record is as follows:

**OFF1,supply,width,length**

- OFF1** is the line descriptor
- supply** is the quantity of the drop
- width** is the width of the drop
- length** is the length of the drop

Following the drop records are the pattern description records.

For each pattern in the file there are five header records. Following the header records are the cut type records.

The five pattern records are as follows:

The first pattern record has the following format:

**PAT1,invn,pat\_type,repeat,cycles**

- PAT1** is the line descriptor
- invn** indicates the sequence number of the sheet stock item used to cut the pattern, It is a string length of 2. with the leading zeros included.
- pat\_type** indicates a one character field indicating the type of pattern as follows:
  - L = Long rip pattern
  - S = Short rip pattern
  - H = Head cut pattern
- repeat** indicates the number of sheets in the book (stack)
- cycles** indicates the number of cycles (books/stacks cut). i.e. If you have 11 sheets, but the book height is 3 sheets, the cycle would be 4 indicating (3 + 3 + 3 + 2). With the NcCenter's current output the cycle is always 1 because the NcCenter pre-groups the materials by maximum book (stack) height.

*NOTE: A long rip pattern indicates that the rip cuts are made parallel to the larger dimension of the sheet. i.e. If the width dimension entered is less than the length dimension entered for the sheet, this indicates that the rip cuts are parallel to the length dimension of the sheet. But, if the width dimension is larger than the length dimension, the rip cuts are made parallel to the width dimension entered.*

*A short rip pattern indicates that the rip cuts are made parallel to the shorter dimension of the sheet. i.e. If the width dimension entered is less than the length dimension for the sheet, this indicates that the rip cuts are made parallel to the width of the sheet. But, if the width dimension is larger than the length dimension, the rip cuts are made parallel to the length dimension.*

*A head cut pattern has the first cut across the width of the sheet (like a short rip) but after the initial head cut the rest of the pattern looks like a long rip. The strips within the head section are identical. The size of the head section is controlled by the cutting rules.*

The second pattern header record has the following format:

**PAT2,yield,uarea,uareah,uareal,cpua**

- PAT2** is the line descriptor
- yield** indicates the percentage of yield
- uarea** indicates the used area per panel

|               |                                                                                                                                              |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>uareah</b> | indicates the used area by high priority parts. *All Planit Parts are high priority so this field always matches the value in <b>uarea</b> . |
| <b>uareal</b> | indicates the used area by low priority parts. *All Planit parts are high priority so this field is always 0.                                |
| <b>cpua</b>   | indicates the cost of the used area per panel                                                                                                |

The third pattern header record has the format below. Each field is right justified to a length of 5.

#### **PAT3,nrcut,nxcut,n3cut,mnxcut**

|               |                                                                         |
|---------------|-------------------------------------------------------------------------|
| <b>PAT3</b>   | is the line descriptor                                                  |
| <b>nrcut</b>  | indicates the number of rip cuts per cycle                              |
| <b>nxcut</b>  | indicates the number of cross cuts per cycle                            |
| <b>n3cut</b>  | indicates the number of 3 <sup>rd</sup> and higher phase cuts per cycle |
| <b>mnxcut</b> | indicates the number of cross cuts in the worst strip in the pattern    |

The fourth pattern header record has the format below. Each field in the record is right justified and is 11 characters in length with the format 'XXXXXXX.YYY'.

#### **PAT4,drcut,dxcut,d3cut**

|              |                                                                        |
|--------------|------------------------------------------------------------------------|
| <b>PAT4</b>  | is the line descriptor                                                 |
| <b>drcut</b> | indicates the total rip cut distance per cycle                         |
| <b>dxcut</b> | indicates the total cross cut distance per cycle                       |
| <b>d3cut</b> | indicates the total of 3 <sup>rd</sup> and higher phase cuts per cycle |

The fifth pattern header record has the following format:

#### **PAT5,ton,toff,con,coff**

|             |                                                       |
|-------------|-------------------------------------------------------|
| <b>PAT5</b> | is the line descriptor                                |
| <b>ton</b>  | indicates the on-line time in minutes for all cycles  |
| <b>toff</b> | indicates the off-line time in minutes for all panels |
| <b>con</b>  | indicates the on-line operating cost                  |
| <b>coff</b> | indicates the off-line operating cost                 |

Following the **PAT5** record will be one of the following record types.

Patterns that have head cuts are divided into two sections. We call these sections the "head section" and the "body section". The cutting instructions (the **CUT1** records) do not change for the head section, they still have codes for rip cuts, cross cuts, and 3<sup>rd</sup> and higher phase cuts. The head section will be preceded by a record that starts with the keyword "**HCUT**", followed by the **CUT1** records. At the end of the head section (or at the start of the body section) you will have a record that starts with the keyword "**BCUT**", followed by the **CUT1** records for this part of the pattern.

#### **HCUT**

The presence of this record means that this pattern has a head section and the **CUT1** records for the head section will follow this record.

Note that this record is OPTIONAL and will NOT appear for patterns that do not have head cuts.

#### **BCUT**

This record will precede the **CUT1** records for the pattern. If the pattern contains head cuts then this record will precede the **CUT1** records for the body section.

Following the above **HCUT** or **BCUT** records are pattern description records, which indicate how to cut the sheet. This record has the following format:

CUT1,level,dist,otype,onum

|       |                                                                                                                                                                                                                                                                                                                                            |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CUT1  | this is the line descriptor                                                                                                                                                                                                                                                                                                                |
| level | indicates the phase number of the cut. It is a two character field with leading zeros included, i.e.,<br><br>01 = rip cut<br><br>02 = cross cut<br><br>03 = 3 <sup>rd</sup> phase cut<br><br>04 = 4 <sup>th</sup> phase cut, etc.                                                                                                          |
| dist  | indicates the dimensional distance of the cut                                                                                                                                                                                                                                                                                              |
| otype | indicates the part type. It is a one character field where:<br><br>0 = Part not produced with cut<br><br>1 = Scheduled part<br><br>2 = Drop part                                                                                                                                                                                           |
| onum  | indicates the sequence number of the size produced with this cut. If additional cuts are required before a cut size is produced with a cut, a zero is supplied for <b>onum</b> . Thus a value of '5' means that this cut results in a finished cut size whose index is '5' in the <b>ORD</b> records (the first <b>ORD</b> record is '1'). |

NOTE: Instructions are not provided for leading trim cuts. They must be determined from the corresponding sheet stock item and the head cut trim setting.

The following indicates the end of the file:

END

Based on CPOUT specification by Pattern Systems International. Revised to meet CADCode specifications. Revised to meet Planit NcCenter specifications.

CPOUT - Mayer

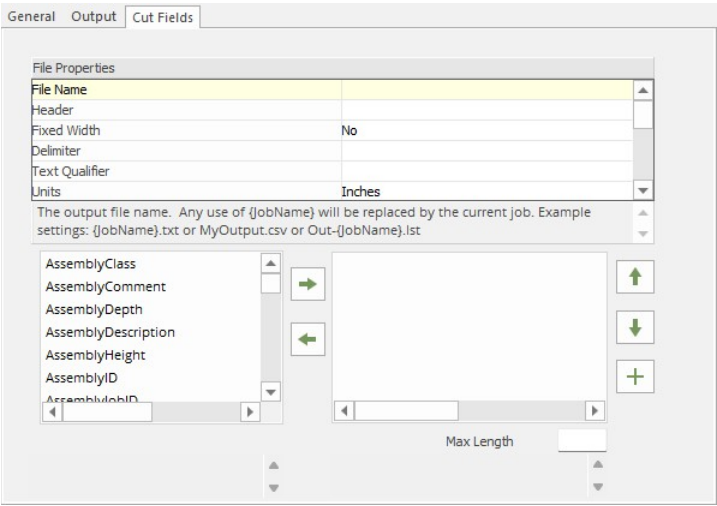
This is a Panel Saw Link.

This link outputs CPOUT file for Mayer beam saws. This output is compatible with Mayer's CPRead 9/99.

Custom CUT File

User configurable ASCII file output of part sizes to be used by 3rd party optimizers.

When this Saw NC Link is selected there will be a new Tab in the Machine Catalog named "CUT Fields".



File Properties Area

There are ten general options available to configure your output:

|                    |           |
|--------------------|-----------|
| File Name          |           |
| Header             |           |
| Fixed Width        | No        |
| Delimiter          |           |
| Text Qualifier     |           |
| Units              | Inches    |
| Unit Precision     | No Limit  |
| Boolean Format     | Yes vs No |
| Date Format        |           |
| Currency Precision | 1         |

**File Name** - This field will allow you to specify a specific or partial file name. You may choose to use {JobName} as a variable. Any use of {JobName} will be replaced by the current job.

Example settings:

{JobName}.txt ► Smith.txt  
MyOutput.csv ► MyOutput.csv  
Out-{JobName}.lst ► Out-Smith.lst

**Header** - The header is optional. If specified it will be placed once at the top of the file. It can be left blank to omit. Any use of \n will be replaced with a carriage return.

Example Setting: ;This is a heading\nThis is the second line\nThis is the 3rd line.

;This is a heading  
  
This is the second line  
  
This is the 3rd line.

**Fixed Width** - This will determine if each field will use a specific number of characters in the output. If the actual data has fewer characters then that specified the remaining characters will be padded with space. If the actual data exceeds the character limit then the value will be truncated.





String Length includes the Text Qualifier. If the Field Length is set to 4 and the String is “ABCDEFGH” the output will be “AB”. If there is no text qualifier the output would be ABCD.

To set the length of the output field first select the field on the right then enter the Fixed Length at the bottom of the list.



If Fixed Width is false then the Text Length indicates the maximum length of any field. This by default can be left at zero, but can be set to a number if you wish to limit the output length. Unlike Fixed Width, the max length does not include the text qualifier. A max length of 4 with the string ABCDEF would result “ABCD”

**Delimiter** - This is an optional single character to separate each field. It is commonly a comma (,).

**Text Qualifier** - This is an optional single character that will surround and text or memo output. Typical values are (") and (').

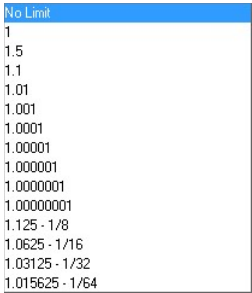


When the output type is not fixed width if the text qualifier character is found in a text or memo field it will be replaced with either “ or ‘ (whichever is NOT the text qualifier character). The example below uses a double quote (") as the text qualifier.

A “sample” string with “double” and ‘single’ quotes: “A ‘sample’ string with ‘double’ and ‘single’ quotes.”

**Units** - The unit can either be Inches or MM. As an option the unit may also be multiplied by a factor of 10, 100, or 1000.

**Units Precision** - This option specifies the unit precision. Value will be rounded to the nearest selection. The unit precision applies to measurements as well as singles and doubles.



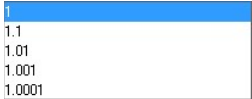
**Boolean Format** - This option allows for multiple Boolean formats.



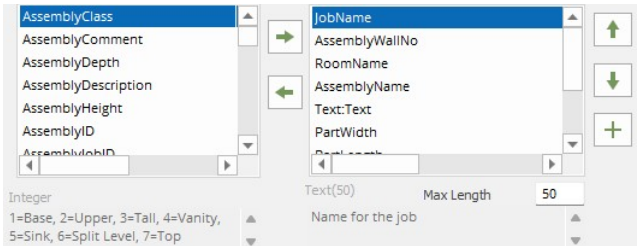
**Date Format** - A string value using (M)onth, (D)ay & (Y)ear as variables. Example settings:

|                       |                            |
|-----------------------|----------------------------|
| M/D/YY                | 10/4/13                    |
| DD-MM-YYYY            | 04-10-2013                 |
| DDD DD-MM-YYYY        | Mon 04-10-2013             |
| DDD, MMM DD/YY        | Mon, Oct 04/13             |
| DDDD, MMMM DD of YYYY | Monday, October 04 of 2013 |

**Currency Precision** - Similar to unit precision, the value will be rounded to the specified precision.



Field Definition Area



The list on the left are all of the Fields that you can include in your output file. Simply select the fields you would like then click the Right Arrow button to add them to the list on the right. Once a field has been moved over to the list on the right it's order can be modified by selecting the Up or Down arrow buttons. Multiple fields can be moved at the same time by 1st selecting multiple fields then selecting the Up/Down buttons.

It may be required to output a value that it not represented in the available data list from CABINET VISION. This type of output is typically not-critical data but is still required by the importing software to fill a space in the output format.

To add a fixed text field click on the Green Plus Button then enter in the text:

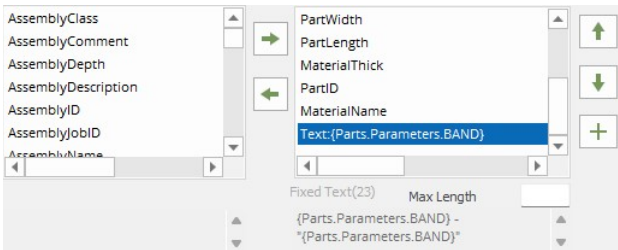


You can also use the Text field to pull any Parameters that are in the **Parts | Parameters** field in the PSNC-CV.mdf database by simply defining them as thus:

{Parts.Parameters.ParamName}

So if you wanted to pull the BAND Paramater the value for the Text field would be.

{Parts.Parameters.BAND}



## CUT File Format

ASCII file output of part sizes to be used by 3rd party optimizers.

| Field # | Description                                                             |
|---------|-------------------------------------------------------------------------|
| 1       | Job number                                                              |
| 2       | Wall number                                                             |
| 3       | Room number                                                             |
| 4       | Cabinet number                                                          |
| 5       | Part quantity                                                           |
| 6       | Part width                                                              |
| 7       | Part width string                                                       |
| 8       | Part length                                                             |
| 9       | Part length string                                                      |
| 10      | Part thickness                                                          |
| 11      | Part thickness string                                                   |
| 12      | Part ID (from Material.cvm file)                                        |
| 13      | Part name                                                               |
| 14      | Material ID (from Material.cvm file)                                    |
| 15      | Material name                                                           |
| 16      | Material list number<br>1 = Panel Stock<br>2 = Board Stock<br>3 = Frame |

|    |                                                                                                                                                                            |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | 4 = Door<br>5 = Drawers                                                                                                                                                    |
| 17 | Material grain default:<br>1 = Yes<br>2 = No                                                                                                                               |
| 18 | Optimize:<br>1 = Yes<br>0 = No                                                                                                                                             |
| 19 | Stock part:<br>1 = Yes<br>0 = No                                                                                                                                           |
| 20 | Waste Factor                                                                                                                                                               |
| 21 | Material Cost                                                                                                                                                              |
| 22 | Material cost unit:<br>1 = Each<br>2 = Lineal Feet<br>3 = Sq. Feet                                                                                                         |
| 23 | Labor (minutes)                                                                                                                                                            |
| 24 | Material type<br>0 = Panel Stock<br>1 = Board Stock<br>2 = Laminate<br>3 = Banding<br>4 = Hinges<br>5 = Drawer Guides<br>6 = Pulls<br>7 = Other<br>8 = Tops<br>9 = Molding |
| 25 | Spare field 2 (decimal)                                                                                                                                                    |
| 26 | Spare field 3 (decimal)                                                                                                                                                    |
| 27 | Spare field 4 (decimal)                                                                                                                                                    |
| 28 | Cabinet name                                                                                                                                                               |
| 29 | Part comment                                                                                                                                                               |
| 30 | Primary part program                                                                                                                                                       |
| 31 | Part banding code (field Edge Band)                                                                                                                                        |
| 32 | Room name                                                                                                                                                                  |
| 33 | Secondary part program                                                                                                                                                     |
| 34 | Primary part program mirrored<br>1 = Yes<br>0 = No                                                                                                                         |
| 35 | Secondary part program mirrored<br>1 = Yes<br>0 = No                                                                                                                       |
| 36 | Primary part program barcode                                                                                                                                               |
| 37 | Secondary part program barcode                                                                                                                                             |
| 38 | Job description                                                                                                                                                            |
| 39 | Interior banding material                                                                                                                                                  |
| 40 | Interior banding color                                                                                                                                                     |
| 41 | Exterior banding material                                                                                                                                                  |
| 42 | Exterior banding color                                                                                                                                                     |
| 43 | Door banding material                                                                                                                                                      |
| 44 | Door banding color                                                                                                                                                         |
| 45 | Part banding code (field Banding)                                                                                                                                          |

|    |                                        |
|----|----------------------------------------|
| 46 | Part banding code (field BandingFlags) |
|----|----------------------------------------|

## Elcon Quadra

This is a Panel Saw Link.

Elcon Beam Saw. Optimization priority must be set to Labor and Eliminate 3rd Phase Cuts must be set True to ensure optimization does not exceed the control's limitations.

## Etalon Saw

Customized version of CPOUT, specific to Etalon.

## Excitech EP

This is a Panel Saw Link.

Saw output for the Excitech EP series control EP330/EP270

## Fimal Concept 350

Saw output for Fimal Concept 350. This is a binary output file.

Due to the limitation of a single cut only representing a single label it is recommended this output should not be booked.

## Fimal P3200

Saw output for Fimal P3200.

## Gabbiani Prima

This is a Panel Saw Link.

Output for Gabbiani Prima/Class/X1000 Control. Optimization priority must be set to Labor to ensure optimization does not exceed the control's limitations.

## Giben Prismatic

This is a Panel Saw Link.

Giben Prismatic 201/301 with GDRIVE controller - AC and AD formats.

## Holz-Her Saw with Kvara Controller

This is a Panel Saw Link.

Output for Kvara Control. Trim values should be set to desired trim plus saw kerf. Optimization priority must be set to Labor to ensure optimization does not exceed the control's limitations.

## Holz-Her Saw with Tria Controller

This is a Panel Saw Link.

Binary output for Tria 4000 Control. Control does not accept 3rd phase cuts so third phase cuts are output as absolutes after the cross-cuts of each strip. Optimization priority must be set to Labor to ensure optimization does not exceed the control's limitations.

## Holzma

This is a Panel Saw Link.

This link outputs the file format for the Holzlink interface for all Holzma saws.

### PATTERN EXCHANGE FILE SPECIFICATION

V1.02.4

Bath Software Research Ltd

04 November 1999

## 1. INTRODUCTION

This document describes a data structure for the exchange of cutting lists and patterns for sheet material between various design programs, optimising packages, and panel saw controllers.

This data structure contains the information that is required for transferring cutting lists to an optimising package and for transferring optimised cutting patterns with label information to a panel saw. The data structure can be created in two formats as follows.

- ASCII comma-separated file (Created by the Planit NcCenter)
- Access database

The data structure consists of 12 record types each with a number of fields. In the ASCII file each record type is represented by a token which begins each line and in the database each record type is a separate table.

Each record type and each field name are in uppercase and use underscores between separate words. No table name or field name is more than 10 characters.

## 2. RECORD TYPES & TABLE NAMES

The 'cutting list' record types are as follows.

**HEADER** - general information describing the complete data structure (or file)

**JOBS** - header data for each job (cutting list or optimised run)

**PARTS\_REQ** - basic requirement details for each item in the cutting list

**PARTS\_INF** - standard information about each part

**PARTS\_UDI** - user-defined information about each part

**BOARDS** - information about each item in the board (stock) list

**MATERIALS** - information about each material type

**NOTES** - other information for a job

The 'post-optimisation' records are:

**OFFCUTS** - record describing each off-cut produced

**PATTERNS** - pattern header records - one for each cutting pattern

**CUTS** - cutting instructions – occur many times per pattern – once for each cut required

**VECTORS** – vector graphics describing the pattern

### 3. FORMAT

The ASCII version of the file uses standard comma-separated format, and has the suffix .PTX (PaTtern eXchange). The main part of the filename could be the job/order number or batch name if the file contains multiple jobs.

Examples:-

01234.PTX            ABC123-1.PTX

Note that the structure allows for the ASCII file to contain more than 1 cutting list or run if necessary, for example it could contain a batch of runs. Note that there may be restrictions on the file name because some controllers will, for example, only accept 5 digits for the job number.

All normal CSV format conventions apply, including optional use of quotation marks around text data. Leading spaces are ignored. Trailing commas (separators) are not necessary. Text fields containing commas must be enclosed in quotes, and commas will be replaced by 'blank' (ASCII code 32) on import.

The format and size restrictions for each field are tabulated in section 18. Note, that the limitations (eg. max length of material code) will vary according to the implementation and specification of the saw.

All 'index numbers' must be integer values, starting at 1 for the first record, and incrementing consecutively up to the maximum specified. Note, in particular that all part, board, pattern and cutting records must contain the appropriate job index number showing which job they relate to.

The Access database version stores each record type in a separate table. The file has the standard extension of MDB.

Examples:

01234.MDB            BATCH32.MDB

### 4. HEADER - GENERAL INFORMATION

**HEADER – VERSION, TITLE, UNITS, ORIGIN, TRIM\_TYPE**

The header record contains descriptive and global information for the job. This record appears as a line in the ASCII file. In the Access database this information will be stored in the database properties.

**VERSION** - File version (set to 1)

**TITLE** - File title

**UNITS** - Measurement mode = 0 (metric), 1(decimal inches).

**ORIGIN** - This field indicates the origin for the VECTOR drawing records. The origin for the CUT records is assumed to be 0 (top left).

- 0 = top to bottom - left to right
- 1 = top to bottom – right to left
- 2 = bottom to top – left to right
- 3 = bottom to top – right to left

**TRIM\_TYPE** - Indicates whether the waste strip/piece is cut first or last. That is, is the fixed trim done on the leading edge or as a final trim?

- 0 = trim waste piece first

1 = trim fixed trim first

#### ASCII & Database examples

PROPERTIES,1,“This is an example”,0,0,1

PROPERTIES TABLE

| VERSION | TITLE              | UNITS | ORIGIN | TRIM_TYPE |
|---------|--------------------|-------|--------|-----------|
| 1       | This is an example | 0     | 0      | 1         |

## 5. JOBS – JOB RECORD

**JOBS, - JOB\_INDEX, NAME, DESC, ORD\_DATE, CUT\_DATE, CUSTOMER, STATUS**

This record contains data about each job contained in the file. These records are optional and in the absence of job records all parts and patterns are assumed to belong to the same job.

**JOB\_INDEX** - Unique index number used to link other records to an appropriate job

**NAME** - Job number/name – reference for job

**DESC** - Job description/title - title of job

**ORD\_DATE** – Date of order (DD/MM/YYYY)

**CUT\_DATE** – Date for cutting/delivery (DD/MM/YYYY)

**CUSTOMER** - Customer code or name

**STATUS** - Status of the job.

0 - not optimised

1 - optimised

2 - optimise failed

Note: there may be a range of other error codes

#### ASCII & Database examples

JOBS,1,ORD1234,SAMPLE JOB – CUSTOMER WOODCO,17/01/1999, 22/01/1999,WOODCO,0

JOBS TABLE

| JOB_INDEX | NAME    | DESC                         | DATE       |
|-----------|---------|------------------------------|------------|
| 1         | ORD1234 | SAMPLE JOB - CUSTOMER WOODCO | 17/01/1999 |

## 6. PARTS\_REQ – PART REQUIREMENT RECORD



**PARTS\_REQ** - **JOB\_INDEX**, **PART\_INDEX**, **CODE**, **MAT\_INDEX**, **LENGTH**, **WIDTH**, **QTY\_REQ**, **QTY\_OVER**, **QTY\_UNDER**, **GRAIN**, **QTY\_PROD**

This record contains data about each different size (or line item) in the cutting list. This record is used to provide details about each part (over and above cut sizes).

**JOB\_INDEX** - Index number used to link this record to other records for this job.

**PART\_INDEX** - Index number to link this record with other associated part records

**CODE** - Part code or description.

**MAT\_INDEX** - Index of material used for this part.

**LENGTH** - Cut length of part shown in appropriate measurement unit

**WIDTH** - Cut length of part shown in appropriate measurement unit

**QTY\_REQ** - number of pieces this size

**QTY\_OVER** - allowed over production

**QTY\_UNDER** - allowed under production.

**GRAIN** - 0=No grain/part can be rotated, 1 = grain/part cannot be rotated.

**QTY\_PROD** - quantity of parts produced by patterns

-

#### ASCII & Database examples

PARTS\_REQ,1,1,SD900X,1,890.0,645.5,50,0,2,0,50

PARTS-REQ TABLE

| JOB_INDEX | PART_INDEX | CODE   | MAT_INDEX | LENGTH | WIDTH | ..... |
|-----------|------------|--------|-----------|--------|-------|-------|
| 1         | 1          | SD900X | 1         | 890.0  | 645.5 | ..... |

## 7. PARTS\_INF – STANDARD PART INFO RECORD

**PARTS\_INF** - **JOB\_INDEX**, **PART\_INDEX**, **DESC**, **LABEL\_QTY**, **FIN\_LENGTH**, **FIN\_WIDTH**, **ORDER**, **EDGE1**, **EDGE2**, **EDGE3**, **EDGE4**, **EDG\_PG1**, **EDG\_PG2**, **EDG\_PG3**, **EDG\_PG4**, **FACE\_LAM**, **BACK\_LAM**, **CORE\_MAT**, **PALLET**, **DRAWING**, **PRODUCT**, **PROD\_INFO**, **PROD\_WIDTH**, **PROD\_HGT**, **PROD\_DEPTH**, **PROD\_NUM**, **ROOM**, **BARCODE1**, **BARCODE2**

This optional record contains standard information about each different size (or line item) in the cutting list. One use of this record is to hold data for label printing.

**JOB\_INDEX** - Index number used to link this record to other records for this job.

**PART\_INDEX** - Index number linking this record with other part records.

**DESC** - A second part description

**LABEL\_QTY** - Number of copies of the label for this part.

0 = no labels for this part

default if not specified =1

**FIN\_LENGTH** - Length of part after edging and trimming

**FIN\_WIDTH** - Width of part after edging and trimming

**ORDER** - Original order/job/work number which part relates to

**EDGE1** - Code or description of edging for bottom (length) edge

**EDGE2** - Code or description of edging for top (length) edge

**EDGE3** - Code or description of edging for left (width) edge

**EDGE4** - Code or description of edging for right (width) edge

**EDG\_PG1** – Program or operation code for bottom (length) edge

**EDG\_PG2** - Program or operation code for top (length) edge

**EDG\_PG3** - Program or operation code left (width) edge

**EDG\_PG4** - Program or operation code for right (width) edge

**FACE\_LAM** - Code/description of laminate material for face (topside) of part

**BACK\_LAM** - Code /description of laminate material for back (underside) of part

**CORE\_MAT** - Code or description of core material

**PALLET** – Pallet layout (stacks in length and width)

**DRAWING** - Name of drawing file, drill program or CNC program for machine centre

**PRODUCT** - Product or cabinet code or template name to which part belongs

**PROD\_INFO** - Description of product or cabinet

**PROD\_WIDTH** - External dimension of product or cabinet

**PROD\_HGT** - External dimension of product or cabinet

**PROD\_DEPTH** - External dimension of product or cabinet

**PROD\_NUM** - Item number of cabinet in room

**ROOM** - Room or floor or group item number for cabinet

**BARCODE1** - data for 1st barcode – as text string

**BARCODE2** - data for second bar code – as text string

**ASCII & Database examples**

PARTS\_INF,1,2,BOTTOM,1,690.0,475.0,ORD1234,GREY-1MMPVC,,  
GREY-1MMPVC,,GREYLAM,GREYLAM,MDF18,BU-SIDE-CNC,BU4DW,  
4 Dwr Base Unit, 600.0, 890.0, 570.0,1,1,0690004750, 0012301  
PARTS-INF TABLE

| JOB_INDEX | PART_INDEX | DESC | LABEL_QTY | FIN_LENGTH | FIN_WIDTH | ..... |
|-----------|------------|------|-----------|------------|-----------|-------|
|           |            |      |           |            |           |       |

|   |   |        |   |       |       |       |
|---|---|--------|---|-------|-------|-------|
| 1 | 2 | BOTTOM | 1 | 690.0 | 475.0 | ..... |
|---|---|--------|---|-------|-------|-------|

## 8. PARTS\_UDI - USER DEFINED PART INFO RECORD

### PARTS\_UDI - JOB\_INDEX, PART\_INDEX, INFO1, INFO2, .... INFO30

This optional record contains user-defined information about each different size (or line item) in the cutting list. One use of this data is for label printing.

**JOB\_INDEX** - Index number used to link this record to other records for this job.

**PART\_INDEX** - Index number linking this record with other part records

### INFO*n*

There are up to 30 information fields. The name of each field is INFO followed by the field number. The fields may be used for any purpose such as customer details, dates, CNC operations, and other items not included in the other part records.

### ASCII & Database examples

PARTS-UDI,1,3,CNC84,Smith & Co,20-APR-1999

PARTS-UDI TABLE

| JOB_INDEX | PART_INDEX | INFO1 | INFO2      | INFO3       | INFO4 | ..... |
|-----------|------------|-------|------------|-------------|-------|-------|
| 1         | 3          | CNC84 | Smith & Co | 20-APR-1999 |       | ..... |

## 9. BOARDS – BOARD RECORD

### BOARDS - JOB\_INDEX, BRD\_INDEX, CODE, MAT\_INDEX, LENGTH, WIDTH, QTY\_STOCK, QTY\_USED, COST, STK\_FLAG

These records contain details of the board/sheet sizes to be used; one record for each different size/material.

**JOB\_INDEX** - Index number used to link this record to other records for this job.

**BRD\_INDEX** - index number linking this record with the **PATTERNS** records for this job.

**CODE** – Board code - usually the stock code for the sheet size.

**MAT\_INDEX** - Index of material used for this part.

**LENGTH** - Size of sheet in appropriate measurement unit.

**WIDTH** - Size of sheet in appropriate measurement unit.

**QTY\_STOCK** - Total number of sheets available – default 99999 (0=none)

**QTY\_USED** - Total number of sheets this size used in patterns

**COST** - Cost per sq. metre or sq. foot according to measurement unit

**STK\_FLAG** – Flag to indicate action if insufficient stock

**ASCII & Database examples**

BOARDS,1,1,WLAM15MM-1,1,2550.0,1525.0,100, 7, 8.50,0

BOARDS TABLE

| JOB_INDEX | BRD_INDEX | CODE       | MAT_INDEX | LENGTH | WIDTH  | ..... |
|-----------|-----------|------------|-----------|--------|--------|-------|
| 1         | 1         | WLAM15MM-1 | 1         | 2550.0 | 1525.0 | ..... |

**10. MATERIALS – MATERIAL RECORDS**

**MATERIALS - JOB\_INDEX, MAT\_INDEX, CODE, DESC, THICK, BOOK, KERF\_RIP, KERF\_XCT, TRIM\_FRIP, TRIM\_VRIP, TRIM\_FXCT, TRIM\_VXCT, TRIM\_HEAD, TRIM\_FRCT, TRIM\_VRCT, RULE1, RULE2, RULE3, RULE4**

These records define the different material types. There should be a least one of these records in data structure. This record is used to pass a detailed material description, the thickness and other parameters which may vary according to material type.

**JOB\_INDEX** - Index number used to link this record to other records for this job.

**MAT\_INDEX** - Unique index of material used to link this record to other records

**CODE** – Material code

**DESC** - Material description

**THICK** - Material thickness in appropriate measurement mode

**BOOK** - Max sheets per book, reflects cutting height of saw

**KERF\_RIP** - Rip saw kerf – in unit of measurement

**KERF\_XCT** - Crosscut saw kerf – in unit of measurement

**TRIM\_FRIP** - Fixed rip trim (**Rear**) – includes saw kerf – amount sheet size is reduced by

**TRIM\_VRIP** - Minimum variable rip trim (**Front**) – includes saw kerf

**TRIM\_FXCT** - Fixed crosscut trim (**Rear**) – includes saw kerf

**TRIM\_VXCT** - Minimum variable cross cut trim (**Front**) – includes saw kerf

**TRIM\_HEAD** - Internal head cut trim – includes saw kerf

**TRIM\_FRCT** - Fixed recut trim (**Rear**) – includes saw kerf

**TRIM\_VRCT** - Minimum variable recut trim (**Front**) – includes saw kerf

**RULE1** - Optimisation rule 1 – cut nesting limit – 1 to 9 (e.g. 3 = allow third phase recuts)

**RULE2** - Optimisation rule 2 – head cuts allowed (0=No, 1 =Yes)

**RULE3** - Optimisation rule 3 – board rotation allowed (short rip) (0=No, 1=Yes)

**RULE4** - Optimisation rule 4 – show separate patterns for duplicate parts (0=No 1=Yes)

The saw kerf and trims are optional. Note that it is assumed that one of the two rip trims will be constant and the other rip trim includes the waste strip. Either (a) the leading edge is trimmed and the waste strip comes out last, or (b) the waste strip is removed by the first rip and the last rip is a constant trim. This assumption also applies to cross cut trims and recut trims.

**ASCII & Database examples**

MAT,1,1,WHITE18,"White laminate -chipboard18mm",18,4,4,8,4,8,10,10,8,8,8,8,8,

4,1,1,1

MATERIALS TABLE

| JOB_INDEX | MAT_INDEX | CODE    | DESC                            | THICK | BOOK | ..... |
|-----------|-----------|---------|---------------------------------|-------|------|-------|
| 1         | 1         | WHITE18 | White Laminate - chipboard 18mm | 18    | 4    | ..... |

**11. NOTES - NOTE RECORDS**

**NOTES - JOB\_INDEX, NOTES\_INDEX, TEXT**

This record is optional and holds any messages or notes that need to be associated with a job, for example customer details, special instructions, etc, or any details that are job related rather than part or material related. As many lines as required can be stored for each job.

**JOB\_INDEX** - index linking note to job

**NOTES\_INDEX** - index storing order of notes

**TEXT** - text of note - Maximum length of text field is 250 characters.

**ASCII & Database examples**

NOTES,1,1,"Customer ref. A1234-0987 – Smith & Jones Cabinets"

NOTES TABLE

| JOB_INDEX | NOTES_INDEX | TEXT                                              |
|-----------|-------------|---------------------------------------------------|
| 1         | 1           | Customer ref. A1234-0987 - Smith & Jones Cabinets |

**12. OFFCUTS – RECORDS FOR OFF-CUTS**

**OFFCUTS - JOB\_INDEX, OFC\_INDEX, CODE, MAT\_INDEX, LENGTH, WIDTH, OFC\_QTY**

This record is optional and can occur once for each different off-cut size per material created by the cutting patterns.

**JOB\_INDEX** - Index number used to link this record to other records for this job.

**OFC\_INDEX** - Unique index number of offcut used to link this record to the CUT record.

**CODE** - Offcut code or description - used to identify offcut.

**MAT\_INDEX** - Index of material used for this offcut. Enables offcuts of similar material composition, thickness and colour, but different size to be grouped together.

**LENGTH** - Length of offcut in appropriate measurement unit

**WIDTH** - Width of offcut in appropriate measurement unit

**OFC\_QTY** – Quantity of this size produced

#### ASCII & Database examples

OFFCUT,1,1,WHITE15-123,2,1450.0, 425.0,1

OFFCUT TABLE

| JOB_INDEX | OFC_INDEX | CODE        | MAT_INDEX | LENGTH | WIDTH | ..... |
|-----------|-----------|-------------|-----------|--------|-------|-------|
| 1         | 1         | WHITE15-123 | 2         | 1450.0 | 425.0 | ..... |

## 13. PATTERNS – PATTERN RECORDS

**PATTERNS - JOB\_INDEX, PTN\_INDEX, BRD\_INDEX, TYPE, QTY\_RUN, CYCLES, MAX\_BOOK, PICTURE**

This record occurs once per pattern. It is used to describe header detail for the pattern, such as board size used, number of sheets to be cut etc.

**JOB\_INDEX** - Index number used to link this record to other records for this job.

**PTN\_INDEX** - Sequential number incrementing by 1 for each pattern record for each job.

**BRD\_INDEX** - Index number from the Boards records.

**TYPE** - determines the direction of the first cut, and the type of pattern

- 0 = rip length first – non-head cut pattern
- 1 = turn board before ripping - non-head cut pattern
- 2 = head cut pattern – head cut across width
- 3 = head cut pattern – head cut along length
- 4 = crosscut only

**QTY\_RUN** - Run quantity – number of sheets to be cut to this pattern

**QTY\_CYCLES** - Number of cycles or books

**MAX\_BOOK** – Maximum number of sheets per book (cutting height)

**PICTURE** – Name of file containing picture of cutting pattern

**ASCII & Database examples**

PATTERNS,1,1,2,0,20,4,5,“Pattern1”

PATTERNS TABLE

| JOB_INDEX | PTN_INDEX | BRD_INDEX | TYPE | QTY_RUN | QTY_CYCLES |
|-----------|-----------|-----------|------|---------|------------|
| 1         | 1         | 2         | 0    | 20      | 4          |

**14. CUTS - CUTTING INSTRUCTIONS**

**CUTS - JOB\_INDEX, PTN\_INDEX, CUT\_INDEX, SEQUENCE, FUNCTION, DIMENSION, QTY\_RPT, PART\_INDEX, QTY\_PART, COMMENT**

These records define each cut for the saw and determine the parts produced by each cut. This is used, for example, so that the correct labels can be printed at the saw in synchronisation with the cutting.

**JOB\_INDEX** - Index number used to link this record to other records for this job.

**PTN\_INDEX** - Index number used to link this record with pattern records

**CUT\_INDEX** - Sequential index number incrementing by 1 for each cut

record for each job

**SEQUENCE** - Cut sequence number indicating order in which cuts are processed by saw

**FUNCTION** - The type of cut:

0 = head cut

1 = rip cut

2 = cross cut

3 = 3rd phase / recut

4 = 4th phase /recut

90,91,92,93 = trim / waste cut corresponding to phase of cut (to override defaults)

**DIMENSION** - The size of cut in measurement units

**QTY\_RPT** - The repeat quantity for this cut

**PART\_INDEX** - 0 if no part produced or part index number in part or offcut records

**QTY\_PARTS** - Quantity of this part produced by this cut for all cycles of this pattern.

**COMMENT** - optional field to store narrative about the cut instruction

-

**Some points about the cut record**

Note that some cuts produce several parts with different item numbers because although the parts may have the same size they will be labelled uniquely. This can occur when cutting multiple sheets in a book where the parts on different have different item numbers or where a crosscut of two strips side-by-side may produce parts with different identities.

The Sequence number allows definition of different parts in stack produced from same cut. Note, the cuts are not listed in cut sequence because it is necessary to nest the 2nd, 3rd and later phase cuts. The SEQUENCE number is optional, and if not given then the cutting sequence should be determined by the saw or a post-processor.

The QTY\_PARTS field allows for the display of the correct part quantities for duplicate parts. In a pattern with run quantity 20, and cut 6 sheets at a time, then there will be 4 cycles or books (3 with 6 sheets and 1 with 2 sheets). Say the first part in the top left corner is a mixture of 14 parts item 1, and 6 parts item 2. The first book at the appropriate cut will produce quantity 6 labels of item 1, the second book also 6 of 1, the third book will produce 2 labels of item 1 and 4 of item 2, and the last book will give 2 of item 2. In this example, the CUTS records would show two cut lines, item 1 quantity 14, and item 2 quantity 6. The saw takes care of counting the cycles.

1

Notes in [] are for information only and not part of the file.

CUTS,1,1,2,4,2,800.0,3,1,14 [*crosscut 500 strips at 800mm x3 >part 1 qty 14*]

CUTS,1,1,4,3,1,200.0,1,0,0,"Rip" *[rip 1 strip 200mm wide]*

CUTS,1,1,5,7,2,1400.0,1,8,20 *[crosscut at 1400mm producing part 8]*

CUTS,1,1,6,7,2, 0,0,X3,20 *and offcut 3]*

CUTS TABLE

| JOB_INDEX | PTN_INDEX | CUT_INDEX | SEQUENCE | FUNCTION | DIMENSION | ..... |
|-----------|-----------|-----------|----------|----------|-----------|-------|
| 1         | 1         | 1         | 1        | 1        | 500.0     | ..... |
| 1         | 1         | 2         | 4        | 2        | 800.0     | ..... |
| 1         | 1         | 3         | 4        | 2        | 0.0       | ..... |
| 1         | 1         | 4         | 3        | 1        | 200.0     | ..... |
| 1         | 1         | 5         | 7        | 2        | 1400.0    | ..... |
| 1         | 1         | 6         | 7        | 2        | 0.0       | ..... |

**VECTORS - JOB\_INDEX, PTN\_INDEX, CUT\_INDEX, X\_START, Y\_START, X\_END, Y\_END**

This table holds an optional description of the pattern as a set of vectors.

**JOB INDEX** - Index number used to link this record to other records for this job



**PTN\_INDEX** - This is an index number used to link this to the PATTERN record

**CUT\_INDEX** - This is an index number to relate the vector to the CUT record

**X\_START** - Start co-ordinate of cut in X (always positive)

**Y\_START** - Start co-ordinate of cut in Y ( “ ” )

**X\_END** - Start co-ordinate of cut in X ( “ ” )

**Y\_END** - Start co-ordinate of cut in Y ( “ ” )

The origin of the drawing is defined in the HEADER record. The x and y positions specify the distance to include the saw kerf, away from origin. So, a 10 mm trim would result in a vector at x=10, where if saw kerf is 4.5, then waste removed is 5.5. The position for cuts producing parts must include all saw kerfs. Note, that unlike the CUT records where all dimensions are relative, in these records all dimensions are absolute values.

ASCII & Database examples

*Notes in [] are for information only and not part of the file.*

VECTORS, 1, 1, 1, 1, 0.0,0.0,3660.0,10.0

VECTORS, 1, 1, 2, 0.0, 315.0,3660.0, 315.0 *[rip 300 wide strip - note absolute dimension]*

VECTORS, 1, 1, 5,5.2,10.0,10.0,315.0

VECTORS TABLE

| JOB_INDEX | PTN_INDEX | CUT_INDEX | X_START | Y_START | X_END  | Y_END |
|-----------|-----------|-----------|---------|---------|--------|-------|
| 1         | 1         | 1         | 0.0     | 0.0     | 3660.0 | 10.0  |
| 1         | 1         | 2         | 0.0     | 315.0   | 3660.0 | 315.0 |
| 1         | 1         | 5         | 5.2     | 10.0    | 10.0   | 315.0 |

16. EXAMPLE OF PATTERN EXCHANGE STRUCTURE

The following example relates to example order batch A123 containing one job number 00011 comprising 6 sample patterns for which printouts are included. The patterns are designed to show a variety of different cutting situations in as few patterns as possible, and therefore are unusual and non-optimal rather than typical examples. Text in italics enclosed in square brackets[] is for information only. The fields have been spaced out for ease of reading.

*Notes in [] are for information only and not part of the file.*

Filename = A123.PTX

*[file starts with header record]*  
HEADER,1,“This is a sample PTX file for batch A123”,0,0,1

*[version1, millimetres, ... ..top-to-bottom, fixed first trim]*  
JOBS,1,00010,“Example run 10”,,,,1

*[data for parts – would also include data for label]*  
PARTS\_REQ,1,1,P1,1, 1200.0, 725.0, 9,0,0,0,9

PARTS\_REQ,1,2,P2,1, 1200.0, 725.0, 8,0,0,0,8  
 PARTS\_REQ,1,3,P3,1, 1272.6, 600.0, 7,0,0,0,7  
 PARTS\_REQ,1,4,P4,1, 790.0, 450.0, 4,0,0,0,4  
 PARTS\_REQ,1,5,P5,1, 580.0, 200.0,20,2,0,0,22  
 PARTS\_REQ,1,6,P6,2, 1400.0, 300.0, 5,0,0,0,5  
 PARTS\_REQ,1,7,P7,2, 650.0, 275.0,14,0,0,0,14  
 PARTS\_REQ,1,8,P8,2, 480.0, 230.0,20,5,0,0,23  
 PARTS\_REQ,1,9,P9,1, 600.0, 200.0, 7,0,0,0,7  
 PARTS\_REQ,1,10,P10,2,480.0, 400.0, 3,0,0,0,3

*[data for board (sheet) sizes]*

BOARDS,1,1,WHLAM15MM-1, 1, 2550.0, 1525.0, 436, 6, 2.540,0  
 BOARDS,1,2,WHLAM15MM-2, 1, 3050.0, 1830.0, 178, 1, 2.430,0  
 BOARDS,1,3,MDF18-97, 2, 2440.0, 1220.0, 371, 3, 4.320,0

*[definition of materials]*

MATERIALS,1,1,WHITE15, "White laminated chipboard 15mm", 15.0, 4, 4.8, 4.8, 10.0, 10.0, 10.0, 10.0, 8.0, 8.0, 8.0, 4, 1, 1, 0  
 MATERIALS,1,2,MDF18, "Medium density fibreboard 18mm", 18.0, 3, 4.8, 4.8, 10.0, 10.0, 10.0, 10.0, 8.0, 8.0, 8.0, 4, 1, 1, 0

*[data for off-cut sizes]*

OFFCUTS,1,1, X00010/0001,1,1830.0,621.0,1  
 OFFCUTS,1,2, X00010/0002,1,1110.4,450.0,1

*[patterns]*

PATTERNS,1,1,1,0,2,1,2 *[Job 1, Pattern 1, Board 1, Long rip, Qty 2]*  
 PATTERNS,1,2,1,1,3,1,3 *[Job 1, Pattern 2, Board 1, Short rip, Qty 3]*  
 PATTERNS,1,3,1,0,1,1,1 *[Job 1, Pattern 3, Board 1, Long rip, Qty 1]*  
 PATTERNS,1,4,2,2,1,1,1 *[Job 1, Pattern 4, Board 2, Head cut, Qty 1]*  
 PATTERNS,1,5,3,2,2,1,2 *[Job 1, Pattern 5, Board 3, Head cut, Qty 2]*  
 PATTERNS,1,6,3,2,1,1,1 *[Job 1, Pattern 6, Board 3, Head cut, Qty 1]*

*[cuts]*

CUTS,1,1, 1,0, 0,2550.0,0,0,0,"MAIN"  
 CUTS,1,1, 2,1,91, 5.2,1,0,0  
 CUTS,1,1, 3,2, 1, 725.0,2,0,0,"RIP" *[2 rips at 725 mm]*  
 CUTS,1,1, 4,4,92, 5.2,1,0,0 *[cross cut trim 5.2 mm]*  
 CUTS,1,1, 5,5, 2,1200.0,2,1,4 *[crosscuts at 1200 yielding part 1]*

CUTS,1,2, 1,0, 0,1525.0,0,0,0,"MAIN"  
 CUTS,1,2, 2,1,91, 5.2,1,0,0 *[rip trim]*  
 CUTS,1,2, 3,2, 1,1200.0,1,0,0,"RIP"  
 CUTS,1,2, 4,4,92, 5.2,1,0,0  
 CUTS,1,2, 5,5, 2, 725.0,2,1,1 *[2 xcuts at 725 yielding part 1 x 1 ...*  
 CUTS,1,2, 6,5, 2, 0.0,0,2,5 *... and part 2 x 5]*  
 CUTS,1,2, 7,3, 1,1272.6,1,0,0,"RIP"  
 CUTS,1,2, 8,7,92, 5.2,1,0,0  
 CUTS,1,2, 9,8, 2, 600.0,1,3,3  
 CUTS,1,2,10,9, 2, 725.0,1,0,0  
 CUTS,1,2,11,10,93, 5.2,1,0,0  
 CUTS,1,2,12,11,3,1200.0,1,2,3 *[recut to 1200mm yielding part 2]*

CUTS,1,3, 1,0, 0,2550.0,0,0,0,"MAIN"  
 CUTS,1,3, 2,1,91, 5.2,1,0,0 *[rip trim]*  
 CUTS,1,3, 3,2, 1, 600.0,1,0,0,"RIP"  
 CUTS,1,3, 4,6, 2,1272.6,1,3,2 *[no trims & 1 xcut giving part 3 x 2]*  
 CUTS,1,3, 5,3, 1, 450.0,1,0,0,"RIP"  
 CUTS,1,3, 6,7,92, 5.2,1,0,0 *[xcut trim]*  
 CUTS,1,3, 7,8, 2, 790.0,1,4,1 *[cross cut producing part 4 x 1]*

CUTS,1,3, 8,9, 2, 630.0,1,0,0  
CUTS,1,3, 9,10,93, 5.2,1,0,0, [recut trim]  
CUTS,1,3,10,11, 3,200.0,1,9,1  
CUTS,1,3 11,12, 3,200.0,1,0,0  
CUTS,1,3,12,13,94, 5.2,1,0,0 [4th phase recut trim]  
CUTS,1,3,13,14, 4,580.0,1,5,1 [4th phase cut to produce part 5]  
CUTS,1,3,14, 9, 2, 0.0,0,X2,1 [cut 9 also produces off-cut 2]  
CUTS,1,3,15, 4, 1,200.0,2,0,0,"RIP"  
CUTS,1,3,16,15,92, 5.2,1,0,0 [xcut trim]  
CUTS,1,3,17,16, 2,580.0,4,5,8  
  
CUTS,1,4, 1, 1, 0,3034.2,1,0,0,"MAIN"  
CUTS,1,4, 2, 2,91, 5.2,1,0,0 [rip trim]  
CUTS,1,4, 3, 3, 1, 600.0,1,0,0,"RIP"  
CUTS,1,4, 4, 8,92, 5.2,1,0,0 [xcut trim]  
CUTS,1,4, 5, 9, 2,1250.0,2,3,2 [part 3]  
CUTS,1,4, 6,11, 2, 200.0,2,0,0  
CUTS,1,4, 7,13,93, 5.2,1,0,0  
CUTS,1,4, 8,14, 3, 580.0,1,9,2 [part 9]  
CUTS,1,4, 9, 4, 1, 450.0,1,0,0,"RIP"  
CUTS,1,4,10,15,92, 5.2,1,0,0 [xcut trim]  
CUTS,1,4,11,16, 2, 790.0,3,4,3  
CUTS,1,4,12,19, 2, 580.0,1,0,0 [crosscut at 580]  
CUTS,1,4,13,20,93, 5.2,1,0,0 [recut trim]  
CUTS,1,4,14,21, 3, 200.0,2,5,2 [3rd phase cut 200 x 2 -> part5 x 2]  
CUTS,1,4,15, 5, 1, 200.0,3,0,0,"RIP"  
CUTS,1,4,16,23,92, 5.2,1,0,0 [xcut trim]  
CUTS,1,4,17,24, 2, 580.0,3,5,9 [part 5 x 9]  
CUTS,1,4,18,27, 2, 630.0,2,9,6 [part 9 x 6]  
CUTS,1,4,19, 1, 0, 621.0,0,X1,1,"HEAD" [producing offcut 1]  
  
CUTS,1,5, 1, 1, 0,1415.0,1,0,0,"MAIN" [head cut at 1415.0]  
CUTS,1,5, 2, 2,91, 5.2,1,0,0 [rip trim]  
CUTS,1,5, 3, 3, 1, 300.0,2,0,0,"RIP" [rip main 300 x 2]  
CUTS,1,5, 4, 7,92, 5.2,1,0,0 [xcut trim]  
CUTS,1,5, 5, 8, 2,1400.0,1,6,4  
CUTS,1,5, 6, 5, 1, 275.0,2,0,0,"RIP"  
CUTS,1,5, 7, 9,92, 5.2,1,0,0 [xcut trim]  
CUTS,1,5, 8,10, 2, 650.0,2,7,8  
CUTS,1,5, 9, 1, 0,1020.2,0,0,0,"HEAD" [start of head section]  
CUTS,1,5,10,12,91, 5.2,1,0,0 [rip trim]  
CUTS,1,5,11,13, 1, 650.0,1,0,0,"RIP"  
CUTS,1,5,12,16,92, 5.2,1,0,0 [xcut trim]  
CUTS,1,5,13,17, 2, 275.0,3,7,6  
CUTS,1,5,14,14, 1, 230.0,2,0,0,"RIP"  
CUTS,1,5,15,20,92, 5.2,1,0,0 [xcut trim]  
CUTS,1,5,16,21, 2, 480.0,2,8,8  
  
CUTS,1,6, 1, 1, 0,1464.6,1,0,0,"MAIN" [head cut at 1464.6]  
CUTS,1,6, 2, 5,91, 5.2,1,0,0 [rip trim]  
CUTS,1,6, 3, 6, 1, 300.0,1,0,0,"RIP" [rip main 300 x 1]  
CUTS,1,6, 4,10,92, 5.2,1,0,0 [xcut trim]  
CUTS,1,6, 5,11, 2,1400.0,1,6,1  
CUTS,1,6, 6, 7, 1, 400.0,1,0,0,"RIP"  
CUTS,1,6, 7,12,92, 5.2,1,0,0 [xcut trim]  
CUTS,1,6, 8,13, 2, 480.0,3,10,3  
CUTS,1,6, 9, 8, 1, 230.0,2,0,0,"RIP"  
CUTS,1,6,10,12,92, 5.2,0,0,0 [trim and cross cuts checkerboard ...  
CUTS,1,6,11,13, 2, 480.0,0,8,6 cut with preceding strip]  
CUTS,1,6,12, 2, 0, 480.0,1,0,0,"HEAD" [head section]

CUTS,1,6,13,16,91, 5.2,1,0,0 [rip trim]  
CUTS,1,6,14,17, 1, 230.0,5,8,5,"RIP"  
CUTS,1,6,15, 3, 0, 230.0,2,0,0,"HEAD" [head sections]  
CUTS,1,6,16,22,91, 5.2,1,0,0 [rip trim]  
CUTS,1,6,17,23, 1, 480.0,2,8,4,"RIP"

17. EXAMPLES

**PATTERN 1**  
Board: WHLAN-15MM-1 2550 x 1525 x 15.0 Material: WHITE15 No. of Boards: 2

Sawkerf: 4.8 Book Height: 2 Cycles: 1  
Rear trim (inc kerf) Rip: 10.0 Cross: 10.0 Retrim (inc kerf): 5.0

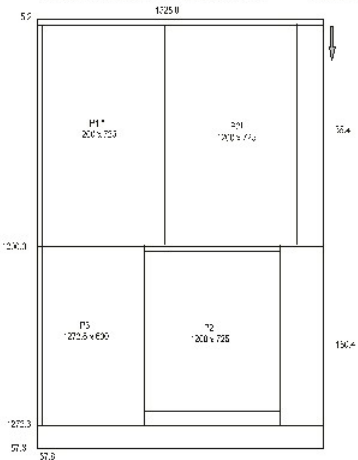
| No | Part | Length | Width | Total<br>Prod | Cut<br>Far | so<br>Brd | Qty<br>Ptn | Still<br>To cut |
|----|------|--------|-------|---------------|------------|-----------|------------|-----------------|
| 1. | P1   | 1200.0 | 725.0 | 9             | NIL        | 4         | 8          | 1               |

ADR/PRG: [41]

| Cut  | Size  | Qty | Part | Cut       | Size   | Qty | Part |
|------|-------|-----|------|-----------|--------|-----|------|
| MAIN |       |     |      | Trim      | 5.2    | 1   |      |
| Trim | 5.2   | 1   |      | Cross cut | 1200.0 | 2   | P1   |
| Rip  | 725.0 | 2   |      |           |        |     |      |

CUTS,1,1, 1,0, 0,2550.0,0,0,0,"MAIN"  
CUTS,1,1, 2,1,91, 5.2,1,0,0  
CUTS,1,1, 3,2, 1, 725.0,2,0,0,"RIP" [2 rips at 725 mm]  
CUTS,1,1, 4,4,92, 5.2,1,0,0 [cross cut trim 5.2 mm]  
CUTS,1,1, 5,5, 2,1200.0,2,1,4 [crosscuts at 1200 yielding part 1]

PATTERN 2  
Board: WHLAM15MM-1 1525 x 2550 x 15.0  
Material: WHITE15  
No. of Boards: 3



Saw kerf: 4.8 Book Height: 3 Cycles: 1  
Rear trim (no kerf) Rip: 10.0 Cross: 10.0 Retrim (no kerf): 5.0

| No | Part | Length | Width | Total Cut so |     | Qty | Qty | Still  |
|----|------|--------|-------|--------------|-----|-----|-----|--------|
|    |      |        |       | Prod         | Far | Brd | Etn | To cut |
| 1. | P1   | 1200.0 | 725.0 | 9            | 8   | -   | 1   | NIL    |
| 2. | P2   | 1200.0 | 725.0 | 8            | NIL | -   | 8   | NIL    |
| 3. | P3   | 1250.0 | 600.0 | 7            | NIL | 1   | 3   | 4      |

ADR/PRG: [41]

| Cut       | Size   | Qty | Part | Cut       | Size   | Qty | Part |
|-----------|--------|-----|------|-----------|--------|-----|------|
| MAIN      |        |     |      | Rip       | 1272.6 | 1   |      |
| Trim      | 5.2    | 1   |      | Trim      | 5.2    | 1   |      |
| Rip       | 1200.0 | 1   |      | Cross cut | 600.0  | 1   | P3   |
| Trim      | 5.2    | 1   |      | Cross cut | 725.0  | 1   |      |
| Cross cut | 725.0  | 1   | P1   | Recut     | 5.2    | 1   |      |
| Cross cut | 725.0  | 1   | P2   | Recut     | 1200.0 | 1   | P2   |

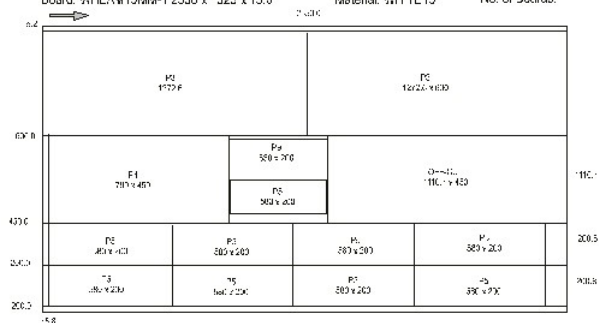
CUTS,1,2, 1,0, 0,1525.0,0,0,0,"MAIN"  
CUTS,1,2, 2,1,91, 5.2,1,0,0 [rip trim]  
CUTS,1,2, 3,2, 1,1200.0,1,0,0,"RIP"  
CUTS,1,2, 4,4,92, 5.2,1,0,0  
CUTS,1,2, 5,5, 2, 725.0,2,1,1 [2 xcuts at 725 yielding part 1 x 1 ...  
CUTS,1,2, 6,5, 2, 0.0,0,2,5 ... and part 2 x 5]  
CUTS,1,2, 7,3, 1,1272.6,1,0,0,"RIP"  
CUTS,1,2, 8,7,92, 5.2,1,0,0  
CUTS,1,2, 9,8, 2, 600.0,1,3,3  
CUTS,1,2,10,9, 2, 725.0,1,0,0  
CUTS,1,2,11,10,93, 5.2,1,0,0  
CUTS,1,2,12,11,3,1200.0,1,2,3 [recut to 1200mm yielding part 2]

## PATTERN 3

Board: VHLAW13MM-1 2550 x 1525 x 15.0

Material: VHL TE15

No. of Boards: 1



Saw kerf: 4.8 Book Height: 1 Cycles: 1

Rear trim (inc kerf) Rip: 10.0 Cross: 10.0 Return (inc kerf): 5.0

| No | Part | Length | Width | Total<br>Prod | Cut so<br>Far | Qty<br>Brd | Qty<br>Rtn | Still<br>To cut |
|----|------|--------|-------|---------------|---------------|------------|------------|-----------------|
| 3. | P3   | 1272.6 | 600.0 | 7             | 3             | 2          | 2          | 2               |
| 4. | P4   | 790.0  | 450.0 | 4             | NIL           | 1          | 1          | 3               |
| 5. | P5   | 580.0  | 200.0 | 22            | NIL           | 9          | 9          | 13              |
| 9. | P9   | 630.0  | 200.0 | 7             | NIL           | 1          | 1          | 6               |

ADR/PRG: [41]

| Cut       | Size   | Qty | Part | Cut       | Size  | Qty | Part |
|-----------|--------|-----|------|-----------|-------|-----|------|
| MAIN      |        |     |      | Recut     | 5.2   | 1   |      |
| Trim      | 5.2    | 1   |      | Recut     | 200.0 | 1   | P9   |
| Rip       | 600.0  | 1   |      | Recut     | 200.0 | 1   |      |
|           |        |     |      | Recut     | 5.2   | 1   |      |
| Cross cut | 1250.0 | 2   | P3   | Recut     | 580.0 | 1   | P5   |
| Rip       | 450.0  | 1   |      | Recut     | 200.0 | 2   |      |
| Trim      | 5.2    | 1   |      | Trim      | 5.2   | 1   |      |
| Cross cut | 790.0  | 1   | P4   | Cross cut | 580.0 | 4   | P5   |
| Cross cut | 630.0  | 1   |      |           |       |     |      |

CUTS,1,3, 1,0, 0,2550.0,0,0,0,"MAIN"

CUTS,1,3, 2,1,91, 5.2,1,0,0 [rip trim]

CUTS,1,3, 3,2, 1, 600.0,1,0,0,"RIP"

CUTS,1,3, 4,6, 2,1272.6,1,3,2 [no trims &amp; 1 xcut giving part 3 x 2]

CUTS,1,3, 5,3, 1, 450.0,1,0,0,"RIP"

CUTS,1,3, 6,7,92, 5.2,1,0,0 [xcut trim]

CUTS,1,3, 7,8, 2, 790.0,1,4,1 [cross cut producing part 4 x 1]

CUTS,1,3, 8,9, 2, 630.0,1,0,0

CUTS,1,3, 9,10,93, 5.2,1,0,0 [recut trim]

CUTS,1,3,10,11, 3,200.0,1,9,1

CUTS,1,3,11,12, 3,200.0,1,0,0

CUTS,1,3,12,13,94, 5.2,1,0,0 [4th phase recut trim]

CUTS,1,3,13,14, 4,580.0,1,5,1 [4th phase cut to produce part 5]

CUTS,1,3,14, 9, 2, 0,0,0,X2,1 [cut 9 also produces off-cut 2]

CUTS,1,3,15, 4, 1,200.0,2,0,0,"RIP"

CUTS,1,3,16,15,92, 5.2,1,0,0 [xcut trim]

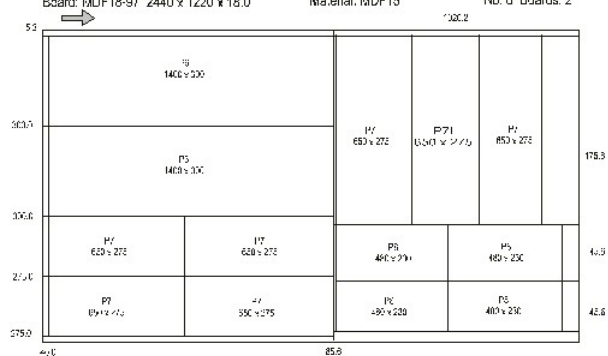
CUTS,1,3,17,16, 2,580.0,4,5,8

| No | Part | Length | Width | Total<br>Prod | Cut so<br>Far | Qty<br>Brd | Qty<br>Ptn | Still<br>To cut |
|----|------|--------|-------|---------------|---------------|------------|------------|-----------------|
| 3. | P3   | 1276.6 | 600.0 | 7             | 5             | 2          | 2          | NIL             |
| 4. | P4   | 790.0  | 450.0 | 4             | 1             | 3          | 3          | NIL             |
| 5. | P5   | 580.0  | 200.0 | 22            | 9             | 13         | 13         | NIL             |
| 9. | P9   | 630.0  | 200.0 | 7             | 1             | 6          | 6          | NIL             |

| ADR/PRG: [41] |        |     |      |           |       |          |
|---------------|--------|-----|------|-----------|-------|----------|
| Cut           | Size   | Qty | Part | Cut       | Size  | Qty Part |
| Head cut      | 3034.2 | 1   |      | Trim      | 5.2   | 1        |
| MAIN          |        |     |      | Cross cut | 790.0 | 3 P4     |
| Trim          | 5.2    | 1   |      | Cross cut | 580.0 | 1        |
| Rip           | 600.0  | 1   |      | Recut     | 5.2   | 1        |
| Trim          | 5.2    | 1   |      | Recut     | 200.0 | 2 P5     |
| Cross cut     | 1276.2 | 2   | P3   | Rip       | 200.0 | 3        |
| Cross cut     | 200.0  | 2   | P9   | Trim      | 5.2   | 1        |
| Recut         | 5.2    | 1   |      | Cross cut | 580.0 | 3 P5     |
| Recut         | 580.0  | 1   | P5   | Cross cut | 630.0 | 2 P9     |
| Rip           | 450.0  | 1   |      |           |       |          |

CUTS,1,4, 1, 1, 0,3034.2,1,0,0,,"MAIN"  
CUTS,1,4, 2, 2,91, 5,2,1,0,0 [*rip trim*]  
CUTS,1,4, 3, 3, 1, 600.0,1,0,0,"RIP"  
CUTS,1,4, 4, 8,92, 5,2,1,0,0 [*xcut trim*]  
CUTS,1,4, 5, 9, 2,1250.0,2,3,2 [*part 3*]  
CUTS,1,4, 6,11, 2, 200.0,2,0,0  
CUTS,1,4, 7,13,93, 5,2,1,0,0  
CUTS,1,4, 8,14, 3, 580.0,1,9,2 [*part 9*]  
CUTS,1,4, 9, 4, 1, 450.0,1,0,0,"RIP"  
CUTS,1,4,10,15,92, 5,2,1,0,0 [*xcut trim*]  
CUTS,1,4,11,16, 2, 790.0,3,4,3  
CUTS,1,4,12,19, 2, 580.0,1,0,0 [*crosscut at 580*]  
CUTS,1,4,13,20,93, 5,2,1,0,0 [*recut trim*]  
CUTS,1,4,14,21, 3, 200.0,2,5,2 [*3rd phase cut 200 x 2 -> part5 x 2*]  
CUTS,1,4,15, 5, 1, 200.0,3,0,0,"RIP"  
CUTS,1,4,16,23,92, 5,2,1,0,0 [*xcut trim*]  
CUTS,1,4,17,24, 2, 580.0,3,5,9 [*part 5 x 9*]  
CUTS,1,4,18,27, 2, 630.0,2,9,6 [*part 9 x 6*]  
CUTS,1,4,19, 1, 0, 621.0,0,X1,1,"HEAD" [*producing offcut 1*]

Board: MDF18-97 2440 x 1220 x 18.0      Material: MDF18      No. of Boards: 2



Saw kerf: 4.8 Book Height: 2 Cycles: 1  
Reartrim (inc kerf) Rip: 10.0 Cross: 10.0 Retrim (inc kerf): 5.0

| No | Part | Length | Width | Total<br>Prod | Cut so<br>Far | Qty<br>Brd | Qty<br>Ptn | Still<br>To cut |
|----|------|--------|-------|---------------|---------------|------------|------------|-----------------|
| 6. | P6   | 1400.0 | 300.0 | 5             | NIL           | 2          | 4          | 1               |
| 7. | P7   | 650.0  | 275.0 | 14            | NIL           | 7          | 14         | NIL             |
| 8. | P8   | 480.0  | 230.0 | 23            | NIL           | 4          | 8          | 15              |

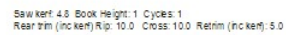
ADR/PRG: [41]

| Cut       | Size   | Qty | Part | Cut       | Size  | Qty | Part |
|-----------|--------|-----|------|-----------|-------|-----|------|
| Head cut  | 1415.0 | 1   |      | HEAD 1    |       |     |      |
| MAIN      |        |     |      | Trim      | 5.2   | 1   |      |
| Trim      | 5.2    | 1   |      | Rip       | 650.0 | 1   |      |
| Rip       | 300.0  | 2   |      | Trim      | 0.2   | 1   |      |
| Trim      | 5.2    | 1   |      | Cross cut | 275.0 | 3   | P7   |
| Cross cut | 1400.0 | 1   | P6   | Rip       | 230.0 | 2   |      |
| Rip       | 275.0  | 2   |      | Trim      | 0.2   | 1   |      |
| Trim      | 5.2    | 1   |      | Cross cut | 480.0 | 2   | P8   |
| Cross cut | 650.0  | 2   | P7   |           |       |     |      |

CUTS,1.5, 1, 1, 0,1415.0,1,0,0,"MAIN" *[head cut at 1415.0]*  
CUTS,1.5, 2,2.91, 5.2,1,0,0 *[rip trim]*  
CUTS,1.5, 3, 3, 1, 300.0,2,0.0,"RIP" *[rip main 300 x 2]*  
CUTS,1.5, 4, 7.92, 5.2,1,0,0 *[xcut trim]*  
CUTS,1.5, 5, 8, 2,1400.0,1,6,4  
CUTS,1.5, 6, 5, 1, 275.0,2,0.0,"RIP"  
CUTS,1.5, 7, 9.92, 5.2,1,0,0 *[xcut trim]*  
CUTS,1.5, 8, 10, 2, 650.0,2,7,8  
CUTS,1.5, 9, 1, 0,1020.2,0,0,0,"HEAD" *[start of head section]*  
CUTS,1.5,10,12,91, 5.2,1,0,0 *[rip trim]*  
CUTS,1.5,11,13, 1, 650.0,1,0,0,"RIP"  
CUTS,1.5,12,16,92, 5.2,1,0,0 *[xcut trim]*  
CUTS,1.5,13,17, 2, 275.0,3,7,6  
CUTS,1.5,14,14, 1, 230.0,2,0.0,"RIP"  
CUTS,1.5,15,20,92, 5.2,1,0,0 *[xcut trim]*  
CUTS,1.5,16,21, 2, 480.0,2,8,8



Board: MDF-9-97 24-0 x 1220 x 9.0      Material: MDF-8      No. of Boards: 1



ADR/PRG: [41]

CUTS,1.6, 1, 1, 0,1464.6,1.0,0,"MAIN" *[head cut at 1464.6]*  
 CUTS,1.6, 2, 5.91, 5.2,1.0,0 *[rip trim]*  
 CUTS,1.6, 3, 6, 1, 300.0,1.0,0,"RIP" *[rip main 300 x 1]*  
 CUTS,1.6, 4,10.92, 5.2,1.0,0 *[xcut trim]*  
 CUTS,1.6, 5,11, 2,1400.0,1.6,1  
 CUTS,1.6, 6, 7, 1, 400.0,1.0,0,"RIP"  
 CUTS,1.6, 7,12.92, 5.2,1.0,0 *[xcut trim]*  
 CUTS,1.6, 8,13, 2, 480.0,3.10,3  
 CUTS,1.6, 9, 8, 1, 230.0,2.0,0,"RIP"  
 CUTS,1.6,10,12.92, 5.2,0.0,0 *[trim and cross cuts checkerboard ...]*  
 CUTS,1.6,11,13, 2, 480.0,0.8,6 *cut with preceding strip]*  
 CUTS,1.6,12, 2, 0, 480.0,1.0,0,"HEAD" *[head section]*  
 CUTS,1.6,13,16.91, 5.2,1.0,0 *[rip trim]*  
 CUTS,1.6,14,17, 1, 230.0,5.8,5,"RIP"  
 CUTS,1.6,15, 3, 0, 230.0,2.0,0,"HEAD" *[head sections]*  
 CUTS,1.6,16,22.91, 5.2,1.0,0 *[rip trim]*  
 CUTS,1.6,17,23, 1, 480.0,2.8,4,"RIP"

file:///C:/Users/celav/AppData/Local/Temp/~hhEA7D.htm

This section summarises the position and format of each field in each record type. The maximum length of each text field is listed in the comment column. The range of acceptable values for other field types is listed where applicable. Please refer to earlier comments for a fuller explanation of fields.

Each field can be categorised by one of the following types.

**DIM** Dimension. Number single. When working in millimetres these range from 0.0 to 9999.9. When working in decimal inches dimensions must range from 0.000 to 999.9

**FLT** Number single. Floating point value.

**IDX** Index. Number integer. These are integer values which are used to link records. For example all data for a particular job must have the same job index.

**INT** Number integer.

**QTY** A long integer used to store quantity. No quantity can be greater than 99999.

**TXT** A text field used to store information

Note that spaces are not allowed in the material code, and any spaces will be converted to an underscore ('\_') on import. Also note that material, part and board codes are converted to upper case on import.

The job records must have unique job index numbers starting at 1, and incrementing consecutively within specified range. The part, board and pattern records must each have their respective index numbers unique within the job, and again be numbered from 1 and incremented consecutively.

### 'HEADER' RECORD

| No. | Name      | Description               | Type | Comment       | MDB            |
|-----|-----------|---------------------------|------|---------------|----------------|
| 1   | VERSION   | File version              | TXT  | Set to 1      | Text           |
| 2   | TITLE     | File title                | TXT  | 25 chars max. | Text           |
| 3   | UNITS     | Measurement mode          | INT  | 0,1           | Number-Integer |
| 4   | ORIGIN    | Pattern origin            | INT  | 0-3           | Number-Integer |
| 5   | TRIM_TYPE | Fixed trim front or rear? | INT  | 0,1           | Number-Integer |

### 'JOBS' RECORD

| No. | Name      | Description      | Type | Comment        | MDB             |
|-----|-----------|------------------|------|----------------|-----------------|
| 1   | JOB_INDEX | Job index        | IDX  | 1-99           | Number-Integer  |
| 2   | NAME      | Job number/name  | TXT  | 8 chars max.   | Text            |
| 3   | DESC      | Job description  | TXT  | 20 chars max   | Text            |
| 4   | ORD_DATE  | Date of order    | TXT  | DD/MM/YYYY     |                 |
| 5   | CUT_DATE  | Date for cutting | TXT  | DD/MM/YYYY     |                 |
| 6   | CUSTOMER  | Customer code    | TXT  | 100 chars max. | Text            |
| 7   | STATUS    | Job status       | INT  | 0,1,2          | Number- Integer |

### 'PARTS\_REQ' RECORD

| No. | Name       | Description          | Type | Comment       | MDB                             |
|-----|------------|----------------------|------|---------------|---------------------------------|
| 1   | JOB_INDEX  | Job index            | IDX  | 1-99          | Number- Integer                 |
| 2   | PART_INDEX | Part index           | IDX  | 1-2000        | Number- Integer                 |
| 3   | CODE       | Part code            | TXT  | 25 chars max. | Text                            |
| 4   | MAT_INDEX  | Material index       | IDX  | 1-2000        | Number- Integer                 |
| 5   | LENGTH     | Part length          | DIM  |               | Number-Single                   |
| 6   | WIDTH      | Part width           | DIM  |               | Number-Single                   |
| 7   | QTY_REQ    | Number of pieces     | QTY  | Max 99999     | Number-Long <a href="#">Int</a> |
| 8   | QTY_OVER   | Max over production  | QTY  | Max 99999     | Number-Long <a href="#">Int</a> |
| 9   | QTY_UNDER  | Max under production | QTY  | Max 99999     | Number-Long <a href="#">Int</a> |
| 10  | GRAIN      | Grain                | INT  | 0,1           | Number-Integer                  |
| 11  | QTY_PROD   | Num pieces produced  | QTY  | Max 99999     | Number-Long <a href="#">Int</a> |

### 'PARTS\_INF' RECORD

| No. | Name       | Description             | Type | Comment      | MDB             |
|-----|------------|-------------------------|------|--------------|-----------------|
| 1   | JOB_INDEX  | Job index               | IDX  | 1-99         | Number-Integer  |
| 2   | PART_INDEX | Part index              | IDX  | 1-2000       | Number-Integer  |
| 3   | DESC       | Second part desc        | TXT  | 50 chars max | Text            |
| 4   | LABEL_QTY  | Label quantity          | QTY  |              | Number-Long Int |
| 5   | FIN_LENGTH | Finished length         | TXT  |              | Text            |
| 6   | FIN_WIDTH  | Finished width          | TXT  |              | Text            |
| 7   | ORDER      | Original order          | TXT  |              | Text            |
| 8   | EDGE1      | Btm length edge code    | TXT  |              | Text            |
| 9   | EDGE2      | Top length edge code    | TXT  |              | Text            |
| 10  | EDGE3      | Left width edge code    | TXT  |              | Text            |
| 11  | EDGE4      | Right width edge code   | TXT  |              | Text            |
| 12  | EDG_PG1    | Bottom edge program     | TXT  |              | Text            |
| 13  | EDG_PG2    | Top edge program        | TXT  |              | Text            |
| 14  | EDG_PG3    | Left edge program       | TXT  |              | Text            |
| 15  | EDG_PG4    | Right edge program      | TXT  |              | Text            |
| 16  | FACE_LAM   | Face laminate           | TXT  |              | Text            |
| 17  | BACK_LAM   | Back laminate           | TXT  |              | Text            |
| 18  | CORE_MAT   | Core material           | TXT  |              | Text            |
| 19  | PALLET     | Pallet layout           | TXT  |              | Text            |
| 20  | DRAWING    | Name of drawing file    | TXT  |              | Text            |
| 21  | PRODUCT    | Product code            | TXT  |              | Text            |
| 22  | PROD_INFO  | Product description     | TXT  |              | Text            |
| 23  | PROD_WIDTH | Product width           | TXT  |              | Text            |
| 24  | PROD_HGT   | Product height          | TXT  |              | Text            |
| 25  | PROD_DEPTH | Product depth           | TXT  |              | Text            |
| 26  | PROD_NUM   | Product number          | TXT  |              | Text            |
| 27  | ROOM       | Room/group              | TXT  |              | Text            |
| 28  | BARCODE1   | Data for first barcode  | TXT  |              | Text            |
| 29  | BARCODE2   | Data for second barcode | TXT  |              | Text            |

**'PARTS\_UDI' RECORD**

| No. | Name       | Description          | Type | Comment      | MDB  |
|-----|------------|----------------------|------|--------------|------|
| 1   | JOB_INDEX  | Job index            | IDX  | 1-99         | Text |
| 2   | PART_INDEX | Part index           | IDX  | 1-2000       | Text |
| 3   | INFO1      | Information field 1  | TXT  | 50 chars max | Text |
| 4   | INFO2      | Information field 2  | TXT  | 50 chars max | Text |
| 5   | INFO3      | Information field 3  | TXT  | 50 chars max | Text |
| 6   | INFO4      | Information field 4  | TXT  | 50 chars max | Text |
| 7   | INFO5      | Information field 5  | TXT  | 50 chars max | Text |
| 8   | INFO6      | Information field 6  | TXT  | 50 chars max | Text |
| 9   | INFO7      | Information field 7  | TXT  | 50 chars max | Text |
| 10  | INFO8      | Information field 8  | TXT  | 50 chars max | Text |
| 11  | INFO9      | Information field 9  | TXT  | 50 chars max | Text |
| 12  | INFO10     | Information field 10 | TXT  | 50 chars max | Text |
| 13  | INFO11     | Information field 11 | TXT  | 50 chars max | Text |
| 14  | INFO12     | Information field 12 | TXT  | 50 chars max | Text |
| 15  | INFO13     | Information field 13 | TXT  | 50 chars max | Text |
| 16  | INFO14     | Information field 14 | TXT  | 50 chars max | Text |
| 17  | INFO15     | Information field 15 | TXT  | 50 chars max | Text |
| 18  | INFO16     | Information field 16 | TXT  | 50 chars max | Text |
| 19  | INFO17     | Information field 17 | TXT  | 50 chars max | Text |
| 20  | INFO18     | Information field 18 | TXT  | 50 chars max | Text |
| 21  | INFO19     | Information field 19 | TXT  | 50 chars max | Text |
| 22  | INFO20     | Information field 20 | TXT  | 50 chars max | Text |
| 23  | INFO21     | Information field 21 | TXT  | 50 chars max | Text |
| 24  | INFO22     | Information field 22 | TXT  | 50 chars max | Text |
| 25  | INFO23     | Information field 23 | TXT  | 50 chars max | Text |
| 26  | INFO24     | Information field 24 | TXT  | 50 chars max | Text |
| 27  | INFO25     | Information field 25 | TXT  | 50 chars max | Text |
| 28  | INFO26     | Information field 26 | TXT  | 50 chars max | Text |
| 29  | INFO27     | Information field 27 | TXT  | 50 chars max | Text |
| 30  | INFO28     | Information field 28 | TXT  | 50 chars max | Text |
| 31  | INFO29     | Information field 29 | TXT  | 50 chars max | Text |
| 32  | INFO30     | Information field 30 | TXT  | 50 chars max | Text |

**'BOARDS' RECORD**

| No. | Name      | Description            | Type | Comment      | MDB                    |
|-----|-----------|------------------------|------|--------------|------------------------|
| 1   | JOB_INDEX | Job index              | IDX  | 1-99         | Number-Integer         |
| 2   | BRD_INDEX | Board index            | IDX  | 1-2000       | Number-Integer         |
| 3   | CODE      | Board code             | TXT  | 25 chars max | Text                   |
| 4   | MAT_INDEX | Material index         | IDX  | 1-2000       | Number-Integer         |
| 5   | LENGTH    | Board length           | DIM  |              | Number-Single          |
| 6   | WIDTH     | Board width            | DIM  |              | Number-Single          |
| 7   | QTY_STOCK | Number of sheets       | QTY  | Max 99999    | Number-Long <u>Int</u> |
| 8   | QTY_USED  | Number of sheets used  | QTY  | Max 99999    | Number-Long <u>Int</u> |
| 9   | COST      | Cost per sq metre/foot | FLT  | 0-9.99       | Number-Single          |
| 10  | STK_FLAG  | Board limit/ratio      | INT  | 0-9          | Number-Integer         |

**'MATERIALS' RECORD**

| Num | Name      | Description                             | Type | Comment      | MDB                             |
|-----|-----------|-----------------------------------------|------|--------------|---------------------------------|
| 1   | JOB_INDEX | Job index                               | IDX  | 1-99         | Number-Integer                  |
| 2   | MAT_INDEX | Material index                          | IDX  | 1-2000       | Number-Integer                  |
| 3   | CODE      | Material code                           | TXT  | 25 chars max | Text                            |
| 4   | DESC      | Material description                    | TXT  | 50 chars max | Text                            |
| 5   | THICK     | Material thickness                      | DIM  |              | Number-Single                   |
| 6   | BOOK      | Max sheets per book                     | QTY  |              | Number-Long <a href="#">Int</a> |
| 7   | KERF_RIP  | Saw <a href="#">kerf</a> (rip)          | DIM  |              | Number-Single                   |
| 8   | KERF_XCT  | Saw <a href="#">kerf</a> (crosscut)     | DIM  |              | Number-Single                   |
| 9   | TRIM_FRIP | Fixed rip trim                          | DIM  |              | Number-Single                   |
| 10  | TRIM_VRIP | Min variable rip trim                   | DIM  |              | Number-Single                   |
| 11  | TRIM_FXCT | Fixed crosscut trim                     | DIM  |              | Number-Single                   |
| 12  | TRIM_VXCT | Min var. crosscut trim                  | DIM  |              | Number-Single                   |
| 13  | TRIM_HEAD | Internal Head trim                      | DIM  |              | Number-Single                   |
| 14  | TRIM_FRCT | Fixed <a href="#">recut</a> trim        | DIM  |              | Number-Single                   |
| 15  | TRIM_VRCT | Min variable <a href="#">recut</a> trim | DIM  |              | Number-Single                   |
| 16  | RULE1     | Optimising rule 1                       | INT  | 1-9          | Number-Integer                  |
| 17  | RULE2     | Optimising rule 2                       | INT  | 0,1          | Number-Integer                  |
| 18  | RULE3     | Optimising rule 3                       | INT  | 0,1          | Number-Integer                  |
| 19  | RULE4     | Optimising rule 4                       | INT  | 0,1          | Number-Integer                  |

**'OFFCUTS' RECORD**

| Num | Name      | Description     | Type | Comment      | MDB                             |
|-----|-----------|-----------------|------|--------------|---------------------------------|
| 1   | JOB_INDEX | Job index       | IDX  | 1-99         | Number-Integer                  |
| 2   | OFC_INDEX | Offcut index    | IDX  | 1-3000       | Number-Integer                  |
| 3   | CODE      | Offcut code     | TXT  | 25 chars max | Text                            |
| 4   | MAT_INDEX | Material index  | IDX  | 1-2000       | Number-Integer                  |
| 5   | LENGTH    | Offcut length   | DIM  |              | Number-Single                   |
| 6   | WIDTH     | Offcut width    | DIM  |              | Number-Single                   |
| 7   | OFC_QTY   | Offcut quantity | QTY  | Max 99999    | Number-Long <a href="#">Int</a> |

**'PATTERNS' RECORD**

| Num | Name       | Description          | Type | Comment     | MDB                             |
|-----|------------|----------------------|------|-------------|---------------------------------|
| 1   | JOB_INDEX  | Job index            | IDX  | 1-99        | Number-Integer                  |
| 2   | PTN_INDEX  | Pattern index        | IDX  | 1-2000      | Number-Integer                  |
| 3   | BRD_INDEX  | Board index          | IDX  | 1-2000      | Number-Integer                  |
| 4   | TYPE       | Pattern type         | INT  | 0-4         | Number-Integer                  |
| 5   | QTY_RUN    | Run quantity         | QTY  |             | Number-Long <a href="#">Int</a> |
| 6   | QTY_CYCLES | Cycle quantity       | QTY  |             | Number-Long <a href="#">Int</a> |
| 7   | MAX_BOOK   | Max sheets per book  | QTY  |             | Number-Long <a href="#">Int</a> |
| 8   | PICTURE    | Pattern picture file | TXT  | 8 chars max | Text                            |

**'CUTS' RECORD**

| No. | Name       | Description         | Type | Comment            | MDB                             |
|-----|------------|---------------------|------|--------------------|---------------------------------|
| 1   | JOB_INDEX  | Job index           | IDX  | 1-99               | Number-Integer                  |
| 2   | PTN_INDEX  | Pattern index       | IDX  | 1-2000             | Number-Integer                  |
| 3   | CUT_INDEX  | Cut index           | IDX  | 1-1000             | Number-Integer                  |
| 4   | SEQUENCE   | Cut sequence        | INT  |                    | Number-Integer                  |
| 5   | FUNCTION   | Cut type            | INT  | 0-4, 90-94         | Number-Integer                  |
| 6   | DIMENSION  | Size of cut         | DIM  |                    | Number-Single                   |
| 7   | QTY_RPT    | Cut quantity        | QTY  |                    | Number-Long <a href="#">Int</a> |
| 8   | PART_INDEX | Part/Offcut Index   | TXT  | 1-2000 or X1-X2000 | Text                            |
| 9   | QTY_PARTS  | Total part quantity | QTY  | Max 99999          | Number-Long <a href="#">Int</a> |
| 10  | COMMENT    | Additional comment  | TXT  | 100 chars max      | Text                            |

**5766 Import PTX file for Parts and Boards into Cut Rite V7.x**

The import/export of part and board records in the Pattern Exchange format (.PTX/MDB) has been modified to allow the use of all the part information (PARTS\_INF) and user defined part information (PARTS\_UDI) records.



When importing pattern exchange files, all PARTS\_UDI records are imported into the corresponding additional information box. System parameter settings for boxes will be observed, so imported text will be truncated to the box length specified in the parameters. If a box is not set the text for that box will be ignored.

The import of PARTS\_INF records is dependant on the necessary information box key letters being set. Each PART\_INF value will be imported into the box with the corresponding key letter. These values will override the PARTS\_UDI setting. The table below summarises each PARTS\_INF record and the corresponding key ‘letter’. Some new characters have been added to allow the use some fields. These characters are marked by a ‘\*’ in the following list:-

| PARTS_INF field | Key letter                                             |
|-----------------|--------------------------------------------------------|
| DESC            | M                                                      |
| LABEL_QTY       | X*                                                     |
| FIN_LENGTH      | E (and D)                                              |
| FIN_WIDTH       | F (and D)                                              |
| ORDER           | B (Order ref.)                                         |
| EDGE1           | G (Bottom edge)                                        |
| EDGE2           | H (Top edge)                                           |
| EDGE3           | I (Left edge)                                          |
| EDGE4           | J (Right edge)                                         |
| EDG_PG1         | 1* (Program / op. code for bottom (length) edge)       |
| EDG_PG2         | 2* (Program or operation code for top (length) edge)   |
| EDG_PG3         | 3* (Program or operation code left (width) edge)       |
| EDG_PG4         | 4* (Program or operation code for right (width) edge)  |
| FACE_LAM        | K (and O) (Front laminate)                             |
| BACK_LAM        | L (and O) (Back laminate)                              |
| CORE_MAT        | Y* (and O) (Code or description of core material)      |
| PALLET          | Z* (Pallet layout (stacks in length and width))        |
| DRAWING         | U (Name of drawing file for machine centre)            |
| PRODUCT         | C (Product code)                                       |
| PROD_INFO       | A (Product information)                                |
| PROD_WIDTH      | 5* (External dimension of product or cabinet)          |
| PROD_HGT        | 6* (External dimension of product or cabinet)          |
| PROD_DEPTH      | 7* (External dimension of product or cabinet)          |
| PROD_NUM        | 8* (Item number of cabinet in room)                    |
| ROOM            | 9* (Room or floor or group item number for cabinet)    |
| BARCODE1        | <* (data for 1 <sup>st</sup> barcode – as text string) |
| BARCODE2        | >* (data for second bar code – as text string)         |

The FIN\_LENGTH and FIN\_WIDTH fields are also used to fill out the combined finished length and width information box (key letter D). The CORE\_MAT, FACE\_LAM and BACK\_LAM are also used to fill out the material combination information box (key letter O).

The export of pattern exchange (transfer mode 11 and 12) also uses the new key letters above. The FIN\_LENGTH and FIN\_WIDTH fields are filled using data from the combined length/width information box if the individual finished length and width boxes are undefined. Similarly, the CORE\_MAT, FACE\_LAM and BACK\_LAM fields are filled with data from the material combination box if the individual core material and laminate boxes are undefined.

Importing Parts and Grain match templates from Pattern Exchange data

Users can now use files in the PTX file format to import grain match template patterns when importing parts/boards. Templates can be imported by adding PATTERNS and CUTS records to the PTX file.

The PATTERNS record should contain valid JOB\_INDEX along with a sequential PTN\_INDEX. The BRD\_INDEX, QTY\_RUN, CYCLES and MAX\_BOOK should be set to ‘1’. The TYPE field should be one of four values.

- 5 = Create master part - divide at saw
- 6 = Create master part - divide at machining centre
- 7 = Cut parts in main pattern
- 8 = Cut parts in separate pattern

Each pattern of type 5,6, 7 or 8 a series of CUTS records are used to represent a template pattern. The CUTS records should contain valid part indexes. Grain matching information for parts will be added to additional information fields provided system parameters have been configured for grain matching additional information.

A MATERIALS record can be used to set parameters for the templates. If this record does not exist, default ‘zero’ values will be used. (e.g. kerf = 0)

On import template patterns are stored in a part list pattern library (<part list filename>.PTL). This pattern library is used instead of the main pattern library when (a) creating a cutting list (b) adjusting optimising results for grain matching (i.e. inserting embedded patterns, appending template patterns). Each pattern in this library is named after it’s index in the pattern exchange file (i.e. the first pattern has an index of ‘1’ and so it’s name is ‘1’).

Users will not have to specify grainmatch info in a PARTS\_UDI record. Instead, the patterns will contain valid part indexes relating to the parts. These will get converted on import to valid grain match info.

Thus the PTX file is like a partial optimisation.

For example a simple two part template (A,B) can be imported from the following PTX file:

```
JOBS,1,PTXGRAIN,,,,,1
PARTS_REQ,1,1,CABDOORS-L,1,400.0,300.0,10,0,0,0,10
PARTS_REQ,1,2,CABDOORS-R,1,400.0,300.0,10,0,0,0,10
MATERIALS,1,1,MFC18-OAK,Prelaminated - Oak
18mm,18.0,0,4.8,4.8,10.0,10.0,10.0,10.0,5.0,0,0,0,0,1,1,0,0
PATTERNS,1,1,1,S,1,1,1,
CUTS, 1, 1, 1, 0, 0, 824.8, 0, 0, 0,MAIN
CUTS, 1, 1, 2, 0, 91, 5.2, 1, 0, 0
CUTS, 1, 1, 3, 0, 1, 300.0, 1, 0, 0,RIP
CUTS, 1, 1, 4, 0, 92, 5.2, 1, 0, 0
CUTS, 1, 1, 5, 0, 2, 400.0, 1, 1, 1
CUTS, 1, 1, 6, 0, 2, 400.0, 1, 2, 1
```

Note the use of the MATERIALS line to import parameters into the part list pattern library; parts 1 & 2 both have a material index of ‘1’.

There are no boards in this file so users will have to add boards after import. In practice other part sizes will also be in the list.

AF 4 June 2001

Pattern Exchange (PTX) changes

**Importing templates from PTX** - Improvements have been made to the import of template patterns from pattern exchange data (PTX). These changes fix a reported problem with recent versions where the fields in the template information dialog are greyed out after import.

Users can use files in the PTX file format to import grain match template patterns when importing parts/boards. CUTS records are used to represent template patterns. Each template must have a PATTERNS record that contains a valid JOB\_INDEX, a PTN\_INDEX relating to the relevant CUTS records and the TYPE field must be set to one of the following values:-

- 5 = Standard grain match pattern – appended to be cut on saw
- 6 = Standard grain match pattern – appended to be ‘cut’ on machine centre
- 7 = Embedded grain match pattern
- 8 = Separate template pattern

The other fields in the PATTERNS record must be present but they are ignored.

A MATERIALS record can be used to set parameters for the templates. If this record does not exist, default ‘zero’ values will be used. (e.g. kerf = 0)

The CUTS records should contain valid part indexes. Grain matching information for parts will be added to additional information fields provided system parameters have been configured for grain matching additional information.

Imported template patterns are stored in a temporary pattern library that will be used for grain matching optimisations after import. The template names will be the pattern indices. Reference: 7527

**Importing PTX patterns with trim cuts producing offcuts/parts** - Improvements have been made to the import of patterns in pattern exchange format. These changes improve the import of patterns containing trim cuts producing either parts or offcuts. Reference: 7511

**Importing part lists from a PTX** - This now follows the same rules as importing a batch of part lists. That is, if the import option is selected from the File menu at the main screen the program provides an option to view the first of the imported files. If the import option is selected from the File tree at the main screen the first part list imported is shown without any preceeding prompt. Reference: 7551

**Import parts only PTX and board list files** - If an import parts of a PTX file is performed and the PTX file does not contain any board information then no BRD files will be produced. The minimum requirement of lines in a PTX file is PARTS\_REQ and MATERIALS lines. Reference: 7572

**Import PTX for parts** - optimising & saw parameters - Importing parts list(s) from a PTX file uses the first available optimising and saw parameters for the part list rather than the Job Name specified in the JOBS record in the PTX file.' Reference: 7583

**Material parameters in PTX** - Modifications have been made to the parameter values created in the pattern exchange data generated by the transfer to saw option (transfer modes 11 and 12). These changes fix the reported problem of previous versions, which did not interpret material parameter differences correctly. Reference: 7597, 7595

Homag España CH03

This is a Panel Saw Link.

This link is for the Homag España CH03 for the Master TRIA 460 Controller. This is an unofficial file format since there is no way to communicate to the saw the cut type for each instruction. For that reason the following rules must be followed to ensure cuts are made in the correct sequence. The 1st rule is that you must optimize for Labor which will limit the output to 3rd phase cuts. The 2nd rule is that the last strip out will be the first strip back in. With a

pattern with rips, cross-cuts and 3rd phase cuts, the rips are cut first. The final rip will be put back into the saw first for cross cuts. The last cross cut will be fed back into the saw first for 3rd phase cuts. The 3rd rule is that there will always be an absolute cut when feeding a new piece of material into the saw.

## Homag España CH03+

This is a Panel Saw Link.

This link is for the Homag España CH03+ with the CNI Controller, model NPS400.

## Homag España CH12

This is a Panel Saw Link.

This link is for the Homag España CH12 with the CNI Controller, model NPS400.

## Kappa Automatic

This link will create text files (.txt) that can be run on the Kappa Automatic series of panel saws. Currently, the ability to output labels to the machine directly has been disabled. To get labeling at the machine, you will need to use a standalone license of xLabel.

## KDT XML

Saw output for the KDT Panel Saw.

## Macmazza On-Board

ASCII file output of part sizes to be used by a Macmazza Saw's On-Board Optimizer.

| Field # | Description                                     |
|---------|-------------------------------------------------|
| 1       | Indicates Piece Importing (Always a value of D) |
| 2       | Material Name                                   |
| 3       | Quantity                                        |
| 4       | Part Length                                     |
| 5       | Part Width                                      |
| 6       | Rotation (X Allowed, = Not Allowed              |
| 7       | Description                                     |
| 8       | Part Program                                    |
| 9       | Upper Edge                                      |
| 10      | Right Edge                                      |
| 11      | Lower Edge                                      |
| 12      | Left Edge                                       |

## Macmazza SEZ2000

This is a Panel Saw Link.

This link is for Macmazza SEZ2000 saws.



## Macmazza Tork MP70

This is a Panel Saw Link.

Optimization priority must be set to Labor to ensure optimization does not exceed the control's limitations.

## Martin T74

This is a Panel Saw Link.

Saw output for the Martin T74 Sliding Table Saw. This link may be valid with the entire Martin T7x series of controls.

## Mayer

This is a Panel Saw Link.

This link is for Mayer saws with the PC5400 controller.

## Mayer PC6200 controller

This is a Panel Saw Link.

This link is for Mayer saws with the PC6200 controller.

## Mayer PC7000 controller

This is a Panel Saw Link.

This link is for Mayer saws with the PC7000 controller.

## Recommended Settings

Strip sort = Descending

Parts in Strip Sort = Descending

If Set origin to bottom left = Yes

Strip sort = Ascending

Parts in Strip Sort = Descending

## NIMAC Ilenia

This is a Panel Saw Link.

Saw output for NIMAC beam saws with the Ilenia controller

## PIOS

This is a Panel Saw Link.

Saw output for the Schelling PIOS control

PNL File

ASCII file output of part sizes to be used by 3rd party optimizers.

PNL File Format

| Field # | Description             |
|---------|-------------------------|
| 1       | Part Quantity           |
| 2       | Cabinet Number          |
| 3       | Material Thickness      |
| 4       | Part Width              |
| 5       | Part Length             |
| 6       | Job Name                |
| 7       | Part Name               |
| 8       | Material Name           |
| 9       | Room Name (Optional)    |
| 10      | Cabinet Name (Optional) |

Scheer DNC - AT

This is a Panel Saw Link.

Saw output for the Scheer DNC - AT Control.

Schelling HPO

*Creates a customized part list for Schelling's High Performance Optimizer. Custom text fields can be setup in the machine parameters. For the complete list of variables, as well as how to format them, please look for 'Schelling HPO' in the S2M Center help files.*

## Parameters

### Remove Line/Position

This option allows you to remove the Positionnumber and Linenumber fields that are put at the end of the F 01 section.

### Output Board Database

This parameter allows you to output additional files for output:

- .pdb
- .pla

These files are added in addition to the .stk file

### Custom Text F[X]

This allows you to provide additional information to the .stk file in the given section. You can use either literal text, or you can pull from a limited list of variables to display information.

A variable is formatted as follows:

{variable name}

As an example, if you wanted to pull the Front Barcode Program to a section, you would do the following:

Custom Text F10: {FrProgram}

In the .stk file, this would give a result similar to:

F 10 r008001

The list of available variables is:

- RunID: Reference to [RunInfo.ID] linking the record to a run
- PartID: Reference to [Parts.ID] linking the record to a part record
- PartRunData.Output: 0x01=Selected, 0x02=PrimaryMachine, 0x04=SecondaryMachine, 0x08=DrillAndDowel, 0x10=TooBigForStdSheet, 0x20=TooBigForMachine
- GroupID: Reference to [Groups.ID] linking the part to a group
- FrProgram: Program name for the part front face operations
- BaProgram: Program name for the part back face operations
- FrMirror: Face operation are in a mirrored work field
- BaMirror: Back operation are in a mirrored work field
- FrBarCode: Front program bar code
- BaBarCode: Back program bar code
- FrWorkZone: Front program work zone
- BaWorkZone: Back program work zone
- LeBarCode: Bar code for horizontal machining on left edge
- RiBarCode: Bar code for horizontal machining on right edge

- ToBarCode: Bar code for horizontal machining on top edge
- BoBarCode: Bar code for horizontal machining on bottom edge
- IntBandBarCode: Bar code for interior banding
- ExtBandBarCode: Bar code for exterior banding
- Image: Binary graphic for this part used in reports
- ImageName: File title of part/label image
- RunMargin: Margin around the part for nesting
- RunPrimaryEdge: Primary part edge that rest against the point-to-point pins (1=left, 2=top, 3=right, 4=bottom)
- RunQty: Quantity of the part in this MachineSet Run
- CutWidth: Part width as cut on machine. Extended size: which includes pre-mill + EDX & EDY
- CutLength: Part length as cut on machine. Extended size: which includes pre-mill + EDX & EDY
- Orientation: 0x01 ArrowLe, 0x02 ArrowRi, 0x04 ArrowUp, 0x08 ArrowDn, 0x10 ArrowFlip, 0x20 Face, 0x40 2nd Program, 0x80 Spare (1st 8 Bits 0xff for Left/Face)(Next 8 bits 0xff00 for Right/Back)(Next 8 bits 0xff0000 for Top)(Next 8 bits 0xff000000 for Bottom)
- BandedWidth: Finished part width after banding. Size include banding
- BandedLength: Finished part length after banding. Size include banding and EDX & EDY
- Parts.ID: Unique Primary Reference
- Parts.JobID: Reference to [Jobs.ID] linking part to a job
- MatDBID: Material identifier
- Parts.Name: Name for the part
- InternalName: Internal part name
- Optimize: Part has been set for optimization
- CNC: Part has been set for CNC output
- Parts.Width: Width of the part + EDX value without banding
- Length: Length of the part + EDY value without banding
- WidthString: Width displayed in output increment and precision
- LengthString: Length displayed in output increment and precision
- NominalWidth: Width of part without banding or EDX value
- NominalLength: Length of part without banding or EDY value
- Parts.Comment: User comment for part
- Qty: Quantity of the part
- OverMake: Quantity that may be over made for this part
- Parts.RoomID: Reference to [Rooms.ID] linking the part to a room
- AssemblyID: Reference to [Assemblies.ID] linking the part to an

- PrimaryEdge: Primary part edge that rest against the point-to-point pins (1=left, 2=top, 3=right, 4=bottom)
- Mirror: Output part to a mirrored zone
- Shaped: Part is shaped
- Outline: Shape in an ASCII comma delimited string. Format:(node count,x1,x2 ,radius1,flags1,.....xN,yN,radiusN,flagsN) Flags:(0x08=VTX\_Radius,
- OutlineToolID: Reference to [Tools.ID] linking a tool to use for the
- GangBore: Part has gang boring
- HorizontalBore: Part has horizontal boring
- PriMachineID: Reference to [Machines.ID] linking this part to the machine used for primary output
- SecMachineID: Reference to [Machines.ID] linking this part to the machine used for secondary output
- HorMachineID: Reference to [Machines.ID] linking this part to the machine used for horizontal edge output
- CutMachineID: Reference to [Machines.ID] linking this part to the machine used for cutting part from a strip
- PrefPriMachineID: Reference to [Machines.ID] linking this part to the preferred machine to use for primary output
- PrefSecMachineID: Reference to [Machines.ID] linking this part to the preferred machine to use for secondary output
- Margin: Margin around part for nesting
- HasBanding: EdgeBanding table contains data for this part
- Banding: Four character banding code (Width1, Length1, Width2,Length2) 0=None, 1=Door edging, 2=Exterior banding,
- BandingCode: Banding code format. Edge types; (L)ength, (W)idth,(R)adius and (A)ngle. Banding types; e(X)terior and (I)nterior. Example:1L15I1R6X1A12X - 1 Length 15" long with Interior, 1 6" Radius with eXterior, 1 Angled edge 12" long with eXterior
- BandingFlags: Solid edge banding flags from a parts origin related to part edge as follows: (edges code\_1....code\_n) Example: 6NNENEN 6 sided part from face clockwise
- FinishFace: Face finish ID from material database
- FinishBack: Back finish ID form material database
- FinishEdge: Edge finish ID form material database
- Parts.Parameters: List of parameters associated with this part
- Composite: A boolean value set true if this part is the finished
- CompositeID: Reference to [Parts.ID] linking this child part to the master composite part
- Assemblies.ID: Unique Primary Reference
- Assemblies.JobID: Reference to [Jobs.ID] linking the assembly to the job
- Number: Number for the assembly
- Assemblies.Name: Name for the assembly
- Assemblies.Width: Width for the assembly
- Height: Height for the assembly
- Depth: Depth for the assembly
- Assemblies.RoomID: Reference to [Rooms.ID] linking the assembly to a room

- WallNo: Wall #
- Class: 1=Base, 2=Upper, 3=Tall, 4=Vanity, 5=Sink, 6=Split
- Type: 1=Standard, 2=Corner, 3=Corner 45, 4=Corner 90, 5=Peninsula, 6=Lazy Susan, 7=Pantry, 8=End, 9=Blind Left, 10=Blind Right, 11=Filler
- Assemblies.Description: Description for the assembly
- Assemblies.Comment: User comment for the assembly
- Assemblies.Parameters: Parameters for the assembly
- RoomName: Name for the room
- Jobs.ID: Unique Primary Reference
- Jobs.Name: Name for the job
- Jobs.Description: Description for the job
- CustName: Customer name as specified in the job
- CustAddress: Customer address as specified in the job
- CustCity: Customer city as specified in the job
- CustState: Customer state as specified in the job
- CustZip: Customer Zip as specified in the job
- CustContact: Customer contact person as specified in the job
- CustEmail: The Customer Email for the current Job.
- CustFax: The Customer Fax Number for the current Job.
- CustPhone: Customer phone number as specified in the job
- CustMobile: The Customer Mobile Phone for the current Job.
- Measure: Date of measuring the job
- Detail: Date of detailing the job
- Production: Date of production for the job
- Delivery: Date of delivery for the job
- Deposit: Date of deposit acceptance for the job
- Install: Date of installation for the job
- FinalPayment: Date of final payment for the job
- Comments: Comments set on the job
- JobFilePath: Data file the job is saved in
- Build: Program build number that produced the data
- CompanyName: Company name on security dongle
- Source: Flag indicating who produced the job (1=Solid, 2=KitchenBuilder, 3=CWareFF, 4=CWare32)
- PncFile: Data file job was imported from.

- RunInfo.ID: A unique internal number to identify the machine output
- Counter: Counter incremented each time output is created
- RunTag: A 3 character string to identify the Run: R01
- RunTime: Time stamp when output was created
- MachineID: Reference to [Machines.ID] indicating what machine was used for output
- RunInfo.Output: A boolean value set true if output needs to be created
- OutputPath: Directory selected for the output files
- ImagePath: Directory for label images
- LabelITFile: Label-IT file with data for this run
- TrueShape: Current output is a geometric nest
- PrefUnfinFace: A boolean flag set true if output occurred with the Mill From Unfinished End When Possible was true

As a note, if more information is added to the PartRunData table, more information will be populated into this list as well.

Variables are case insensitive, meaning that {FrProgram} works just like {frprogram} or {FrPrOgRaM}. You can also use multiple variables in one section and mix in literal strings as well:

{FrProgram} / {FrBarCode}

The user has 19 sections that they may enter fixed text/variables into the output (F10 – F29)

Notes for Grain Matching

When using the ‘Close grain match gaps larger than x’ preference in the S2M Center (under the Geometry tab), you may get additional cut passes for doors that sit under drawers, or in other situations. This output will use the Kerf width (Cross and Rip) to determine what the gaps between doors and drawer should be and you can adjust those values to get the output you need to remove those additional passes.

SCM MaestroCUT

This link will create a series of .xml files that can be used with the SCM MaestroCUT line of saws.

Following are the Link Parameters:

1. Campo[xx] – [string]: This allows the user to define variable text that can be used for labeling in the link
2. XMix Output Directory – [Path]: this allows the user to define a custom output directory for the .xmix files

The Campo[xx] is actually a series of parameters that start at Campo22 and end at Campo64. The user can enter variable text and literal text, or even a combination of the two. For example:

Part Size: {WidthString} x {LengthString}

This could output the following text:

Part Size: 6 x 21

Following are the valid Variables to use:

Each variable should be enclosed in a set of curly braces ({ }) otherwise the text will be considered a string literal.

| Name    | Type         | Size |
|---------|--------------|------|
| ID      | Long Integer | 4    |
| JobID   | Long Integer | 4    |
| MatDBID | Long Integer | 4    |

| Name             | Short Text   | 50  |
|------------------|--------------|-----|
| InternalName     | Short Text   | 25  |
| Optimize         | Yes/No       | 1   |
| CNC              | Yes/No       | 1   |
| Width            | Double       | 8   |
| Length           | Double       | 8   |
| WidthString      | Short Text   | 12  |
| LengthString     | Short Text   | 12  |
| NominalWidth     | Double       | 8   |
| NominalLength    | Double       | 8   |
| Comment          | Long Text    | N/A |
| Qty              | Integer      | 2   |
| OverMake         | Integer      | 2   |
| RoomID           | Long Integer | 4   |
| AssemblyID       | Long Integer | 4   |
| PrimaryEdge      | Byte         | 1   |
| Mirror           | Yes/No       | 1   |
| Shaped           | Yes/No       | 1   |
| Outline          | Long Text    | N/A |
| OutlineToolID    | Long Integer | 4   |
| GangBore         | Yes/No       | 1   |
| HorizontalBore   | Yes/No       | 1   |
| PriMachineID     | Long Integer | 4   |
| SecMachineID     | Long Integer | 4   |
| HorMachineID     | Long Integer | 4   |
| CutMachineID     | Long Integer | 4   |
| PrefPriMachineID | Long Integer | 4   |
| PrefSecMachineID | Long Integer | 4   |
| Margin           | Double       | 8   |
| HasBanding       | Yes/No       | 1   |
| Banding          | Short Text   | 4   |
| BandingCode      | Short Text   | 50  |
| BandingFlags     | Short Text   | 50  |
| FinishFace       | Short Text   | 25  |
| FinishBack       | Short Text   | 25  |
| FinishEdge       | Short Text   | 25  |
| Parameters       | Long Text    | N/A |
| Composite        | Yes/No       | 1   |
| CompositeID      | Long Integer | 4   |
| PartID           | Long Integer | 4   |
| MatName          | Short Text   | 60  |
| Thick            | Double       | 8   |
| Grain            | Yes/No       | 1   |
| JobName          | Short Text   | 50  |
| JobDesc          | Short Text   | 255 |



|            |              |    |
|------------|--------------|----|
| CustName   | Short Text   | 50 |
| Number     | Long Integer | 4  |
| MaterialID | Long Integer | 4  |
| PatNo      | Integer      | 2  |
| FrProgram  | Short Text   | 8  |
| BaProgram  | Short Text   | 8  |
| FrWorkZone | Short Text   | 10 |
| BaWorkZone | Short Text   | 10 |
| FrBarCode  | Short Text   | 20 |
| BaBarCode  | Short Text   | 20 |

SCM Ottimo

This is a Panel Saw Link.  
PTX file format for SCM Gabianni Ottimo Optimizer.

SCM PTX

This is a Panel Saw Link.  
Saw output for SCM PTX.

SCMI Seziona

This is a Panel Saw Link.  
SCMI Sigma / Galaxy Series using Seziona Control.

SCMI Sigma 65

This is a Panel Saw Link.  
SCMI Sigma 65 for the Master S124 Controller

SCMI WinCut

This is a Panel Saw Link.  
SCMI Sigma / Galaxy Series using WinCut Control

SCMI x915

This is a Panel Saw Link.  
This link is for the SCMI x915 with the Alfa A and Alfa C Controls.

## SCMI z32

This is a Panel Saw Link.

SCMI z32 for the Master 2A Controller.

## SCMI z45

This is a Panel Saw Link.

This link is for the SCMI z45 for the Master 2A and Master Plus Controls.

## Selco XML

This is a Panel Saw Link.

This link is for the Selco OptiPlanning XML format.

## CNC Output Path Explanation

There are 3 different types of output paths you can produce with Intelli-CAM. They come from a combination of the tool catalog (Centerline vs. Offset Path) and the machine catalog (Tool Center vs. Tool Radius)

1. Centerline - Tool Center NO G41/G42
2. Centerline - Tool Center G41/G42
3. Offset - Tool Radius

### CENTERLINE - TOOL CENTER NO G41/G42:

The outline path of the part is larger than the part by the tool radius. This path is to be followed by the center of the tool.

### CENTERLINE - TOOL CENTER G41/G42:

The outline path is identical to Centerline - Tool Center NO G41/G42. The difference is it triggers to look up tool compensation values on the machines control.

Tool comp is important when you start sharpening tools. When you sharpen a tool it gets smaller. Tool comp allows you to inform the control that the bit has changed size and to compensate for the change.

**For example:** If you use a ½” router bit (.25 radius) and as a result of the bit being sharpened it has a new size (.23 radius). You need to account for this tool size change somewhere.

>> With Centerline - Tool Center NO G41/G42 you would go back to the Tool Catalog in the S2M Center and enter the new size of the tool (.46 diameter) and re-output the program. Otherwise your part would cut out larger than it should be.

>> With Centerline - Tool Center G41/G42 you can enter a value of .02 at the control (.25 - .23 = .02) and the control will offset the tool path by .02 and correctly cut out the part.

It boils down to user preference whether the sharpened tool size is accounted for in the S2M Center or at the machine although you have advantages at the machine since if you need to swap bits you can compensate for different sizes on the fly rather than having to re-output from the S2M Center.

### OFFSET - TOOL RADIUS:

This path follows the outline of the part exactly. Thus if the router were to follow the path centerline, the part would be too small. This path must be compensated for at the control. If you are using a 1/2 bit, the control will need an offset of .25 (radius). If you sharpen the bit you would need to change the comp value at the control to (.23). In either case the control will offset the path of the tool by the tool radius resulting in the part being milled to the correct size.

## CncRunXXX.txt Text File Format

The CNCRun.txt file is placed in the Output directory for Primary and Secondary output and displays information about what was output. Following is the format specification for the file:

^Job : Job Name  
^Description : Job Description  
^Date : Monday, Jun 26 2000  
^Parts : 1

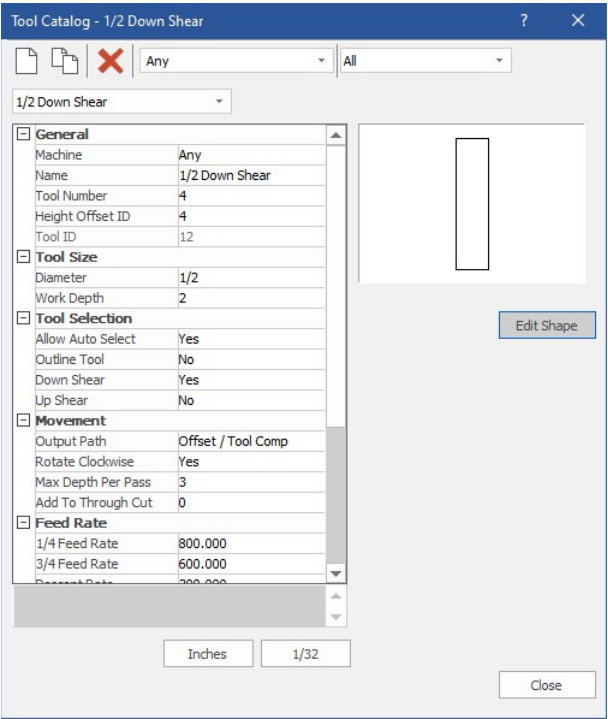
r01b0001.dxf,Bottom(BO),304.00,889.00,19.05,3/4 2s White Melamine,5,1,0,B,0,

Field 1 - Program Name  
Field 2 - Part Name (Internal Part Name)  
Field 3- Part Width  
Field 4- Part Length  
Field 5- Material Thickness  
Field 6- Material Name  
Field 7- Cabinet Number  
Field 8- Part Qty  
Field 9 - Material Grain (0 – False, 1 – True)  
Field 10- Program Face (F – Front, B – Back)  
Field 11- Material Face-Dependent (0 – False, 1 – True)  
Field 12 - Machine Tag Code  
Field 13 - Machine Name



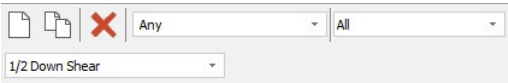
### Overview of the Tool Catalog

The [Tool Catalog](#) is used in creating and defining tools for use in your CNC operations. The Tool Catalog also allows an unlimited quantity of tools, of various types, in any combination, to be assigned to a user-defined "Tool Set".



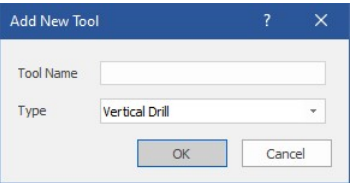
The following is a brief introduction to the Tool Catalog's features:

Toolbar



New

Allows you to create a New Tool.



There are 11 possible Tool Types you can create in the Tool Catalog:

- 1. **Vertical Drill** - Used for Boring Part Faces/Backs.

2. **Horizontal Drill** - Used for Boring into Part Edges. Will always be output as a Secondary Operation.
3. **Flat Router** - Covers the wide range of Standard Straight router bits.
4. **Ball Router** - Used for 3 dimensional routing and carving also used for routing slots with rounded bottoms or rounded inside corners (Flutes).
5. **Bull Nose Router** - Typically used for decorative bottom cutting.
6. [Tool Set](#) - A method for linking one or multiple tools together to perform a single operation. The primary purpose for a Tool Set is for the Raised Panel detail for MDF Doors. A couple of other examples may be to preset a router bit used to create a bull nose to drop at the correct depth and offset for the bull nose. Another example may be to use a flat router to cut out a section then follow with a shaped router bit to alleviate extra wear on the shaped router bit.
7. [Dovetail Tool Set](#) - Allows the definition of a Dovetail Tool as well as the Male and Female toolpath movements required.
8. **Saw** - Typically used for Grooving. A common accessory on Point-to-Point Machines. Also an Aggregate option on most CNC Routers.
9. **5-Axis Saw** - Typically used for cutting Miters on Parts.
10. **V Router Bit** - A conical bit used for machining Miters in 2 ½ axis.
11. [Shaped Router](#) - Allows definition of Shaped Router bits.

**Copy**

Allows you to Copy the current Tool.

**Delete**

Allows you to Delete the current Tool.

**Tool Display Filter**

This option allows you to control which Tools are displayed by Machine assignment. The Any option will show Tools which are available for [Any Machine](#). Selecting a Machine from the list will show Only Tools that are assigned to that Machine and/or Tools available for that Machine (Tools available for Any Machine).

**Tool Type Filter**

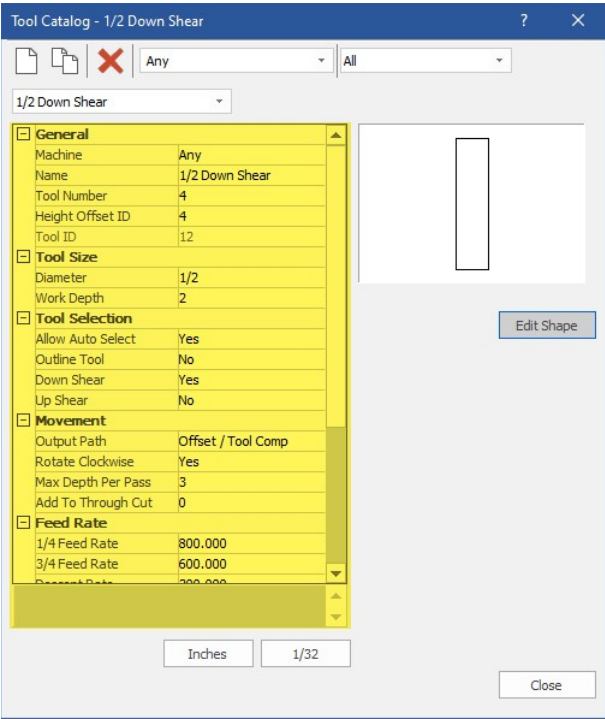
This option allows you to control which Tools are available by Tool Type.

**Tool Selector**

This option allows you to select which Tool is active for editing here.

## Tool Properties

See the [Tool Catalog Setup](#) topic for more information.



S2M CENTER for ALPHACAM Tool Catalog Option

When opening the Tool Catalog from inside ALPHACAM there is an additional Tool in the toolbar.

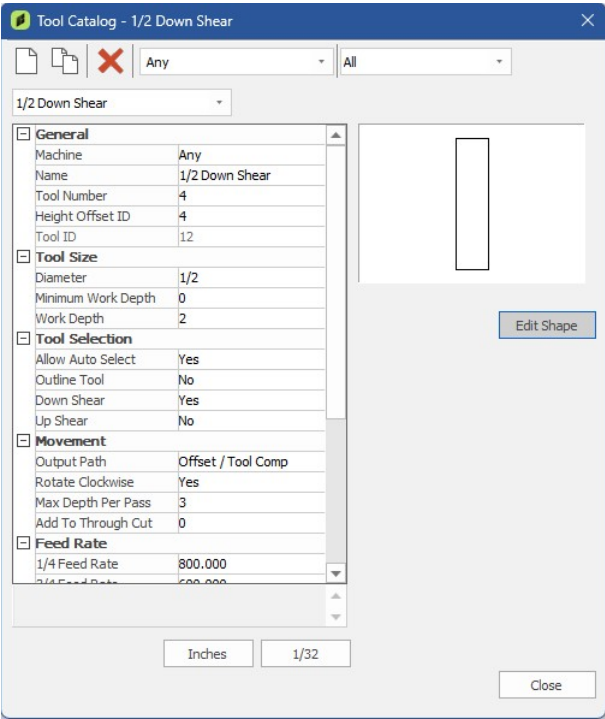
**Import Tool**  - Allows you to browse to your ALPHACAM Tools and Import selected ALPHACAM Tools into the S2M CENTER Tool Catalog.

Tool Catalog

Each Tool, when selected, will display a window to the left with the properties of that particular Tool listed. Each of these properties are referred to as a "Field". The properties in the Tool Catalog are of greater significance when outputting with Intelli-CAM. With Intelli-CAM ALL fields MUST have valid values.



The following covers the Tool Properties Tab for Standard Tools. See the [Standard Tool Shape](#) Topic for information on Shaping Standard Tools, the [Dovetail Tool Set](#) Topic for creating Dove Tail Tools, and the [Tool Sets](#) Topic for creating Tool Sets.



The following is a description of each field including a reference to the importance of that field keyed by the number of stars preceding the item. Three stars (★★★) indicate the field is of critical importance, while one star, (★) indicates the fields of lesser importance.

General Tool Properties

★★★Machine - All Tools

Machine

Any

Select the Machine to assign this Tool to. This list also includes three standard choices plus a list of each individual Machine you have setup in your [Machine Catalog](#).

Any

Unavailable

Multiple

Chop Saw

Drill and Dowel

Hauncher

Nest Router

Part Router

Point to Point

Router Set

Saw Set

The three standard choices are:

**Any** - Indicates that the active Tool is available to any Machine in your Machine Catalog.

**Unavailable** - Indicates that the Tool is unavailable (Not in Use).

**Multiple** - Indicates that the Tool is used on Multiple specific Machines in your Machine Catalog. After selecting this option you will be presented with a list of all of your Machines and you can simply select which Machines this Tool is available for.

| General                |          |
|------------------------|----------|
| Machine                | Multiple |
| Sheet - Nest Router    | No       |
| Part - Part Router     | No       |
| Part - Point to Point  | No       |
| Edge - Drill and Dowel | No       |
| Trim - Chop Saw        | No       |
| Other - Hauncher       | No       |
| Other - Router Set     | No       |
| Other - Saw Set        | No       |

◦ ■Name - All Tools

Enter a Name to identify the Tool. The name you give is for your reference only. In some posts the tool name will appear in a comment in the NC code next to the tool change.

Name

1/2 Down Shear

◦ ■■■Tool Number - All Tools

Critical for all tools except drills in a gang drill head. This is the ID that Intelli-CAM will use to match the Tool to the Machine in the Machine Catalog to the actual Tool on the Machine. For drills used in a gang drill head this value is ignored.

Tool Number

4



Some Machines (Post) do not use the Tool Number but instead use the Tool Name to uniquely identify the Tool.

◦ ■■■Height Offset ID - All Tools

This is typically the same value as the tool number or the tool number + 100. This number refers to a preset value on the controller that defines the distance from the tip of the tool to the top of the spoil board or top of the Material. This is a value set by the CNC [operator](#). It references the different heights of the tools above the work surface (*In some posts this value is ignored*).

Height Offset ID

4

◦ ■Tool ID - All Tools

This is a unique ID Number assigned by the S2M Center. This value can be called from within your design package to assign a specific Tool to a specific Operation. *For Example, within Cabinet Vision, in a User Created Standard you can use the Parameter ToolID equal to the Tools Tool ID number to make an assignment.*

Tool ID

12

Tool Size Properties

◦ ■■Diameter/Kerf - All Tools

The S2M Center will use this size to match tools to operations. In addition it will determine for dado's if they are a single pass, rectangle, or pocket. This diameter should match the diameter of the tool on the machine - however there are some instances where you may not want them to represent the actual size. One example; if you always wanted your dadoes to create a rectangle or single pass regardless of the actual size of the tool on the Machine. In this case you would enter a Tool Diameter that matched the dado width for a single pass or was smaller than the dado width for a rectangle. See the document on [Tool Selection Logic](#) for more information. With a Saw Tool the Diameter field defines the actual Diameter of the Saw.

Diameter

1/2

Kerf

9/32

◦ ■■Minimum Work Depth - All Tools

Defines minimum work depth the tool can be used to perform.



Minimum Work Depth

0

◦ **Work Depth /Length - All Tools**

Defines the maximum cutting length of the tool. Typically, this is for a visual in the Tool Shape Tab. This information is not normally used by the Post Processor as Tool Heights are typically set up at the Machine.

Work Depth

2



Minimum Work Depth and Work Depth work together to set an optimum range of material thicknesses for the tool. For example let's say you want to cut your 16mm and 19mm thick materials with your 9.5mm Compression Bit and any material which is over 20mm thick you want to cut with your 12mm Compression bit. In this case your tools would be setup with a Minimum Work Depth and Work Depth as below:

| Router Bit        | Minimum Work Depth | Work Depth |
|-------------------|--------------------|------------|
| 9.5mm Compression | 7mm                | 19.9mm     |
| 12mm Compression  | 20mm               | 34mm       |

◦ **Cone Angle - V Router Bit**

Defines the Cone Angle for the V Router Bit.

Cone Angle

90

◦ **Cone Angle - V Router Bit**

Defines the Side Wall height of the V Router Bit.



Sample Miter Values and Required Cone Angles:

- 80 or -80 degree miter requires a 20 degree cone angle
- 70 or -70 degree miter requires a 40 degree cone angle
- 60 or -60 degree miter requires a 60 degree cone angle
- 50 or -50 degree miter requires a 80 degree cone angle
- 45 or -45 degree miter requires a 90 degree cone angle
- 40 or -40 degree miter requires a 100 degree cone angle
- 30 or -30 degree miter requires a 120 degree cone angle
- 20 or -20 degree miter requires a 140 degree cone angle

◦ **Shank Length - 5-Axis Saw**

Defines the Shank Length for a 5-Axis Saw.

Shank Length

1 7/8

◦ **Corner Radius - Bull Nose Routers**

Specifies the Corner Radius for Bull Tools as shown in the figure below.

Corner Radius

1/8

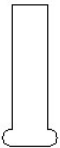


◦ **■Lollipop - Ball and Bull Nose Routers**

This field is used to display the Lollipop Tool Shape for Ball and Bull Routers. The figure below represents the Bull Tool above with Lollipop set to Yes.

Lollipop

No



◦ **■Shank Diameter - Ball and Bull Nose Routers, 5-Axis Saw, V Router Bit**

This field is used to Display the Graphic for the Tools shank properly.

Shank Diameter

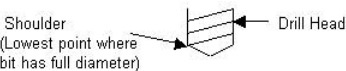
1/2

◦ **\*\*\*Drill to Shoulder - Vertical Drills**

This will determine the final depth of any drill hole. Be cautious setting this value to true since the drill tip has the potential to drill deeper than the shoulder. This is a good option to use on a drill bit with a sharp drill cone angle and it is imperative the hole diameter at depth specified is the full diameter of the bit. This is related to DRILL CONE ANGLE below.

Drill To Shoulder

False

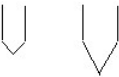


◦ **\*\*\*Drill Cone Angle - Vertical Drills**

This number is only important if you are drilling to the shoulder. You have the option in Intelli-CAM when drilling, to reference the drill bit by the tip or by the shoulder. A value of 0° or 180° here sets the drill tip and drill shoulder at the same value. The smaller the drill cone angles the greater the gap between the drills tip and shoulder. On most standard drill bits the drill cone angle is 118°.

Drill Cone Angle

0



A 90-degree point, and a 45-degree point shown respectively. (Typical is approximately 118)

Tool Selection Properties

◦ **\*\*\* Allow Auto Select**

This option will keep the S2M Center from evaluating and assigning this Tool automatically using [Automatic Tool Selection logic](#). This Tool will still be used if it is specifically assigned to an Operation via your CAD application.

Allow Auto Select

Yes

◦ **\*\*\*Allow Auto Select on Stop Dado - Saw**

Allow Auto Select on Stop Dado No

This option, when enabled, will allow a Saw to be auto-selected for Stop Dados. Typically this is unwanted as the saw will travel to its midpoint and overshoot the dado to maintain full depth along the dados length.

▫ ■■Outline Tool - Flat Routers

The Outline tool is the tool used to cut all perimeter cuts in a nest. It is necessary for one and **ONLY ONE** tool to be set as the Outline tool. See the document on [Tool Selection Logic](#) for more information.

Outline Tool No

▫ ■Down Shear/Up Shear - Flat Routers

This field assists the S2M Center in selecting the correct Tool for an operation depending on whether the Material properties requires the minimization of Face Chipping, Back Chipping, or both.

|            |     |
|------------|-----|
| Down Shear | Yes |
| Up Shear   | No  |

▫ ■Through - Vertical Drills

This value will assist the S2M Center in selecting the right Tool for the Job. If you have two identical Tools except for this value it will decide which Tool to use based on if the operation goes all the way through the Material or not.

Through No

▫ ■Diameter Tolerance - Vertical and Horizontal Drills

This value allows a Diameter mismatch tolerance in case your Tool Diameter differs from the Geometry you would like to drill with the Tool. The most common use for this would be if you define your Drill Diameter in Metric but your geometry may be defined in Imperial.

Diameter Tolerance 0

▫ ■Direction - Saws and Horizontal Drills

This option restricts a Horizontal Drill to moving in a certain direction to match the actual tool/machine restrictions. This will assist the S2M Center in selecting the correct tool for an operation. Some machines are orientated with the width along the long axis, while others are orientated with the width along the short axis. This allows Intelli-CAM to meet each requirement. Allows saws, for instance, to operate only on the "X" axis. If no tool can be found to create the operation it will produce an error message during output.

Direction Any

Tool Movement Properties

▫ ■■Output Path - Saws, Flat, Ball, Bull Nose, Shaped Routers, 5-Axis Saw

This will determine if the path of the tool is described from the center or the edge of the tool. This, typically, should be set to "Offset" for Intelli-CAM output, Intelli-CAM favors the "Offset" path. "Offset" is the preferred output for the greatest level of accuracy after tools have been sharpened. However, only some output methods support offset paths. For these output methods the tool path should be set to "Center Line". Check for your Machine's capabilities.

Output Path Offset / Tool Comp

▫ ■Rotate Clockwise - All Tools

This field in addition to the Material field "Climb Cut" will determine the Tool path direction. Think of the router bit as a wheel. If the path the wheel is following is the same direction, as a wheel would follow if rotated that direction, it is a "Climb Cut" otherwise, it is a "Conventional Cut". An inside cut will always be in the opposite direction of an outside cut. A "Conventional Cut" will always be in the opposite direction of a "Climb Cut". A simple rule is: If the Material is set for a "Climb Cut" an outside pass would be in the same direction of Tool rotation. Here is a chart to keep straight all the options: Remember the analogy to the wheel that will help keep the matter, including inside, outside, etc... straight.

Rotate Clockwise Yes

| Tool Rotation | Material Climb Cut |               | Material Conventional Cut |               |
|---------------|--------------------|---------------|---------------------------|---------------|
|               | Outside            | Inside (Dado) | Outside                   | Inside (Dado) |
|               |                    |               |                           |               |

|                   |     |     |     |     |
|-------------------|-----|-----|-----|-----|
| Clockwise         | CW  | CCW | CCW | CW  |
| Counter-Clockwise | CCW | CW  | CW  | CCW |

◦ ■■■Max Depth Per Pass - Vertical Drills, Flat, Ball, Bull Nose, and Shaped Routers

For some CNC output, this will split an operation into multiple instructions so that a tool will make multiple passes to achieve the final depth.

**This field should be set to 0 for a drill being used in a gang drill head unless your Post has been modified to support it.** Setting this value to 0 will ensure the correct routine is called in the post for the gang drill head. This is a good option to use with large drill bits and narrow router bits where the depth of cut may require excess strain on the tool. For Drill tools any value other than 0 will call a Pecking rather than a Bore Cycle in your Post.

Max Depth Per Pass



Drill operations terminology:

1. **Bore:** The tool makes the complete operation in one cycle.
2. **Pecking:** The tool creates the bore by making a series of plunges, each one deeper than the previous one, until the desired depth is reached.



If you receiving a "Pecking Not Allowed" or "No Pecking Allowed" message during output then that means your Post does not support Pecking. You can contact Cabinet Vision to see if your Post can be modified to support Pecking.

◦ ■■Add to Through Bore/Cut - Flat Routers and Vertical and Horizontal Drills

This value is very important; the machine operator can be very particular about this option. If the value is kept at the default (0) it may be a little difficult to separate the parts on your 1<sup>st</sup> nested sheet. Adding a small value allows for a cut slightly deeper than the thickness of the sheet thusly allowing cleaner part separation from the original sheet, this will leave a trace of the path in the spoil board.

Add To Through Cut

## Tool Nesting Properties

◦ ■ Nest Gap Padding - V Router Bit

There are many factors when determining the spacing required for adjacent parts when allowing miters on a nest. The thickness of the material, the angle of the tool, the through cut adjustment and the angles of the previous and next edge to the edge being mitered. All these factors are combined to determine the gap required for an adjacent part. This value allows you to add extra spacing. This value would be similar to the "Pad" in the Machine Catalog Part Spacing settings.

Nest Gap Padding

## Tool Feed Rate Properties

◦ ■■■Feed Rate - Flat, Ball, Bull Nose, Shaped Routers and Vertical Drills

This determines how fast the Machine will try to move the Tool through the Material. This is often a difficult field to find values for since we lack a standard for feed rate. The value represents Inches per minute, or MM per Minute. This can be even further confusing since some posts expect a unit but then convert the unit in the post. For example the MultiCAM post expects Inches Per Minute but converts the final output to Inches per Second. *As this process progresses expect to see a standardized unit for feed rate. Until we get there you may have to experiment with this output.* In addition the material will have an effect on the final feed rate. When testing take into account the % feed rate property in the Material Catalog.

Feed Rate

◦ ■■■1/4 Feed Rate - Saws, 5-Axis Saw

This determines how fast the Machine will try to move the Tool through the Material at 1/4" Deep.

1/4 Feed Rate

◦ **3/4 Feed Rate - Saws, 5-Axis Saw**

This determines how fast the Machine will try to move the Tool through the Material at 3/4” Deep.

|               |         |
|---------------|---------|
| 3/4 Feed Rate | 200.000 |
|---------------|---------|

◦ **Descent Rate - Flat, Ball, Bull Nose, and Shaped Routers**

This will have the same unit of measurement as feed rate.

|              |       |
|--------------|-------|
| Descent Rate | 2.000 |
|--------------|-------|

◦ **Spindle Speed - Flat, Ball, Bull Nose, Shaped Routers and Vertical Drills**

This is the speed at which the tool will spin. The unit is almost always RPM - Rotations per minute. Setting a value of 0 here will usually cause an error in the NC Code. A valid range would be about 4000 to 24000. It would depend on the Machine and Tool's recommended specifications. Some Machines will ignore this number and use a preset number for a Tool.

|               |          |
|---------------|----------|
| Spindle Speed | 4000.000 |
|---------------|----------|

4 and 5 Axis Router Setup

◦ **A or B Axis Rotation +/- - Flat, Ball, Bull Nose, Shaped Routers, Vertical Drills, 5-Axis Saw**

Amount tool can rotate on the A or B axis. A value of +/-90 allows a rotation of 180 degrees. Set to zero if your Machine does not have an A or B Axis.

|                          |    |
|--------------------------|----|
| A or B Axis Rotation +/- | 90 |
|--------------------------|----|

◦ **C Axis Rotation +/- - Flat, Ball, Bull Nose, Shaped Routers, Vertical Drills, 5-Axis Saw**

Amount tool can rotate on the C axis. A value of +/-180 allows a rotation of 360 degrees. Set to zero if your Machine does not have a C Axis.

|                     |     |
|---------------------|-----|
| C Axis Rotation +/- | 180 |
|---------------------|-----|

Tool CAD Properties

◦ **Pen - All Tools**

Defines a pen color when viewing DXF output in AutoCAD® or other 3rd Party CAD/CAM applications.

|     |         |
|-----|---------|
| Pen | 1 - Red |
|-----|---------|

◦ **Layer - All Tools**

This is the Layer Name Operations utilizing this Tool will pass to your CAD/CAM package. You can use variables to define a layer name.

|       |  |
|-------|--|
| Layer |  |
|-------|--|

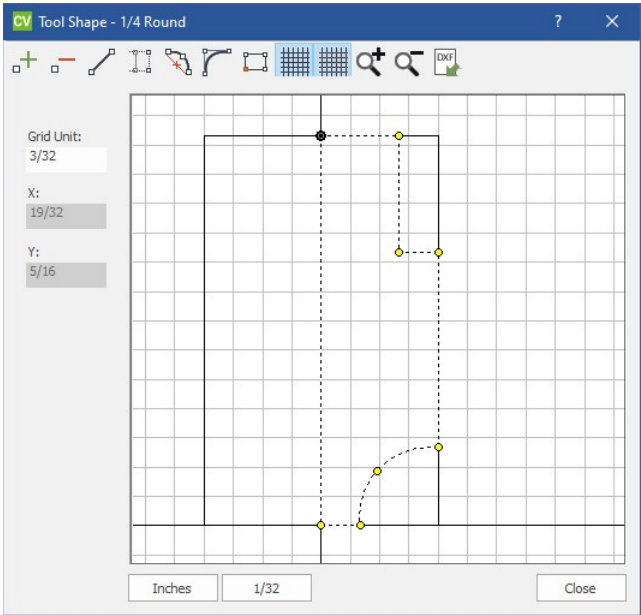


Valid variables include:

- {TS} = Tool Size
- {DZ} = Tool Depth
- {ZP} = Tool Z Position (Measured from work piece top/front face.)
- {ZB} = Tool Z Position (Measured from work piece bottom/back face.)
- {TC} = Tool Comp
- {TN} = Tool Number
- {PN} = Tool Pen Number

- {FD} = Tool Feed Rate
- {DR} = Tool Descent Rate.

Standard Tool Shape



The Tool Shape toolbar has the following options:

 **Add Points**

Use this tool to Add new Points to your Tool's Shape.

 **Delete Points**

Use this tool to remove Points.

 **Move Points**

Use this tool to Move Points.

 **Parallel Line**

Use this Tool to move parallel points at the same time.

 **Arc Line**

Use this tool to pull an Arc.

 **Radius Point**

Use this tool to enter a radius value for a selected point.

 **Reference Point**

Reset the reference point on the object. This changes the Reference Point temporarily during editing.

 **Show Grid**

This tool allows you to turn on/off the display of the Grid.

 **Snap to Grid**

This tool turns on the option to Snap to the Grid lines.

 **Zoom In**

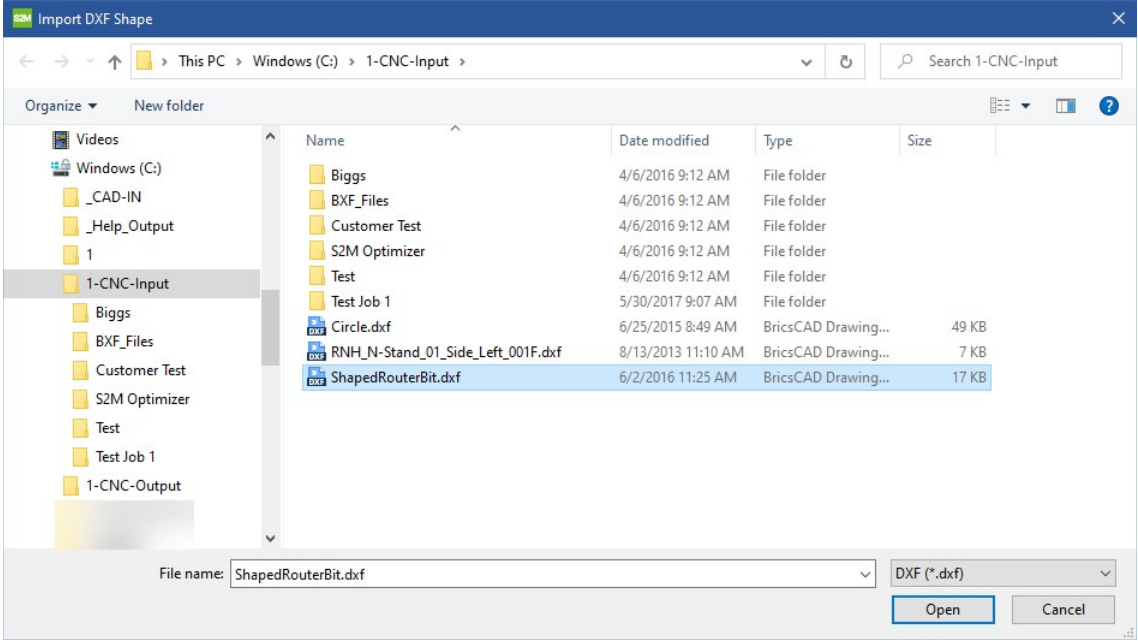
This tool allows you to Zoom In.

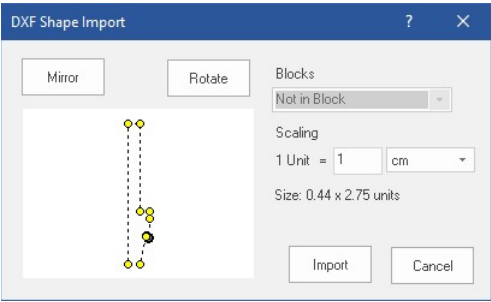
 **Zoom out**

This tool incrementally Zooms Out your image.

 **Import DXF**

Use this tool to browse to and Import a DXF of the Tool shape.





Prior to Import you can, if needed, Mirror, Rotate, and/or Scale the DXF.

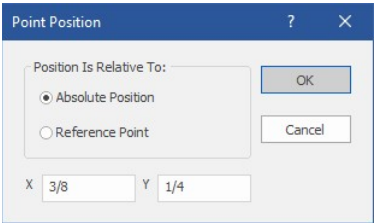
Grid Unit:  
0.09375

**Grid Unit Box**

Allows for resetting the pattern size/snap precision of the Grid.



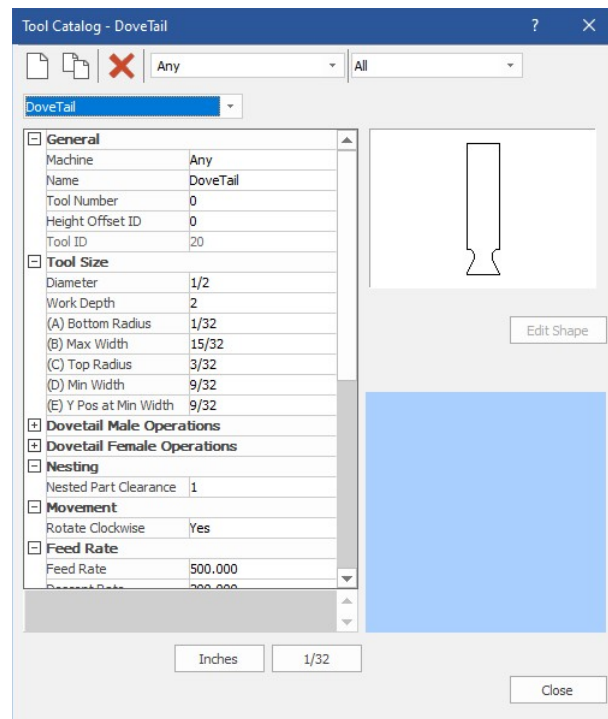
Right Clicking on any point will allow precision placement of the selected Point in either Absolute Dimensions or from the current Reference (Green Dot) Point. The Absolute Position is referenced from the Cross Point of the Main Grid.



## Dovetail Tool Set

Other than the standard Tool Properties there are several Tool Properties specific to the Dovetail Tool Sets. Each of these is preceded by a letter which is visually defined on the Tool Shape tab.





## Tool Size



Items preceded by a letter designation reference the area noted with that letter in the illustration on the lower right of the Tool Catalog window.

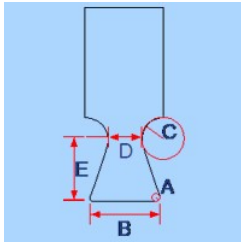
- **Diameter**

The S2M Center will use this size to match tools to operations. In addition it will determine for dado's if they are a single pass, rectangle, or pocket. This diameter should match the diameter of the tool on the machine; however there are some instances where you may not want them to represent the actual size. One example; if you always wanted your dados to create a rectangle or single pass regardless of the actual size of the tool on the Machine. In this case you would enter a Tool Diameter that matched the dado width for a single pass or was smaller than the dado width for a rectangle. See the topic on [Automatic Tool Selection Logic](#) for more information.

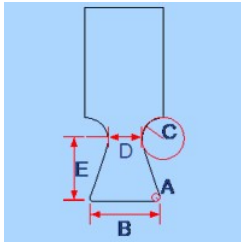
- **Work Depth**

This field is used to Display the Graphic for tools. It holds no other importance.

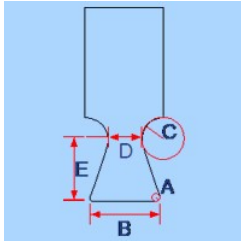
- **(A) Bottom Radius** - Sets the bottom radius of the Dovetail Tool as highlighted below. For proper fit it is important you have a tool no larger than this that can perform the Male cut.



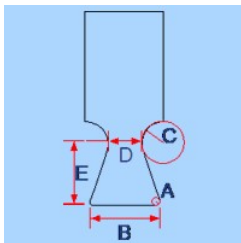
- (B) Max Width



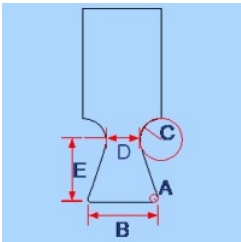
- (C) Top Radius



- (D) Min Width



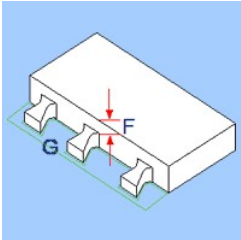
- (E) Y Pos at Min Width



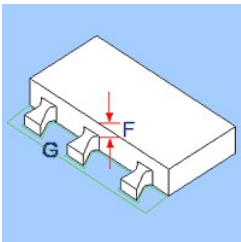
Dovetail Male Operations

|                          |                |
|--------------------------|----------------|
| Dovetail Male Operations |                |
| (F) Rabbet Depth         | 0.250          |
| (F) Rabbet Tool          | N/A            |
| (G) Male Outline Tool    | 1/8 Down Shear |

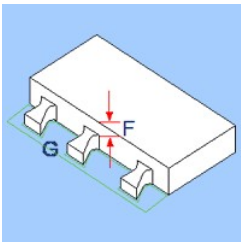
- (F) Rabbet Depth



- (F) Rabbet Tool



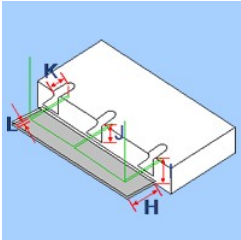
- (G) Male Outline Tool



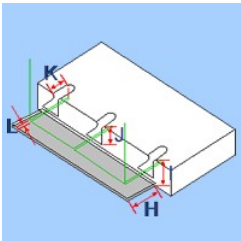
Dovetail Female Operations

| Dovetail Female Operations |      |
|----------------------------|------|
| (H) Pocket Width           | 5/8  |
| (I) Pocket Depth           | 1/2  |
| (H) Pocket Tool            | N/A  |
| (J) Plunge Depth           | 3/8  |
| (K) Cut Depth              | 1/4  |
| (L) Cut Adjustment         | 0    |
| (L) Tail Width Extension   | 1/32 |

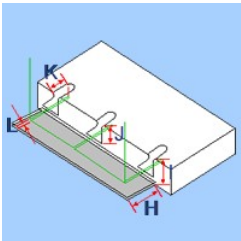
• (H) Pocket Width



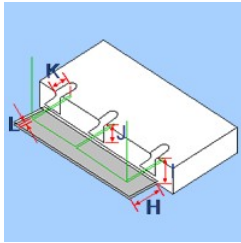
• (I) Pocket Depth



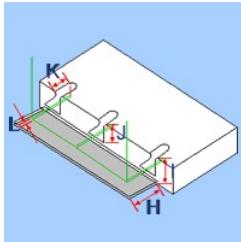
• (H) Pocket Tool



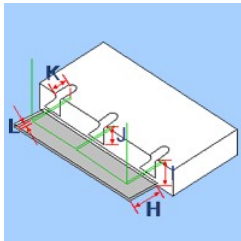
• (J) Plunge Depth



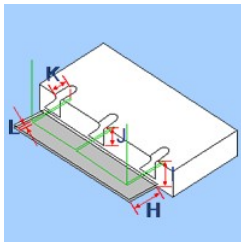
- **(K) Cut Depth**



- **(L) Cut Adjustment** - Allows you to adjust the Width of the Female Operation in order to adjust the fit. This setting will not affect the size of the Male Operation.

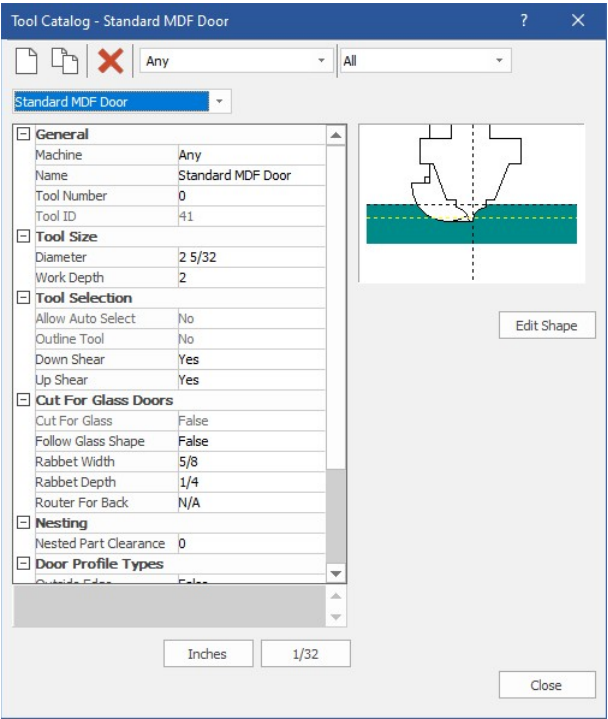


- **(L) Tail Width Extension** - Adjusts the path for both the Male and Female and allows you to widen your actual Dovetail larger than the actual Tool. This is not a "Fudge" value for making the Female Operation wider for the Male Dovetails. See (L) Cut Adjustment above for fit adjustment.



## Tool Sets

Tool Sets give the ability to assign multiple Tools to a single Route path with one selection. You have the ability to assign each Tools offset, depth, and other items in relation to a predefined Route Path.



Following are the sections in the Properties section for Tool Sets:

General

- **Machine** - Assigns the selected Tool Set to a particular Machine(s) from the Machine Catalog.  
 Machine
- **Name** - A name you enter to easily identify the Tool Set.  
 Name
- **Tool Number** - This is typically 0 as each individual Tool within the Tool Set dictates the Tool used by the machine. However, if you want to assign a Tool Set to a DXF Layer on imported DXF Parts you can use this as the Tool Number for identifying the Tool Set. It is recommended in that case that you set this to some unique Tool Number.  
 Tool Number
- **Tool ID** - This field is not editable. Its value is a unique number automatically assigned to each Tool/Tool Set. This value can be called from within your design package to assign a specific Tool to a specific Operation. *For Example, within Cabinet Vision, in a User Created Standard you can use the Parameter ToolID equal to the Tools Tool ID number to make an assignment.*  
 Tool ID

Tool Size

- **Diameter** - Specifies the Diameter of the Tool Set. Establishes logic for Automatic Tool Selection. Only valid if Allow Auto Select is enabled.

Diameter

1/2

- **Work Depth** - Specifies the Working Depth of the Tool Set.

Work Depth

2

Tool Selection

- **Allow Auto Select** - This option allows you to exclude the Tool Set from [Automatic Tool Selection Logic](#). This field will only be available if one of the Tools within the Tool Set has a depth reference of Operation Depth.

Allow Auto Select

No

- **Outline Tool** - This option allows you to establish the Tool Set as an Outline Tool. This field will only be available if one of the Tools within the Tool Set has a depth reference of Bottom of Material or Operation Depth.

Outline Tool

No

- **Down Shear** - Establishes logic for Automatic Tool Selection. Only valid if Allow Auto Select is enabled.

Down Shear

Yes

- **Up Shear** - Establishes logic for Automatic Tool Selection. Only valid if Allow Auto Select is enabled.

Up Shear

Yes

Cut For Glass Doors

- **Cut For Glass** - This field is not editable. Its value will change depending on whether you have defined a Tool in your Tool Set to "Cut / Glass".

Cut For Glass

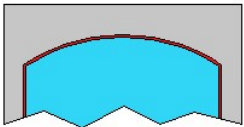
False

- **Follow Glass Shape** - This option defines whether or not you would like your Tool Set, if set to cut glass, to follow the shape of the opening or to setup to pocket out the opening for a piece of square glass. In the images below the Rabbet on the Backside for the Glass to fit into is represented in red.

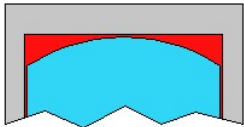
Follow Glass Shape

False

Follow Glass Shape = True



Follow Glass Shape = False



- **Rabbet Width** - Defines the Width of the Rabbet on the Backside for the Glass to fit into.

Rabbet Width

5/8

- **Rabbet Depth** - Defines the Depth of the Rabbet on the Backside for the Glass to fit into.

|              |     |
|--------------|-----|
| Rabbet Depth | 1/4 |
|--------------|-----|

- **Router for Back** - Defines the Route Tool to use to route the Rabbet on the Backside for the Glass to fit into.

|                 |     |
|-----------------|-----|
| Router For Back | N/A |
|-----------------|-----|

Nesting

- **Nested Part Clearance** - This sets the space in between Doors and Door Parts on a nested sheet when the selected Tool Set is used. This value is added to the Part Clearance set on your Nesting Machine. For example you set a 2” Nested Part Clearance (NPM) and ½” Part Spacing on the Machine, the space between two parts with a NPM of 2” will be 4 ½”. Between a part with a NPM and one without, the spacing will be 2 ½”.

|                       |   |
|-----------------------|---|
| Nested Part Clearance | 1 |
|-----------------------|---|

Door Profile Types

- **Outside Edge** - Defines whether the selected Tool Set will be available as an Outside Edge Profile.

|              |       |
|--------------|-------|
| Outside Edge | False |
|--------------|-------|

- **Inside Edge** - Defines whether the selected Tool Set will be available as an Inside Edge Profile.

|             |       |
|-------------|-------|
| Inside Edge | False |
|-------------|-------|

- **Raised Panel Detail** - Defines whether the selected Tool Set will be available as an Raised Panel Detail.

|                     |       |
|---------------------|-------|
| Raised Panel Detail | False |
|---------------------|-------|

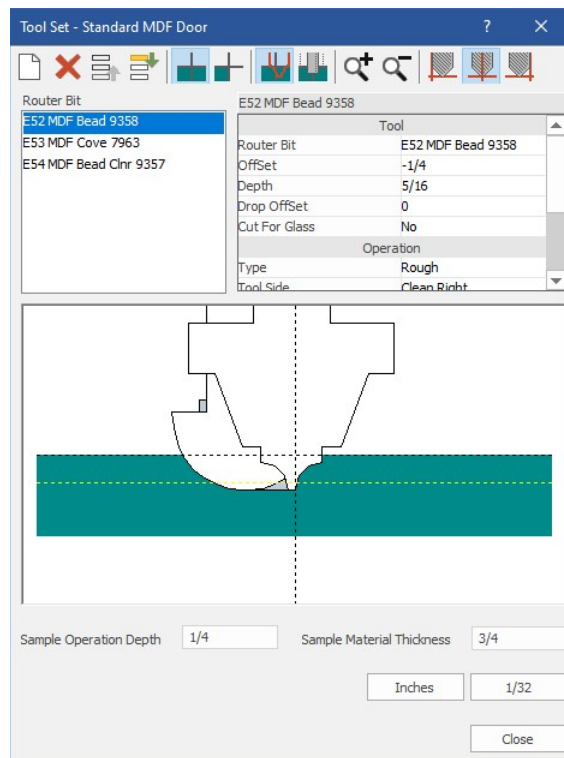
- **Route** - Defines whether the selected Tool Set will be available as an Route Pattern.

|       |      |
|-------|------|
| Route | True |
|-------|------|

Edit Shape

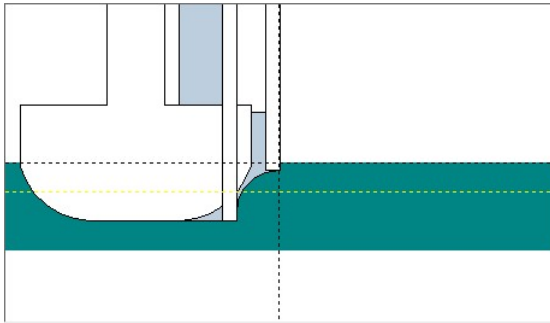
Edit Shape



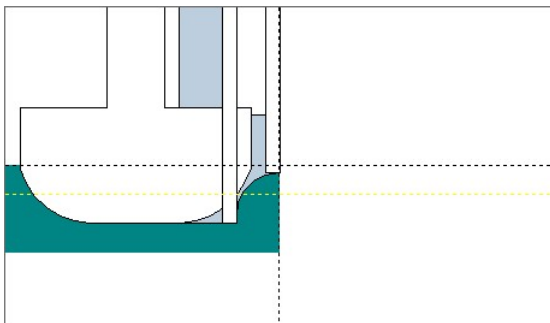


When defining a Tool Set's Shape you are actually defining the Tools and Offsets used to define the Tool Set. Since this information will transfer to your CNC Machine, the accuracy of the information you enter here is paramount.

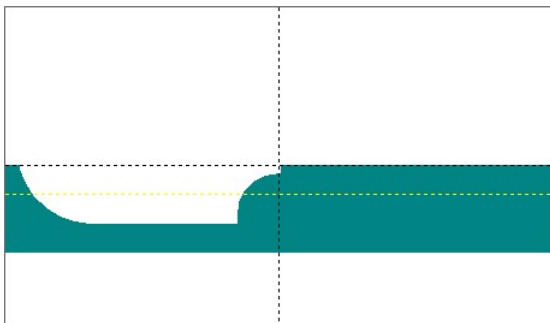
- **Add** - Adds a new Tool to the Tool Set.
- **Delete** - Removes the selected Tool from the Tool Set.
- **Move Up** - Moves the selected Tool Up one level for each click.
- **Move Down** - Moves the selected Tool Down one level for each click.
- **On Center** - This option displays the Tool Set as it will follow a path defined in the Door, such as a Route Pattern or Template Door Route.



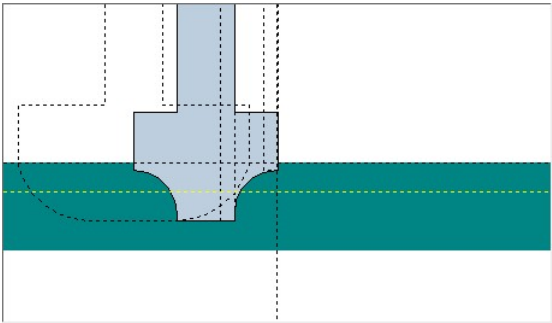
-  **On Edge** - This option displays the Tool Set as it will follow a path defined on the Edge of a Door.



-  **Show Tool** - This option toggles the Tools display on/off in the Graphic sample. This can make it a little easier to see your final result.



-  **Show Only Selected Tool** - Will allow you to only see the currently active tool.



- **Zoom In** - Allows you to Zoom In on the Sample Shape.
- **Zoom Out** - Allows you to Zoom Out on the Sample Shape.
- **Reference From Left** - Establishes that the Offset Dimension for all Tools in this Tool Set are referenced from the Left Side of the Tool Path.
- **Reference From Center** - Establishes that the Offset Dimension for all Tools in this Tool Set are referenced from the Center of the Tool Path.
- **Reference From Right** - Establishes that the Offset Dimension for all Tools in this Tool Set are referenced from the Right Side of the Tool Path.

Following are the options for Tools added to your Tool Set:

**Tool Properties**

|               | Tool           |
|---------------|----------------|
| Router Bit    | 1/8 Down Shear |
| Offset        | -7/16          |
| Depth         | 1/2            |
| Drop Offset   | 0              |
| Cut For Glass | No             |

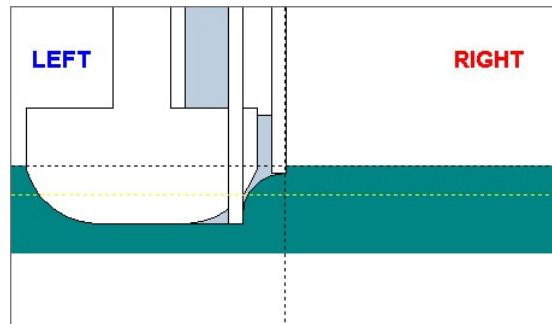
- Router Bit - This drop down allows you to select which route Tool, defined in the Tool Catalog, to use in the Tool Set.
- Offset - The path a Tool Set follows is defined by a single line or path. This setting determines the offset Left or Right the Tool should be from the line or path.
- Depth - This value determines the depth the Tool should cut within the Tool Set.
- Drop Offset - This value is for use with under-cut type bits. This value instructs Solid how far away from the edge the center of the Tool needs to be prior to entering the Part to be cut.
- Cut For Glass - Select this option if you want the selected Tool to define the Glasses position if you are defining a Glass Door Tool Set. This will generate a backside Rabbet for your glass. See the Rabbet definition above.

**Operation Properties**

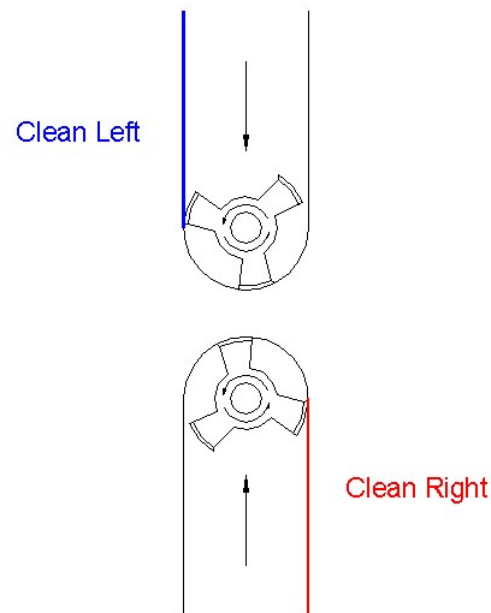
The Operation Properties presented here vary depending upon the selected Operation Type. Following are the available Operation Types and their associated Properties:

- **Type - Rough** - A standard, and the most common, Operation Type.
  - Tool Side - This option sets the side of the Path the Bit is following that has bit pressure focus. See diagrams below.

Looking at the Tool Set below the Raised Panel Cutter is set to Clean Left, this is because the Left Side of this operation requires a Finish pass and we would want the Left Side to be the cleanest possible cut. The other three Tools are set to Clean Right, this is because the Right Side of these operations requires Finish passes and we would want the Right Side to be the cleanest possible cut. See the second diagram below for more information.



In the diagram below if you will notice the direction of cut in combination with tool rotation places the bits cutting focus on the specified side of the cut or the entry of the cut. Whereas the opposite edge is the exit of the cut and therefore would be more likely to have tear out than the entry.



- **Type - Engrave** - Engrave is a special Operation Type designed for Engraving.



Make sure to use a V Router Bit for the Engrave Operation Type the path of the Engrave is calculated from the shape of the Tool. A Flat Router would not create a proper path.

- - Tool Side - [Previously Covered](#).
  - Start Point - Standard Operations start in the Middle, however, when engraving typically you would start in the corner.
- **Type - Pocket** - Pocket is a special Operation Type designed for making Pocket Cuts.

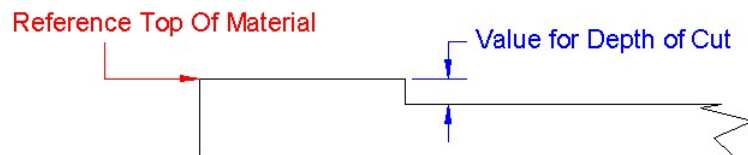
- **Type - Clean Up** - Clean Up is a special Operation Type designed to Clean Up Corners after a Large Diameter Tool Pass. Simply specify the Tool Radius to Clean Up after. Notice in the animation below that the Clean Up Cutter comes in after the initial pass and only cleans up in the Corners rather than making a complete set of passes to clean up.
- **Type - Vertical & Horizontal Slats** - Vertical and Horizontal Slats are special Operation Types designed to Cut Slats allowing the creation of Shaker Style MDF doors. Specify the Distance from the Route Path to the Start/End of Slat; this keeps the Engraver used for the Slats from overlapping your perimeter. There are two options for the Slat Spacing "Optimal" which will vary the Slat Spacing but not exceed the Offset value and "Fixed" will use a Fixed value for the Slat Spacing based on the Offset value. Simply specify the Distance between the Slats in the Offset Column.0
- **Type - Finish** - The Finish Operation Type, similar to the Clean Up Operation Type, is designed to allow you to assign a Finish Tool to clean up inside radiuses on the outside of a tool path. There are a few very IMPORTANT things to note when utilizing this Operation Type.
  - The Outline Tool option is disabled unless the depth reference is Bottom of Material or Operation Depth
  - A toolset will not be available for auto-select unless at least one tool is selected for Operation Depth
  - Set the Nest Clearance to 0 unless you need extra clearance
  - **You have to manually set the tool size.**
  - The outline path should have a clean left edge.

The animation below demonstrates the path of a Finish Operation Type.
- **Type - Profile** - The Profile Operation Type, similar to the Clean Up Operation Type, is designed to allow you to assign a Profile Tool to clean up inside and outside radiuses on a profiled toolpath.
  - **Maximum Depth Per Cut** - This is the depth of cut used if the depth calculated by the Maximum Error exceeds this value. In most cases the default value should be sufficient.
  - **Maximum Error** - This value is used to calculate the depths of cut and determines the surface finish. The depth of cut is calculated by moving the tool out to the Maximum Error then down till the contact point meets the slope. In most cases the default value should be sufficient.
  - **Clean Both Sides** - This determines if you want both Outside and Inside Corners cleaned. When disabled it only cleans Outside Corners.

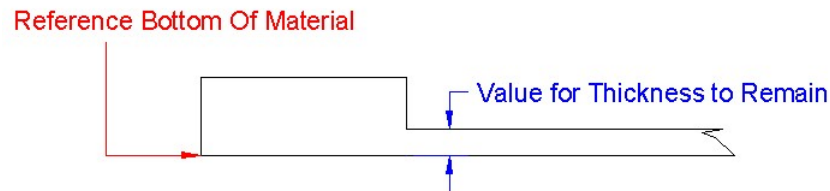
## Reference Properties

- **Depth** - The Fixed Depth selection allows you to specify the Depth of the Cut from the Top of the Material being cut by a positive amount. Material + Depth allows you to specify the Depth of cut from the Bottom of the Material by a Negative number. See Diagrams below.

**Fixed Depth** is a Positive Depth Value Measured from the Top of the Material this allows the Depth to stay consistent with a Material thickness change.



**Material + Depth** is a Negative Thickness Value Measured from the Bottom of the Material; this allows the remaining thickness from the cut to stay consistent with a Material thickness change.



**Operation + Depth** gets its depth from the defined Operation. Use the Sample Operation Depth at the lower left to see how your Tool Set will be applied utilizing this option.

## Enhanced Tool Selection Logic for CNC Output

The following represents the automatic tool selection logic, which, the S2M CENTER employs when parts are output to CNC and is organized by operation type. This, in part, is made possible by the many properties that can be assigned to a particular tool in the Tool Catalog. If you are unfamiliar with the properties that can be assigned to particular tools in the Tool Catalog, it may be best to review the information presented in the [Tool Catalog Setup](#) topic in this help file before reviewing the information below.

### **Part Outline Operation** - See Chart #6 at the end of this topic

Search tools for flat routers with Outline = "True"

1. If found then select tool in this order
  1. Up / Down Shear bit that matches sheet requirements
  2. Compression bit
  3. Down Shear bit
  4. Up Shear bit
  5. "Expansion" bit (Up Shear & Down Shear = "False")
2. No match – **ERROR**

### **Vertical Hole Operation** - See Chart #1 at the end of this topic

Search tools for vertical drill bit with exact size match

1. If found then select tool in this order
  - Through bore setting matches hole requirements 1st drill with size match
  - Check if Drill Bit is in Gang Drill Head.
    - If found - Check if hold position is in gang drill head limits
  - If no match – **WARNING** – Search flat routers for size match
    - If found then select tool in this order to create hole in one drop
      1. Down Shear bit
      2. Compression bit
      3. Up Shear bit
      4. "Expansion" bit (Up Shear & Down Shear = "False")
  - If no exact size match then search for flat router with diameter smaller than hole size
    - If found then select tool in this order to create pocket to match hole size
      1. Down Shear bit
      2. Compression bit
      3. Up Shear bit
      4. "Expansion" bit (Up Shear & Down Shear = "False")

2. No match – **ERROR**

### **Horizontal Hole Operation** - See Chart #2 at the end of this topic

Search horizontal drill bits for a size and direction match

1. If found then select that horizontal drill
2. No match - **ERROR**

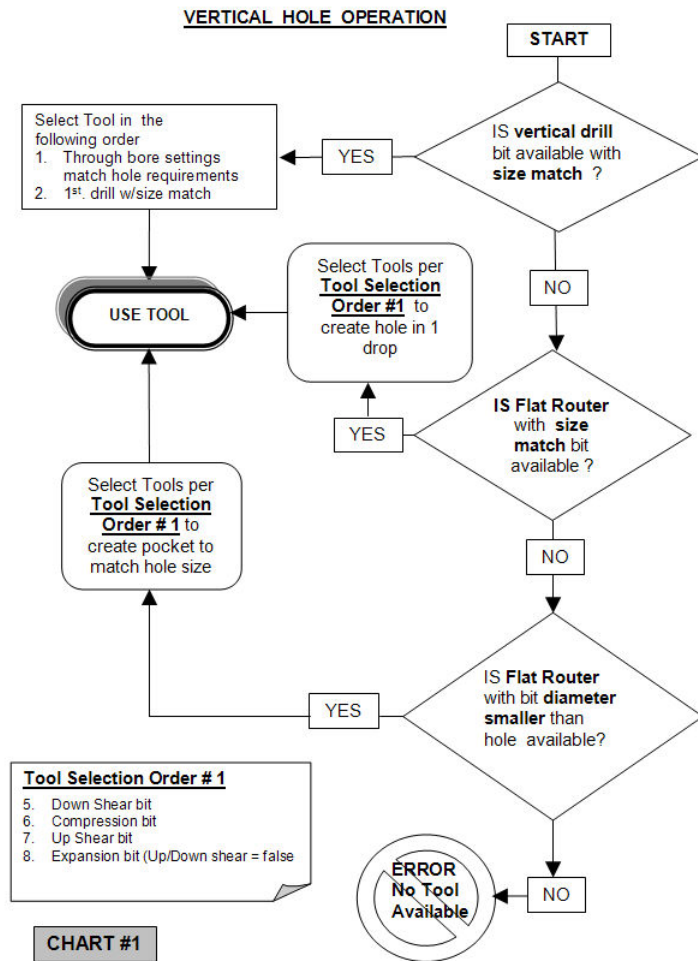
### **Dado Operation** - See Chart #3 and Chart #4 at the end of this topic

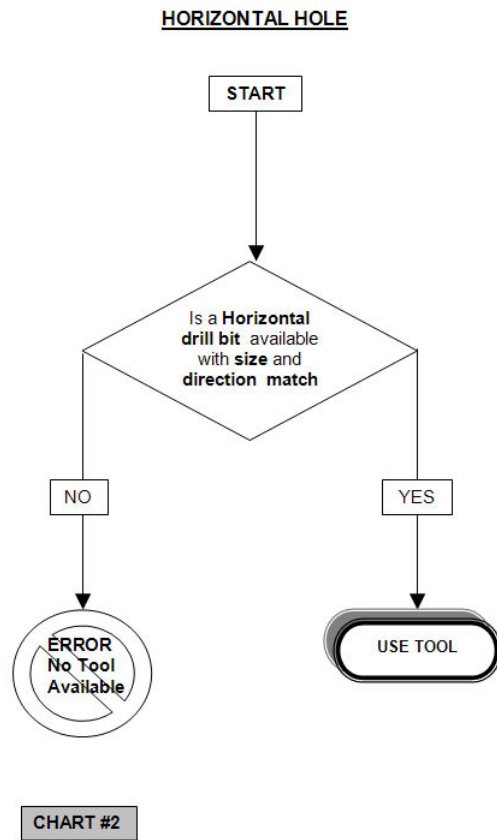
1. If dado extends up to or past the edges of the part – Use saw
2. Search saws for size match
  - If direction is OK then use tool
  - If direction is opposite then swap dado direction and use tool
  - If direction is not usable then search for saw with smaller kerf
3. If no size match then use saw with smaller kerf
  - If direction is OK then use tool and create multiple passes
  - If direction is opposite then swap dado direction and use tool with multiple passes
  - If direction is not usable – **WARNING** – Search for router
4. If no saw match Use router
  - Search flat routers for size match and Up Shear / Down Shear match (Will Select Down Shear for a Dado)
5. If found then create single path using router
  - If no size match then search for flat router with smaller diameter than dado width
  - If found and tool diameter is greater than or equal to  $\frac{1}{2}$  dado width then create 4-pass route using tool selected in this order
    1. Up / Down shear bit that matches sheet requirements
    2. Down Shear bit
    3. Compression bit
    4. "Expansion" bit (Up Shear & Down Shear = "False")
    5. Up Shear bit
  - If found and tool diameter <  $\frac{1}{2}$  dado width then create multi-pass route (starting from the outside and then moving inward concentrically) using tool in this order
    1. Up / Down shear bit that matches sheet requirements
    2. Down Shear bit
    3. Compression bit
    4. "Expansion" bit (Up Shear & Down Shear = "False")
    5. Up Shear bit
6. No match – **ERROR**

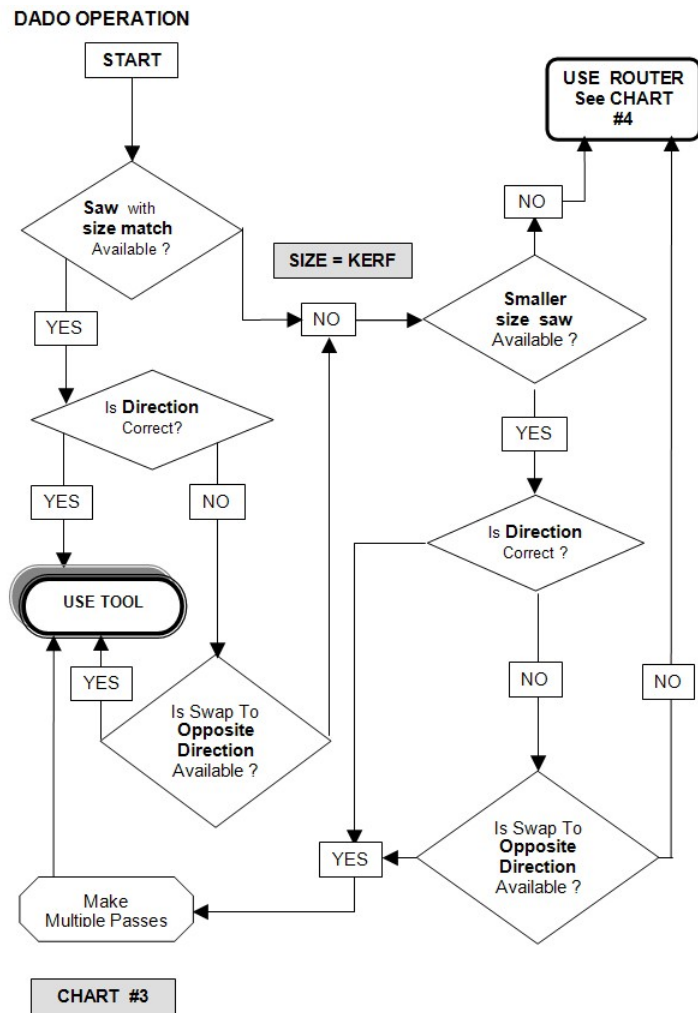
## **Cutout / Pocket Route Operation** - See Chart #5 at the end of this topic

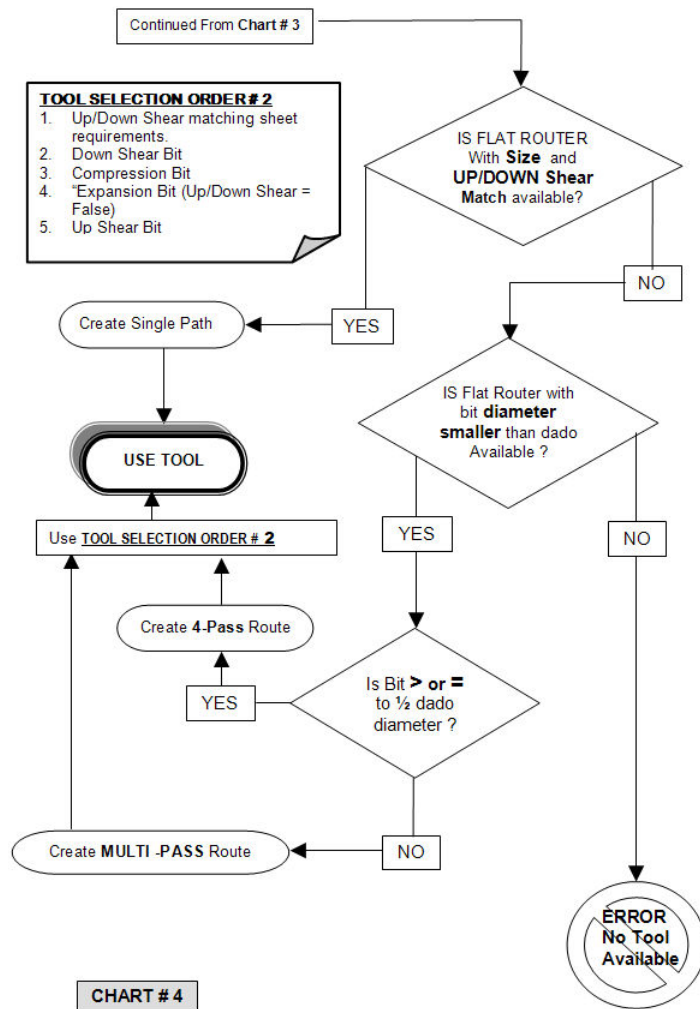
1. Search flat routers for size match and Up Shear / Down Shear match
  - If found then create single path using tool
  - If no size match then search for smaller diameter flat router with Up Shear / Down Shear match
    1. If found and tool diameter is greater than or equal to  $\frac{1}{2}$  dado width then create 4-pass route
    2. If found and tool diameter  $< \frac{1}{2}$  dado width then create multi-pass route
      - If no match found then search for any flat router with smaller diameter then the route operation
      - If found and tool diameter is greater than or equal to  $\frac{1}{2}$  dado width then select tool in this order to create 4-pass route
        - Cutout route operation
          1. Compression bit
          2. Down Shear bit
          3. Up Shear bit
          4. "Expansion" bit (Up Shear & Down Shear = "False")
        - Pocket route operation
          1. Down Shear bit
          2. Compression bit
          3. "Expansion" bit (Up Shear & Down Shear = "False")
          4. Up Shear bit
      - If found and tool diameter  $< \frac{1}{2}$  dado width then select tool in this order to create multi-pass route
        - Cutout route operation
          1. Compression bit
          2. Down Shear bit
          3. Up Shear bit
          4. "Expansion" bit (Up Shear & Down Shear = "False")
        - Pocket route operation
          1. Down Shear bit
          2. Compression bit
          3. "Expansion" bit (Up Shear & Down Shear = "False")
          4. Up Shear bit
2. No match – **ERROR**

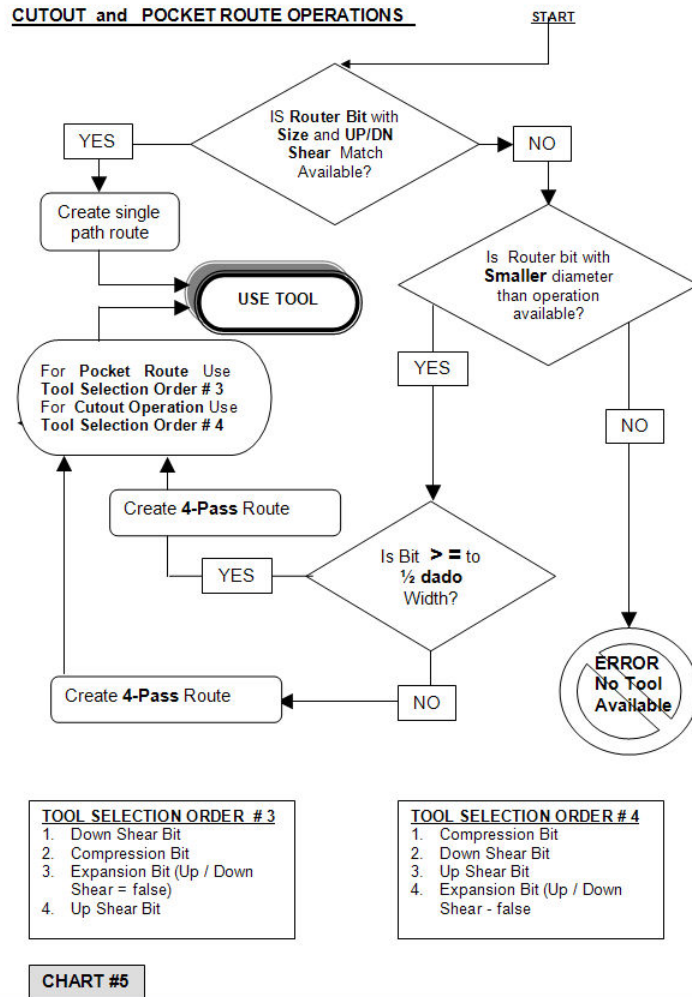


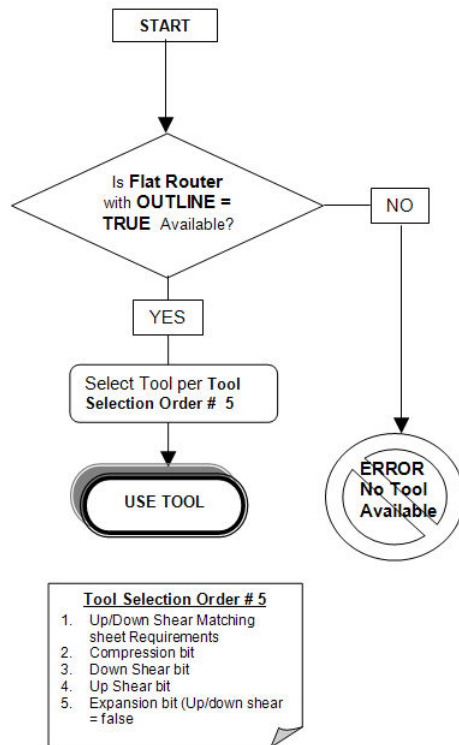










**PART OUTLINE OPERATIONS****CHART #6****☒ Import DXF**

Import DXF Parts allows you to bring in Parts you have created in other CAD packages into the S2M CENTER.



To Import a DXF Part:

1. Click the Select  button.
2. Browse to the location of your DXF Part, click on it, and click Open. See the [Other CAD Importers](#) topic for other import options.

- ☒ You can use standard Windows Ctrl + Click or Shift + Click combinations to select multiple DXF's here.



3. Click the Open button.

Once your DXF Parts are displayed on the screen they ready to be added to your Job by clicking Save. However, there are several options prior to Adding the Part to the Job you may want to modify. First, your Parts will appear on the Viewing Screen to the Left.



By selecting each Layer Name in the list at the upper Right of the screen you can see that Operation highlight in the preview screen you will also be able to see below how the S2M CENTER has translated the Layer Names.

- ☒ Each identified Part or set of Operations will be displayed, when selected, either in Blue or in Red. If the Part or Operation is displayed in Red it means the S2M CENTER has been unable to determine the Operation Type. This typically means that the Layer has not been defined correctly. You can either edit the Layer Scheme for the S2M CENTER or edit the Layer Name within the DXF from this area.

## Edit DXF Layer Scheme



This is where you define how your Layers coming from your CAD package are defined.



### Toolbar



There are 5 options in the toolbar:

- ☒ **Layer Scheme Selector** - Allows you to select which DXF Layer Scheme to use.

- ☒ **New Scheme** - Allows you to create a new DXF Layer Scheme.

- ☒ **Copy Scheme** - Allows you to copy the current DXF Layer Scheme.

- ☒ **Rename Scheme** - Allows you to rename the current DXF layer Scheme.

- ☒ **Delete Scheme** - Allows you to delete the current DXF Layer Scheme.

There are three sections in the Advanced Screen:

**Layers** - There are 12 options in the Layers Section.

1. Part - Specifies which Layer defines the Parts boundary.
2. Outline - Specifies which Layer defines the Parts Outline.
3. Vertical Bore - Specifies which Layer defines Vertical Boring Operations.
4. Horizontal Bore - Specifies which Layer defines Horizontal Boring Operations.

5. Left Horizontal Bore - Specifies which Layer defines Left Horizontal Boring Operations.
6. Right Horizontal Bore - Specifies which Layer defines Right Horizontal Boring Operations.
7. Front Horizontal Bore - Specifies which Layer defines Front Horizontal Boring Operations.
8. Back Horizontal Bore - Specifies which Layer defines Back Horizontal Boring Operations.
9. Route - Specifies which Layer defines Route Operations.
10. Dado - Specifies which Layer defines Dado Operations.
11. Pocket - Specifies which Layer defines Pocket Operations.
12. Edge Miter - Specifies which Layer defines an Edge Miter.

**Keywords** - There are 8 options in the Keywords Section.

1. Tool Number - Specify the character that represents your Tool Number. Typically it is the letter T.
2. Operation Depth - Specify the character that represents your Operation Depth. Typically it is the letter D.
3. Operation Width - Specify the character that represents your Operation Width. Typically it is the letter W.
4. Offset Left - Specify the character that represents if your Operation needs to machine with a Left Cutter Offset (Comp). Typically it is the letter L.
5. Offset Right - Specify the character that represents if your Operation needs to machine with a Right Cutter Offset (Comp). Typically it is the letter R.
6. Tool Inside - Specify the character that represents if your Operation needs to be machined on the Inside.
7. Tool Outside - Specify the character that represents if your Operation needs to be machined on the Outside.
8. Secondary Face - Identifies that the Operation is on the Secondary Face. Without this designation on an operation the assumption is the Operation is on the Primary Face. This allows you to define machining on both faces of the Part from one DXF drawing.

**Miscellaneous** - There are 3 options in the Miscellaneous Section.

1. Decimal Separator - Specify what character you use as a decimal separator. Typically it is either the letter p or an underscore \_.
2. Part Face Up - Specify what character you use to indicate the Part is Face Up. By default it is assumed to be Back Up.
3. Use Thickness as Depth - Set this to True if your CAD geometry includes a true thickness representation.

☒ If there is a duplicate Layer name or Keyword a notification occurs next to the OK button and you will be prompted to fix the problem before exiting, copying or selecting another scheme.

☒ The Part Layer should (but is not required) contain a rectangle representing the extent of the part. The Outline Layer should hold the part shape.

## Layer Breakdown



The LeftEnd.dxf Imported above includes five Layers. Let's break them down:



☒ Layer Names are case sensitive.

Dadoes can be defined three different ways, as shown below, to be interpreted correctly by the S2M CENTER.



**Single Line with No Width** – The S2M CENTER will define the Dado to the Right of your drawn Line and the Width must be defined in the Layer Name.



**Single Line with Width** – The S2M CENTER will define the Dado from the Center of the Line out based on the Width of the line.



**Rectangle Polyline** – The S2M CENTER will define the on the inside of the rectangle based on the Offset defined in your layer name.



## Edit DXF Layer Names



This utility allows you to edit the Layer name passed by the DXF. Simply type the new layer name(s) you wish to use in the corresponding New Layer Name field.



## Known Limitations

Although the DXF Import reads a DXF file there are some differences between an output DXF and Input DXF that should be noted.

1. One DXF file per part. It is common to have a Partition output both a front and back program. However since each DXF file is assigned to a part the DXF file should contain both the face and the back operations rather than having them split between two DXF files.
2. No mirrored parts. The DXF files input should not be mirrored. Currently the application will only rotate part 90 degrees or flip the part over if necessary.
3. Operation size is assumed to be the bounding box of the operation.

For example, if you output the tool path for a dado as centerline, the geometry for the dado is smaller than the actual dado required since the operation will grow by the tool diameter.



4. A DXF file can define quite an assortment of CAD entities that the S2M CENTER can not process. If the S2M CENTER can not process an entity it will be ignored and a warning will be issued. The following entities are supported by the S2M CENTER:
  - Arc
  - Circle
  - Line
  - LWPOLYLINE
  - POLYLINE
  - SEQEND
  - VERTEX

## Other CAD Importers

The DXF Import utility has some predefined DXF Layer Schemes you can utilize depending upon what CAD application you are using to generate your DXF's. There are six pre-defined Layer Schemes provided which are setup to match your CAD applications default output however each can be modified to suit your needs.

### Default

A generic sample you can modify to suit your custom needs. This one and any new ones you create will be this style and require that you set the Open File Type to DXF files (\*.dxf) when importing your DXF's.



[Blum Dynaplan \(BXP\)](#)

[KAB-NX](#)

[KCDw](#)

[Pattern Systems](#)

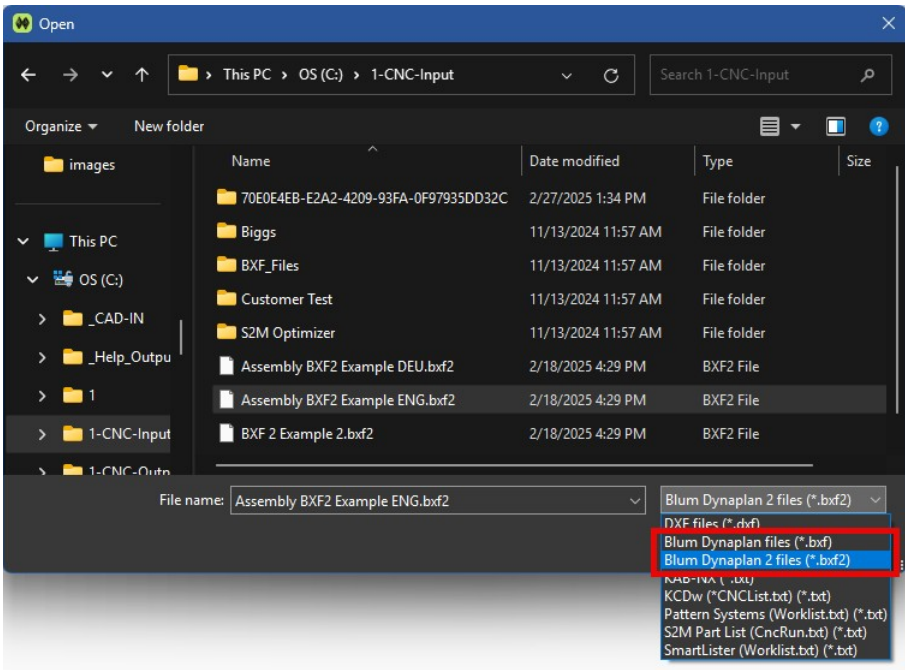
[S2M Part List](#)

[SmartLister](#)

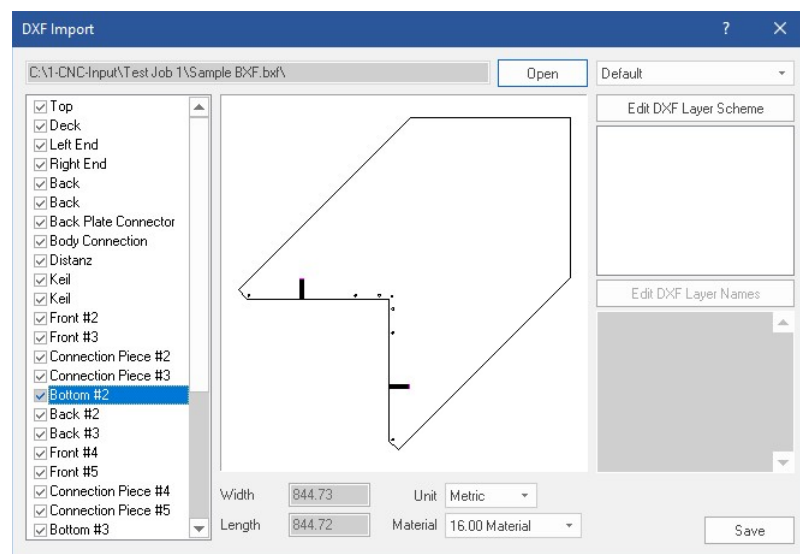
# Blum Dynaplan

If you are using Blum Dynaplan or the Blum Product Configurator to create the BXF files you want to process in the S2M CENTER then you are not required to have a DXF Scheme selected as we can read all machining information directly from the file.

Simply select the Blum Dynaplan files (\*.bxf) or Blum Dynaplan 2 files (\*.bxf2) for your file type then select and open your BXF file.



All Parts within the BXF file will be listed in the DXF Import area.



The material list will be filled with materials that match the part size. If there is no match a material will be created to match the part size.

You can then choose which parts you would like to import or import all the parts by selecting the "Save" button.

## BXFExt.ini



The BXFExt.ini file is only required for the files with BXF extension, BXF2 does not require this ini file.

The BXFExt.ini is located in the installation extraction folder at \Support\BlumDynaplan\ and should be copied to the \S2M 2025\ folder. The BXFExt.ini file serves two purposes, localization and part orientation.

### Localization

The data in the BXF file is in German, along with the part names. The .ini file can be used to change the part to recognizable part names. Note: The same part might have different names or may be in a numbered sequence. The following is a sample:

```
[LocalPartNames]
LinkeSeitenwand - Seitenwand=Left End
LinkeSeitenwand=Left End
Rueckwand=Back
*Rueckwand_2=Back #2
*Rueckwand_3=Back #3
```

### Orientation

The parts in a BXF file do not have a typical orientation like the S2M CENTER is used to, they are simply shapes in space. The orientation is determined by the size and position of the part. The part's face is based on the part's location in the cabinet. By default all part faces point out. However there may be some parts, like a back, where the part should be oriented face in. To override the normal orientation of a part use the [PartFaceTowardCenter] section of the BXFExt.ini.



Parts listed in this section can be the actual part name or the localized name. Be sure to account for parts in a numbered series. Any non-zero value (such as 1) will change the part face towards the center of the cabinet.

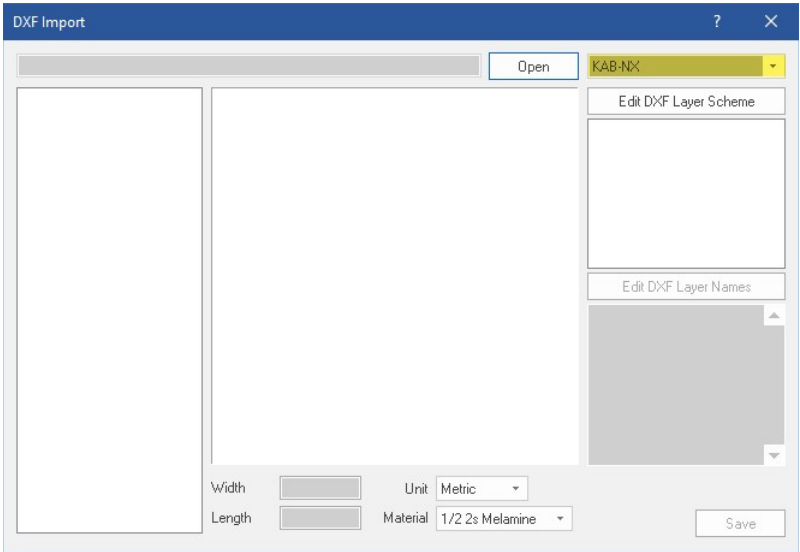
The following is a sample:

```
[PartFaceTowardCenter]
Rueckwand=1
Back=1
Back #2=1
Back #3=1
Top=0
```

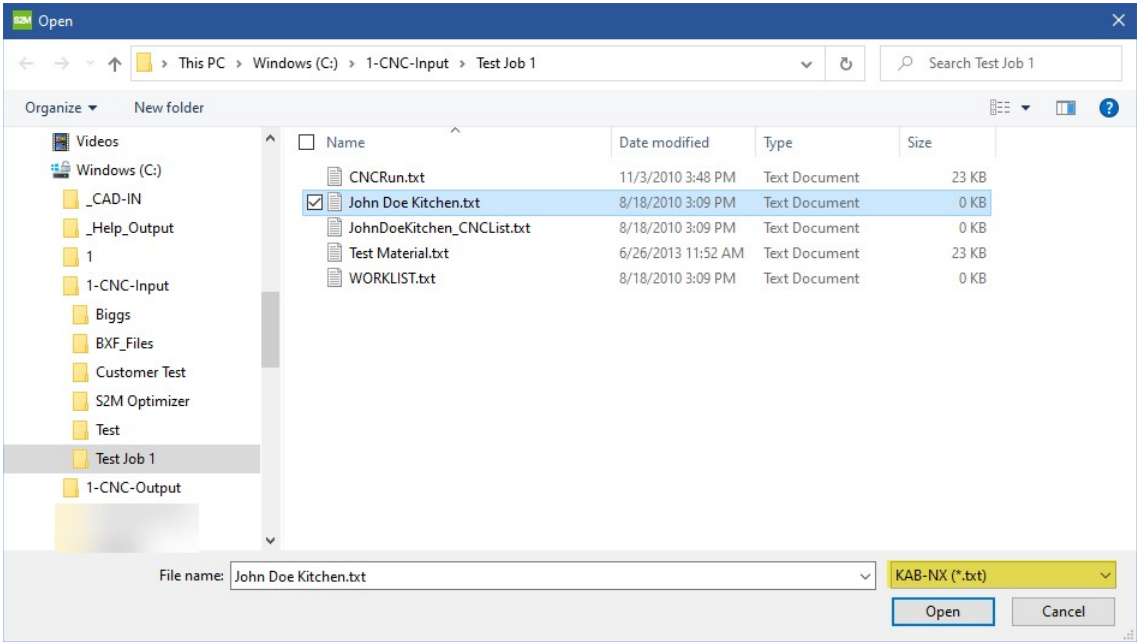
KAB-NX

To use the KAB-NX CAD Import Link you will need to get the Import Link License added to your License by from your local S2M CENTER supplier.

If you are using KAB-NX to create the DXF files you want to process in the S2M CENTER then this is the Scheme you need to have selected prior to importing KAB-NX DXF files.



When you output your DXF's from KAB-NX it will create a *JobName.txt* file which is what you would use to Import the DXF files into the S2M CENTER. Simply change the File Type in the Open Dialog to KAB-NX (\*.txt).



Following is the format for the KAB-NX Part List text file.

**KAB-NX Part List**

**JobName.txt File Structure**

**Typical Part Entry**

End\_Left\_0\_f.dxf, End\_Left, 12, 30, 0.75, 3/4 Melamine, 1, 1, 1, 0, F

**Field Contents**

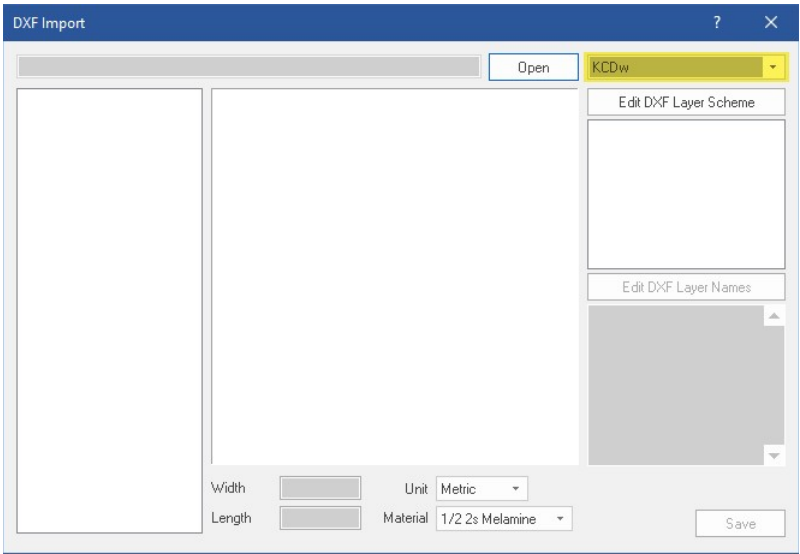
- 1 DXF filename
- 2 Part description
- 3 Part width (x)
- 4 Part length (y)
- 5 Part thickness (zz)
- 6 Part material name
- 7 Cabinet or unit number
- 8 Quantity to be generated
- 9 Rotation flag (0=no 1=yes)
- 10 Part Mirrored (0=no 1=yes)

11 Part Face (F=Front B=Back )

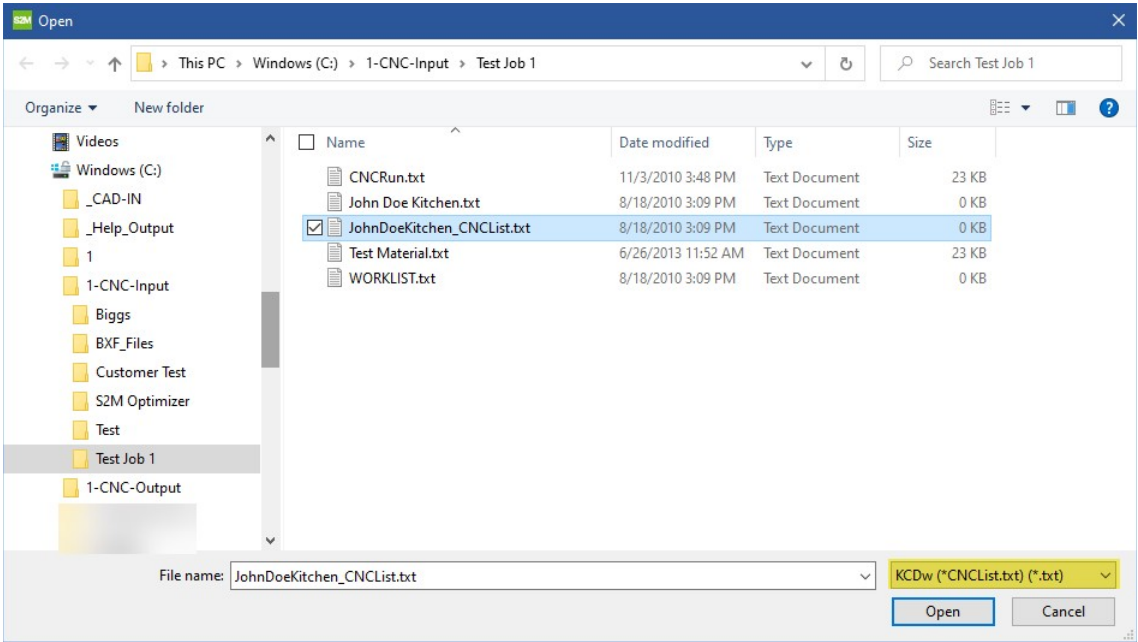
# KCDw

To use the KCdw CAD Import Link you will need to get the Import Link License added to your License by from your local S2M CENTER supplier.

If you are using KCDw to create the DXF files you want to process in the S2M CENTER then this is the Scheme you need to have selected prior to importing KCDw DXF files.



When you output your DXF's from KCDw it will create a *JobName\_CNCList.txt* file which is what you would use to Import the DXF files into the S2M CENTER. Simply change the File Type in the Open Dialog to KCDw (\*CNCList.txt).



Following is the format for the KCDw Part List text file.

**KCDw Part List**

**JobName\_CNCList.txt File Structure**

**Typical Part Entry**

End\_Left\_0\_f.dxf, End\_Left, 12, 30, 0.75, 3/4 Melamine, 1, 1, 1, 0, F

**Field Contents**

- 1 DXF filename
- 2 Part description
- 3 Part width (x)
- 4 Part length (y)
- 5 Part thickness (zz)
- 6 Part material name
- 7 Cabinet or unit number
- 8 Quantity to be generated
- 9 Rotation flag (0=no 1=yes)
- 10 Part Mirrored (0=no 1=yes)

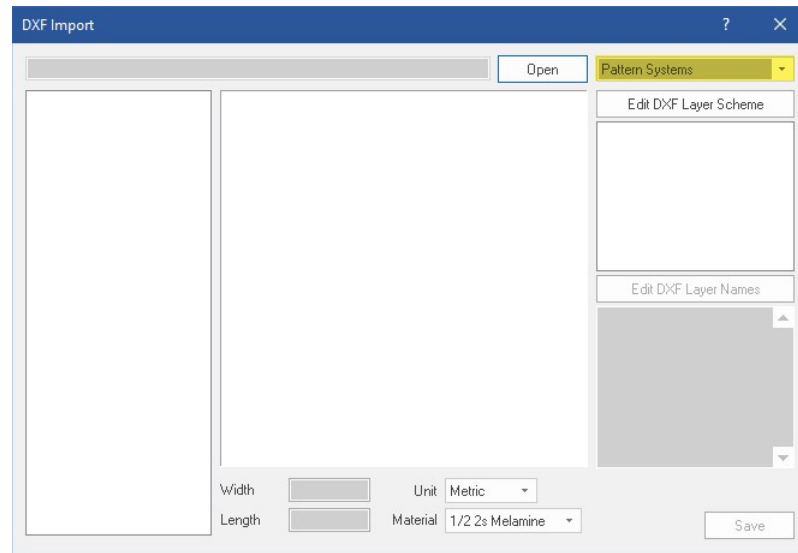


11 Part Face (F=Front B=Back )

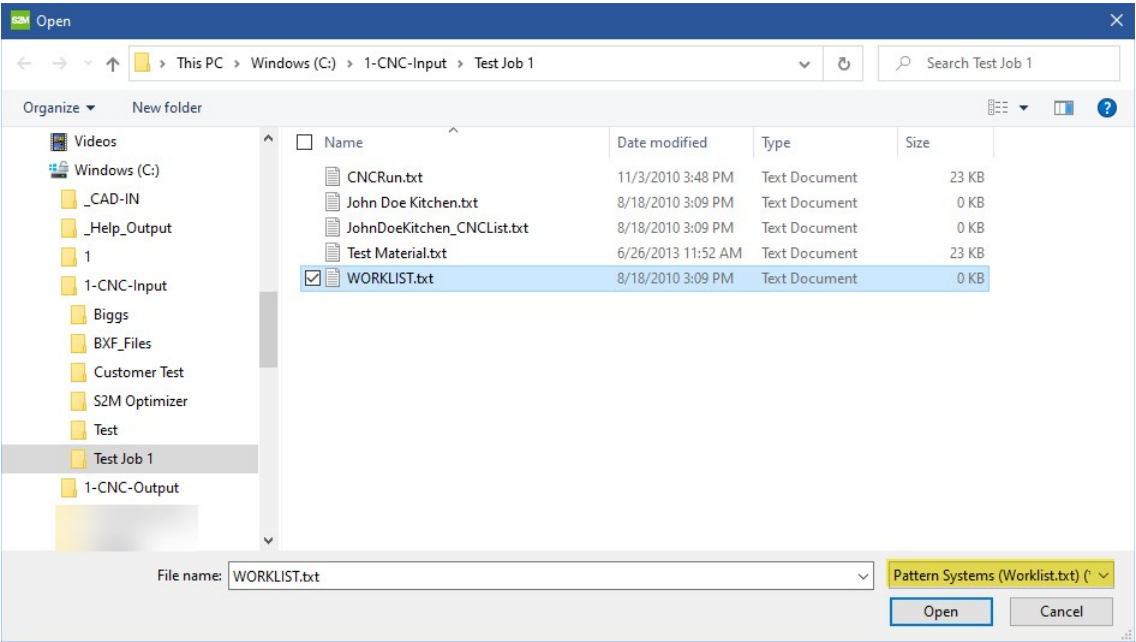
## Pattern Systems

To use the Pattern Systems CAD Import Link you will need to get the Import Link License added to your License by from your local S2M CENTER supplier.

If you are using Pattern Systems to create the DXF files you want to process in the S2M CENTER then this is the Scheme you need to have selected prior to importing Pattern Systems DXF files.



When you output your DXF's from Pattern Systems it will create a WORKLIST.txt file which is what you would use to Import the DXF files into the S2M CENTER. Simply change the File Type in the Open Dialog to Pattern Systems (Worklist.txt).



Following is the format for the Pattern Systems Part List text file.

**Pattern Systems Part List**

**WORKLIST.txt File Structure**

**Typical Part Entry**

C:\1-CNC-Input\Test Job 1\End\_Left\_0\_f.dxf, 3/4 Melamine,1,1,"SD, End\_Left", JohnDoeKitchen-1,580,U,U,U,U,U,U

**Field Contents**

- 1 Full Path to DXF File
- 2 Part material name
- 3 Rotation flag (0=no 1=yes)
- 4 Quantity to be generated
- 5 Part description
- 6 Job Name-Part cabinet number
- 7 Part ID number to link to jobrpt.mdb file
- 8 User Defined Input 1 (Optional)
- 9 User Defined Input 2 (Optional)
- 10 User Defined Input 3 (Optional)

- 11 User Defined Input 4 (Optional)
- 12 User Defined Input 5 (Optional)
- 13 User Defined Input 6 (Optional)

## S2M Part List

To use the S2M Part List CAD Import Link you will need to get the Import Link License added to your License by from your local S2M CENTER supplier.

If you are using a 3rd Party CAD Application, other than those predefined for the S2M CENTER, to create the DXF files you want to process then the S2M Part List is the Scheme you need to have selected prior to importing DXF files.



When you output your DXF's from your CAD Application you can make the Batch Import of these DXF's more seamless by creating a CncRun.txt file (See the Creating an S2M Part List topic for more information). Simply change the File Type in the Open Dialog to S2M Part List (CncRun.txt).



### S2M Part List      CncRun.txt File Structure

Following is the format for the S2M Part List text file.

#### Typical Header Layout

^Job : Test Job 1  
^Description: My First Test Job  
^Date: 08-18-2010 (01:03 PM)  
^Parts: 5  
^Unit: IN  
**Banding,E,Oak EB,Honey,.01**



The Banding listing is only necessary for defining Banding coming from Cabinet Vision if you are mixing an Imported List of files with existing Cabinet Vision Parts. This section is no longer part of the S2M Part List Creator application and would need to be added manually. The S2M Part List Creator only defines S2M CENTER Material Catalog Banding through [Banding Indicators](#).

#### Header Layout Field Contents

- Job: Text
- Description :Text
- Date: Date
- Parts: Indicates Number of Part Line Items in the File
- Unit: Can be either IN (Inches) or MM (Metric)
- 1 Banding** (the word "Banding" must be the first field when defining Banding)
- 2 Banding Code** (A single letter code you use to identify the Edgeband used to build the Banding Flag described below)
- 3 Banding Material**
- 4 Banding Color
- 5 Banding Thickness**

**Typical Part Entry**

LeftEnd.dxf, End Left, 12, 30, 0.75, 3/4 Melamine, 1, 1, 0, b, 0,EEEE, 1,.5,.5

**Part Entry Field Contents**

1 DXF filename

2 Part description

3 Part width (x)

4 Part length (y)

5 Part thickness (z)

6 Part material name

7 Cabinet or unit number

8 Quantity to be generated

9 Grain flag (0=no, can rotate, 1=yes, cannot rotate)



The Grain flag is by Material which means a Grain and Non-Grained Material must exist and be assigned for Parts where this setting is different.

10 Face Identifier (b=Back Face f=Front Face)

11 Face Dependency (0=non face dependent, 1=face dependent)

12 Banding Flag (Optional field containing the banding indicators for each edge.)



See the [Banding Flag](#) section of the Designing Label-IT Labels topic for more information.

13 Grain Match Group (Leave at 0 (or blank) if not grain matched. This number can begin at an index of 1.)



When imported this number may be modified so it doesn't conflict with any parts that already have a grain match group.

14 Grain Match Position X (Establishes the X Position of the Part within the Grain Match Group)



This field is required to have a valid value if Grain Match Group has a number greater than 0.

15 Grain Match Position Y (Establishes the Y Position of the Part within the Grain Match Group)



This field is required to have a valid value if Grain Match Group has a number greater than 0.

**Importing Parts with no DXF File**

The CAD Import routine does not require a DXF file too import a part. If no DXF file is specified it will assume the part has no operations and will import it as a rectangle.

**Creating an S2M CENTER Part List**

The S2M Part List Creator application has been supplied in the Installation extraction folder which can be used to easily create the CncRun.txt file to simplify the process of bringing your DXF's into the S2M CENTER. The installation file is located in the Installation extraction folder at \Support\PartListCreator\. Simply double click Setup.exe to install the Part List Creator on your computer.

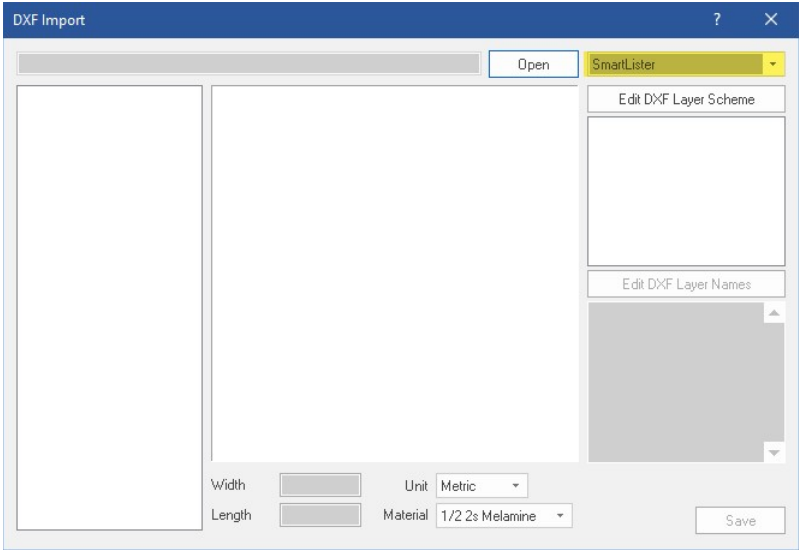
S2M CENTER Part List Creator

Open the S2M Part List Creator and see its Help file for information on its use.

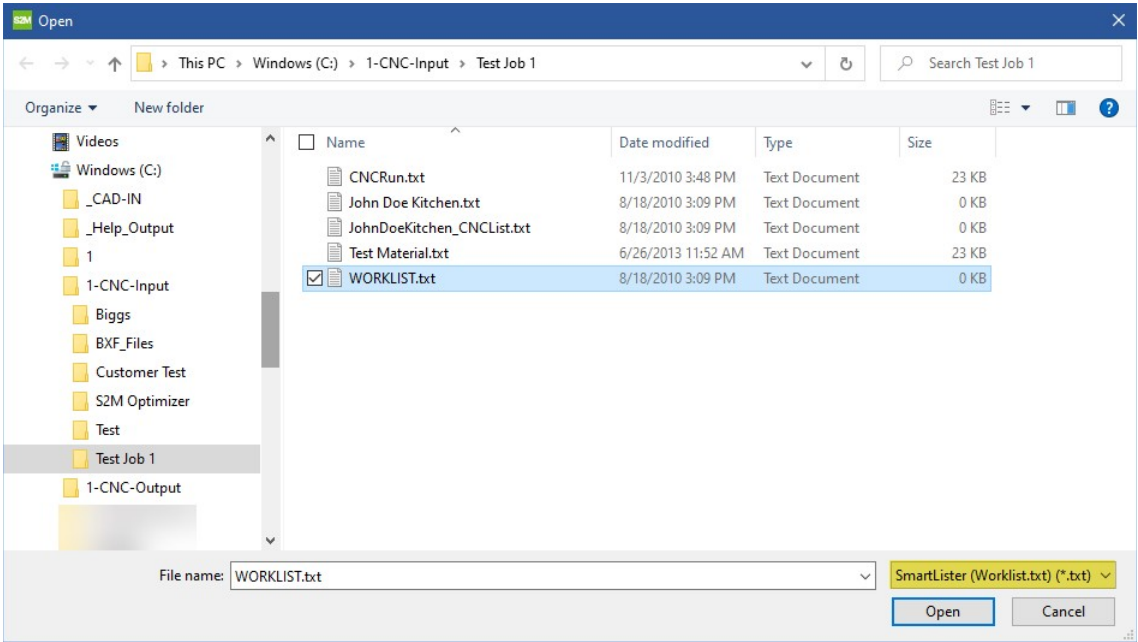
SmartLister

To use the SmartLister CAD Import Link you will need to get the Import Link License added to your License by from your local S2M CENTER supplier.

If you are using SmartLister to create the DXF files you want to process in the S2M CENTER then this is the Scheme you need to have selected prior to importing SmartLister DXF files.



When you output your DXF's from SmartLister it will create a WORKLIST.txt file which is what you would use to Import the DXF files into the S2M CENTER. Simply change the File Type in the Open Dialog to SmartLister (Worklist.txt).



Following is the format for the SmartLister Part List text file.

**SmartLister Part List**

**WORKLIST.txt File Structure**

**Typical Part Entry**

C:\1-CNC-Input\Test Job 1\End\_Left\_0\_f.dxf, 3/4 Melamine,1,1,"SD, End\_Left", JohnDoeKitchen-1,???,EB,EB,EB,EB

**Field Contents**

- 1 Full Path to DXF File
- 2 Part material name
- 3 Rotation flag (0=no 1=yes)
- 4 Quantity to be generated
- 5 Part description
- 6 Job Name-Part cabinet number
- 7 Not used
- 8 Edgeband Info 1 (Optional)
- 9 Edgeband Info 2 (Optional)
- 10 Edgeband Info 3 (Optional)

11 Edgeband Info 4 (Optional)

☒ **S2M for ALPHACAM**

If you own ALPHACAM Standard or higher Versions between Version 2016 R1(64-bit) and 2022.0 or AC:Core in Version 2022.1 or higher then you have access to the S2M CENTER from within ALPHACAM.

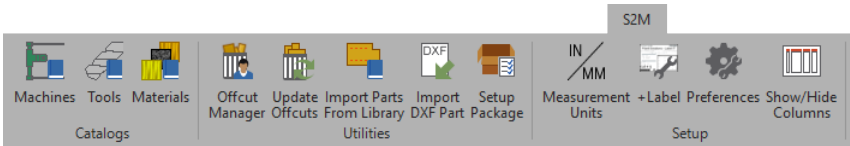
☒ The Documentation in this Help system assumes a working knowledge of ALPHACAM.

[Click one of the areas below for more information](#)



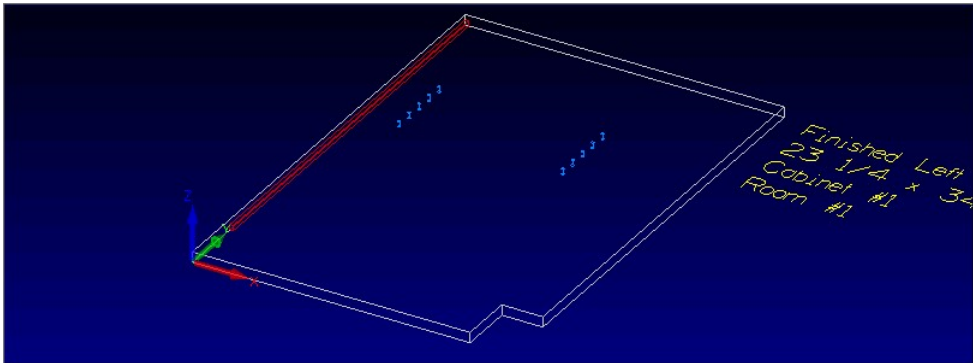
**S2M Tab**

[Click the options below for more information](#)



**Part Edit Screen**

This area allows modification of Parts created by your Design/Manufacturing software. You have full ALPHACAM modification capability in this area. The Documentation in this Help file assumes a working knowledge of ALPHACAM. Make sure to use the [S2M Edit Tools...](#)



**Parts List**

To display the Parts List simply click the Show Parts List button  in the sidebar.

|                                                                                   | Sel | Qty | Description          | Length | Width    | Over | CabNo | Material         | Banding | Ignore Grain                | Opt Machine | CNC Machine | D&D Machine | Cut Machine | Job           | F ^ |
|-----------------------------------------------------------------------------------|-----|-----|----------------------|--------|----------|------|-------|------------------|---------|-----------------------------|-------------|-------------|-------------|-------------|---------------|-----|
|  | ... | 1   | Finished Left End    | 34 1/2 | 23 1/4   | 0    | 1     | 3/4 2s Ash Ply   | 6NNNNNN | <input type="checkbox"/> No |             |             |             |             | Job - Default |     |
|  | ... | 1   | Unfinished Right End | 34 1/2 | 23 1/4   | 0    | 1     | 3/4 2s Birch Ply | 6NNNNNN | <input type="checkbox"/> No |             |             |             |             | Job - Default |     |
|  | ... | 1   | Unfinished Back      | 29 3/4 | 29       | 0    | 1     | 1/4 2s Birch Ply | 4NNNN   | <input type="checkbox"/> No |             |             |             |             | Job - Default |     |
|  | ... | 1   | Nailer               | 28 1/4 | 4        | 0    | 1     | 3/4 2s Birch Ply | 4NNNN   | <input type="checkbox"/> No |             |             |             |             | Job - Default |     |
|  | ... | 1   | Top Stretcher        | 13 3/4 | 4        | 0    | 1     | 3/4 2s Birch Ply | 4NNNN   | <input type="checkbox"/> No |             |             |             |             | Job - Default |     |
|  | ... | 1   | Top Stretcher        | 13 3/4 | 4        | 0    | 1     | 3/4 2s Birch Ply | 4NNNN   | <input type="checkbox"/> No |             |             |             |             | Job - Default |     |
|  | ... | 1   | Top Stretcher        | 13 3/4 | 4        | 0    | 1     | 3/4 2s Birch Ply | 4NNNN   | <input type="checkbox"/> No |             |             |             |             | Job - Default |     |
|  | ... | 1   | Top Stretcher        | 13 3/4 | 4        | 0    | 1     | 3/4 2s Birch Ply | 4NNNN   | <input type="checkbox"/> No |             |             |             |             | Job - Default |     |
|  | ... | 1   | Deck                 | 28 1/4 | 23       | 0    | 1     | 3/4 2s Birch Ply | 4NNNN   | <input type="checkbox"/> No |             |             |             |             | Job - Default |     |
|  | ... | 1   | Front Toe            | 30     | 4 3/4    | 0    | 1     | 3/4 2s Ash Ply   | 4NNNN   | <input type="checkbox"/> No |             |             |             |             | Job - Default |     |
|  | ... | 1   | Partition            | 29     | 22 31/32 | 0    | 1     | 3/4 2s Birch Ply | 6NNNNNN | <input type="checkbox"/> No |             |             |             |             | Job - Default |     |
|  | ... | 1   | Adjustable Shelf     | 13 1/2 | 16 31/32 | 0    | 1     | 3/4 2s Birch Ply | 4NNNN   | <input type="checkbox"/> No |             |             |             |             | Job - Default |     |
|  | ... | 1   | Adjustable Shelf     | 13 1/2 | 16 31/32 | 0    | 1     | 3/4 2s Birch Ply | 4NNNN   | <input type="checkbox"/> No |             |             |             |             | Job - Default |     |
|  | ... | 1   | Finished Left End    | 30     | 11 1/4   | 0    | 4     | 3/4 2s Ash Ply   | 4NNNN   | <input type="checkbox"/> No |             |             |             |             | Job - Default |     |

The same Right Click Options are available here as in the normal [S2M CENTER](#).

## Sidebar

There are three Primary Areas in the S2M for ALPHACAM sidebar:

S2M CENTER 2025



Optimizer Parameters

Add Offcuts

Filter Parts

Optimize

Machines

Machine Sets

ManualYes

Saw SetNo

Router SetYes

Primary Machine

MachineNest Router

Secondary Machine

MachinePart Router

Pre-BandNo

Drill and Dowel Machine

MachineDrill and Dowel

Pre-BandNo

Chop Saw

MachineChop Saw



## Toolbar

There are 8 options in the toolbar:

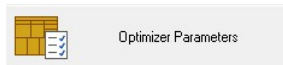


1.  Open a New PNC File - Allows you to open a previously exported PNC file. This will overwrite the currently open Job.
2.  Import a PNC File into Current Job - Allows you to add the contents of a previously exported PNC file into the current Job.
3.  Save Current Job as a PNC File - Allows you to Save the current Job as a PNC file for later use.
4.  Print - Allows printing of the Parts List.
5.  [Edit Part Library](#) - Allows you to manage your S2M CENTER Part Library.
6.  Show Par/Material List - Allows you to toggle on/off the Part List/Material List.
7.  [Materials List](#) - Displays the Material List for the Current Job.
8.  [Parts List](#) - Displays the Part List for the Current Job.

## Main Buttons

There are 4 Main Buttons:

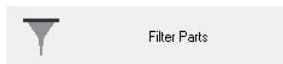
1. [Optimizer Parameters](#) - Same as in the Standard S2M CENTER.



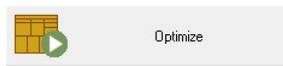
2. [Add Offcuts](#) - Same as in the Standard S2M CENTER.



3. [Filter Parts](#) - Same as in the Standard S2M CENTER.



4. [Optimize](#) - Similar to the Standard S2M CENTER.



## Machines

There 4 main Machine Types that are selectable in this area.

|                         |                 |
|-------------------------|-----------------|
| Machines                |                 |
| Machine Sets            |                 |
| Manual                  | Yes             |
| Router Set              | No              |
| Saw Set                 | No              |
| Primary Machine         |                 |
| Machine                 | Nest Router     |
| Secondary Machine       |                 |
| Machine                 | Parts Router    |
| Pre-Band                | No              |
| Drill and Dowel Machine |                 |
| Machine                 | Drill and Dowel |
| Pre-Band                | No              |
| Chop Saw                |                 |
| Machine                 | Chop Saw        |

1. [Primary Machine](#)
2. [Secondary Machine](#)
3. [Drill and Dowel Machine](#)
4. [Chop Saw](#)

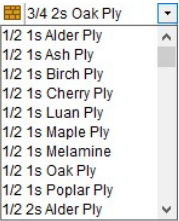
## Material List

To display the Materials List simply click the Materials List  tool in the sidebar toolbar.

| Material                                                                                           | Length | Width | Cost | Qty | Grain                                   |
|----------------------------------------------------------------------------------------------------|--------|-------|------|-----|-----------------------------------------|
|  3/4 2s Ash Ply   | 96     | 48    | 0.00 | 999 | <input checked="" type="checkbox"/> Yes |
|  3/4 2s Birch Ply | 96     | 48    | 0.00 | 999 | <input checked="" type="checkbox"/> Yes |
|  1/4 2s Birch Ply | 96     | 48    | 0.00 | 999 | <input checked="" type="checkbox"/> Yes |
|  3/4 Solid Ash    | 96     | 48    | 0.00 | 999 | <input checked="" type="checkbox"/> Yes |
|                   |        |       |      |     | <input type="checkbox"/> No             |



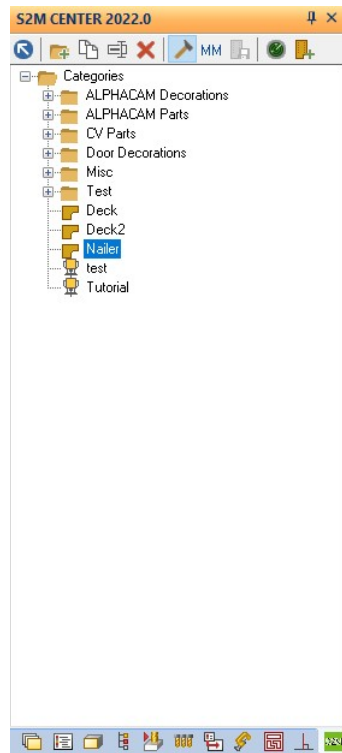
Unlike the Standard S2M CENTER, when in the Material List in S2M for ALPHCAM you can add or change any Material to any existing Material in your [Material Catalog](#).



The same Right Click Options are available here as in the [Standard S2M CENTER](#).

## Edit Part Library

Allows you to Edit Parts in your Part Library with ALPHACAM.



## Edit Part Toolbar Tools

There are ten tools in the Edit Part toolbar:



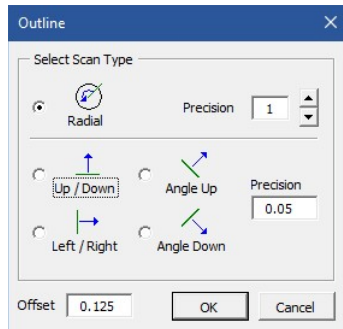
1. **Return** - Returns you to the main sidebar.
2. **Create New Category** - Allows creation of a new Part Category in the Tree below.
3. **Copy Selected Item** - Makes a copy of the selected Part/Category.
4. **Rename Selected Item** - Allows you to Rename the selected Part/Category.
5. **Delete Selected Item** - Allows you to Delete the selected Part/Category.
6. **Build** - Each time you select a Part from the Parts List ALPHACAM draws (Builds) that Part on the screen. If you de-select the Build button it stops this which allows you to move around the Categories a bit quicker.
7. **Display In Metric** - Displays your Parts in Metric.



ALPHACAM, unlike your Design/Manufacturing software, does not handle Unit of Measure changes on the fly. Therefore, when you Save a Part to the Part Library there are two Parts Saved. One Part is in Imperial Units and the Other is in Metric Units. When clicking the MM Tool it displays the parts "Alter Ego".

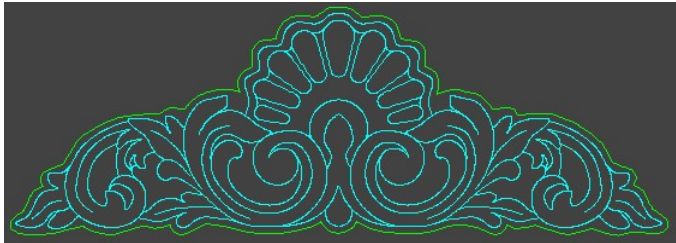
8. **Save Changes to Current Part** - Allows changes made to the current ALPHACAM Part to be Saved to the Library overwriting the existing Part.

9. **Create Part Outline** - Defines the extents of an ALPHACAM Part. This is required if you are going to save an Alphacam Part into the Part Library. This is also very helpful if you are going to utilize these Parts in your Design/Manufacturing software.



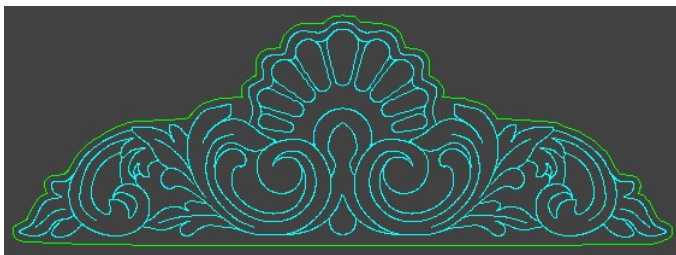
Depending upon your Parts Shape you will want to experiment with each of these Scan types to get the results you desire. Below are examples of each scan types results on the same Part.

#### Radial



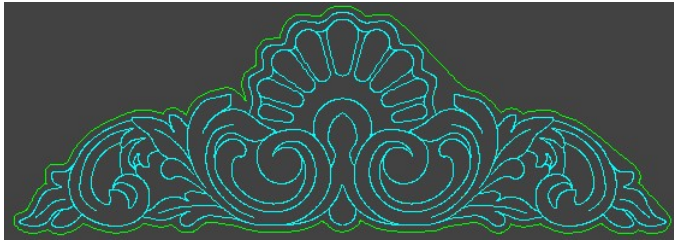
The Radial Type seems to follow the shape pretty closely.

#### Up / Down



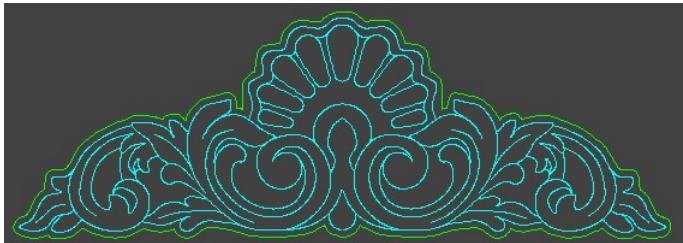
The Up/Down Type does well except at the Bottom of this Part.

#### Angle Up



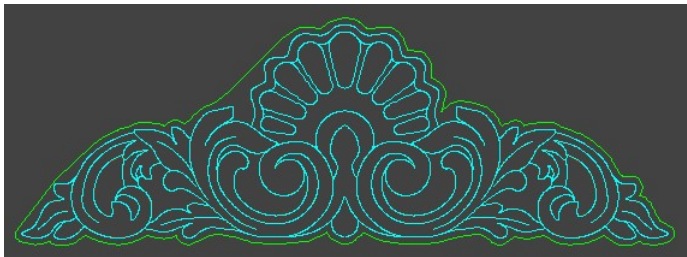
The Angle Up Type has a problem on the Right Side of this Part.

#### Left / Right



The Left/Right Type seems to follow the shape pretty closely.

#### Angle Down



Angle Down Type has a problem on the Left Side of this Part.

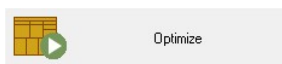
As you can see in the preceding images the Radial or Left / Right scan option would be the best choice. As stated earlier it really depends on the shape of the Part the Outline is being generated for. Experiment!

10.  **Add Part to Library** - Adds the current Part to the Part Library.

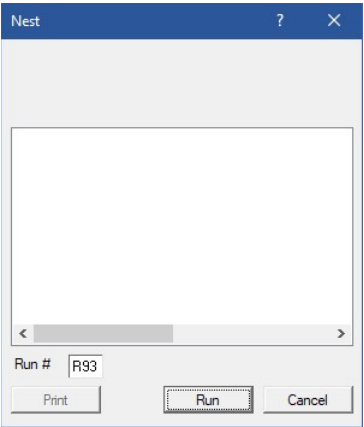


Make sure that Parts you Add to the S2M CENTER from ALPHACAM are set to 0 as Top of Material and Depths are Negative Value equal to actual Depths for the Parts to be processed properly in the S2M Center. This must be setup this way **regardless** of whether you are set up in your Machine Catalog as "Mic at Spoilboard" or not.

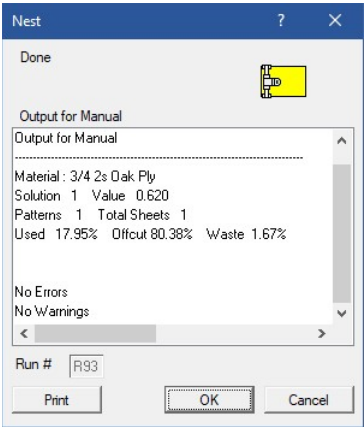
## Optimize



The Optimize button, once depressed, will begin the Nesting process. Once selected the Optimizing Screen will appear.



- 1. Click on the Run button to begin the Optimization process.



- 2. Once the Optimizing is completed the Completed section will display each Material optimized with the total number of sheets, the Used percentage, the Offcut percentage, and the Waste percentage. Also displayed in the Completed section are any [Tool selection](#) Errors or Warnings. You can at this point Print the completed section by clicking the Print button or by clicking the OK button you can accept the results and move on to the [Patterns Section](#) of the ALPHACAM S2M CENTER.

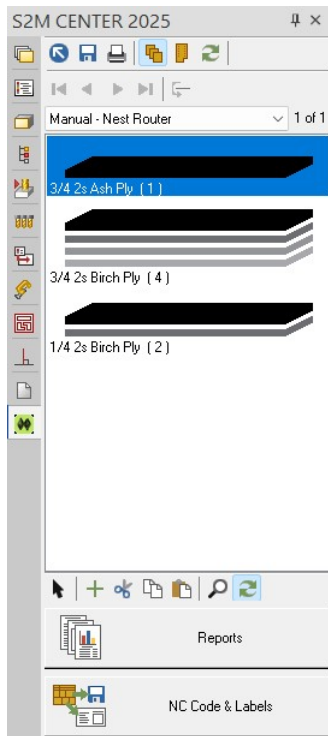
## S2M Center Patterns Section

[Click the sidebar or pattern display area below for more information](#)



## Sidebar

## Cut Patterns View



There are three Primary Areas in the sidebar:

### Toolbar

There are 11 tools available in the main toolbar:



- **Return** - Will return you to the previous sidebar.
- **Save Job State** - Allows you to Save a Job State (SNC) File.
- **Print** - Allows you to Print the Pattern Layouts.
- **Sheet View** - Shows the Sheet stacks below.
- **Part View** - Shows the Parts programs below.
- **Rebuild Sheet** - Rebuilds the active Sheet Pattern.
- **First** - Goes to the first Sheet in the selected stack.

- ◀ **Previous** - Goes to the Previous Sheet in the selected stack.
- ▶ **Next** - Goes to the Next Sheet in the selected stack.
- ▶ **Last** - Goes to the Last Sheet in the selected stack.
- 📄 **Show 6th Face** - Will show the 6th Face of your sheet if 6th Face Nesting is active and 6th Face operations exist.

**Cut Patterns Area**

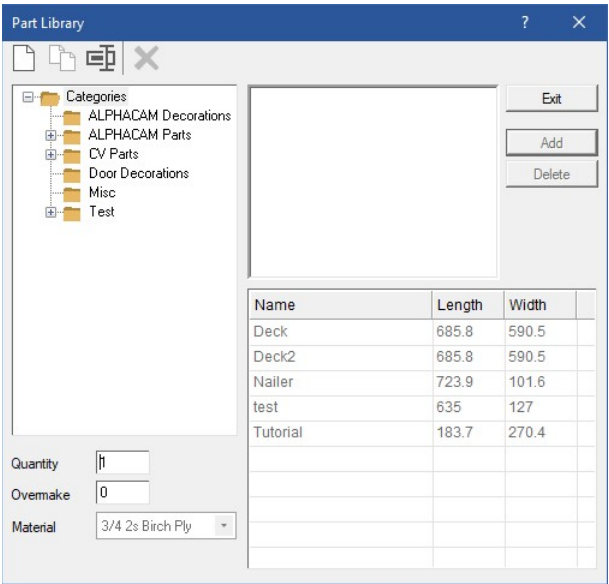
Select the stacks in the list to view.



There are seven tools available in the Cut Patterns View toolbar:



- 🖱️ **Select A Part On The Current Sheet** - This tool allows you to select a Part on the active Pattern in the Drawing Area.
- ➕ **Add Part (s) from Library** - This tool allows you to add a Part from the Part Library to the active Pattern in the Drawing Area.



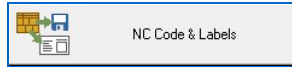
- ✂️ **Cut Selected Part** - Allows you to Cut the selected Part to the Clipboard for pasting on the active sheet or an alternate sheet.
- 📄 **Copy Selected Part** - Allows you to Copy the selected Part to the Clipboard for pasting on the active sheet or an alternate sheet.
- 📄 **Paste Part** - Paste the Part stored on the Clipboard to the active sheet.
- 📄 **Show Part List** - Shows a Part List of all Parts on the active sheet.



-  **Automatically Rebuild after Cut or Paste** - When active this tool will automatically rebuild a sheet when a Part is Cut from or Pasted to the active sheet.

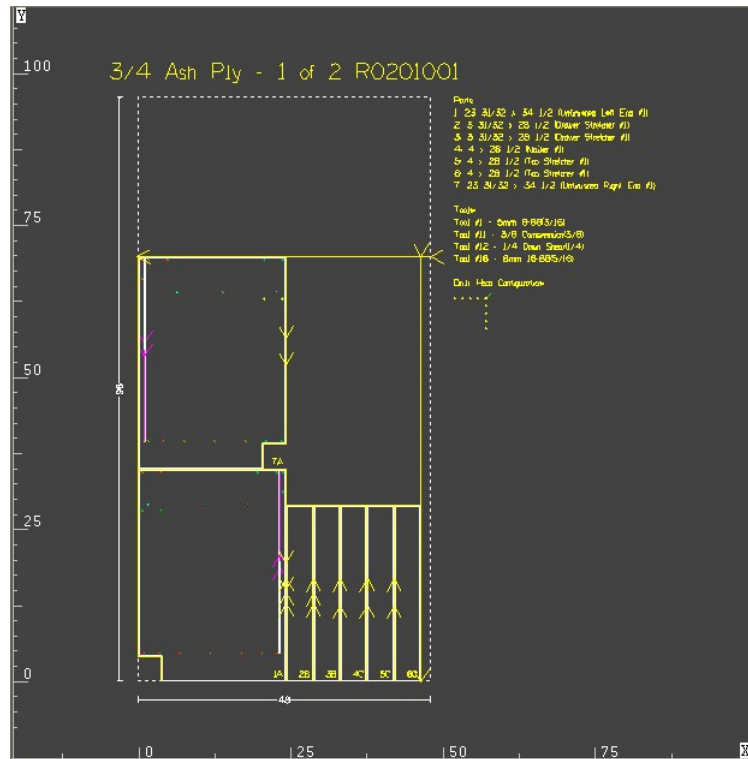
### Sidebar Buttons

There is two sidebar buttons available here, click them below for more information.



## Sheet Display Area

Allows Viewing and Modification, via any ALPHACAM Tools, of your Nested Output. Along with other NC Specific Information.



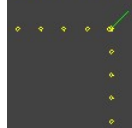
## Parts:

- 1: 23 31/32 x 34 1/2 (Unfinished Left End #1)
- 2: 3 31/32 x 28 1/2 (Drawer Stretcher #1)
- 3: 3 31/32 x 28 1/2 (Drawer Stretcher #1)
- 4: 4 x 28 1/2 (Nailer #1)
- 5: 4 x 28 1/2 (Top Stretcher #1)
- 6: 4 x 28 1/2 (Top Stretcher #1)
- 7: 23 31/32 x 34 1/2 (Unfinished Right End #1)

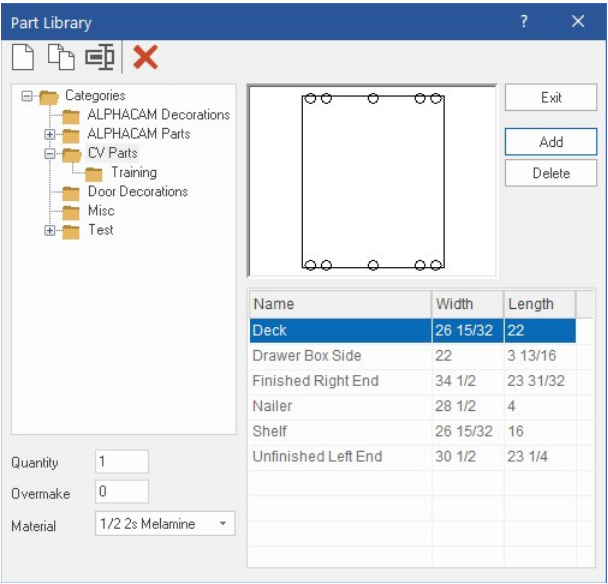
## Tools:

- Tool #1 - 5mm 8-88(3/16)
- Tool #11 - 3/8 Compression(3/8)
- Tool #12 - 1/4 Down Shear(1/4)
- Tool #16 - 8mm 16-88(5/16)

## Drill Head Configuration:

**Part Library**

The Add Library Parts option allows you to add Parts you have saved into your Part Library into your current Job. To Add a Part simply highlight it in the list and click the Add button. There are several other options here.



Toolbar



There are 4 options in the toolbar:

1. **New** - Creates a new Folder.
2. **Copy** - Copies the selected folder and its contents.
3. **Rename** - Allows you to rename the selected folder.
4. **Delete** - Deletes the selected Folder. Be careful once a folder is deleted it can not be restored.

Other Options

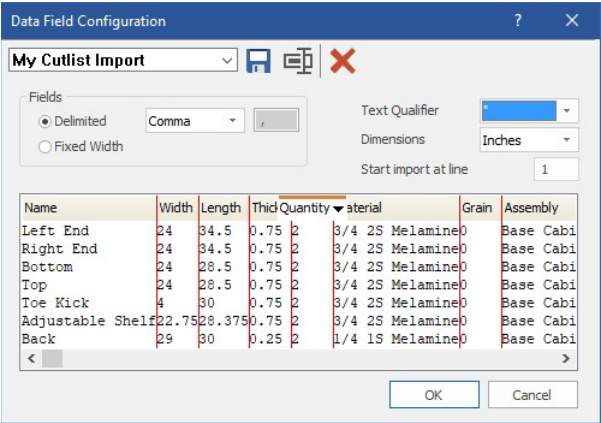
- **Quantity** - Prior to clicking Add you can set a Quantity for the currently selected Part.
- **Overmake** - This field allows you to set how many extra of these Parts can be made if they will fit on a sheet. It will try to get these without a adding an additional sheet to your Nest. If an additional Sheet is required to get them it will not produce them.
- **Material** - Select which Material you want the selected part out of.
- **Orientation** - If the Material you selected is defined as a one sided Material this option allows you to tell the S2M CENTER which Face you want the Primary Operations performed on.



Import Cutlist

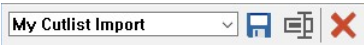
Import Cutlist allows you to bring in a list of Parts in a CSV format for Nesting/Optimization. You can use your favorite spreadsheet application or design application to create a CSV that can be imported directly into the S2M CENTER.

This utility examines the text file that you are importing and helps you ensure that the data is imported in the way that you want. You can use this Utility to define what each field in your Text file represents so it is imported properly.



Toolbar

There are four options in the toolbar:



**My Cutlist Import** Select Configuration

Allows you to select which of your Saved data field configurations to use.

**Save**

Allows you to Save the current configuration for re-use.

**Rename**

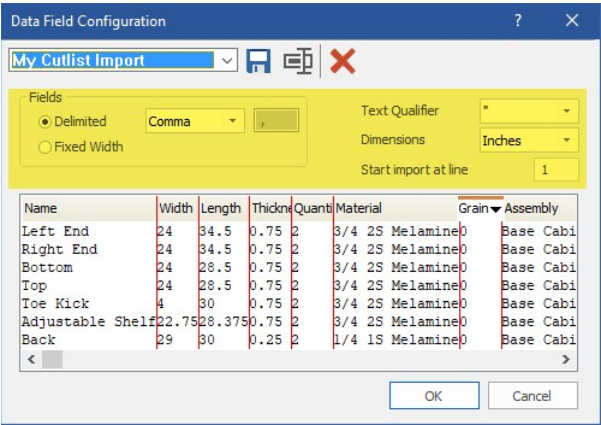
Allows you to rename the currently selected configuration.

**Delete**

Allows you to delete the currently selected configuration.

General Options

There are four general options available to configure your Text file:



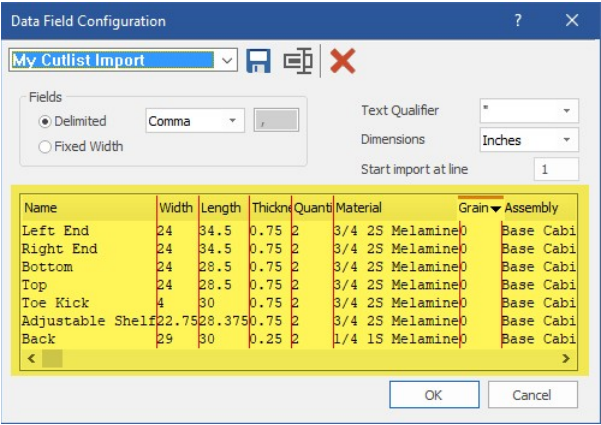
**Fields** - This option allows you to establish how the fields in the Text file are defined. If items in the text file are separated by tabs, colons, semicolons, spaces, or other characters, select Delimited; then select what the delimiter is (Tab, Comma, Semicolon, Space, Other). If all of the items in each column are the same length, select Fixed width.

**Text Qualifier** - Select the character that encloses values in your text file. When Solid encounters the text qualifier character, all of the text that follows that character and precedes the next occurrence of that character is imported as one value, even if the text contains a delimiter character. For example, if the delimiter is a comma (,) and the text qualifier is a quotation mark ("), "Tuscaloosa, Alabama" is imported into one field as Tuscaloosa, Alabama.

**Dimensions** - Select whether the file you are importing is Inches or Metric.

**Start Import at Line** - Type or select a line number to specify the first line of the data that you want to import.

Import Preview Area



In this area you will see all of the columns that the S2M Optimizer has available for your Text file. Click the column header of the column(s) that you do not want Ignored and set its type. There are eleven options:

|                 |
|-----------------|
| Ignore          |
| Name            |
| Width           |
| Length          |
| Thickness       |
| Quantity        |
| Material        |
| Grain           |
| Assembly        |
| Assembly Number |
| Banding         |

**Ignore** - Will ignore this column when importing.

**Name** - The Part's Name/Description.

**Width** - The Part's Width.

**Length** - The Part's Length.

**Thickness** - The Part's Thickness.

**Quantity** - The Part's Quantity.

**Material** - The Material the Part will be cut from.

**Grain** - Establishes if the Part is Grained or not.

**Assembly** - The Name/Description of the Assembly the Part is used in.

**Assembly Number** - The Assembly Number of the Assembly the Part is used in.

**Banding** - The [Banding Flag](#) for the Part. The Banding Flag should only include Banding Codes you have defined either in your design package (typically upper case letters) or within the S2M CENTER Material Catalog (lower case letters). 0 (zero) would equal an Edge with no Banding.

## Help



Help gives you access to the following options:



Accesses this Help System. For information on how to use the S2M CENTER Help file see the [Using Help](#) topic.



Access the Nexus Community web site. You must have an Internet Connection setup to use the Nexus Community web site.



Access the Training Options for the S2M CENTER. You must have an Internet Connection setup to view the Training Options.



This option allows you to start a Team Viewer desktop sharing session with an S2M CENTER Technician. Use this option when instructed by your S2M CENTER Technician and provide the displayed ID and Password.



## Zoom Group

The Zoom Group has three options:



### Window

The Zoom Window option is used for increasing the size of an area or Part on your screen to see a detail or examine a condition. By clicking on the Zoom Window tool, then clicking in the center of the area you would like to Zoom In to, and, while holding the mouse button down, dragging the cursor diagonally across the item you wish to Zoom In on, finally releasing the mouse button at the end of the area, your screen will Zoom In.

You can also change the Window from Center out to Corner to Corner in [Preferences | Tools](#) Tab.



### Out

The Zoom Out option isn't a click and drag task, instead, simply clicking the Zoom Out option will incrementally Zoom the drawing out.



### All

The Zoom All option will restore the object size to the original size and re-center the object in the process.



## View Mode

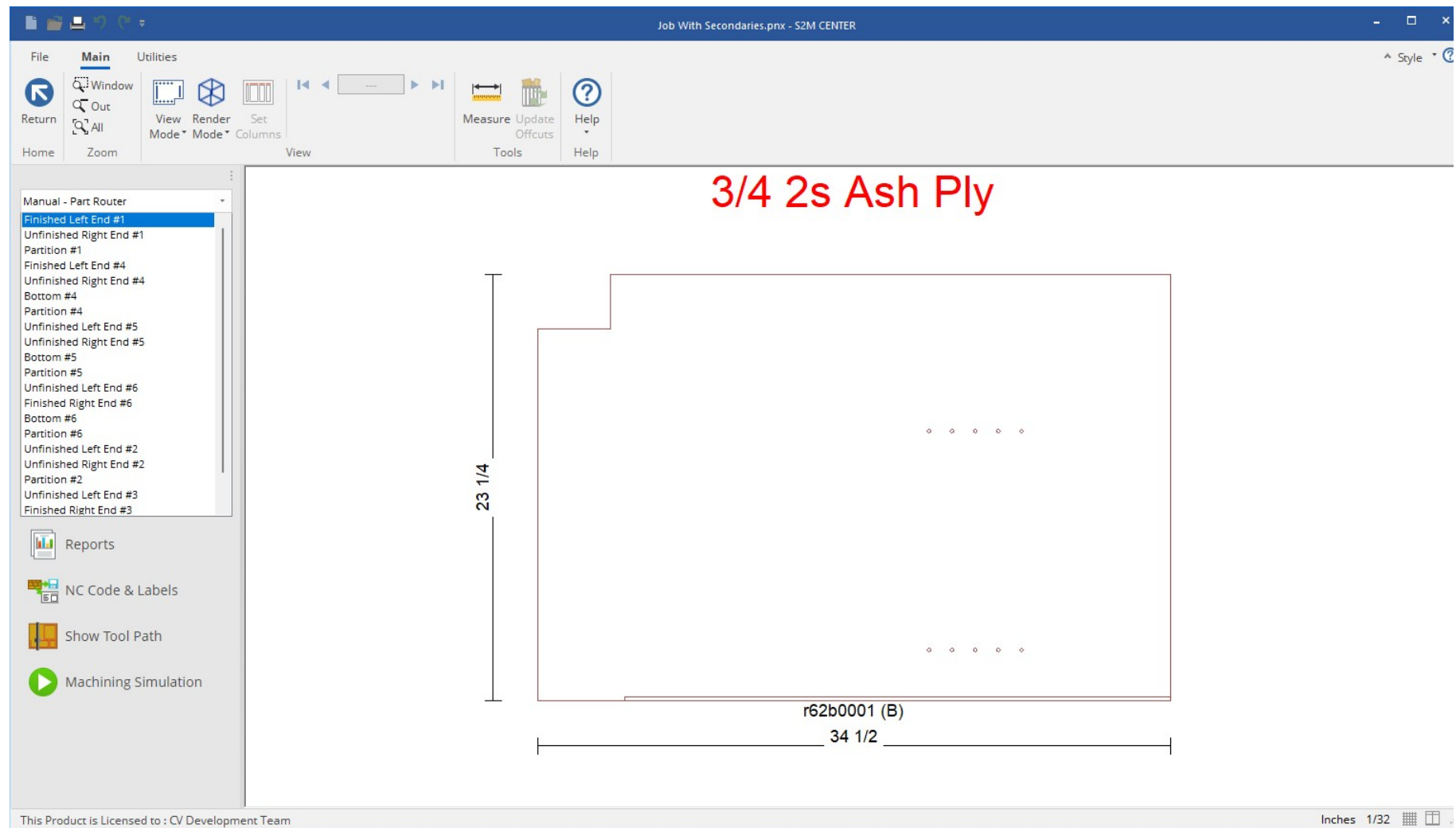


There are two options in View Mode:



### Part View

This is the Default View when only outputting to a Secondary Machine. The Part View displays the Materials and Parts list in the sidebar and the selected Part and it's machining in the Pattern/Part Display Area.



### Right Click Menu

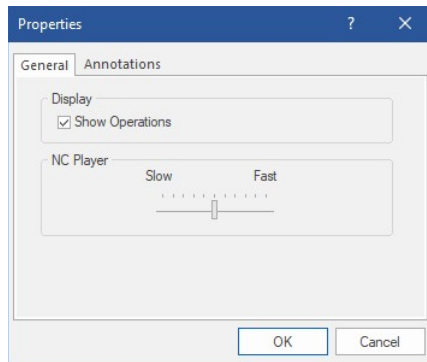


**To Drawings** - Only available with CABINET VISION. This option will send the current View as a Drawing Scene to the CABINET VISION Drawings Area.



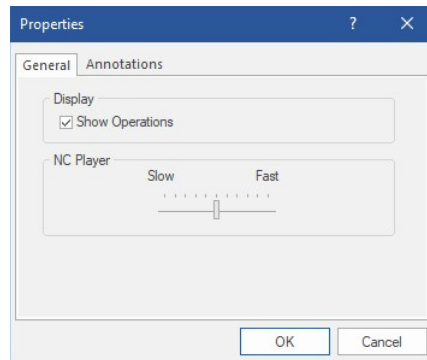


**Properties** - Clicking on the Properties command brings up the following window.



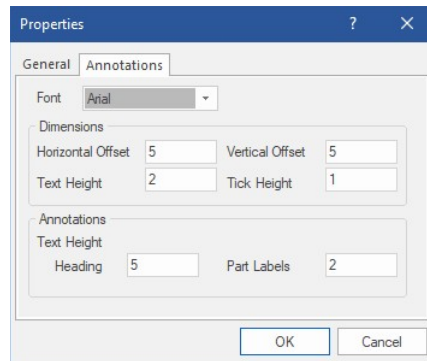
### General Tab

There are two main options on the General Tab:



1. You can control whether Operations are visible on the Pattern printouts.
2. You can, via the slide bar, control the speed of the CNC Player when using Show Tool Path in the Patterns section of the S2M CENTER.

### Annotations Tab



On the Annotations Tab there are seven main options:

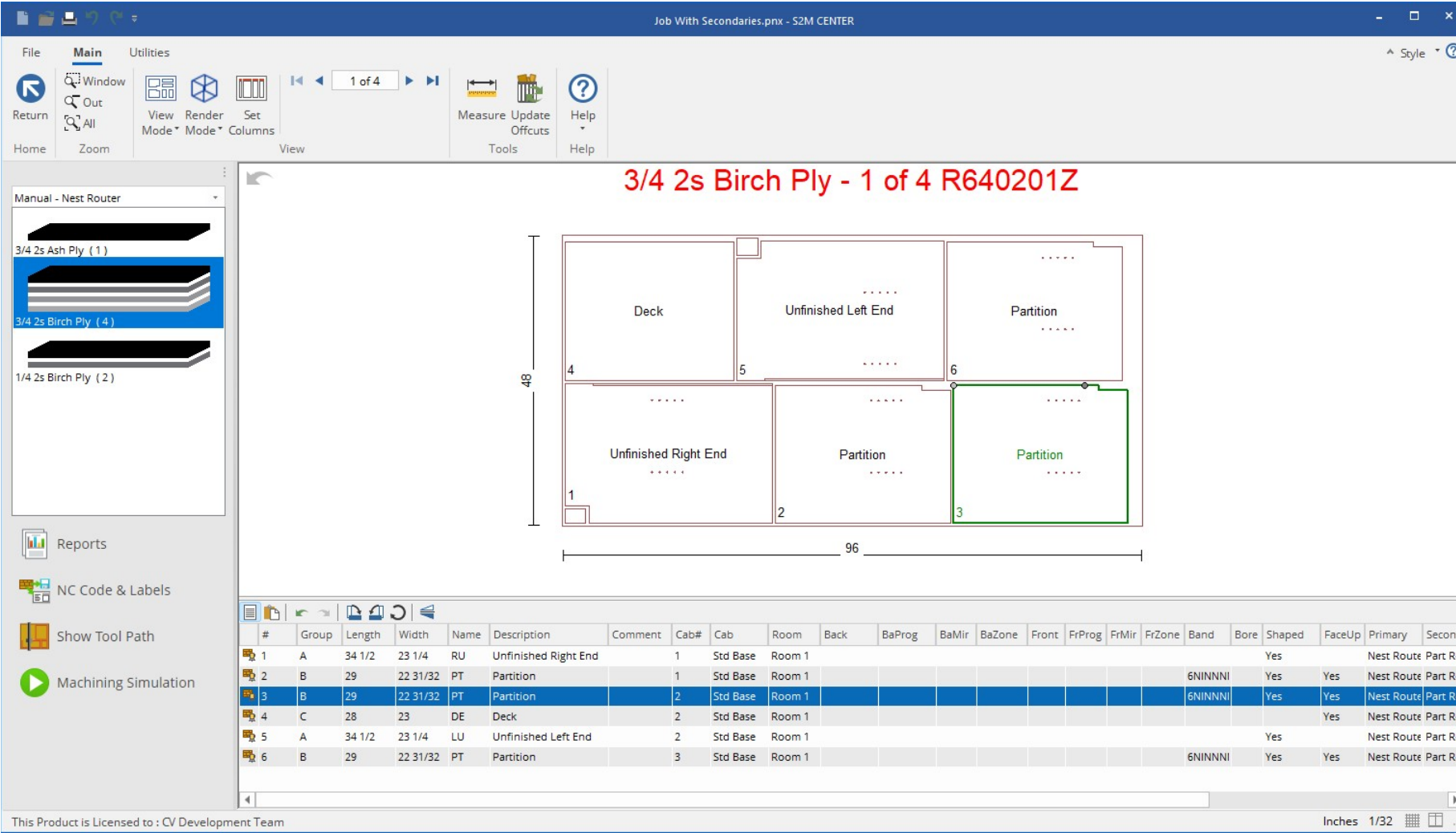
1. **Font** - You can control the displayed Font.
2. **Dimensions Horizontal Offset** - You can control the Horizontal Offset of your Dimensions.
3. **Dimensions Vertical Offset** - You can control the Vertical Offset of your Dimensions.
4. **Dimensions Text Height** - You can control the Text Height in your Dimensions.
5. **Dimensions Tick Height** - You can control the Tick Height in your Dimensions.
6. **Annotations Heading Text Height** - You can control the Text Height in the Heading Annotations.
7. **Annotations Part Labels Text Height** - You can control the Text Height in your Part Labels (Labels here refers to the Text marking your Part on the Sheet not the Label Reports you Print.).



Pattern View

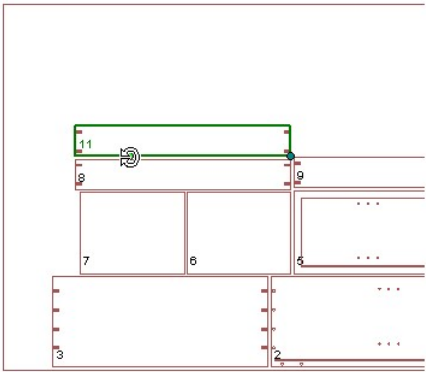
## Pattern View

This is the Default View when Optimizing or Nesting. The Pattern View displays the Patterns list in the sidebar and the selected Pattern and it's machining in the Pattern/Part Display Area.



This area displays your Sheet/Pattern layouts.

If a part can be rotated (not saw output, not a grain match part/block, not viewing the 6th face). A point of origin (dark green) and a rotation point (light green) will appear on the part. When the mouse is near the rotation point the cursor will change to indicate the part can be rotated.



Click the mouse once to start rotation, click again to finish.



Whenever the rotated part is near 0, 90, 180, or 270, the part will snap to that angle.

The lower Section is the Part List and Clipboard area. The Parts on the active sheet are listed in the lower section of the screen. There are several columns of information which you can sort the list by clicking on the column heading. A Part selected from either the Pattern are or Part list area will cause that Part to be highlighted in both Views.

Any items Cut or Copied via a Part Right Click will be listed on the clipboard . Simply drag any Part in the Clipboard to the Sheet where you want it placed.

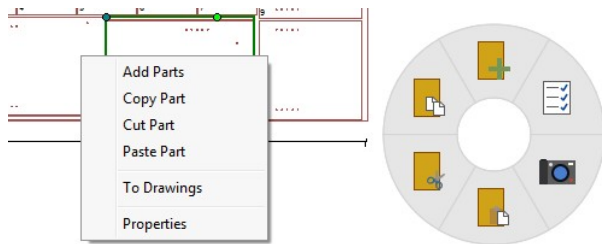
|  | Qty | Pas... | Over | Name | Description | Comment | Length | Width    | Thick | Material         | CabNo | Cabinet  |
|--|-----|--------|------|------|-------------|---------|--------|----------|-------|------------------|-------|----------|
|  | 1   | 0      | 0    | DE   | Deck        |         | 28 1/2 | 23 31/32 | 3/4   | 3/4 2s Birch Ply | 1     | Std Base |

In the Parts List/Clipboard area you have several tools you can use to modify Parts within the Nest:



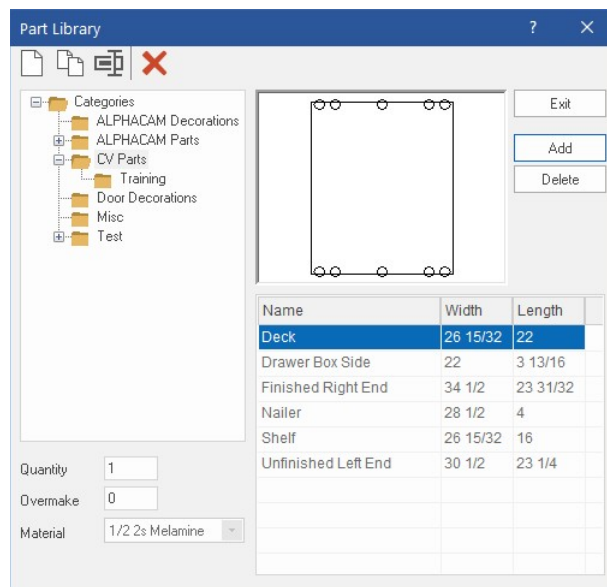
- Part List View** - Shows all Parts on the current Pattern in the list view.
- Clipboard View** - Shows a list of all [Copied Parts](#).
- Show 6th Face/Show 1st Face** - Allows you to flip the Pattern over to see the 6th Face/1st Face operations if there is a 6th Face option on the selected Pattern.
- Rotate Clockwise** - Rotates the selected Part clockwise 90 degrees
- Rotate Counter Clockwise** - Rotates the selected Part counter clockwise 90 degrees.
- Rotate 180** - Rotates the selected Part 180 degrees.
- Flip Part** - This option will flip the selected Part. This action will typically create either a secondary program, modify the secondary program, or modify the 6th face of the sheet.

**Right Click Menu**



There are several options in the Sheet Display Area right click menu.

1.  **Add Parts** - Brings up the Part Library and allows you to add a Part from it to the current sheet.



Simply select any Part you have saved into your Part Library enter a Length, Width, Quantity, and Overmake and click Add.

2.  **Copy Part** - This option copies a Part to the clipboard for adding to the current sheet or any other sheet within the Job. (Even a different Material and/or Thickness.)
3.  **Cut Part** - This option removes a Part to the clipboard for adding to the current sheet or any other sheet within the Job. (Even a different Material and/or Thickness.)
4.  **Paste Part** - This option will take the last part that was either copied or cut and place it on the Current Sheet.
5.  **To Drawing** - Will send the current screen to the Cabinet Vision Drawings Area as a Drawing Scene.
6.  **Properties** - Clicking on the Properties command brings up the following window.

**If using an Optimization Machine:**

**Properties** ? X

General Annotations

Display

Dimension Style

- ☒ Width x Length
- ☐ Length x Width

Dimension Type

- ☐ Parts
- ☐ Legend
- ☒ None

☒ Show Operations

☒ Show Part Shapes

☒ Show Part Sizes

☒ Show Part Name

☐ Set origin at Bottom Left corner

Zoom Mode

- ☒ From Corner
- ☐ From Center

☐ Reverse Scroll Wheel zoom direction

OK Cancel

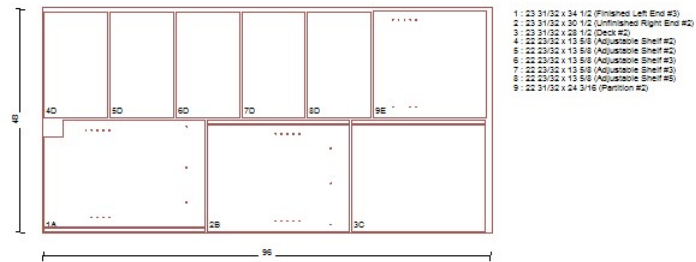
1. **Dimension Style** - You can control whether the output Pattern printouts Dimension Style is "Width x Length" or "Length x Width".
2. **Dimension Type** - You can control if you want Dimensions to display on the Parts or in a Legend on the side of your printouts.

3/4 Ash Ply - 1 of 5 R3501001



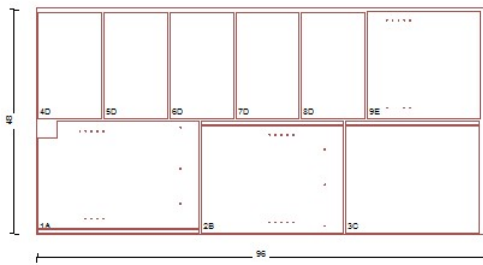
file:///C:/Users/celav/AppData/Local/Temp/~hhEA7D.htm

## 3/4 Ash Ply - 1 of 5 R3501001



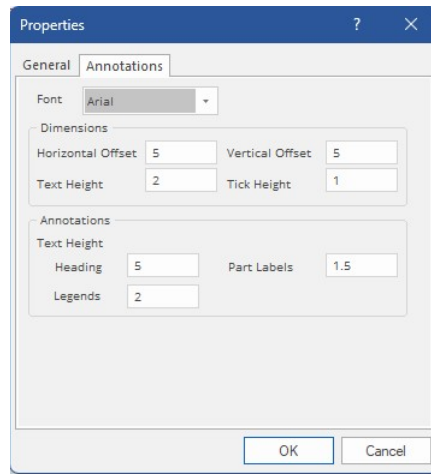
None

## 3/4 Ash Ply - 1 of 5 R3501001



3. **Show Operations** - You can control whether Operations are visible on the Pattern printouts.
4. **Show Part Shapes** - You can control whether Part Shapes are visible on the Pattern printouts.
5. **Show Part Sizes** - You can control whether Part Sizes are shown on the Parts on the Patterns.
6. **Show Part Name** - You can control whether Part Names are shown on the Parts on the Patterns.
7. **Set Origin at Bottom Left Corner** - You can set the Origin at the Bottom Left corner on your printouts.
8. **From Corner/From Center** - You can control if, when using the Zoom Window tool, whether the origin of the Zoom window begins From the Corner of where you pick or the Center.
9. **Factor** - Determines the percentage the Scroll Wheel will Zoom the view In and Out.
10. **Reverse Scroll Wheel Zoom Direction** - Determines which direction the Scroll Wheel Zooms In and Out.

Annotations Tab



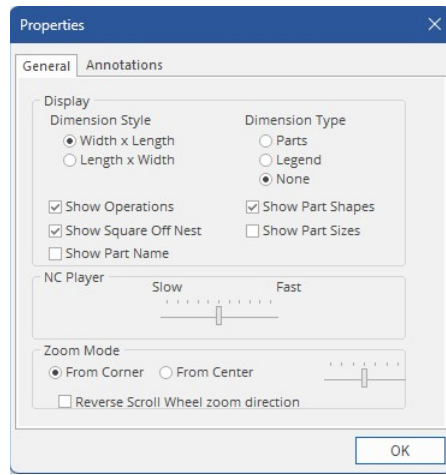
On the Annotations Tab there are seven main options:

1. **Font** - You can control the displayed Font.
2. **Dimensions Horizontal Offset** - You can control the Horizontal Offset of your Dimensions.
3. **Dimensions Vertical Offset** - You can control the Vertical Offset of your Dimensions.
4. **Dimensions Text Height** - You can control the Text Height in your Dimensions.
5. **Dimensions Tick Height** - You can control the Tick Height in your Dimensions.
6. **Annotations Heading Text Height** - You can control the Text Height in the Heading Annotations.
7. **Annotations Part Labels Text Height** - You can control the Text Height in your Part Labels (Labels here refers to the Text marking your Part on the Sheet not the Label Reports you Print.).
8. **Annotations Legends Text Height** - You can control the Text Height in the Legend.

**If using a Nesting Machine:**

**General Tab**



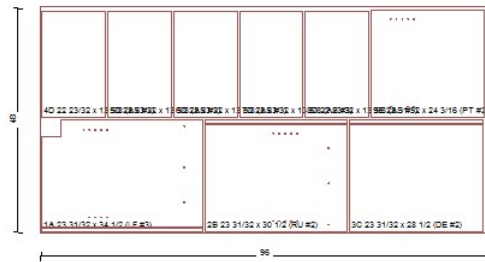


On the General Tab there are eight main options:

1. **Dimension Style** - You can control whether the output Pattern printouts Dimension Style is "Width x Length" or "Length x Width".
2. **Dimension Type** - You can control if you want Dimensions to display on the Parts or in a Legend on the side of your printouts.

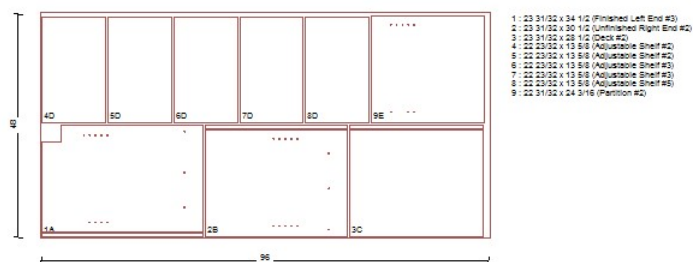
#### Parts

3/4 Ash Ply - 1 of 5 R3501001



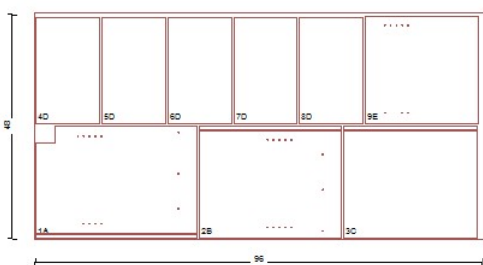
#### Legend

## 3/4 Ash Ply - 1 of 5 R3501001



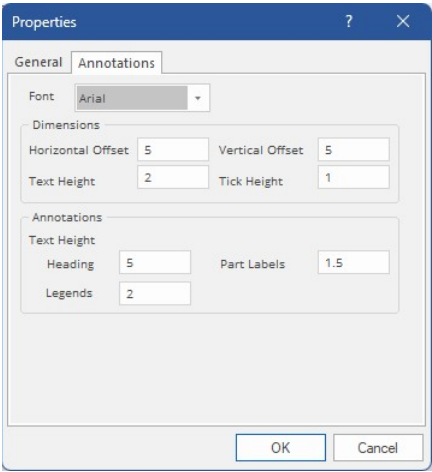
None

## 3/4 Ash Ply - 1 of 5 R3501001



3. **Show Operations** - You can control whether Operations are visible on the Pattern printouts.
4. **Show part Shapes** - You can control whether Part Shapes are visible on the Pattern printouts.
5. **Show Square Off Nest** - Shows Square off nest operations
6. **Show Part Sizes** - You can control whether Part Sizes are shown on the Parts on the Patterns.
7. **Show Part Name** - You can control whether Part Names are shown on the Parts on the Patterns.
8. **NC Player** - You can, via the slide bar, control the speed of the CNC Player when using Show Tool Path in the Patterns section of the S2M CENTER.
9. **From Corner/From Center** - You can control if, when using the Zoom Window tool, whether the origin of the Zoom window begins From the Corner of where you pick or the Center.
10. **Factor** - Determines the percentage the Scroll Wheel will Zoom the view In and Out.
11. **Reverse Scroll Wheel Zoom Direction** - Determines which direction the Scroll Wheel Zooms In and Out.

Annotations Tab



On the Annotations Tab there are seven main options:

- 1. **Font** - You can control the displayed Font.
- 2. **Dimensions Horizontal Offset** - You can control the Horizontal Offset of your Dimensions.
- 3. **Dimensions Vertical Offset** - You can control the Vertical Offset of your Dimensions.
- 4. **Dimensions Text Height** - You can control the Text Height in your Dimensions.
- 5. **Dimensions Tick Height** - You can control the Tick Height in your Dimensions.
- 6. **Annotations Heading Text Height** - You can control the Text Height in the Heading Annotations.
- 7. **Annotations Part Labels Text Height** - You can control the Text Height in your Part Labels (Labels here refers to the Text marking your Part on the Sheet not the Label Reports you Print.).
- 8. **Annotations Legends Text Height** - You can control the Text Height in the Legend.



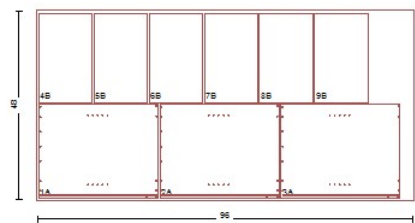
Render Mode **Render Mode**

There are three Render Mode options:



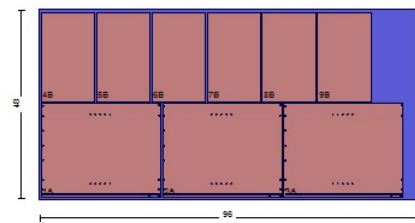
Wire Mode **Wire Mode**

The Wire Mode option allows you to view your Pattern in a line (wire frame) view.



 Fill Mode **Fill Mode**

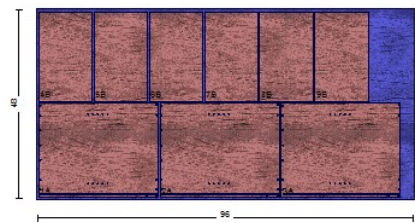
The Fill Mode option allows you to view your Pattern in a filled (basic color) view.



The green color represents offcut/waste. The blue color represents trim. The Brown color represents the components being Optimized/Nested.

 Texture Mode **Texture Mode**

The Render Mode option allows you to view your Pattern in a render (color's and textures) view.



The green color represents offcut/waste. The blue color represents trim. The Brown color represents the components being Optimized/Nested.

 **Pattern Navigation Option**

 **First Page**

Clicking the First Page option will display the first Pattern the Pattern View.

 **Previous Page**

Clicking the Previous Page option will display the previous Pattern the Pattern View. Alternatively you can use the  key on your keyboard.

2 of 5

### Current Page Display

The Current Page Display option displays the current Pattern as well as the total number of patterns in the selected group (material). Clicking in here and changing the first number will allow you to type a specific Pattern number to navigate to.

### Next Page

Clicking the Next Page tool will display the next Pattern in the Pattern View. Alternatively you can use the  key on your keyboard.

### Last Page

Clicking the Last Page tool will display the last Pattern in the Pattern View.

  
Measure

## Measure

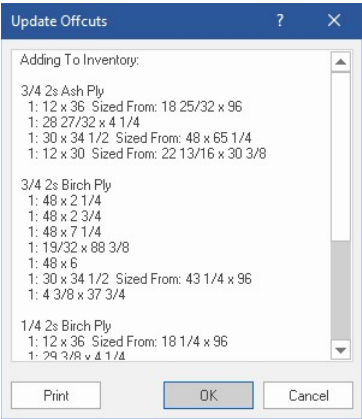
The Measure option, used in conjunction with [Snaps](#), is available for use in determining the dimension of, or distance between Parts.

|             |                                       |
|-------------|---------------------------------------|
| Horizontal: | <input type="text" value="0"/>        |
| Vertical:   | <input type="text" value="0"/>        |
| Distance:   | <input type="text" value="0"/>        |
| Angle:      | <input type="text" value="0.00"/>     |
| Position ^  |                                       |
| X:          | <input type="text" value="-49 3/4"/>  |
| Y:          | <input type="text" value="18 23/32"/> |

  
Update Offcuts

## Update Offcuts

When used from the Patterns View the Update Offcuts option will take all Offcuts in your Patterns and Add them to the [Offcut Manager](#).



You can click Print here, allowing you to communicate to your Machine Operator what needs to be placed in Offcut Inventory.

## Utilities Tab

[Click the options below for more Information](#)

### At Parts and Materials View



When accessing the S2M CENTER from within CABINET VISION there are 2 possible additional options in the Utilities area.

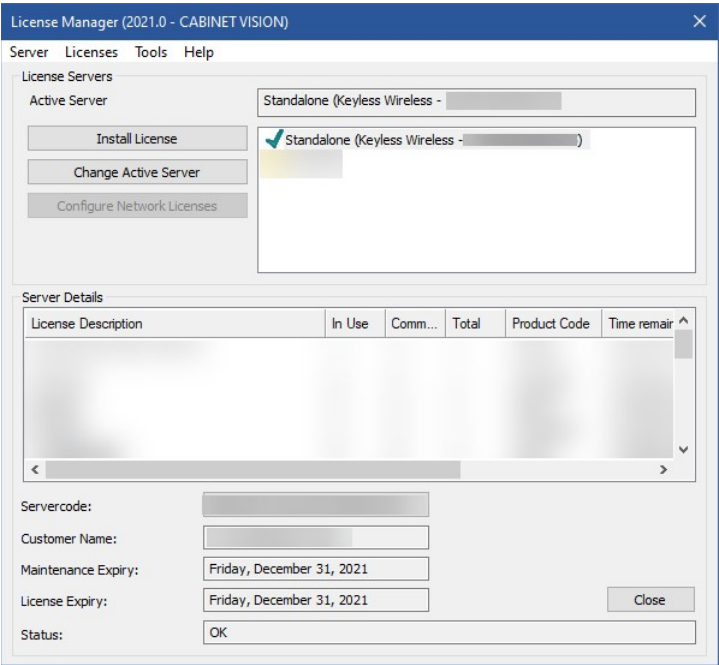


### At Patterns and Parts View



## License Manager

This option opens the License Manager.



For more information see the License Manager Help file.

## Material Catalog

When first entering the Material Catalog in the S2M CENTER you will see a screen similar to the one below with no Materials.



If you are using S2M CENTER stand-alone and do not run CABINET VISION simply click the [New Material](#) tool in the toolbar to start creating your Materials.

If you are using CABINET VISION, the first thing you will want to do is choose your Materials too display. To do this click the Show Materials from CV  tool in the toolbar. Upon entering the Show Materials from CV tool you will be presented with the following screen.



Place a check mark to each Material you would like to display in the list.



Simply click Show once you have checked off those Materials you would like to Import. You should notice that once you click Import each Material you selected is removed from the list. This will make it easier to check if you missed any.



Simply click Close to return to the Material Catalog. You will notice it is now populated with your Materials.

[Click one of the highlighted areas for more Information](#)



**Material List**

You can sort this list easily by clicking the Header for each column. If you want to sort by Width simply click the Width header, etc...



| Name                 | Width   | Length  | Thick  | Grain | Cost   |
|----------------------|---------|---------|--------|-------|--------|
| 1/2 1s Alder Ply     | 48.0000 | 96.0000 | 0.5000 | Yes   | \$0.00 |
| 1/2 1s Ash Ply       | 48.0000 | 96.0000 | 0.5000 | Yes   | \$0.00 |
| 1/2 1s Birch Ply     | 48.0000 | 96.0000 | 0.5000 | Yes   | \$0.00 |
| 1/2 1s Cherry Ply    | 48.0000 | 96.0000 | 0.5000 | Yes   | \$0.00 |
| 1/2 1s Luan Ply      | 48.0000 | 96.0000 | 0.5000 | Yes   | \$0.00 |
| 1/2 1s Maple Ply     | 48.0000 | 96.0000 | 0.5000 | Yes   | \$0.00 |
| 1/2 1s Melamine      | 48.0000 | 96.0000 | 0.5000 | No    | \$0.00 |
| 1/2 1s Oak Ply       | 48.0000 | 96.0000 | 0.5000 | Yes   | \$0.00 |
| 1/2 1s Poplar Ply    | 48.0000 | 96.0000 | 0.5000 | Yes   | \$0.00 |
| 1/2 2s Alder Ply     | 48.0000 | 96.0000 | 0.5000 | Yes   | \$0.00 |
| 1/2 2s Ash Ply       | 48.0000 | 96.0000 | 0.5000 | Yes   | \$0.00 |
| 1/2 2s Birch Ply     | 48.0000 | 96.0000 | 0.5000 | Yes   | \$0.00 |
| 1/2 2s Cherry Ply    | 48.0000 | 96.0000 | 0.5000 | Yes   | \$0.00 |
| 1/2 2s Luan Ply      | 48.0000 | 96.0000 | 0.5000 | Yes   | \$0.00 |
| 1/2 2s Maple Ply     | 48.0000 | 96.0000 | 0.5000 | Yes   | \$0.00 |
| 1/2 2s Melamine      | 49.0000 | 97.0000 | 0.5000 | No    | \$0.00 |
| 1/2 2s Oak Ply       | 48.0000 | 96.0000 | 0.5000 | Yes   | \$0.00 |
| 1/2 2s Poplar Ply    | 48.0000 | 96.0000 | 0.5000 | Yes   | \$0.00 |
| 1/2 Baltic Birch Ply | 60.0000 | 60.0000 | 0.5000 | Yes   | \$0.00 |

Material Properties

Let's take a look at the Material Properties area.

| Imported From CV  |                  |
|-------------------|------------------|
| Name              | 1/2 1s Alder Ply |
| Description       | Alder            |
| Type              | Panel            |
| Grain             | Yes              |
| Cost              | \$0.00           |
| Size              |                  |
| Width             | 48.000           |
| Length            | 96.000           |
| Thick             | 0.500            |
| Face Properties   |                  |
| Face Surface      | Blank            |
| Back Surface      | Alder            |
| Edge Surface      | Particle Board   |
| Clean Face        | Yes              |
| Clean Back        | Yes              |
| Tool Movement     |                  |
| Climb Cut         | No               |
| Feed rate %       | 100 %            |
| Spindle rate %    | 100 %            |
| MaxDepth Per Pass | 0.000            |

The first thing you should make note of is that the Product the Material was imported is notated in the header.

| Imported From CV |
|------------------|
|------------------|

There are four sections in the Material option section:

- **Product Properties** - There are five options here:

| Imported From CV |                  |
|------------------|------------------|
| Name             | 1/2 1s Alder Ply |
| Description      | Alder            |
| Type             | Panel            |
| Grain            | Yes              |
| Cost             | \$0.00           |

- **Name** - Enter a unique name to identify the Material here.
- **Description** - Enter a Material description here.
- **Type** - Allows you to set the Materials Type.
- **Grain** - Select Yes/No to choose whether or not to limit part rotation on the Material due to grain.
- **Cost** - Enter the Cost of this Material at the specified size here.

- **Size** - There are three options here:

| Size   |        |
|--------|--------|
| Width  | 48.000 |
| Length | 96.000 |
| Thick  | 0.500  |

- **Width** - Enter the Materials sheet Width here.
- **Length** - Enter the Materials sheet Length here.
- **Thick** - Enter the Materials Thickness here.

- **Face Properties** - There are five options here:

| Face Properties |                |
|-----------------|----------------|
| Face Surface    | Blank          |
| Back Surface    | Alder          |
| Edge Surface    | Particle Board |
| Clean Face      | Yes            |
| Clean Back      | Yes            |

- **Face Surface** - Description of the Materials Face Surface.
- **Back Surface** - Description of the Materials Back Surface.
- **Edge Surface** - Description of the Materials Edge Surface.
- **Clean Face** - If Yes, the S2M Center will automatically select the proper Tool to minimize the blowout on the Material's Face.
- **Clean Back** - If Yes, the S2M Center will automatically select the proper Tool to minimize the blowout on the Material's Back.

- **Tool Movement** - There are four options here:

| Tool Movement     |       |
|-------------------|-------|
| Climb Cut         | No    |
| Feed rate %       | 100 % |
| Spindle rate %    | 100 % |
| MaxDepth Per Pass | 0.000 |

- **Climb Cut** - Use this option to specify whether to utilize a Climb or Conventional Cut. Typically, when cutting Hardwoods you will receive the cleanest edge utilizing a Climb Cut. When cutting Plywoods you will receive the cleanest edge utilizing a Conventional Cut.
- **Feed Rate %** - A percentage override to the Feed Rate for the Outline Tool when used with this Material.

- **Spindle Speed %** - A percentage override to the Spindle Speed for the Outline Tool when used with this Material.
- **Max Depth Per Pass** - An override to the Max Depth per Pass for the Outline Tool when used with this Material.

## Material Toolbar

[Click one of the buttons below for more information](#)



All Materials

## Material List Filter

This tool allows you to filter the Material List display showing only Materials in the selected Category.

All Materials

Banding

Board Stock

Closet Rods

Guides

Hinges

Laminate

Misc

Molding

Panel Stock

Pulls

Tops

Wire Baskets

## New Material

When clicking this tool it will create a new Material line item at the bottom of the [Material List](#) named New Material as shown in the figure below.

|              |        |        |        |    |        |
|--------------|--------|--------|--------|----|--------|
| New Material | 0.0000 | 0.0000 | 0.0000 | No | \$0.00 |
|--------------|--------|--------|--------|----|--------|

Once the new Material has been added to the Material List simply change its Properties in the [Material Properties](#) section.

| S2MCenter         |              |
|-------------------|--------------|
| Name              | New Material |
| Description       |              |
| Type              | Panel        |
| Grain             | No           |
| Cost              | \$0.00       |
| Size              |              |
| Width             | 0.000        |
| Length            | 0.000        |
| Thick             | 0.000        |
| Face Properties   |              |
| Face Surface      |              |
| Back Surface      |              |
| Edge Surface      |              |
| Clean Face        | Yes          |
| Clean Back        | Yes          |
| Tool Movement     |              |
| Climb Cut         | Yes          |
| Feed rate %       | 100 %        |
| Spindle rate %    | 100 %        |
| MaxDepth Per Pass | 0.000        |

 **Copy Material**

When clicking this tool it will create a copy of the selected Material and place it directly below it in the [Material List](#) as shown in the figure below.

| Name                 | Width   | Length  | Thick  | Grain | Cost   |
|----------------------|---------|---------|--------|-------|--------|
| 1/2 1s Alder Ply     | 48.0000 | 96.0000 | 0.5000 | Yes   | \$0.00 |
| 1/2 1s Alder Ply (1) | 0.0000  | 0.0000  | 0.0000 | No    | \$0.00 |

Once the copied Material has been added to the Material List simply change its Properties in the [Material Properties](#) section.

| S2M Center        |                      |
|-------------------|----------------------|
| Name              | 1/2 1s Alder Ply (1) |
| Description       |                      |
| Type              | Panel                |
| Grain             | No                   |
| Cost              | \$0.00               |
| Size              |                      |
| Width             | 0.000                |
| Length            | 0.000                |
| Thick             | 0.000                |
| Face Properties   |                      |
| Face Surface      |                      |
| Back Surface      |                      |
| Edge Surface      |                      |
| Clean Face        | Yes                  |
| Clean Back        | Yes                  |
| Tool Movement     |                      |
| Climb Cut         | Yes                  |
| Feed rate %       | 100 %                |
| Spindle rate %    | 100 %                |
| MaxDepth Per Pass | 0.000                |



Note that the copied Material's type is S2M CENTER. This means that it is separated from your CABINET VISION or Cabnetware Material Database and will not be available there. If you wanted this new Material available in both the S2M CENTER and your Design/Manufacturing product you would want to create it in your Design/Manufacturing product and Import it using the [Import Materials](#) Tool in the S2M CENTER.

 **Delete Material**

When clicking this tool it will Delete the selected Material from the [Material List](#). You can restore the Material, if it was an Imported Material, at anytime by using the [Synchronize Materials](#) Tool.

 **Show Materials from CV**

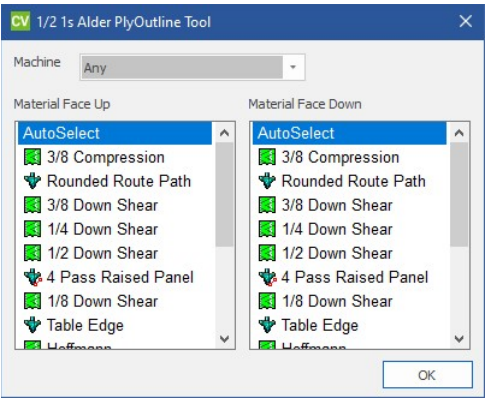
This tool will allow you to Show Materials from your CABINET VISION Material Manager.



Select which Materials you would like to show then click the Show button.

### Select Outline Tool

This Tool allows you to assign specific Tools as the Outline Tool for the selected Material in the [Material List](#).



-  You can, with the Machine selector, also assign different Outline Tools based on which Machine the Material is being Machined on.
-  Also note that you can control which Tool is used depending in whether the Material Face is Up or the Material Face is Down on your Machine.

### Show Advanced Tool Strategy

When selecting this tool the Material List will be split and a Tool Selection screen will be added to the bottom. This option not only allows you to assign specific Tools to your Materials but once selected, allows you to control the Feed, Descent, Speed (RPM), and Max Depth when utilizing the selected Tool with the selected Material.

Material Catalog

All Materials

| Name                 | Width   | Length  | Thick  | Grain | Cost   |
|----------------------|---------|---------|--------|-------|--------|
| 1/2 1s Alder Ply     | 48.0000 | 96.0000 | 0.5000 | Yes   | \$0.00 |
| 1/2 1s Alder Ply (1) | 0.0000  | 0.0000  | 0.0000 | No    | \$0.00 |
| 1/2 1s Ash Ply       | 48.0000 | 96.0000 | 0.5000 | Yes   | \$0.00 |
| 1/2 1s Birch Ply     | 48.0000 | 96.0000 | 0.5000 | Yes   | \$0.00 |
| 1/2 1s Cherry Ply    | 48.0000 | 96.0000 | 0.5000 | Yes   | \$0.00 |
| 1/2 1s Luan Ply      | 48.0000 | 96.0000 | 0.5000 | Yes   | \$0.00 |
| 1/2 1s Maple Ply     | 48.0000 | 96.0000 | 0.5000 | Yes   | \$0.00 |
| 1/2 1s Melamine      | 48.0000 | 96.0000 | 0.5000 | No    | \$0.00 |
| 1/2 1s Oak Ply       | 48.0000 | 96.0000 | 0.5000 | Yes   | \$0.00 |
| 1/2 1s Poplar Ply    | 48.0000 | 96.0000 | 0.5000 | Yes   | \$0.00 |
| 1/2 2s Alder Ply     | 48.0000 | 96.0000 | 0.5000 | Yes   | \$0.00 |
| 1/2 2s Ash Ply       | 48.0000 | 96.0000 | 0.5000 | Yes   | \$0.00 |
| 1/2 2s Birch Ply     | 48.0000 | 96.0000 | 0.5000 | Yes   | \$0.00 |
| 1/2 2s Cherry Ply    | 48.0000 | 96.0000 | 0.5000 | Yes   | \$0.00 |
| 1/2 2s Luan Ply      | 48.0000 | 96.0000 | 0.5000 | Yes   | \$0.00 |

| Tool                                     | Diam   | Length | Feed | Descent | Speed | Max Depth |
|------------------------------------------|--------|--------|------|---------|-------|-----------|
| <input type="checkbox"/> Saw             | 0.2500 | 2.0000 | 600  | 0       | 12000 | 0.0000    |
| <input type="checkbox"/> 5 Axis Saw      | 4.0000 | 2.0000 | 200  | 0       | 12000 | 3.0000    |
| <input type="checkbox"/> Lamello Saw     | 3.9528 | 1.8898 | 200  | 100     | 12000 | 3.0000    |
| <input type="checkbox"/> Test 5 Axis Saw | 4.0000 | 2.0000 | 200  | 0       | 12000 | 3.0000    |
| <input type="checkbox"/> 1/2 Down Shear  | 0.5000 | 2.0000 | 600  | 300     | 18000 | 3.0000    |
| <input type="checkbox"/> 1/4 Down Shear  | 0.2500 | 2.0000 | 600  | 300     | 18000 | 3.0000    |
| <input type="checkbox"/> 1/8 Down Shear  | 0.1250 | 2.0000 | 200  | 150     | 18000 | 0.2000    |
| <input type="checkbox"/> 3/8 Compression | 0.3750 | 3.0000 | 600  | 300     | 18000 | 1.0000    |
| <input type="checkbox"/> 3/8 Down Shear  | 0.3750 | 2.0000 | 600  | 300     | 18000 | 3.0000    |

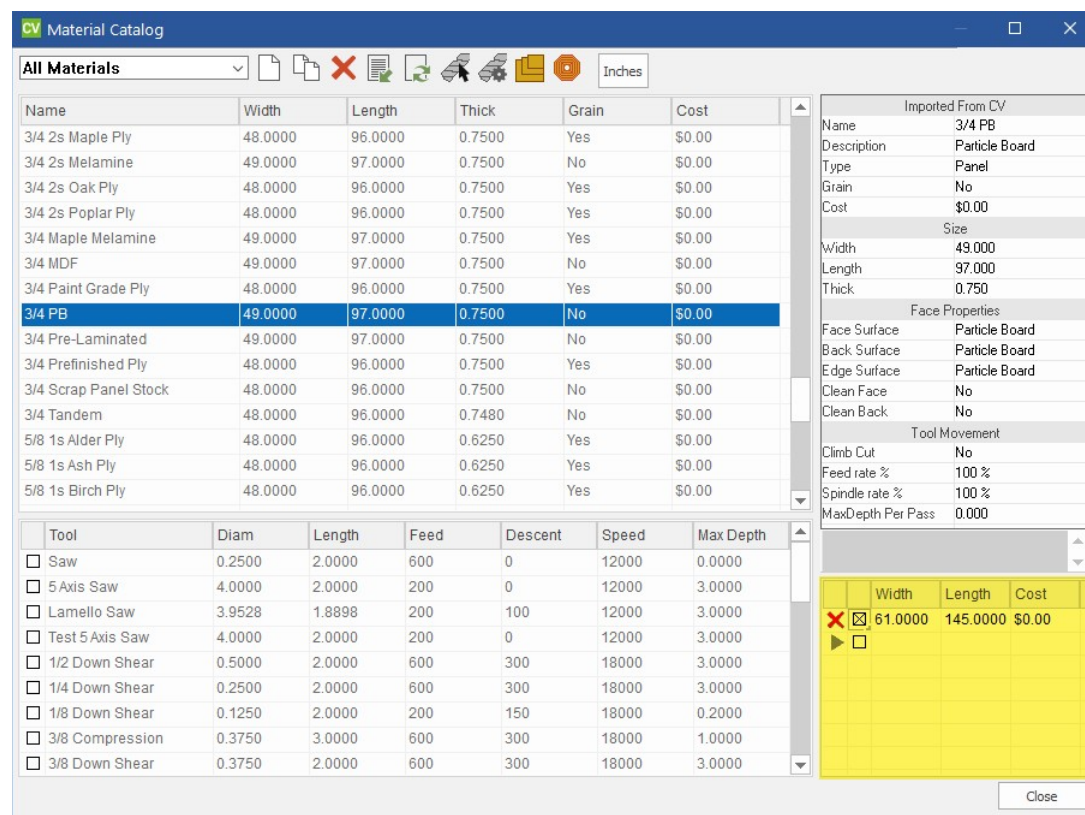
S2MCenter

|                   |                      |
|-------------------|----------------------|
| Name              | 1/2 1s Alder Ply (1) |
| Description       |                      |
| Type              | Panel                |
| Grain             | No                   |
| Cost              | \$0.00               |
| Size              |                      |
| Width             | 0.000                |
| Length            | 0.000                |
| Thick             | 0.000                |
| Face Properties   |                      |
| Face Surface      |                      |
| Back Surface      |                      |
| Edge Surface      |                      |
| Clean Face        | Yes                  |
| Clean Back        | Yes                  |
| Tool Movement     |                      |
| Climb Cut         | Yes                  |
| Feed rate %       | 100 %                |
| Spindle rate %    | 100 %                |
| MaxDepth Per Pass | 0.000                |

Close

Show Additional Material Sizes

This tool, see highlighted area in the figure below, allows you to specify additional sheet sizes for the selected Material.



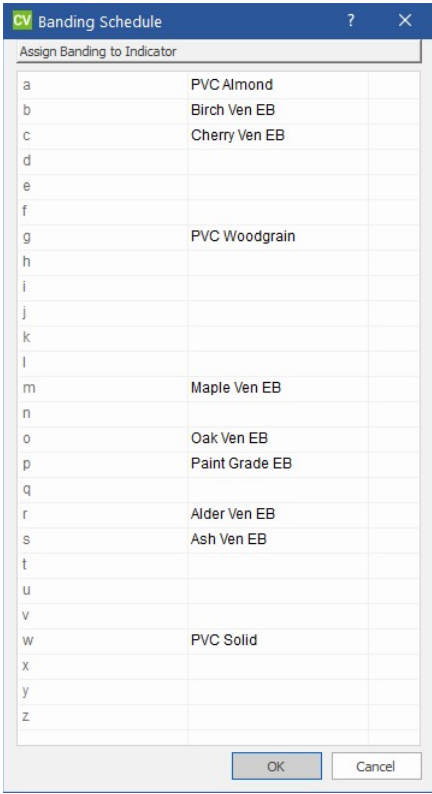
By clicking the green arrow to the left of the list you can add a new Sheet Size. Simply enter its Width, Length, and Cost and you are ready to go.

[illegible]

Enabling/Disabling using the option in column two determines whether the additional sheet size is available for Optimization/Nesting.

## Setup Banding Indicators

This option allows you to tie any Banding Material in your Material Catalog to a Banding Indicator.



Inches **Change Unit of Measurement**

This tool is a toggle button for whether you would like to work in millimeters or inches in the Material Catalog.



+Label Setup **Designing +Label Labels**

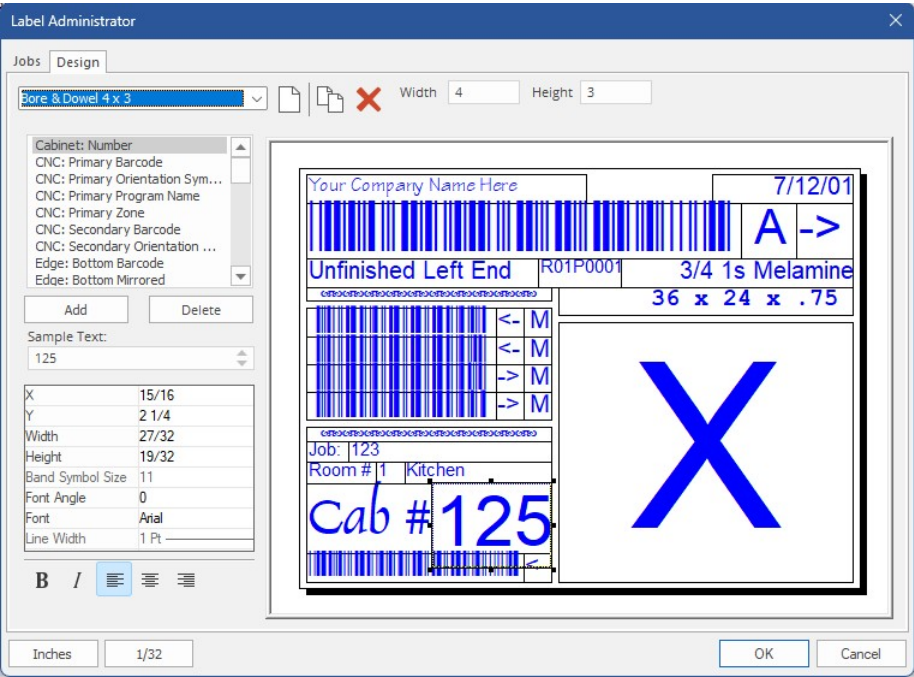


**Accessing the Label Designer**

To access the Label Designer follow these steps:

1. Click on **Utilities Tab | +Label Setup**.
2. In the Label Administrator click on the Design Tab.

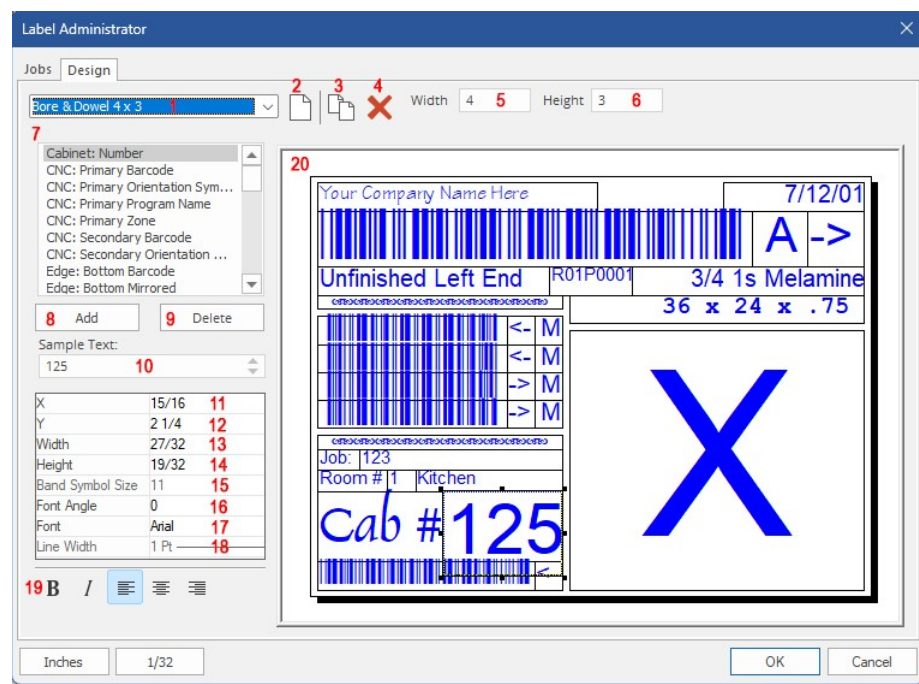




 Make sure that once you have created your Label Design that you go into the [Machine Catalog Setup](#) Tab for your Sheet Router/Saw and select this Label as the active Label.

**Label Designer Feature List**

The features of the Label Designer are denoted with the red numbers in the graphic below, followed by a brief description of each.

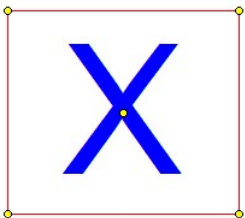


1. Name of Label currently active in the Label Designer.
2. Use this button to create a new Label.
3. Use this button to copy the active Label.
4. Use this button to delete the active Label. You will be prompted whether or not to delete the Label.
5. Use this field to define the Width of the active Label.
6. Use this field to define the Height of the active Label.
7. This list shows all the fields being utilized on the active Label.
8. Use this button to add a new field to the active Label. See the complete list of the fields available with a brief description of the fields content later in this topic.
9. Use this button to delete a selected field on the active Label.
10. This is where you can enter sample text to see what the Label will look like with actual data in place. This is also the area where you will enter permanent text for the text field.
11. This is where you enter the X (Left to Right) location of a selected field on the Label. This value references the top left corner of the field to the top left corner of the Label.
12. This is where you enter the Y (Up and Down) location of a selected field on the Label. This value references the top left corner of the field to the top left corner of the Label.
13. This is where you specify the width of a selected field.
14. This is where you specify the height of a selected field.
15. This is where you specify the Banding Symbol Size included in a Part Graphic.
16. This is where you set the Rotation of the selected field.

17. This is where you specify the Font for the selected field. Typically the 3 of 9 Barcode font is used for bar codes.
18. This is where you specify the line width for the Part: Graphic.
19. This set of tools are used to define text attributes. The following are available:
- 1. Bold
  - 2. Italic
  - 3. Justification (Left/Center/Right)
20. This is the workspace for the active Label.
-  The Unit of Measure buttons   allow you to establish the Unit of Measure and Precision used on +Label Output.

Field Modification Options

- The graphic below shows a highlighted field. There are different methods to move and size a field.
- 1. You can use the fields X and Y to move the field.
  - 2. You can use the fields Width and Height to stretch or shrink the field.
  - 3. Alternatively you may grab the center handle (Yellow Dot) to move the field or the (4) corner handles (Yellow Dots) to stretch or shrink the field.



Field Definitions

| Field Name                  | File Description                                                                                                  |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------|
| Text                        | Simply enter the desired text into the “Sample Text” box. (Text and/or <a href="#">Variables</a> )                |
| 2D Barcode Data Matrix      | This will generate a 2D Barcode based on what is entered in Sample Text. (Text and/or <a href="#">Variables</a> ) |
| 2D BarCode PDF417           | This will generate a 2D Barcode based on what is entered in Sample Text. (Text and/or <a href="#">Variables</a> ) |
| 2D BarCode QR ISO           | This will generate a 2D Barcode based on what is entered in Sample Text. (Text and/or <a href="#">Variables</a> ) |
| AC Custom Part Data 01 - 25 | Displays Custom Part Data as defined when using ALPHACAM to create the xLabel output                              |
| Banding: Code               | Displays the parts edge band code. (See below for definition)                                                     |
| Banding: Colors             | Displays the Banding Colors.                                                                                      |
| Banding: Colors + Materials | Displays the Banding Colors and Materials.                                                                        |
| Banding: Exterior Barcode   | Displays the Barcode for the Exterior Banding                                                                     |
| Banding: Flags              | Displays the parts edge band flags (See below for definition)                                                     |
| Banding: Internal Barcode   | Displays the Barcode for the Interior Banding                                                                     |
| Banding: List               | Displays all Banding used on the Part. See below for more information.                                            |
| Banding: Materials          | Displays the Band Materials.                                                                                      |
| Cabinet: Comment            | Displays the Cabinet Comment                                                                                      |
| Cabinet: Depth              | Displays the Cabinet Depth                                                                                        |
| Cabinet: Description        | Displays the Cabinet Description                                                                                  |

|                                          |                                                                                                          |
|------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Cabinet: Height                          | Displays the Cabinet Height                                                                              |
| Cabinet: Name                            | Displays the Cabinet Name                                                                                |
| Cabinet: Number                          | Displays the Cabinet Number                                                                              |
| Cabinet: Width                           | Displays the Cabinet Width                                                                               |
| CNC: Panel Layout Part Index             | Displays the Part's Index on the panel.                                                                  |
| CNC: Primary Barcode                     | Displays the primary CNC barcode (See below for definition)                                              |
| CNC: Primary Barcode Machine Tags        | Displays the tag for the primary barcode machine                                                         |
| CNC: Primary Face Up / Down              | Displays the primary Face Up / Down position                                                             |
| CNC: Primary Machine                     | Displays the machine that will perform the primary machining                                             |
| CNC: Primary Mirrored                    | Displays if the part is a mirrored part for the primary machine                                          |
| CNC: Primary Orientation Symbol          | Symbol Graphic displaying the orientation of the part on the primary machine. (See below for definition) |
| CNC: Primary Program Name                | Displays the primary program name                                                                        |
| CNC: Primary Program Name Machine Tags   | Displays the primary program name machine tags.                                                          |
| CNC: Primary Zone                        | Displays the zone in which to place the part on the primary machine                                      |
| CNC: Primary Zone Machine Tags           | Displays the tags for the primary zone machines..                                                        |
| CNC: Secondary Barcode                   | Displays the secondary CNC barcode (See below for definition)                                            |
| CNC: Secondary Barcode Machine Tags      | Displays the tag for the secondary barcode machine.                                                      |
| CNC: Secondary Face Up / Down            | Displays whether the part should be machined face up or face down.                                       |
| CNC: Secondary Machine                   | Displays the machine that will perform the secondary machining                                           |
| CNC: Secondary Mirrored                  | Displays if the part is a mirrored part for the secondary machine                                        |
| CNC: Secondary Orientation Symbol        | Graphic displaying the orientation of the part on the secondary machine. (See below for definition)      |
| CNC: Secondary Program Name              | Displays the secondary program name                                                                      |
| CNC: Secondary Program Name Machine Tags | Displays the secondary program name machine tags.                                                        |
| CNC: Secondary Zone                      | Displays the zone in which to place the part on the secondary machine                                    |
| CNC: Secondary Zone Machine Tags         | Displays the tags for the secondary zone machines.                                                       |
| CNC: Workflow                            | Displays the <a href="#">workflow string</a> for your Machine Set(s).                                    |
| Customer: Address                        | Displays the Customer Address                                                                            |
| Customer: City                           | Displays the Customer City                                                                               |
| Customer: Contact                        | Displays the Customer Contact                                                                            |
| Customer: Email                          | Displays the Customer Email                                                                              |
| Customer: Fax                            | Displays the Customer Fax number                                                                         |
| Customer: Mobile                         | Displays the Customer Mobile number                                                                      |
| Customer: Name                           | Displays the Customer Name                                                                               |
| Customer: Phone                          | Displays the Customer Phone number                                                                       |
| Customer: State                          | Displays the Customer State                                                                              |
| Customer: Zip                            | Displays the Customer Zip (Postal) Code                                                                  |
| CW Banding Edge1 Color                   | Indicates Edge Color #1 for Cabnetware Edgebanding                                                       |
| CW Banding Edge1 EdgeIndicator           | Edge Indicator for Edge #1 for Cabnetware Edgebanding                                                    |
| CW Banding Edge1 Indicator               | Indicates Edge #1 L Or W for Cabnetware Edgebanding                                                      |
| CW Banding Edge1 Material                | Indicates Edge Material #1 for Cabnetware Edgebanding                                                    |
| CW Banding Edge1 Thickness               | Indicates Thickness Color #1 for Cabnetware Edgebanding                                                  |
| CW Banding Edge2 Color                   | Indicates Edge Color #2 for Cabnetware Edgebanding                                                       |
| CW Banding Edge2 EdgeIndicator           | Edge Indicator for Edge #2 for Cabnetware Edgebanding                                                    |
| CW Banding Edge2 Indicator               | Indicates Edge #2 L Or W for Cabnetware Edgebanding                                                      |
| CW Banding Edge2 Material                | Indicates Edge Material #2 for Cabnetware Edgebanding                                                    |
| CW Banding Edge2 Thickness               | Indicates Thickness Color #2 for Cabnetware Edgebanding                                                  |
| CW Banding Edge3 Color                   | Indicates Edge Color #3 for Cabnetware Edgebanding                                                       |
| CW Banding Edge3 EdgeIndicator           | Edge Indicator for Edge #3 for Cabnetware Edgebanding                                                    |
| CW Banding Edge3 Indicator               | Indicates Edge #3 L Or W for Cabnetware Edgebanding                                                      |
| CW Banding Edge3 Material                | Indicates Edge Material #3 for Cabnetware Edgebanding                                                    |

|                                   |                                                                                                              |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------|
| CW Banding Edge3 Thickness        | Indicates Thickness Color #3 for Cabnetware Edgebanding                                                      |
| CW Banding Edge4 Color            | Indicates Edge Color #4 for Cabnetware Edgebanding                                                           |
| CW Banding Edge4 EdgeIndicator    | Edge Indicator for Edge #4 for Cabnetware Edgebanding                                                        |
| CW Banding Edge4 Indicator        | Indicates Edge #4 L Or W for Cabnetware Edgebanding                                                          |
| CW Banding Edge4 Material         | Indicates Edge Material #4 for Cabnetware Edgebanding                                                        |
| CW Banding Edge4 Thickness        | Indicates Thickness Color #4 for Cabnetware Edgebanding                                                      |
| CW Primary Edge Color             | Indicates Primary Edge Color for Cabnetware Edgebanding                                                      |
| CW Primary Edge EdgeIndicator     | Primary Edge EdgeIndicator for Cabnetware Edgebanding                                                        |
| CW Primary Edge Indicator         | Indicates Primary Edge L Or W for Cabnetware Edgebanding                                                     |
| CW Primary Edge Material          | Indicates Primary Edge Material for Cabnetware Edgebanding                                                   |
| CW Primary Edge Thickness         | Indicates Primary Edge Thickness for Cabnetware Edgebanding                                                  |
| Edge: Bottom Barcode              | Displays the program to run on the bottom edge of the part with a Drill & Dowel machine.                     |
| Edge: Bottom Barcode Machine Tags | Displays the machine tag for the program to run on the bottom edge of the part with a Drill & Dowel machine. |
| Edge: Bottom Mirrored             | Displays if the program for the bottom edge of the part is mirrored.                                         |
| Edge: Bottom Orientation Symbol   | Displays the orientation of the part in the Drill & Dowel machine to machine the bottom edge.                |
| Edge: Left Barcode                | Displays the program to run on the left edge of the part with a Drill & Dowel machine.                       |
| Edge: Left Barcode Machine Tags   | Displays the machine tag for the program to run on the left edge of the part with a Drill & Dowel machine.   |
| Edge: Left Mirrored               | Displays if the program for the left edge of the part is mirrored.                                           |
| Edge: Left Orientation Symbol     | Displays the orientation of the part in the Drill & Dowel machine to machine the left edge.                  |
| Edge: Right Barcode               | Displays the program to run on the right edge of the part with a Drill & Dowel machine.                      |
| Edge: Right Barcode Machine Tags  | Displays the machine tag for the program to run on the right edge of the part with a Drill & Dowel machine.  |
| Edge: Right Mirrored              | Displays if the program for the right edge of the part is mirrored.                                          |
| Edge: Right Orientation Symbol    | Displays the orientation of the part in the Drill & Dowel machine to machine the right edge.                 |
| Edge: Top Barcode                 | Displays the program to run on the top edge of the part with a Drill & Dowel machine.                        |
| Edge: Top Barcode Machine Tags    | Displays the machine tag for the program to run on the top edge of the part with a Drill & Dowel machine.    |
| Edge: Top Mirrored                | Displays if the program for the top edge of the part is mirrored.                                            |
| Edge: Top Orientation Symbol      | Displays the orientation of the part in the Drill & Dowel machine to machine the top edge.                   |
| Job: Comment                      | Any Job Comments you entered for the job will be reflected in this field                                     |
| Job: Company Name                 | Displays the Company Name                                                                                    |
| Job: CVJ File                     | Displays the name of the Cabinet Vision Job file used to generate the Label-IT output                        |
| Job: Delivery Date                | Displays the Delivery Date for the Job                                                                       |
| Job: Deposit Date                 | Displays the Deposit Date for the Job                                                                        |
| Job: Description                  | The Job description you entered for the job will be reflected in this field                                  |
| Job: Detail Date                  | Displays the Detail Date for the Job                                                                         |
| Job: Final Payment Date           | Displays the Final Payment Date for the Job                                                                  |
| Job: Install Date                 | Displays the Install Date for the Job                                                                        |
| Job: Measure Date                 | Displays the Measure Date for the Job                                                                        |
| Job: Name                         | The Name of the job, this information comes from the Job Number field in Job Properties                      |
| Job: PNC File                     | Displays the PNC File name for the Job                                                                       |
| Job: Production Date              | Displays the Production Date for the job                                                                     |
| Material: Back Texture            | Displays the Materials Back Texture                                                                          |
| Material: Cost                    | Displays the Materials Cost                                                                                  |
| Material: Edge Texture            | Displays the Materials Edge Texture                                                                          |
| Material: Face Dependent          | Displays the Materials Face Dependency                                                                       |
| Material: Face Texture            | Displays the Materials Face Texture                                                                          |
| Material: Feed Percent            | Displays the Materials Feed Percent                                                                          |
| Material: Length                  | Displays the Materials Length                                                                                |
| Material: Name                    | Displays the Materials Name                                                                                  |
| Material: Thick                   | Displays the Materials Thickness                                                                             |
| Material: Width                   | Displays the Materials Width                                                                                 |

|                                       |                                                                                                                                                   |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Part Size: Length                     | Displays the length of the part                                                                                                                   |
| Part Size: Length x Width             | Displays the length and width of the part. (36 x 24)                                                                                              |
| Part Size: Length x Width x Thickness | Displays the length, width, and thickness of the part. (36 x 24 x ¾)                                                                              |
| Part Size: Thickness                  | Displays the thickness of the part.                                                                                                               |
| Part Size: Width                      | Displays the width of the part                                                                                                                    |
| Part Size: Width x Length             | Displays the width and length of the part. (24 x 36)                                                                                              |
| Part Size: Width x Length x Thickness | Displays the width, length, and thickness of the part. (24 x 36 x ¾)                                                                              |
| Part: Back Finish                     | Displays the Part Back Finish                                                                                                                     |
| Part: Comment                         | Any Part Comments you entered for the part will be reflected in this field                                                                        |
| Part: Edge Finish                     | Displays the Part Edge Finish                                                                                                                     |
| Part: Face Finish                     | Displays the Part Face Finish                                                                                                                     |
| Part: Graphic                         | This field displays a Graphic image of the part including machining and edge banding instructions. (See below for details)                        |
| Part: Image Name                      | Displays the Part Image Name for Real-time Labeling                                                                                               |
| Part: Material                        | Displays the material the part is made of                                                                                                         |
| Part: Name                            | Displays the name of the part                                                                                                                     |
| Part: Position on Sheet               | Displays a graphic of the Nest pinpointing the location of the Part within it.                                                                    |
| Part: Short Name                      | Displays the short name of the part. (This name comes from the part catalog, this is the short descriptor used to ID parts in UCS's and formulas) |
| Room: Name                            | Displays the Room Name for the Part.                                                                                                              |
| Room: Number                          | Displays the Room Number for the Job.                                                                                                             |
| Time: Long Date                       | Displays the long date. (March 28, 2002)                                                                                                          |
| Time: Long Date + Time                | Displays the long date and time.(March 28, 2002 05:34 PM)                                                                                         |
| Time: Short Date                      | Displays the short date. (3/28/02)                                                                                                                |
| Time: Short Date + Time               | Displays the short date and time.(3/28/02 05:34 PM)                                                                                               |
| Time: Time                            | Displays the time. (05:34 PM)                                                                                                                     |
| Wall: Number                          | Displays the Wall Number                                                                                                                          |

Changes in Version 6

In Version 6 the Fields utilizing Banding Letter were removed to be replaced with Banding: List. The existing fields on your Labels were converted to new filter fields.

Banding: B Material is now {M}[Filter:B]

The new variables are:

|     |    |             |
|-----|----|-------------|
| {I} | Or | {Indicator} |
|     |    | {Tag}       |
| {C} | Or | {Color}     |
| {M} | Or | {Material}  |
| {T} | Or | {Thick}     |

Now you can format the banding field anyway you like using those variables:

I.e.: {T} – {M}: in the color of {C}

In addition you can add either a complex filter [Filter:E|I|D] or a Simple Filter [Filter:B]

This allows for superior flexibility and optimizing label real estate.

Specifying Edge Banding on your Label

Banding: Code

The following characters can and may be displayed in the Banding: Code:

|  |  |
|--|--|
|  |  |
|--|--|

|   |                  |
|---|------------------|
| L | Length           |
| W | Width            |
| E | Exterior Banding |
| I | Interior Banding |
| D | Door Banding     |
| A | Arc/Radius       |

Here are some examples of simple codes:

**1LE** - Band 1 Length with Exterior Band

**2LE1WI** - Band 2 Lengths with Exterior Band and 1 Width with Interior Band

For shaped parts when there is more than 1 side, you will see a dimension represented to indicate which edge gets banding applied.

Imagine a 34” end panel with a 4” toe notch:

**1LE301LI** - Band the 30" Length with Exterior and the Full Length with Interior

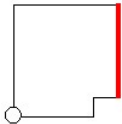
A part with a radius may look something like this:

**1WE241A6E** - Band the 24" Width and the 6" Arc/Radius with Exterior band

**Banding: Flag**

Another alternative to the Banding: Code is the Banding: Flag. The banding flag begins with the number of sides for a part followed by what each side will be banded with.

|     |                    |
|-----|--------------------|
| N   | No Banding         |
| E   | Exterior Banding   |
| I   | Interior Banding   |
| D   | Door Banding       |
| B-Z | Misc (B-Z) Banding |



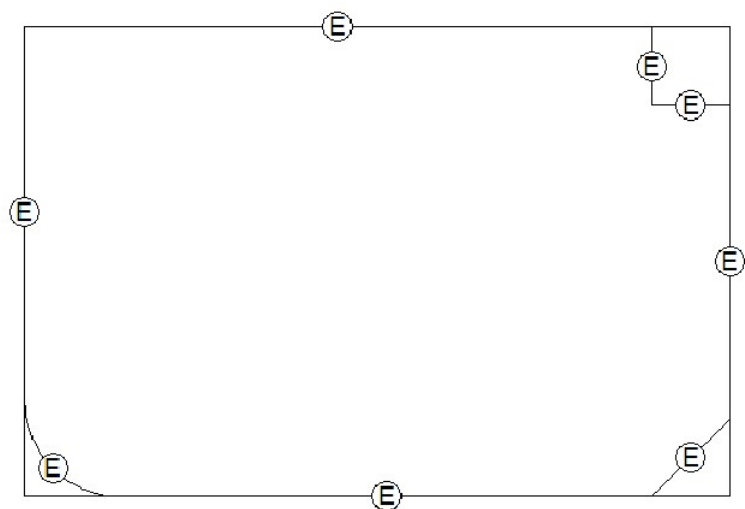
**6NNENNN** - In this sample part, starting from the reference point and going clockwise you get None, None, Exterior, None, None, None.

This type of coding can be quite confusing on complicated parts. The code and/or flag for the part below (24” x 36”) with an edited shape would be:

**CODE:** 1W19E1W5E1L27E1A5E1W16E1L4E1W4E

**FLAG:** 8EEEEEEEE

To simplify the edge banding instructions to machine operators, you can use the Part: Graphic field on your Label. Each edge of the part will be marked with either an E (Exterior), I (Interior), or D (Door), as demonstrated in the graphic below. With this method you would not need to display either the Banding: Code or the Banding: Flag fields.



### Primary & Secondary Programs

There is a difference in the definition of Primary and Secondary programs depending on the types of machinery you utilize.

#### Cell Based Manufacturing (Saw & Point-to-Point)

In the Cell Based Manufacturing model you cut parts to size on a Saw and then machine on your Point-to-Point. In this situation the Primary program is run first on the Back (Interior) side of the part. After this the Secondary program is run on the Face (Exterior) side of the part. This is fairly straightforward.

In Cell Based manufacturing you would need to specify both a Primary program/bar-code as well as a Secondary program/bar-code to include all machining necessary on your Label.

#### Nested Based Manufacturing

In the Nested Based Manufacturing model you typically utilize one machine to do all of your face machining operations. When nesting in the S2M CENTER the program number that is generated is for a whole sheet of parts (Sheet program) therefore no individual program number is generated for the Back (Interior) sides of the part. In this case the Primary program is for the Face (Exterior) side of the part. In this method of production there would be no Secondary program since both faces of the parts are being covered by the Sheet program and then by the Primary program.

In Nested Based manufacturing you would only need to specify a Primary program/bar-code to include all machining necessary on your Label.

#### Variables for Text and/or 2D BarCode PDF417 Fields

Here is a list of variables that can be used in the Text or 2D Barcode fields. To be evaluated properly the variables must be enclosed in brackets {}. This variables correspond to the Field definitions detailed earlier in this topic. If you are in a Text or 2D Barcode field you can type the opening bracket { and a list of these variables will be given to you to choose from.

{Job.CompanyName}

{Job.CustAddress}

{Job.CustCity}

{Job.CustContact}

{Job.CustEmail}

{Job.CustFax}

{Job.CustMobile}



{Job.CustName}  
{Job.CustPhone}  
{Job.CustState}  
{Job.CustZip}  
{Job.DateDelivery}  
{Job.DateDeposit}  
{Job.DateDetail}  
{Job.DateInstall}  
{Job.DateMeasure}  
{Job.DatePayment}  
{Job.DateProduction}  
{Job.PNC}  
{Material.BackTexture}  
{Material.Cost}  
{Material.Description}  
{Material.EdgeTexture}  
{Material.FaceDep}  
{Material.FaceTexture}  
{Material.FeedPercent}  
{Material.Length}  
{Material.Name}  
{Material.Thick.F4} - The Material's thickness, formatted as a 4 character fixed width field (i.e.  $\frac{3}{4}$ " = 0.75, 19mm = 19.0)  
{Material.Thick}  
{Material.Width}  
{Part.BackFinish}  
{Part.BandBColor}  
{Part.BandBMaterial}  
{Part.BandCColor}  
{Part.BandCMaterial}  
{Part.BandCode}  
{Part.BandedLength}  
{Part.BandedWidth}  
{Part.BandDoorColor}

{Part.BandDoorMaterial}  
{Part.BandExtBarcode}  
{Part.BandExtColor}  
{Part.BandExtMaterial}  
{Part.BandFColor}  
{Part.BandFlag}  
{Part.BandFMaterial}  
{Part.BandGColor}  
{Part.BandGMaterial}  
{Part.BandHColor}  
{Part.BandHMaterial}  
{Part.BandIntBarcode}  
{Part.BandIntColor}  
{Part.BandIntMaterial}  
{Part.BandJColor}  
{Part.BandJMaterial}  
{Part.BandKColor}  
{Part.BandKMaterial}  
{Part.BandLColor}  
{Part.BandLMaterial}  
{Part.BandMColor}  
{Part.BandMMaterial}  
{Part.BandOColor}  
{Part.BandOMaterial}  
{Part.BandPColor}  
{Part.BandPMaterial}  
{Part.BandQColor}  
{Part.BandQMaterial}  
{Part.BandSColor}  
{Part.BandSMaterial}  
{Part.BandTColor}  
{Part.BandTMaterial}  
{Part.BandUColor}

{Part.BandUMaterial}  
{Part.BandVColor}  
{Part.BandVMaterial}  
{Part.BandWColor}  
{Part.BandWMaterial}  
{Part.BandXColor}  
{Part.BandXMaterial}  
{Part.BandYColor}  
{Part.BandYMaterial}  
{Part.BandZColor}  
{Part.BandZMaterial}  
{Part.CabComment}  
{Part.Cab.Depth}  
{Part.Cab.Desc}  
{Part.CabHeight}  
{Part.CabName}  
{Part.CabNo}  
{Part.CabWidth}  
{Part.Comment}  
{Part.CutLength}  
{Part.CutWidth}  
{Part.EBBarcode}  
{Part.EBBarcodeTags}  
{Part.EdgeFinish}  
{Part.ELBarcode}  
{Part.ELBarcodeTags}  
{Part.ERBarcode}  
{Part.ERBarcodeTags}  
{Part.ETBarcode}  
{Part.ETBarcodeTags}  
{Part.FaceFinish}  
{Part.FaceUp}  
{Part.ID}

{Part.ImageName}

{Part.JobComment}

{Part.JobCVJ}

{Part.JobDesc}

{Part.JobID}

{Part.JobName}

{Part.JobRunNo}

{Part.Length.F7} - The part's length formatted as a 7 character fixed width field (i.e. 22 ½" = 022.500, 571.5mm = 0571.50)

{Part.Length}

{Part.Mirror}

{Part.Name.F3} - The part's name formatted as a 3 character fixed width field, which will be the first 3 characters of the part's description (i.e. Finished Left End = Fin) while offcuts will provide a value of '000'.

{Part.Name}

{Part.NominalLength}

{Part.NominalWidth}

{Part.Offcut}

{Part.PanelLayoutIndex.F2} - The Part's index on a panel, formatted as a 2 character fixed width field (i.e. 1 = 01, 12 = 12)

{Part.PanelLayoutIndex} - The part's index on a panel.

{Part.PBarcode}

{Part.PBarcodeTags}

{Part.PEdge}

{Part.PFaceUp}

{Part.PMachine}

{Part.PMirror}

{Part.PProgram}

{Part.PProgramTags}

{Part.PZone}

{Part.PZoneTags}

{Part.RoomName}

{Part.RoomNo}

{Part.Rotated}

{Part.SBarcode}

{Part.SBarcodeTags}

{Part.SFaceUp}

{Part.ShortName}

{Part.SMachine}

{Part.SMirror}

{Part.SProgram}

{Part.SProgramTags}

{Part.SZone}

{Part.SZoneTags}

{Part.WallNo}

{Part.Width.F7} - The part's width formatted as a 7 character fixed width field (i.e. 22 ½" = 022.500, 571.5mm = 0571.50)

{Part.Width}

{WorkOrder} - rGUID value in RunInfo table utilized by CV Paperless Traveller to identify the Work Order.

An example would be defining a 2D Barcode to call up a Biesseworks Parametric Template. For the Sample Text you would enter something similar to this (see Biesseworks documentation for specifics on the format).

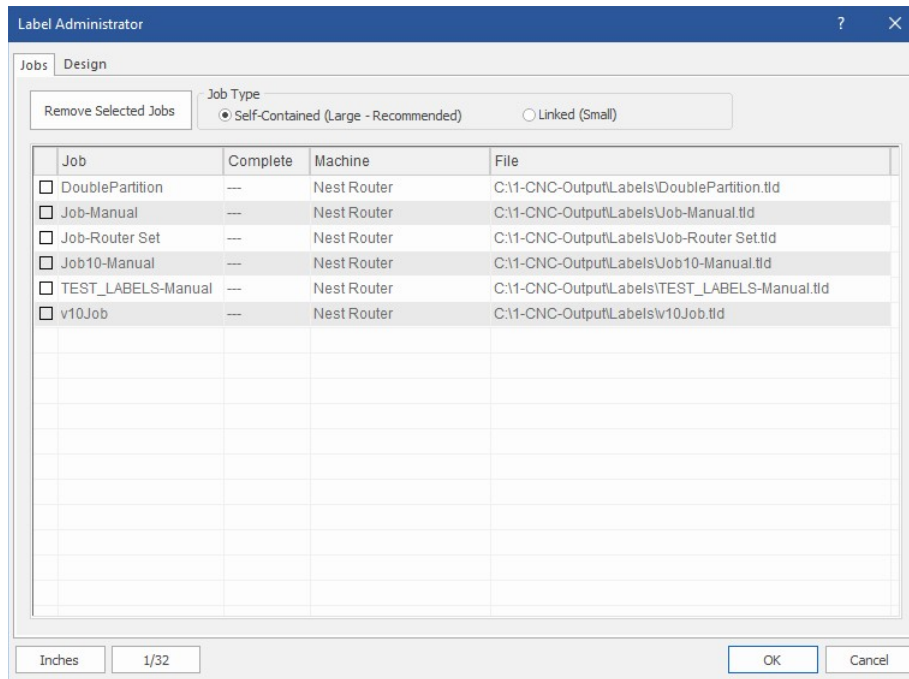
`{part.name}.bpp01CUSTSTR=1;1;100;100;;0;0$WIDTH={part.width} LENGTH={part.length}$`

## Generating +Label Output



+Label output is all generated on the S2M CENTER station. The process is outlined below:

1. Click on **Utilities Tab** | **+Label Setup**.
2. The first tab upon entering +Label Setup is Jobs. This screen shows currently active and inactive +Label jobs. You can remove any of these jobs from the system by simply clicking the check box beside the job and clicking the "Remove Selected Jobs" button.



 You can also select the Job Type from the Jobs Tab. The first option is "Linked (Small)" this +Label Job type requires the [psnc.mdb database](#) file be located in the same folder that you are outputting the +Label Job to. The second option "Self-Contained (Large)" creates a stand-alone +Label file that can be used on any +Label station and does not require a machine database for support. Although a bit larger it is the most stable +Label job file as it contains your Label designs and requires no linking back to the S2M CENTER.

 An additional capability is the ability to control the Unit of Measure for the +Label. Click the Unit of Measure buttons at the lower left to control this.

- Click OK to exit +Label Setup.
- Optimize your jobs as you normally would.
- Once on the Pattern screen, output your NC instructions as normal.
- Enable the "Output to +Label" option. Type in the name of the +Label job. You may want to give the job a name that will tie back to your job name or number. This is the name the +Label [operator](#) will select to start processing the job.

Output Data

Output Tasks

Manual

Create Output

Yes

Nest Router

C:\1-CNC-Output\Nest Router\

Parts Router

C:\1-CNC-Output\Parts Router\

Drill and Dowel

C:\1-CNC-Output\Drill\_Dowel\

Chop Saw

C:\1-CNC-Output\Chop Saw\

Other

No

Other

C:\1-CNC-Output\Hauncher\

Hoffman

No

Hoffman

C:\1-CNC-Output\Hauncher\

+Label

Output to +Label

Yes

+Label Job Name

Job-Manual

+Label Output Path

C:\1-CNC-Output\Labels

Label Images

Output to Label Images

No

Image Output Path

C:\1-CNC-Output\Labels

Save Job State

Create Output File

No

SNC File Name

Job 07-28-2020 1551

SNC Output Path

C:\1-CNC-Input\

Run

Cancel

7. Click on Run.
- You have now successfully output to the +Label Station. See the Topic on "[Using +Label in Production](#)" to learn how to access and use this output.

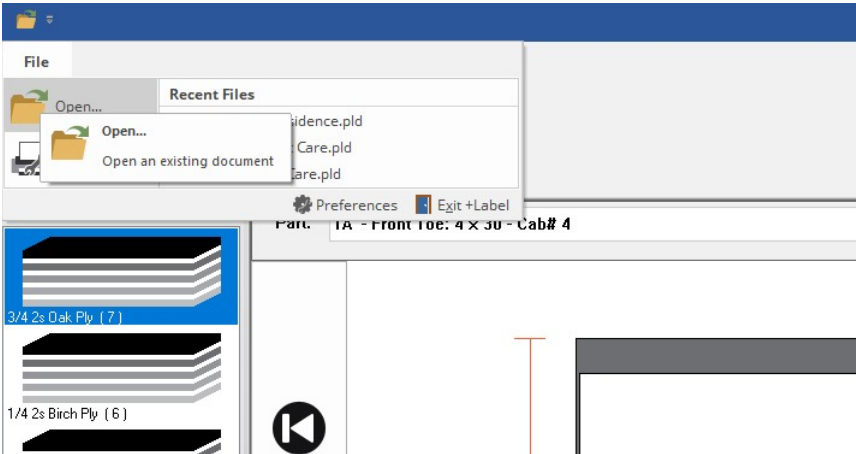
## Using the +Label App in Production



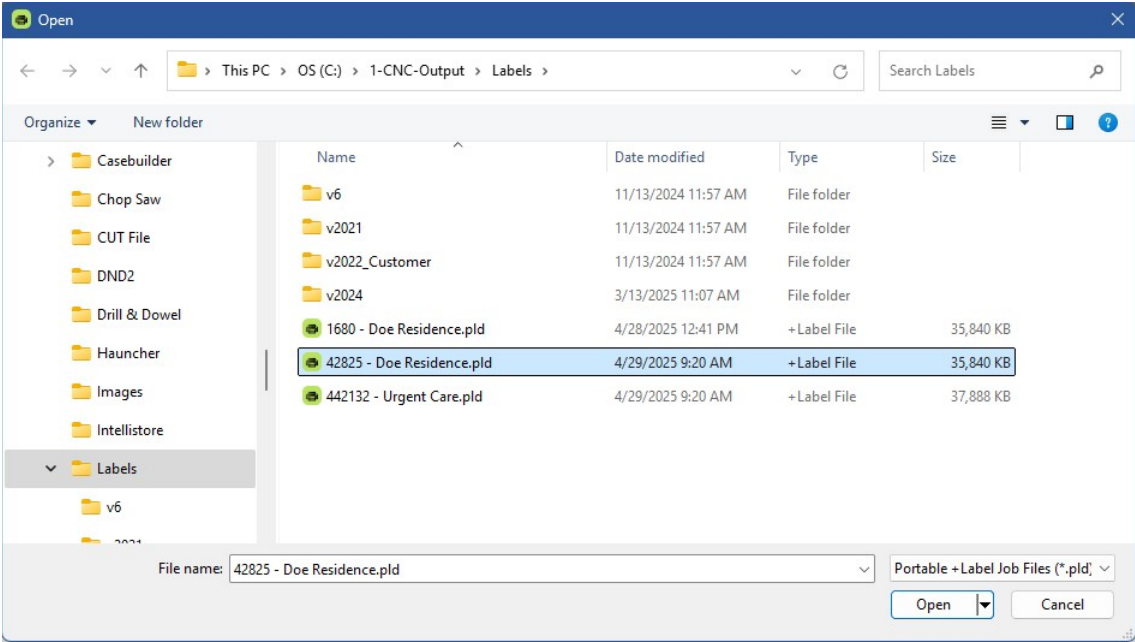
### Selecting a Job

Selecting a Job in +Label app is fairly straightforward:

1. Click on the File tab and click Open.



2. Browse the Job you want to open, select it then click Open.



The following File Types can be opened by the +Label App.

- Legacy +Label Job Files (\*.ltd)
- Linked +Label Job Files (\*.lld)
- Portable +Label Job Files (\*.pld)
- Completed Linked +Label Job Files (\*.lld.done)
- Completed Portable +Label Job Files (\*.pld.done)



- TLD - Legacy +Label Job files (Pre-2024.1)
- LLD - Linked +Label Job Files
- PLD - Portable +Label Job Files
- LLD.done - Completed Linked +Label Job Files
- PLD.done - Completed Portable +Label Job Files



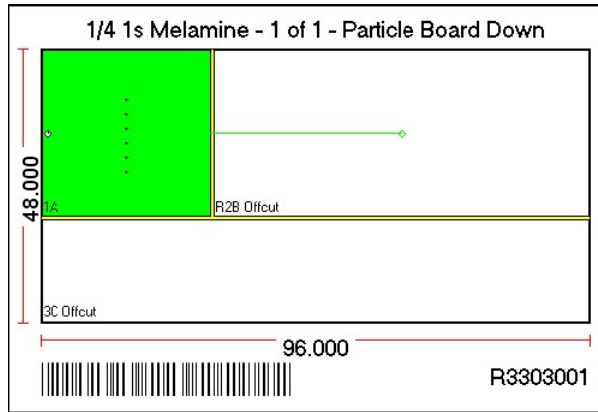
You can open a specific +Label job via command line as follows. c:\Program Files\Hexagon\CABINET VISION\S2M 2025\pnclabelit.exe "C:\Users\me\Desktop\Banding.pld"

## +Label App Feature List



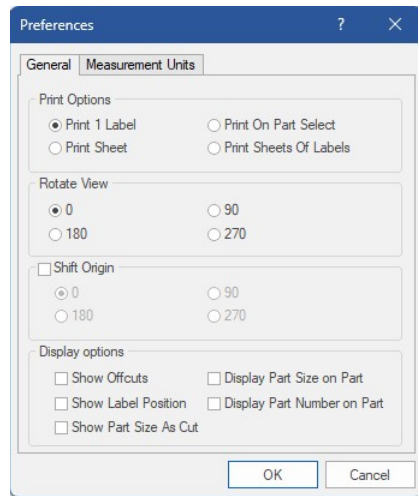
1. Name of the Job currently open in the +Label app.
2. Clicking the **View** option allows you to View the Label for the currently selected Part.
3. **Laser** - Turns on/off the Laser Pattern Generator. See the [Running A Laser](#) topic for more information.
4. **Mark Job Complete** - This option does two things:
  - a. When selected this option will remove the Job from being visible at the Label-IT station.
  - b. This option at the same time will mark the Label-IT job Done on the S2M CENTER station in the front office.
5. **1 Label** - Prints the selected label.
6. **On Part Select** - Prints the selected label.
7. **Sheet** - Will print Labels for all Parts on the active Sheet.
8. **Sheet of Labels** - Will print sheets of labels for the entire job.
9. Patterns included in the selected Job.
10. Current selected part.
  - a. Selected parts that have not had labels printed for them are highlighted in **green**.
  - b. Parts that are not selected and that have not had labels printed for them are white.
  - c. Selected parts that have had labels printed for them are highlighted **red**.
  - d. Parts that are not selected that have had labels printed for them are highlighted in **blue**.
  - e. This field also has a drop down arrow which will show you the Parts below the current Part if you are Booking (Stack Cuts) using a Panel Saw.
11. The Composites button  allows you to view either the sub-part or the composite part. The icon will indicate the current view. Any time a Sub-Part label is printed the composite part label will be printed as well. The button will be disabled if the selected Part is not part of a Composite.

12. Selected Patterns material. This text will also indicate how to "load" one sided Materials on the machine like the [Pattern Print](#) outs from the Pattern area.



13. Selected Sheet's location and quantity.
14. Clicking the |< button moves you to the first Sheet/Label in the selected Pattern.
15. Clicking the >| button moves you to the last Sheet/Label in the selected Pattern.
16. Clicking the < button moves you to the previous Sheet/Label in the selected Pattern. You can also simply swipe right for the previous Part/Label if you have touch screen.
17. Clicking the > button moves you to the next Sheet/Label in the selected Pattern. You can also simply swipe left for the next Part/Label if you have touch screen.
18. Clicking the **Print** option prints the Label or Sheet, depending on your Preference, for the currently selected part.
19. Sheet program bar-code.
20. Sheet program name.
- a. This name matches the name of the program generated by Intelli-CAM for the Router.

## Preferences

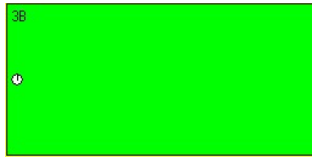


Following are the options in Preferences:

- **Print Options** - Allows you to choose how you want to Print. You have the following options:
  - **Print 1 Label** - If you have a Part selected clicking the Print option will print its Label.
  - **Print On Part Select** - Simply selecting a Part will print its Label.
  - **Print Sheet** - Will print Labels for all Parts on the active Sheet when clicking the Print option.
  - **Print Sheets of Labels** - Will print sheets of labels for the entire job.
- **Rotate View** - Allows you to Rotate the Pattern view to match the Orientation of your Machine.
- **Shift Origin** - Allows you to shift the Origin of your Parts on the Nest. Utilizing Rotate View and Shift Origin in conjunction will allow you to exactly match the Patterns output from the S2M CENTER.
- **Display options** - Allows you to choose extra items you may want displayed on-screen. You have the following options:
  - **Show Offcuts** - A toggle to Hide or Show Offcuts.
  - **Display Part Size on Part**
  - **Show Label Position** - This option will put the [Label position symbol](#) as defined in S2M CENTER Preferences on the Pattern View as well as the Part Graphic on the Label; if your Label is designed to display the Part Graphic.
  - **Display Part Number on Part**
  - **Show Part As Cut**

## Label Placement

Placing your Labels oriented properly on the part is extremely important in your success with Label-IT. There is a built in feature to help insure that you orient the Label properly onto your part. The graphic below shows a part with a small dot. This dot represents how the Label needs to be oriented on the part. If the Label is placed properly on the part then the machine operators can look at simple graphics on the Label that will instruct them on how to orient the part on the machine as well as which edges to edge band and which edges receive boring on your Drill & Dowel machine. The small line, within the circle, points to the top of the Label.



The Label Placement denotation is only displayed on the Label-IT stations screen. Proper Label Placement is important on Parts that receive Secondary operations so the Parts can be oriented correctly on the Secondary Machine. This means it is important that the person responsible for Label Placement has the Label-IT station in his work area.

There are four possibilities:



**Label Up** - Label would be oriented on the part as follows

|                                                                                            |                |        |
|--------------------------------------------------------------------------------------------|----------------|--------|
| Job Name:                                                                                  | Job            | #1     |
| Room:                                                                                      | Room 1         |        |
| Material:                                                                                  | 3/4 Oak Ply B2 |        |
| 22 31/32 x 28 1/2                                                                          |                |        |
| Drill & Dowel Left Edge                                                                    |                | Bottom |
|           |                |        |
| Drill & Dowel Right Edge                                                                   |                |        |
|           |                |        |
| Flip Side 0 r30m0004                                                                       |                |        |
| LabelIT  |                |        |



**Label Down** - Label would be oriented on the part as follows:



Label Left - Label would be oriented on the part as follows:

Job Name: Job

Room: Room 1

Material: 3/4 Oak Ply B2

#1

22 31/32 x 28 1/2

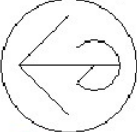
Bottom

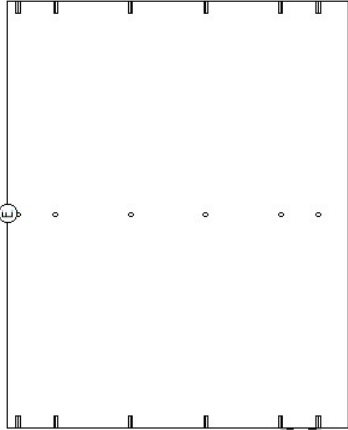
Drill & Dowel Left Edge

Drill & Dowel Right Edge

Flip Side 0 r30m0004

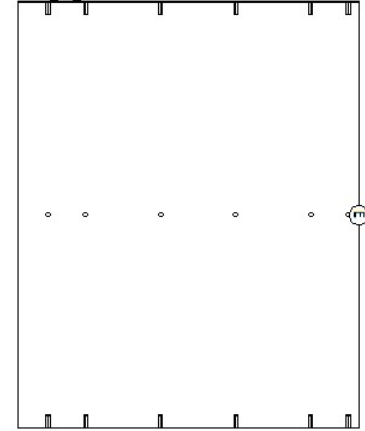
LabelIT





 Label Right - Label would be oriented on the part as follows:

|                          |                   |          |
|--------------------------|-------------------|----------|
| Job Name:                | Job               | #        |
| Room:                    | Room 1            | 1        |
| Material:                | 3/4 Oak Ply B2    |          |
|                          | 22 31/32 X 28 1/2 |          |
| Drill & Dowel Left Edge  |                   | Bottom   |
| Drill & Dowel Right Edge |                   |          |
| Flip Side                | 0                 | r30m0004 |
| Label IT                 |                   |          |

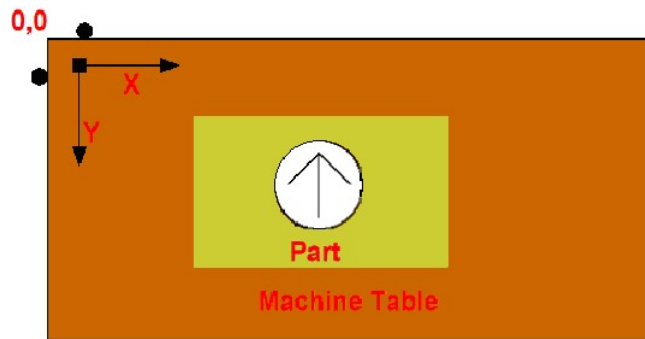


The diagram shows a rectangular machine table with a grid of pins. A part is shown on the table, with a circled arrow pointing to the back placement pins. The part is labeled 'Bottom'.

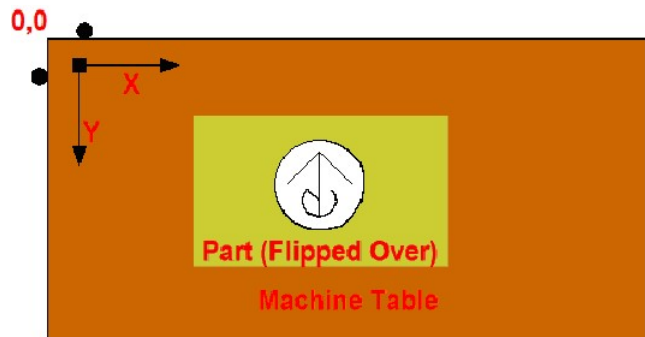
### Orientation Symbols

Orientation symbols are extremely useful in informing the machine [operator](#) how to orient Part on your CNC/P2P Machine. There are two different symbols utilized to express the parts orientation.

The symbol below (Black & White Encircled Arrow) points to the back placement pins on the machine, typically along the X-axis. The below graphic representing the machine table and a part demonstrates this.



The only difference between the (2) symbols is that one below instructs you to flip the part over prior to placing it on the machine.



Additional Key-words: Metric, Imperial, MM , Units, Measure, Inches

## Running A Laser



There are several Laser Pattern Generators on the market such as Carter Laser Products, Z-Laser, and SL Laser.

These Lasers can display the Nest Pattern on your sheet stockpile so that Labels can be pre-applied to your sheet stock before loading into your machine for processing. All of these use an HPGL file. To generate HPGL output in the form of the file Laser.hgl for the Laser to read you will need to add a String Value in the Registry.

```
[HKEY_CURRENT_USER\SOFTWARE\Hexagon\CABINET VISION\S2M 2025\Label IT]
LaserRun="C:\Cabinet Vision\RunLaser.bat"
```



Make sure to backup your Windows Registry prior to making changes. It is highly recommended to not modify your Windows Registry unless you are familiar with the process. Windows and/or S2M CENTER corruption could occur by making an invalid entry.

You can name the bat file anything you like. In the bat file you will need to add code to initialize the laser. This is typically one or two lines of code but since it is different between lasers it is not hard coded. Also in the bat file you will want to make reference to the "Laser.HGL" file located in the \S2M 2025 directory. Below is a sample BAT file:

```
MODE COM1:9600,N,8,1,P
```



```
copy c:\Cabinet Vision\S2M 2025\laser.hgl com1
```

When you select the option Turn Laser ON, the program will create the Laser.HGL file then call then RunLaser.bat file. The Laser.HGL file is output whenever you navigate to a new sheet. At this point it will draw all the part outlines on the sheet. It is also reset every time you print a label. In this case it shows just the outline of the part when the label is to be applied. The concept is you can print the labels one at a time before the sheet is milled then start the router.

## Real-Time Labeling

When using Real-Time Labeling on your Machine (On Board Machine Labeling Solution). Following are the available Label Applicator Machines:

### None

Use this option if you are not driving the Label Applicator from the S2M CENTER.

### Anderson

A proprietary output produced for Anderson Machines. The Anderson output consists of the following elements:

1. The NC Code for the nest
2. An \*.lpf file for each NC Code file. This indicates the label position, based on the center of the label for each part and a rotation flag (1=0, 2=90, 3=180, 4=270). The Anderson machine has the ability to rotate the label 90 degrees so all labels images are output rotated either 0 or 180. The rotation flag then instructs the machine if it should rotate an additional 90 to cover the 90 and 270.
3. An \*.csv file for each NC Code file. The \*.csv file contains additional part information that cab be place on the label. Look to the .CSV File section of this topic for a field map.
4. An \*.zpl file for each part that contains the part graphic.

The \*.anc (Nc Code) \*.lpf and \*.csv file should all have the same file name. For example: r0201001.arb, r0201001.lpf, r0201001.csv

### Biesse

A proprietary output produced for Biesse Machines. The Biesse output is much simpler than the Anderson. Unlike the Anderson there is not additional part data that needs to be output; they will rely completely on the Label IT image.

The .arb (Biesse Nc Code Files) will be modified after they are processed by adding an addition section for the label position and zpl names as follows:

```
[CUSTOMSEZ01]
<LABEL>
PEZZO=1
X=100
Y=200
STRCUST=C:\LBL\
FILERIS=R01P0001.ZPL
PEZZO=2
X=200
Y=200
STRCUST=C:\LBL\
FILERIS= R01P0002.ZPL
</LABEL>
```

%

[]

In addition the [PARAMETRI01] section will need to be modified to include:

```
ETC=1 KE=1024
```

#### **Excerpt from Biesse Setup Documentation**

ETC parameter must be set to 1 when the panel labeling must be performed using the information written in section CUSTOMSEZ01. If the parameter is set to 0, the PC will ignore the information contained in this section, even if it contains some values. When the **ETC** parameter is set to **1**, it affects the value **KE**, which changes from **0** to **1024**. Section **CUSTOMSEZ01** contains the information regarding the labeling operation. This section must starts with <LABEL> and ends with </LABEL>. Each label to be sticked must be described in the program and the description must contain the following information:

```
PIECE
```

```
X
```

```
Y
```

```
STRCUST
```

```
FILERIS
```

The field **PIECE** is a progressive number of the label, i.e.: PEZZO=1 Field **X** is the position, along the X axis, where the label must be sticked (its number is in the **PIECE** field). It refers to the reference point of the applicator (the offset set in the field max\_lung of the tool is added and shows the label used), for example: X=123,75 Field **Y** is the position, along the Y axis, where the label must be sticked (its number is in the **PIECE** field). It refers to the reference point of the applicator (the offset set in the field max\_diam of the tool is added and shows the label used), for example: Y=361,84 The field **STRCUST** must show the complete path where the **watcher** will find the label file to print, i.e.: STRCUST=C:\ETC\ The field **FILERIS** must show the label file name that the watcher will find to print, i.e.: STRCUST=LABEL01.PRN

## **Biesse CIX**

A proprietary output produced for Biesse Machines using CIX.

## **Robotic Automation**

A proprietary output produced for Robotic Automation Machines.

## **S2M Labeling**

A generic format exported by the S2M CENTER format which can be used by some machines. Following are the specifications for the S2M Labeling Output.

There are 3 files output using this solution:

### **.LPT**

This is a reinterpretation of the data in the .PLP (which is essentially raw label position) to the position based on the machine's Point of Origin. For that reason the .LPT file is probably more useful than the .PLP file since no data needs to be transformed.

### **.PLP**

The .PLP is a self-documented text file. Here is a sample:

```
Planit Label Positions - Version 3.0 Format: PanelLayoutID, PartID, XPos, YPos, Rotation x 90
```

```
1,8,26.7248031496063,-6.250000000000003,0
```

```
3,36,55.4496061496063,-6.250000000000003,0
```

```
2,35,26.7248031496063,-6.250000000000003,0
```

.CSV

|    |                      |    |                    |    |                  |
|----|----------------------|----|--------------------|----|------------------|
| 1  | PartIndex            | 32 | PartHorMachineID   | 63 | CustAddress      |
| 2  | PartID               | 33 | PartCutMachineID   | 64 | CustCity         |
| 3  | PartQty              | 34 | PartHasBanding     | 65 | CustState        |
| 4  | PartName             | 35 | PartBanding        | 66 | CustZip          |
| 5  | PartInternalName     | 36 | PartBandingCode    | 67 | CustContact      |
| 6  | PartWidthString      | 37 | PartBandingFlags   | 68 | CustEmail        |
| 7  | PartLengthString     | 38 | PartIntBandBarCode | 69 | CustFax          |
| 8  | PartWidth            | 39 | PartExtBandBarCode | 70 | CustPhone        |
| 9  | PartLength           | 40 | PartFinishFace     | 71 | CustMobile       |
| 10 | PartZplName(0,90)    | 41 | PartFinishBack     | 72 | DateMeasure      |
| 11 | PartZplName(180,270) | 42 | PartFinishEdge     | 73 | DateDetail       |
| 12 | PartComment          | 43 | PartImageName      | 74 | DateProduction   |
| 13 | PartPrimaryEdge      | 44 | RoomID             | 75 | DateDelivery     |
| 14 | PartMirror           | 45 | RoomName           | 76 | DateDeposit      |
| 15 | PartFrProgram        | 46 | AssemblyID         | 77 | DateInstall      |
| 16 | PartBaProgram        | 47 | AssemblyNumber     | 78 | DateFinalPayment |
| 17 | PartFrMirror         | 48 | AssemblyName       | 79 | MaterialID       |
| 18 | PartBaMirror         | 49 | AssemblyWidth      | 80 | MatDBID          |
| 19 | PartFrBarCode        | 50 | AssemblyHeight     | 81 | MatPatNo         |
| 20 | PartBaBarCode        | 51 | AssemblyDepth      | 82 | MatPartIndex     |
|    |                      |    |                    |    |                  |

|    |                    |    |                     |    |                |
|----|--------------------|----|---------------------|----|----------------|
| 21 | PartFrWorkZone     | 52 | AssemblyWallNo      | 83 | MatPatIndex    |
| 22 | PartBaWorkZone     | 53 | AssemblyClass       | 84 | MatName        |
| 23 | PartLeBarCode      | 54 | AssemblyType        | 85 | MatWidth       |
| 24 | PartRiBarCode      | 55 | AssemblyDescription | 86 | MatLength      |
| 25 | PartToBarCode      | 56 | AssemblyComment     | 87 | MatThick       |
| 26 | PartBoBarCode      | 57 | JobID               | 88 | MatGrain       |
| 27 | PartShaped         | 58 | JobName             | 89 | MatTextureFace |
| 28 | PartGangBore       | 59 | JobDescription      | 90 | MatTextureBack |
| 29 | PartHorizontalBore | 60 | JobComments         | 91 | MatTextureEdge |
| 30 | PartPriMachineID   | 61 | JobFilePath         |    |                |
| 31 | PartSecMachineID   | 62 | CustName            |    |                |

## .ZPL File

A .zpl file is basically NC Code for a Zebra printer. It has commands such as move to this X,Y point, and print “Job Name” with this font. Move to this X,Y point, draw a line 60 points wide...

One element of a ZPL file is an image. An image imbedded within a ZPL file indicates a block of points then indicates which points in that block will be black. We have observed 4 different methods for describing the image. The image format can be set/modified in the registry with the following settings:

HKEY\_CURRENT\_USER\Software\Cabinet Vision\NcCenter\_12\CncOutput

**ZPL\_Format** - Valid values are “Anderson”, “Andi”, “Biesse”, 0,1,2,3

If the integrator needs to modify the ZPL\_Format he should set the ZPL\_Format to -1 then use the following settings:

**ZPL\_AllowCount** - Do not use the raw format, use the count method to compress file size

**ZPL\_AllowComma** - Use a comma to separate each line of the ZPL file.

**ZPL\_AllowColon** - Used with allow comma, use a colon to indicate a repeat of the previous line.

**ZPL\_AllowCRLF** - Use a carriage return, line feed after each line of the ZPL file.

**ZPL\_AllowRotate** - Rotate the image 0, 90, 180 or 270 degrees to match the nested part.

## Template.ZPL

Since the ZPL file is not just the image it is the full set of instructions to the zebra printer the you can setup a template to embed your graphic data into your existing file. If a template.zpl file is found in the output directory, it will be used in the following manner:

```

^FX Output from Planit ^FS
[IMAGE.DATA]
^XA
^DF[IMAGE.NAME]^FS
^PRC
...
^XZ
^FX End of job
^XA
^XF[IMAGE.NAME]
^XZ

```

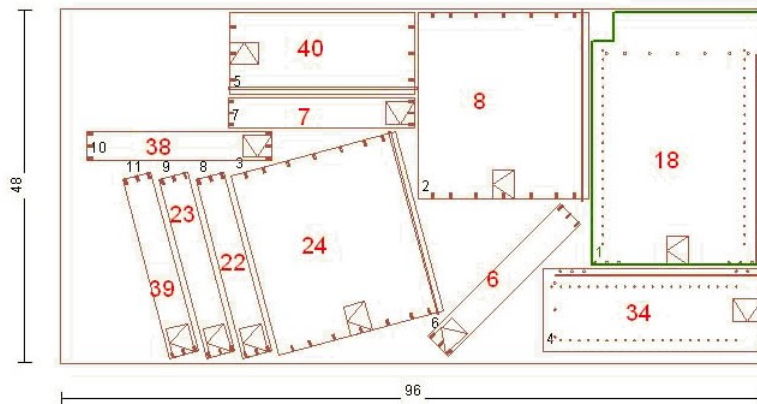
The highlighted areas will be replaced with the image data from the S2M CENTER while the rest of the ZPL file is untouched.



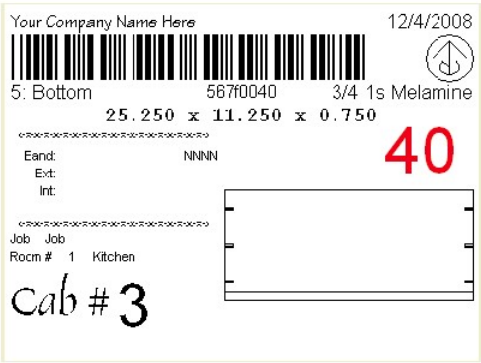
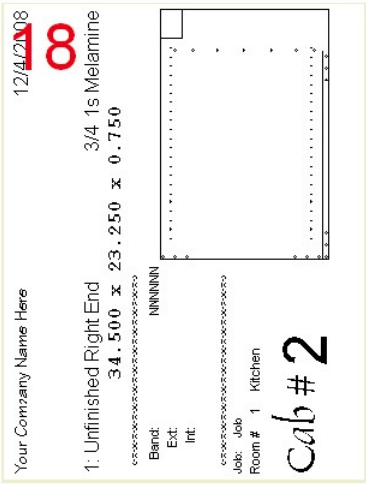
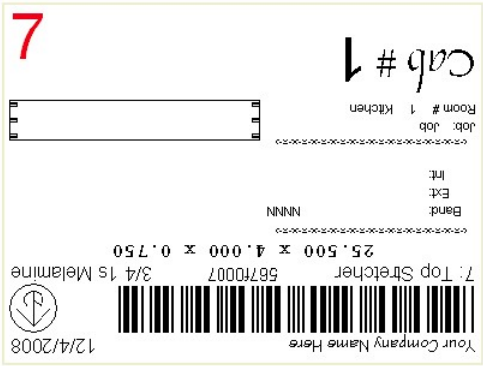
For rotated ZPL output the program will look for the template0.zpl, template90.zpl, template180.zpl, or template270.zpl. This will allow for different label layouts for rotated output. If the rotated template cannot be found it will look for the template.zpl.

## Rotated Images

3/4 1s Melamine - 1 of 1 56704001



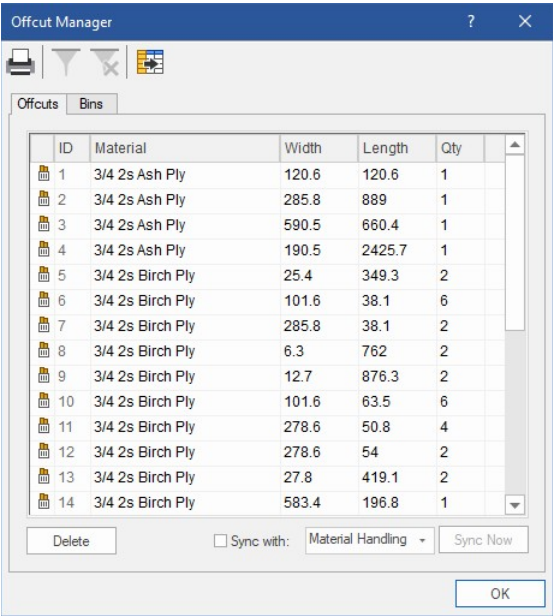
Notice the sample labels printed for the nest above are pre-dash;rotated to match the part rotation:



### Offcut Manager

The Offcut Manager is used to manage the reuse of Offcuts. Offcuts are considered any scrap/waste from an Optimized Sheet, equal to or larger than the size set in the [Offcuts](#) Tab in Optimizer Parameters or the [Offcuts](#) Tab in Material Properties.

### Offcuts Tab



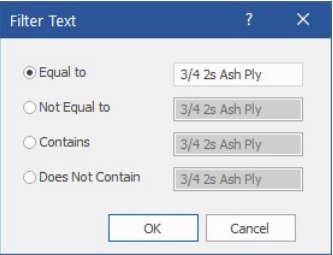
The Offcuts Tab display any Offcuts available for use in Jobs.

If you are using a Material Handling System you can also select to Sync with it here.

Following are the tools available in the toolbar:



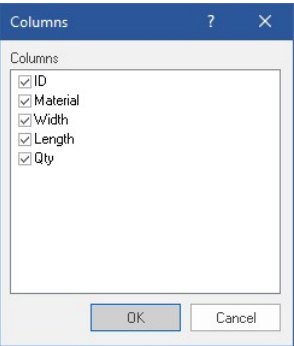
- Print** - Will Print the list of Offcuts.
- Filter** - Will allow you to Filter the Offcuts based on the field you select (Material, Width, Length, Qty) on the following conditions:



- Equal to
- Not Equal to
- Contains
- Does Not Contain

**Remove Filter** - Will Remove the active Filter to show all Offcuts.

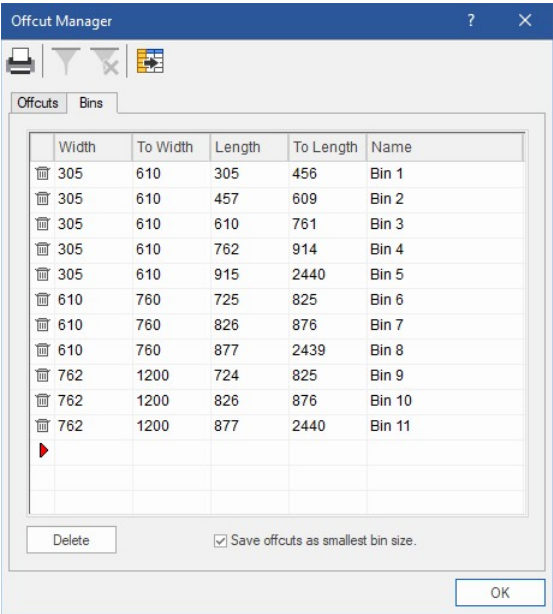
**Set Columns** - Allows you to enable/disable and reorganize the columns.



In addition to controlling which items display you can also control the order in which the columns appear. Simply click and drag the item you want to move into its new position.

Bins Tab

 In previous Versions of the S2M CENTER it was a requirement to setup Bins for Offcuts. This is now not a requirement; all Offcuts, equal to or larger than the size set in the [Offcuts](#) Tab in Optimizer Parameters or the [Offcuts](#) Tab in Material Properties will now be stored when utilizing Update Offcuts from the Patterns View.



The best way to explain Bins in the Offcut Manager is by example:

I would like to save any Off Cuts that I can get Standard Ends out of. Below are my standard End Sizes:



| Upper Cabinets | Standard Base Cabinets | Deep Base Cabinets |
|----------------|------------------------|--------------------|
| 300mm x 300mm  | 610mm x 725mm          | 760mm x 725mm      |
| 300mm x 457mm  | 610mm x 825mm          | 760mm x 825mm      |
| 300mm x 610mm  | 610mm x 875mm          | 760mm x 875mm      |
| 300mm x 760mm  |                        |                    |
| 300mm x 915mm  |                        |                    |

To save each of these sizes I would setup my Offcut Bins as below:

| Part Size | Length | To Length | Width | To Width |
|-----------|--------|-----------|-------|----------|
| 300 x 300 | 300    | 456       | 300   | 609      |
| 300 x 457 | 457    | 609       | 300   | 609      |
| 300 x 610 | 610    | 759       | 300   | 609      |
| 300 x 760 | 760    | 914       | 300   | 609      |
| 300 x 915 | 915    | 2440      | 300   | 609      |
| 610 x 725 | 725    | 824       | 610   | 759      |
| 610 x 825 | 825    | 874       | 610   | 759      |
| 610 x 875 | 875    | 2440      | 610   | 759      |
| 760 x 725 | 725    | 824       | 760   | 1220     |
| 760 x 825 | 825    | 874       | 760   | 1220     |
| 760 x 875 | 875    | 2440      | 760   | 1220     |

If we look at the first bin it is setup to capture Offcuts that I can cut my 300 x 300 Upper ends from. The way the Bin is setup with any Offcuts that are sized between 300 x 300 to 609 x 456 it is going to assign for this Bin. Basically if I go to this Bin it is going to have parts that range in a size that I can get my 300 x 300 parts out of. The parts in this bin maybe larger, say 575 x 400, but the rule I am trying to establish is that it can be utilized in a future Optimization to get a 300 x 300 Upper end panel out of.

The bins are setup to contain Offcuts of at least the minimum specified size - in other words they are used to group Offcuts into similar sized panel stock. If you have a number of Offcuts in a bin and want to add them to the optimization, the smallest size of the bin dimension is used for the material size. The bin is basically saying "I have these pieces of material that are at least this size". When you add Offcuts to the bins, it does not store the size of each part that gets added to the bin, it just increases a quantity of parts that are in a particular bin. So when adding the parts back to the Optimizer it uses the smallest dimensions of the bin. In other words a bin is a set of parts that are AT LEAST THIS SIZE even though some may have originally been bigger.

You can also Print and Set Columns on the Bins tab.



## Setup Package

The Setup Package utility is a means to transfer specific objects from one computer running S2M CENTER to another.

### Exporting a Setup Package

The Setup Package feature allows you to create a single file that can store and transfer multiple custom items for import into another S2M CENTER user's or S2M CENTER Tech's system.

The following list of items can be added to a package for transfer:

**Labels (Label-IT Labels or Saw Graphic designs only)**

**Machines**

**Tools**

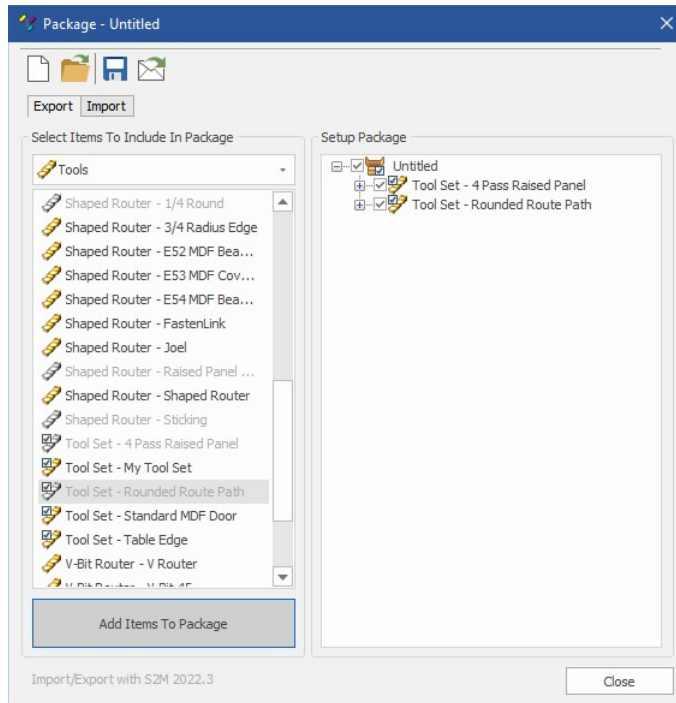
- If you select a Tool Set it will automatically import all Tools associated with that Tool Set.

## Materials

## Library Parts

This topic explains how to export a package and send it to Tech Support or another S2M CENTER user.

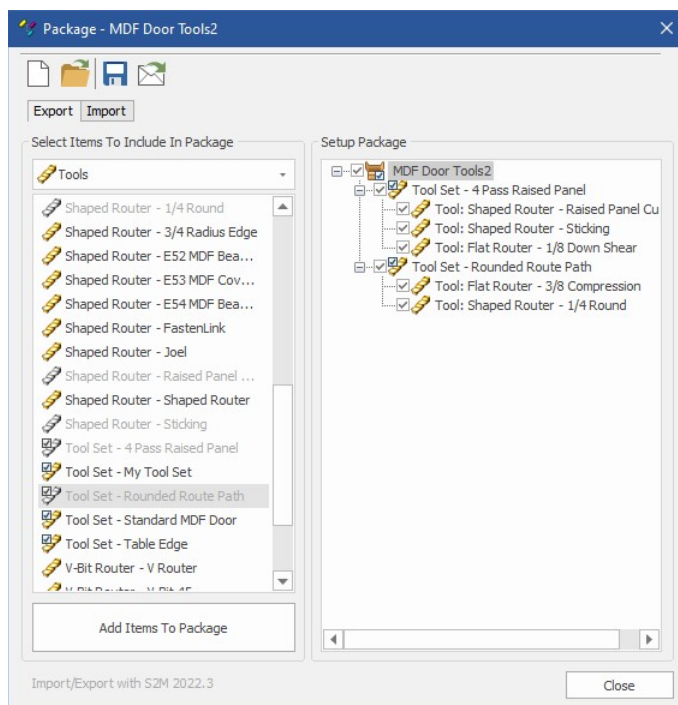
1. Go to the Utilities Tab | Setup Package
2. Select the item or items you would like to include in the Package from the list on the left side of the screen. After highlighting an item you want to add to the package simply click the "Add Items To Package" button in the lower left corner. There is no limit to the number of items that can be added to a package.



3. After selecting the items to add to the package click on the "Save Current Package" tool in the toolbar.



In the graphic below we selected two Tool Sets and the Packager automatically imported the all the individual Tools that make up each Tool Set.

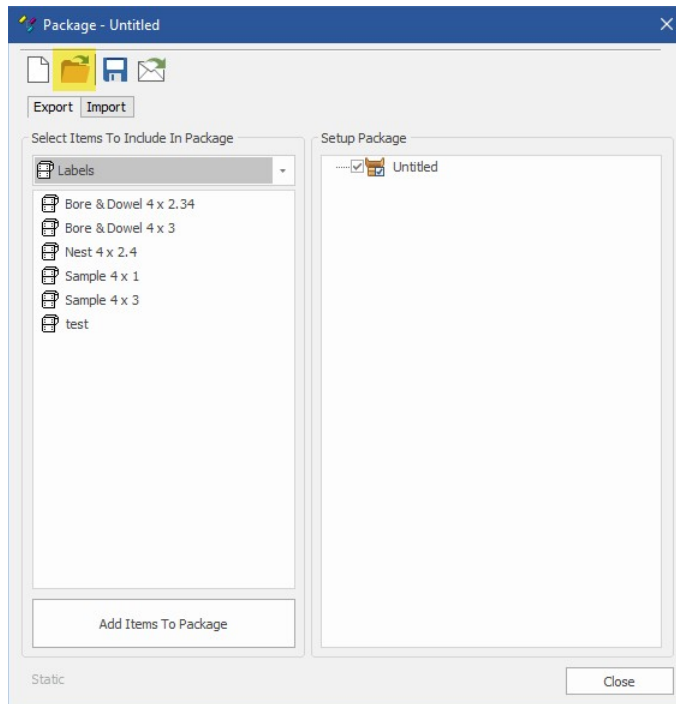


4. Select a location to save the Package and type in a Name for the Package then select "Save".
5. You may now either close the Setup Package screen or, if you use MS Outlook or Outlook Express, you can click on the "E-Mail Current Package" tool in the top toolbar. This will attach the Package to your e-mail and allow you to e-mail the SetupPackage to Tech Support or to another S2M CENTER User.
6. You may now close the Setup Package screen. You have successfully exported a Package from the S2M CENTER.

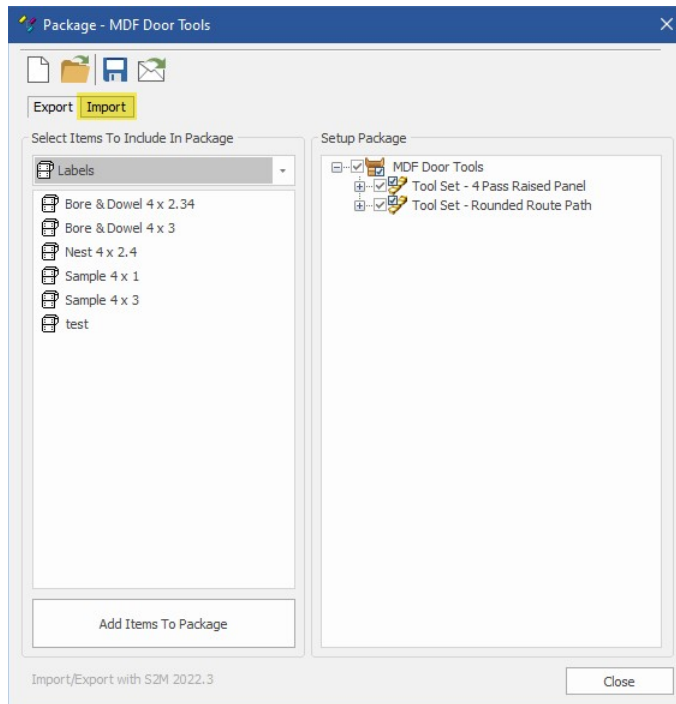
## Importing a Setup Package

This topic explains how to import a package that has been sent to you by Tech Support or another S2M CENTER user.

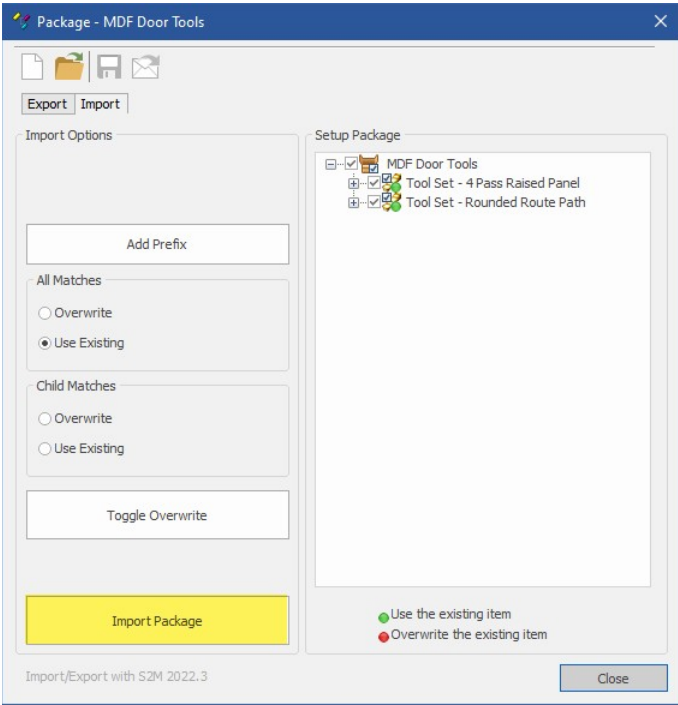
1. Go to **Utilities Tab | Setup Package**
2. Click the "Open An Existing Package" tool in the toolbar.



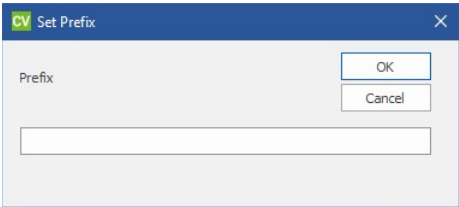
3. Select the package file from the location where it is being stored and click Open.
4. Click the Import tab.



5. Click the "Import Package" button.



 Prior to importing you can optionally have all the names of the objects imported from the Package by setting a Prefix using the Add Preix button on the Import tab.



**Overwriting Existing Items**

Following are the options that allow you to choose how to Overwrite items that already existed in your system.

All Matches

☐ Overwrite

☒ Use Existing

Child Matches

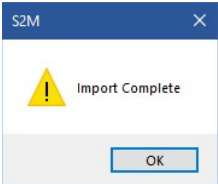
☐ Overwrite

☐ Use Existing

Toggle Overwrite

- **All Matches - Overwrite** - Will overwrite all existing items in your database with the contents of the Package.
- **All Matches - Use Existing** - Will use all the existing items in your database rather than overwrite them with the Package contents.
- **Child Matches - Overwrite** - Will overwrite all existing children of items in the Package that match those in your database.
- **Child Matches - Use Existing** - Will use all the existing children in your database rather than overwrite them with the Package contents.
- **Toggle Overwrite** - Allows you to Overwrite existing items by selecting them individually. This will cause items in the Package to either overwrite the existing item or to use the existing item.

6. Click on OK for the Import Complete notice.

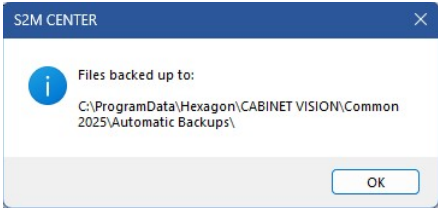


7. You may now close the Package screen you have successfully imported a Package into the S2M CENTER.



## Backup Database

This option backs up the primary databases and your posts to the location specified in the pop up once it is completed.



## Import NC and

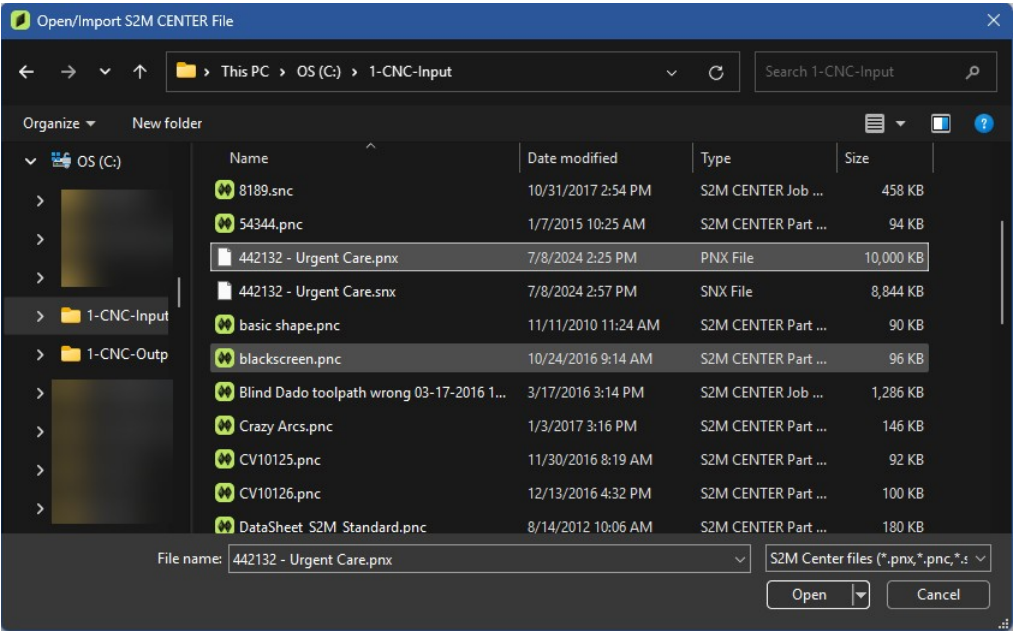


## Export PNX



## Import NC

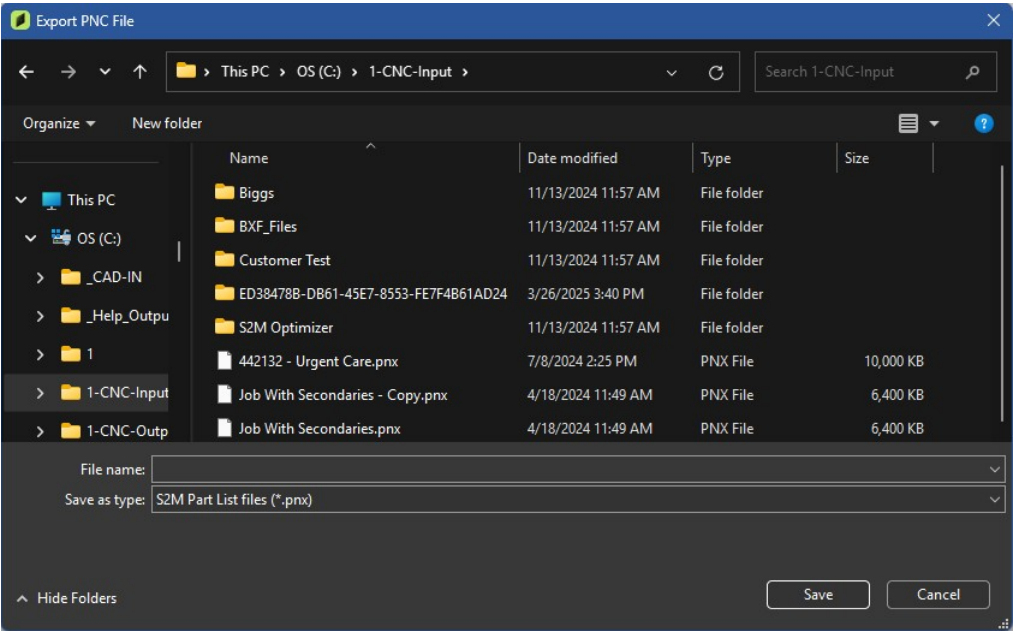
Import NC allows you to build a single Job from multiple NC Imports. You can import PNX, PNC, SNX, or SNC files.



## Export PNX

Creates a PNX file of the Open Job.





## Export Job State

This option will allow you to Save the current Job State as a .SNX file that can be reopened later in the S2M CENTER.